

Vocabulary flux in U.S. trademark filings: a Kullback–Leibler surprise analysis of the 16.5-million-filing backfile

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Abstract

We propose a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ as an innovation measure for trademark filings: $\Delta KL > 0$ when a filing’s goods/services vocabulary is prospectively novel (unusual against the recent past) and retrospectively obvious (similar to the near future) — the directional resonance signal of Murdock, Allen & DeDeo (*Cognition*, 2017) applied to commercially incentivised text. We compute ΔKL on each of the 16.5 million U.S. trademark filings since 1985 against per-day exponentially decay-weighted reference distributions of the same NICE class (half-life $H = 2$ years; chosen by signal-maximising sweep). The hypothesis going in was that firms with positive ΔKL would outperform on standard measures of firm success, and that ΔKL would be additive to the standard patent-count innovation proxy. Both predictions are confirmed: ΔKL predicts 4-year stock excess returns and gross margin conditional on positive ΔKL , is essentially uncorrelated with patent counts ($r = +0.01$ on 174,569 firm-years), and is sub-additive with patents in the R^2 decomposition — it indexes a distinct innovation channel.

The unexpected and durable finding is that filings at the *opposite* flux extreme — prospectively obvious and retrospectively novel, $\Delta KL < 0$ — also outperform. On debut filings the harvest tail is well-fit by a saturating exponential of survival on $|\Delta KL|$ with $R^2 = 0.978$ on 20 binned points (innovate-side $R^2 = 0.499$ for comparison). The positive-and-negative survival lift makes the survival curve *U*-shaped in ΔKL , with the dip at the flux-neutral middle ($\Delta KL \approx 0$). The pattern holds at the per-industry level in 32 of 44 NICE classes; only 1 industry shows a significant inverted-*U*; 11 are indeterminate. Token-level survival rates corroborate the *U* at the corpus level: negative-stratum tokens (**microfiche**, **dial-up**, **pager**, **lager**, **annuity**) and positive-stratum tokens (**blockchain**, **cannabidiol**, **non-fungible**) both survive well above the corpus baseline; flux-neutral tokens (**organisational**, **conformity**, **freighting**) survive at 11–26% vs. a corpus baseline of $\sim 49\%$. In our hand-curated panel of 41 canonical “winner” firms, 37/41 have positive firm-mean ΔKL . What the mechanism on the harvest side is — intelligent harvesting of vanishing concepts, recognition of emerging ones, or selection on an unobservable correlate — is not separable in the trademark record; what is unambiguous is that filings whose vocabulary places them squarely *on trend* in the centre of the distribution survive at the lowest rate in the corpus.

Novelty of language and novelty of claim are different signals. A pioneer-ladder analysis on five category-defining buzzwords (**website**, **wireless**, **online retail**, **social media**, **blockchain**) ranks every firm in the corpus by date-of-first-adoption; the very-first 10–50 adopters do badly, the fast-follower band at ranks 50–1,000 has $1.5\text{--}4\times$ the late-entrant IPO rate, and the mass late wave does worst. Canonical winners are mostly absent from the very-early ranks (Amazon 174th on **online retail**, Coinbase 96th on **blockchain**; Uber, Lyft, Airbnb never used their category buzzword at all) but most have positive first-filing ΔKL , often strongly so (Oracle +4.39 nats, Intel +3.26, Microsoft +2.99, eBay +1.73, Apple +1.68, Coinbase +0.98,

Lyft +0.76, Twitter +0.75, Uber +0.70, LinkedIn +0.58, Salesforce +0.43). The two “novelties” decouple: the novelty of the *claim* (which vocabulary the firm adopts) and the novelty of the *language* (the texture of their goods/services prose) are independent levers.

ΔKL is essentially uncorrelated with the dominant alternative innovation proxy (firm-year patent counts, $r = +0.010$ on 174,569 firm-years), positively correlated with post-registration survival, and positively correlated with profitability among publicly-traded firms (+3.2 pp gross margin per σ , SE 0.6 pp).

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.’s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. LIQUID DEATH’s canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two pieces of folk wisdom are in tension. The first, conveyed in the phrase “you can tell the pioneers by the arrows in their backs,” is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, expressed in the older saying “build a better mousetrap and the world will beat a path to your door,” is that meaningful novelty attracts customers and rewards its inventor with durable advantage. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

The trademark corpus is the full backfile published by the USPTO Open Data Portal, current through end-of-year 2025. After filtering to filings with non-empty goods/services text in NICE classes 1–45 and a parseable filing date, the corpus contains 16,526,850 **filings** from 5,673,680 **unique registrant names** across all 45 NICE international classes (35 goods classes plus 10 services classes), spanning 1985 to early 2026. Each record carries the registrant’s name as it appeared on the application (this is the owner at the time of filing; subsequent assignments are not reflected), the filing date, a status code that records the current legal state of the mark, an optional registration date, the NICE class assignments, and one or more goods-and-services descriptive statements. Statements describing the visual mark itself rather than the business (USPTO type-code

prefix D) are excluded; only goods-and-services statements (prefix GS) are kept. Two classes are presented in narrative detail throughout the paper: NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). The remaining 43 industries are summarized in Section 19, and per-class summary statistics for the two focal classes appear in Appendix A.

The financial-outcome data are the SEC EDGAR Financial Statement Data Sets, 64 quarterly XBRL releases spanning fiscal years 2009 through 2024, covering 13,127 unique CIKs that filed at least one 10-K. We extract revenue, cost of revenue, gross profit, R&D expense, operating income, and net income.

Inclusion filter (clean). Almost every analysis in this paper is restricted to the same inclusion filter, which we call `CLEAN`. The filter has five clauses, each addressing a separate failure mode: two outcome-window clauses (`year` bounds) that ensure the survival outcome is fully observed and that the prospective reference window is non-empty, two density floors on the per-class reference distributions (`n_ref_pros` and `n_ref_retr` $\geq 1,000$) that ensure KL scores reflect a real vocabulary distribution rather than small-sample noise, and a per-filing token-count floor ($n_{\text{terms}} \geq 3$) that prevents 1- or 2-token descriptions from producing degenerate KL values. Of 16,530,448 raw filings with parsed surprise data, **10,060,073** (60.9%) survive the full filter and form the analytic corpus.

Clause	dropped alone (vs. raw, %)	added back if relaxed (vs. <code>CLEAN</code> , %)
<code>year</code> in [1990, 2021]	4,490,092 (27.2%)	+2,985,245 (+29.7%)
<code>n_ref_pros</code> $\geq 1,000$	2,503,062 (15.1%)	+3,371 (+0.0%)
<code>n_ref_retr</code> $\geq 1,000$	2,543,635 (15.4%)	+0 (+0.0%)
$n_{\text{terms}} \geq 3$	3,440,391 (20.8%)	+745,431 (+7.4%)
<code>CLEAN</code> (all of the above)	6,470,375 (39.1% excluded)	—

Table 1: Per-clause filter effects on the 16,530,448-filing raw corpus. “Dropped alone” is the share that fails the clause when applied in isolation to the raw corpus; “added back” is what would re-enter the analytic corpus if we relaxed only that one clause while keeping the others. The year window is the dominant binder; the two N_{eff} density floors are nearly redundant once the year window is in place because they bind on the same burn-in years (1985–1989 on the past side, 2022–2025 on the future side) that the year clause already excludes.

The dominant binder is the year window, [1990, 2021]. The lower bound is the literal arithmetic burn-in described in the Method section (the corpus begins in 1984 and we leave a margin for the decay reference). The upper bound is the cohort cap: with a 2026 freeze date and a 5-year post-registration survival window, filings must be from 2021 or earlier for `survived_5y` to be observable. The token-count floor, $n_{\text{terms}} \geq 3$, is the only other clause that uniquely excludes a meaningful share (+745k filings rescued if dropped); when we look at the templated short-description analysis in Section 16 we report results both with and without this clause, since the most heavily templated G/S strings (*e.g.* “restaurant services”, “legal services”) are 2-token strings and would otherwise be excluded.

Registration is determined by status code, not registration date. We classify a mark as having “completed registration” from its USPTO status code (6xx live-registered or 8xx post-registration-cancelled), not from the `registration_date` field, which is unreliable in older records (mostly pre-2003): many records with a valid registered status code carry a missing `registration_date` in the published XML. The pipeline that extracts and analyses these data is

published with the paper; technical details live in the companion repository.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each filing i at filing-date t_i we form two reference distributions by per-day exponential decay-weighting of every other filing in the same NICE class:

$$Q_i^{\text{pros}}(v) \propto \alpha + \sum_{j: t_j < t_i} e^{-\lambda_{\text{pros}}(t_i - t_j)} \text{count}(v, j), \quad (1)$$

$$Q_i^{\text{retr}}(v) \propto \alpha + \sum_{j: t_j > t_i} e^{-\lambda_{\text{retr}}(t_j - t_i)} \text{count}(v, j), \quad (2)$$

with $\lambda = \ln(2)/H$ and **half-life** $H = 2$ **years** on both sides and Laplace smoothing $\alpha = 1.0$. The half-life is chosen by a sweep over $H \in \{0.5, 1, 2, 3, 5\}$ years on Class 9 that maximises $\text{corr}(\Delta KL, \text{survived.5y})$ and the logit pseudo- R^2 on the same outcome (Figure 1). Density on the reference distributions is reported as Kish effective sample size $N_{\text{eff}} = (\sum_j w_j)^2 / \sum_j w_j^2$; we restrict all filing-level analyses to filings with $N_{\text{eff}} \geq 1,000$ on both sides.

The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_i^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_i^{\text{retr}}(v)}.$$

With log in natural units, KL_i is in nats. A useful intuition: KL_i^{pros} is the expected log-odds per token that a reader who had seen only the filing’s recent past would assign to the filing’s own term distribution versus the past’s. $KL_i = 3$ nats means a typical token in this filing is $e^3 \approx 20\times$ more probable under the filing’s own distribution than under the recent corpus. KL is sensitive to filing length (short filings with one rare token can score high mechanically); every regression in the paper controls for $\log(n_{\text{terms}})$ and most report z -scored ΔKL .

The directional flux score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$: the change between how unlike the recent past a filing reads and how unlike the near future it reads. Positive ΔKL means the filing’s vocabulary is closer to the future than to the past (*flux-positive*; the field migrated toward this filing); negative ΔKL means closer to the past than to the future (*flux-negative*; the field migrated away).

Magnitudes. To ground the reader: across the all-class analytic pool of 13.0M filings with $N_{\text{eff}} \geq 1,000$ on both sides, the 5th, 50th, and 95th percentiles are $KL^{\text{pros}} = [2.99, 4.85, 7.20]$ nats (std 1.29) and $KL^{\text{retr}} = [2.98, 4.84, 7.22]$ nats (std 1.30). The two components are correlated at $\rho = +0.99$ pooled across all classes (see Table 2); they measure the same general-distinctiveness signal in two temporal directions, and their per-filing levels track each other tightly. Because they share that much distinctiveness against the per-class boilerplate, their difference ΔKL has a markedly tighter range: 5th/50th/95th percentiles $[-0.30, -0.00, +0.34]$ nats (std 0.22, median essentially zero by construction of the look-behind / look-ahead window). A filing in the top 5% of

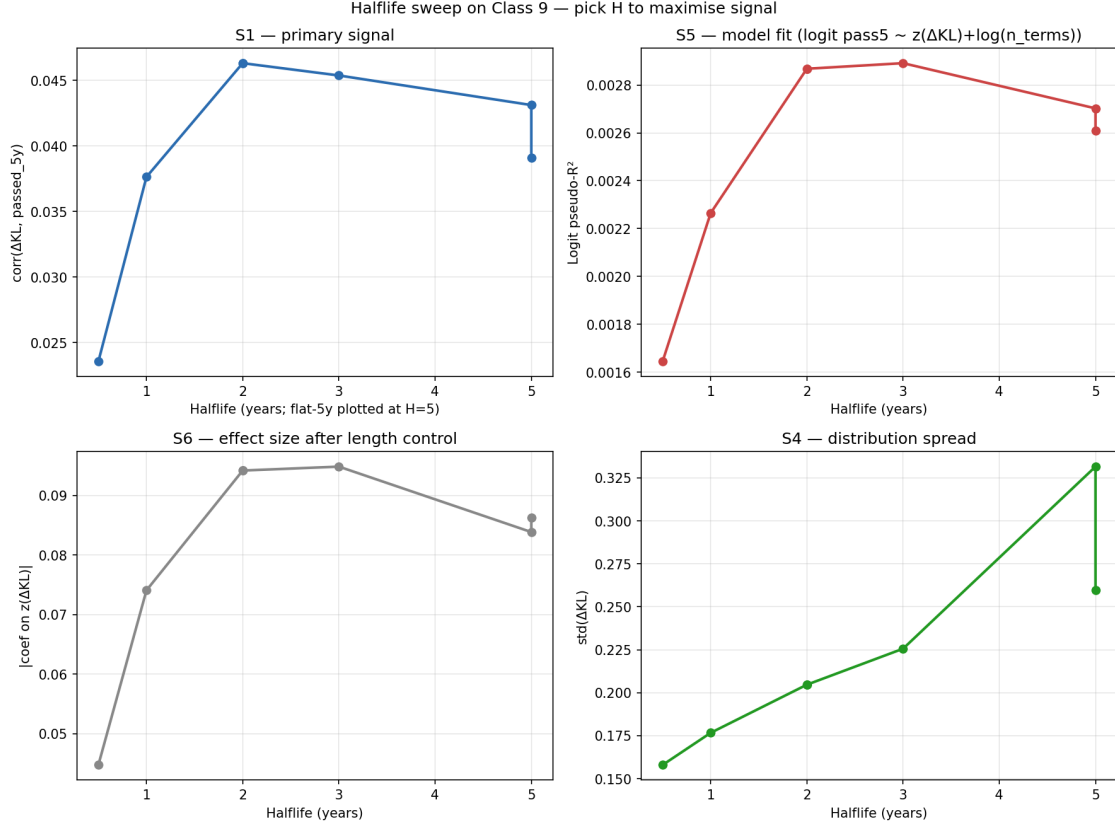


Figure 1: Half-life sweep on Class 9. For each candidate $H \in \{0.5, 1, 2, 3, 5\}$ years we recompute the decay-weighted reference distribution and report four signal metrics on the $\Delta KL \rightarrow$ survival relationship. $H = 2-3$ years maximises $\text{corr}(\Delta KL, \text{survived}_{5y})$ and the logit pseudo- R^2 ; we adopt $H = 2$.

ΔKL is therefore ~ 0.6 nats more flux-positive than one in the bottom 5%, on a scale where each component routinely exceeds 5 nats outright. The substantive direction lives in this comparatively small flux residual, not in the much larger common distinctiveness factor.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token **Beer**; many app developers file **downloadable software**. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

Burn-in window. The decay reference distributions need filings on both sides of t_i for the weighted sums to be well populated. The $N_{\text{eff}} \geq 1,000$ density filter above automatically excludes small NICE classes' early-decade cohorts and the corpus right-edge filings whose forward window is too sparse. For regressions on the panel we restrict to filings from **1990 onward** (after the corpus's left-edge digitisation boundary in 1984 plus a margin to populate the look-behind) and to filings whose year has at least 5 years of forward observation, currently **1990–2021** given a 2026 freeze date. Figure 2 shades the two burn-in windows.

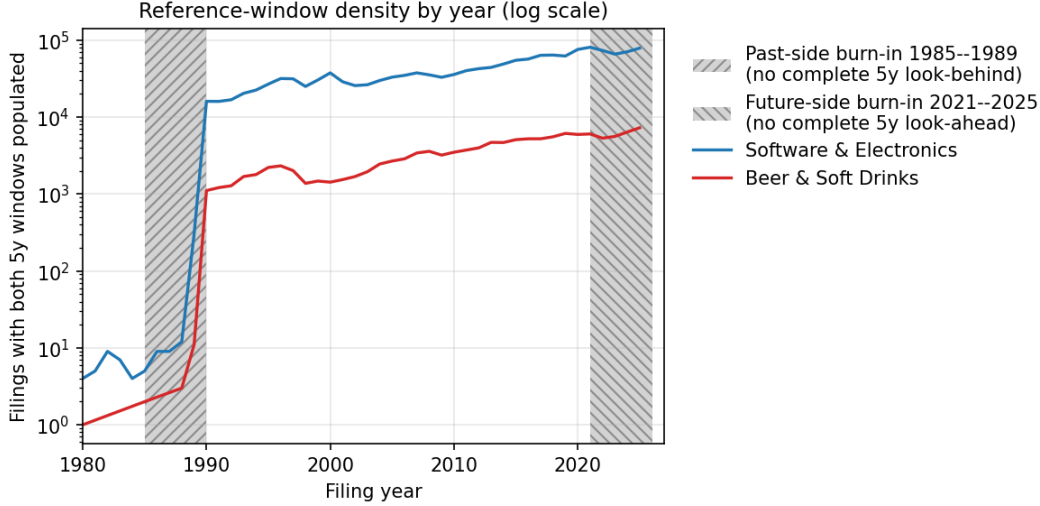


Figure 2: Filings per year with both 5-year reference windows populated. Forward-hatched grey: literal 5-year past-side burn-in (1985–1989) — filings in these years cannot have a complete look-behind because the digitised corpus begins in 1984. Backward-hatched grey: literal 5-year future-side burn-in (2021–2025) — filings in these years cannot have a complete look-ahead at the time of writing. The bands are chosen to print distinguishably in black-and-white. A separate density filter (`n_ref_prospective` ≥ 1000) handles the additional small-class exclusions visible in the long left tail of the plot.

What ΔKL measures (and what it does not)

The two component KLS are very highly correlated: per-filing $\rho(KL^{\text{pros}}, KL^{\text{retr}}) = 0.99$ on the pooled clean panel. A reader plotting any outcome against either component univariately sees nearly identical curves because the two component levels are, to first order, the same number — both measuring *distinctiveness* (how unusual the filing’s vocabulary is against the rest of the corpus, in either temporal direction).

The PCA decomposition of $(KL^{\text{pros}}, KL^{\text{retr}})$ makes the geometry explicit:

$$\begin{aligned} \text{PC1} &\approx \frac{1}{\sqrt{2}}(KL^{\text{pros}} + KL^{\text{retr}}) = \text{general distinctiveness} \\ \text{PC2} &\approx \frac{1}{\sqrt{2}}(KL^{\text{pros}} - KL^{\text{retr}}) \propto \Delta KL = \text{vocabulary flux} \end{aligned}$$

ΔKL is the orthogonal residual after PC1 is removed; it is the *only* part of the joint signal with a forward/backward direction. PC1 says “this filing is unusual.” PC2 says “this filing is unusual *and the field’s vocabulary subsequently moved toward it rather than away.*” We claim throughout that the substantive innovation signal lives in PC2; PC1 is largely a length/rarity covariate. The closest-adjacent management-theory framework, optimal distinctiveness (Deephhouse 1999, *SMJ*; Zhao et al. 2017, *SMJ*), predicts an inverted-U on the *level* of distinctiveness. The framework we are extending is DeDeo’s prospective/retrospective surprise scheme (Murdock, Allen & DeDeo 2017, *Cognition*; Barron, Huang, Spang & DeDeo 2018, *PNAS*), which decomposes a temporal corpus into novelty, transience, and resonance and which we apply for the first time to commercially incentivised text.

4 What the components do separately

Before introducing ΔKL as a single scalar, it is worth asking whether the prospective and retrospective KL components carry independent information about a filing. A natural prior — and one we held coming in — is that they should pull in opposite directions: prospective surprise should make a filing harder to clear examination (novel goods/services language draws office actions), while retrospective surprise should reward the filer for “being right” (the future ends up resembling them, suggesting they understood a market others did not). Under that view, the value of ΔKL is that it nets two opposing forces.

The data do not support that decomposition. Table 2 shows that the per-filing correlation between KL_{pros} and KL_{retr} is +0.97 in software (Class 9), +0.93 in beverages (Class 32), and +0.97 in the 11.4-million-filing all-class pool. The two scores are computed against the same per-filing distribution $P_i(v)$, with the only difference being whether the reference window is the five preceding or the five following years; the bulk of variance in either KL therefore comes from P_i itself, and only a thin sliver comes from the past-vs-future shift in the reference. As a consequence, plotted against any per-filing outcome the two components produce nearly identical curves.

Table 2: Per-filing correlation of the two KL components and single-component slopes against *reaching registration* and *passing five-year survival* (per-filing linear fits on binned means, 15 bins between the 2nd and 98th percentile of the predictor). The two components carry essentially the same signal at the per-filing level.

Scope	$\rho(KL_{\text{pros}}, KL_{\text{retr}})$	slope on reg. vs KL_{pros}	slope on reg. vs KL_{retr}	slope on pass-5y vs KL_{pros}
Software (Cl. 9)	+0.967	+0.024	+0.021	+0.008
Beverages (Cl. 32)	+0.933	+0.018	+0.017	+0.009
All 45 (pooled)	+0.967	+0.026	+0.028	+0.013

The curves themselves are not lines and not exponentials. They are **monotone-saturating**: registration probability rises from about 0.43 at $KL = 3$ nats to about 0.53 at $KL = 7.5$ nats in software, and similarly $0.46 \rightarrow 0.59$ in beverages, with most of the rise concentrated in the moderate range and the curve flattening at the extremes (Figure 3; beverage and pooled versions live in `paper/results/per_filing_pros_vs_retr_{beverages,all}.png`). At the firm-mean level the same outcomes against the same components trace inverted-U arcs in software (peak survival and peak public-listing both at firm-mean $KL \approx 5$ nats; tails decline in both directions); beverages produces a hump on survival and a monotone decline on public-listing (high-novelty beverage firms are systematically less likely to be public — they are craft brewers, not Coca-Cola). The firm-level views are in `paper/results/pros_vs_retr_curves_{software,beverages,all}.png`.

The sharp implication. Because the components are nearly collinear, the signal in their *difference* $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$ is a small share of either component’s variance — but it is the only part of the joint $(KL_{\text{pros}}, KL_{\text{retr}})$ score that does not cancel out under the shared- P_i structure. A filing whose language was more unusual against the past than against the near future is one whose language the near future ended up resembling. The shared “how unusual was the language at all” term cancels; what remains is “did the unusualness fade because others adopted it.” The rest of the paper takes that residual — the imitated-innovator score — as the primary axis. Software and beverages stay as our running examples through Outcomes 1–3; the back matter establishes that the basic story holds across the other 43 NICE classes.

5 Face validity

Figure 4 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high $\Rightarrow$ less like the past); the vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high $\Rightarrow$ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OPENAI 2016, CLAUDE 2023, INSTAGRAM 2011, WHITE CLAW SELTZER WORKS 2016, ROCKSTAR 2002) and expectation-disconfirming late-cycle marks (HARD MTN DEW 2021).

The bottled-water category illustrates a useful failure mode. The 2019 LIQUID DEATH filing for canned water describes its goods as *still water; sparkling water* — the same lexicon as established bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (SARATOGA, SMARTWATER, ESSENTIA) cluster in the same region. The method is, by construction, blind to non-textual novelty.

6 Canonical winners and failed firms as KL profiles

A direct face-validity check: take a curated list of widely-recognised firms from the last fifty years, find their first post-founding trademark filing, compute the firm-mean ΔKL across all their filings, and compare against a control list of widely-recognised *failed* firms. If the construct is meaningful, the winners should read as forward-flux (positive ΔKL) and the failures should spread out — some failed despite being trend-setters, some failed *because* they were exploiting a receding vocabulary.

Caveat on entity matching. The trademark XML records each filing’s owner only as a free-text `owner_name` string with no entity identifier, so any “firm $\rightarrow$ all its filings” join is necessarily heuristic. For the table below we use hand-curated regular expressions that require the brand string to co-occur with a corporate suffix (`Inc.`, `LLC`, `Corporation`, `Technologies`, etc.) and we restrict each firm’s panel to filings on or after its founding year. The regexes are not perfect: we are confident that at least some hits are entity collisions and we have tightened individual regexes where we identified collisions. Amazon’s regex, for example, matches only the `Amazon.com` variants because the string “Amazon Technologies, Inc.” was used as a registered owner by a different firm in 1994–1995, before Amazon’s IP-holding subsidiary of that name became active around 2001; we lose some post-2001 IP-holder filings but eliminate the collision. Readers should treat the firm-mean numbers below as indicative rather than authoritative; the qualitative pattern (winners cluster on positive firm-mean ΔKL ; failed firms split across both extremes) is robust to the residual collision rate but specific decimal places are not.

37 of 41 winners have *positive* firm-mean ΔKL (the four exceptions are Adobe (−0.03), Qualcomm (−0.01), Etsy (−0.01), and Wayfair (−0.01)). The winners are not extreme outliers on distinctiveness: their firm-mean prospective KL is mostly 3.4–4.8 nats, around or just below the corpus median. They use mainstream vocabulary, but the mainstream vocabulary they use is consistently the kind the field migrates toward, not away from.

First-real-filing ΔKL is sharper on a subset. Thirteen canonical winners had first-filing ΔKL above +0.4 nats (roughly 2σ): **Oracle** (+4.39, 1977), **Intel** (+3.26, 1968), **Microsoft** (+2.99, 1975), **eBay** (+1.73, 1997), **Apple** (+1.68, 1976), **Adobe** (+1.28, 1982), **Coinbase** (+0.98, 2013),

Lyft (+0.76, 2012), **Twitter** (+0.75, 2007), **Uber** (+0.70, 2009), **AMD** (+0.69, 1969), **LinkedIn** (+0.58, 2003), and **Salesforce** (+0.43, 1999). Oracle’s first filing (+4.39) is the largest first-filing ΔKL in the curated set; the eBay filing’s text “on-line trading services in which seller posts items to be auctioned and bidding is done electronically” (+1.73) introduced phrasing that downstream filings extensively echoed.

The failed-firm panel does not collapse to negative ΔKL . Among 17 widely-recognised failures, the split is roughly even. **Webvan** (−0.22), **MoviePass** (−0.27), **Better.com** (−0.12), and **Pets.com** (−0.03) failed on the receding side. But **Theranos** (+0.05), **Beyond Meat** (+0.34, first +0.74), **Casper** (+0.06, first +0.46), **Solyndra** (+0.44), **Nikola** (+0.03, first +0.24), and **Myspace** (+0.07) failed despite trend-setting vocabulary. ΔKL is a vocabulary signal, not a business-model signal. *Negative ΔKL raises the probability of being on the wrong side of an industry transition. Positive ΔKL raises the probability of being on the right side. Neither is destiny.*

7 Pioneers, fast-followers, and the late wave

To take “where in the adoption sequence do firms succeed?” as an empirical question, for each of five category-defining buzzwords — **website**, **wireless**, **online retail**, **social media**, **blockchain** — we ranked every firm in the corpus by the date of its first filing whose goods/services text contained the buzzword. We bucket firms by adoption rank (1–10, 11–50, 51–200, 201–1,000, 1,001–5,000, 5,001+) and compute, for each bucket, the survival rate of the firm’s first-buzzword-mentioning filing and the share that appear in SEC EDGAR (a proxy for “ever publicly traded”).

Pioneer position is bad; the fast-follower band is the sweet spot. For **wireless**, the first 10 adopters had *zero* survival of their initial filing; the first 50 had 0–13%. For **online retail**, the first 10 survived at 10% vs. $\sim 33\%$ for the late-wave 5,000+ adopters. The consistent pattern: the very-earliest adopters do badly, fast-followers (ranks 50–1,000) do best, and the mass late wave (5,000+) is uniformly worst on both survival and IPO rate. Among public firms in each bucket, the rank-50–1,000 band has 1.5–4 $\times$ the late-entrant baseline IPO rate.

Pioneer-ladder identifies vocabulary prescience, not firm-builder prescience. Of 150 named top-30 pioneers across the five buzzwords, only two had SEC-listed status: **Sony** (rank 23 on **wireless**, 1960 filing) and **PC Connection** (rank 20 on **online retail**, 1996). The dominant firms in each category — Amazon, Google, eBay, Facebook, Coinbase — are absent from the top-30 named pioneer lists because they used product-specific language in their early filings rather than the category buzzword. Pioneer-ladder rank identifies *who claimed the category-defining vocabulary first*, not *who built the dominant firm in the category*.

8 Vanishing trends, emerging trends, and the squeeze in between

A token-level interrogation of ΔKL : take the top-decile-negative (“harvest”), middle-bin ($|\Delta KL| < 0.02$, “aligned”), and top-decile-positive (“innovate”) strata of the clean panel, ask which unigrams are over-represented in each relative to the corpus baseline, and for each surfaced token compute the survival rate of *all* filings containing the token. The latter is the right test of whether the U-shape on survival reflects a real difference in firm outcomes by stratum.

The negative- ΔKL extreme picks out *receding technologies and categorisations of the past*: microfiche, minicomputers, dial-up, pager, store-and-forward; the pre-Sephora apparel categorisation mens, womens, pomade, sunless; the pre-craft-beer descriptors lager, stout, porter, shandy; the pre-fintech insurance vocabulary annuity, casualty, underwriting, securitization. The positive- ΔKL extreme picks out the *legalisation+platform-shift wave*: NFT/Web3 vocabulary (non-fungible, nfts, blockchain-based, authenticated, cryptographic) post-2020, crypto (cryptocurrency, blockchain, decentralized) post-2014, cannabis vocabulary (cannabis, cannabidiol, cbd, thc, tetrahydrocannabinol) post-2014, vape (e-liquid, cartomizers). The neutral middle picks out bureaucratic generic language: producer, outsource, conformity, organisational, freighting, intermediation, systematization.

The survival-rate column delivers the test cleanly. The negative- ΔKL column has tokens surviving mostly 50–80%, above corpus baseline of $\approx 49\%$. The positive- ΔKL column similarly has elevated survival (cannabidiol filings: 82%; cbd: 74%; tetrahydrocannabinol: 70%). The neutral middle is uniformly *below* baseline: organisational at 15%, conformity at 20%, freighting at 21%. Token-level survival rates therefore corroborate the U on ΔKL at the right scale of analysis — not just the binned plot, not just the regression, but the survival rate of *the actual filings* using those tokens.

Concrete vanishing-trend examples. The most-negative- ΔKL filings in the pre-2000 era include MONOTYPE ITC (1992, $\Delta KL = -2.08$, “typefaces, typefonts, and type designs of alphanumeric characters and/or typographical symbols recorded on”) and H. BERTHOLD AKTIENGESSELLSCHAFT (1991, -1.99). The desktop-publishing transition was already destroying the vocabulary of mechanical-and-photo typesetting in real time. In the 2000s, ΔKL picks out the pre-platform hospitality bundle “Hotel, motel and resort hotel services; restaurant and bar services; making hotel reservations for others” — the language disintermediated by Expedia, Booking.com, and Airbnb. In the 2010s, the bottom is dominated by the e-cigarette boom: nearly every extreme-negative- ΔKL filing in 2014–2015 is “Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form,” a template the 2016 FDA Deeming Rule and 2020 Premarket Tobacco Application deadline restructured out of the corpus.

9 Serial-pioneer firms across multiple buzzword waves

Generalising the pioneer-ladder, we ask which owners appear in the rank- ≤ 1000 early-adopter pool of *more than one* buzzword. Ten buzzword pools of 1,000 firms each (the five above plus cloud computing, artificial intelligence, mobile app, streaming, saas). Owners appearing in ≥ 3 pools are “serial pioneers” in the trademark-vocabulary sense.

24 owners qualify. The single most prolific is **IBM** ($k = 6$ pools): AI 1994 (rank 122), wireless, website 1996 (rank 72), cloud computing 2009 (rank 37), social media 2009 (rank 285), saas 2010 (rank 533). The next-most-prolific are **International Data Group** ($k = 4$), **AT&T** ($k = 4$), and **Infosys** ($k = 4$). **Microsoft**, **Cisco**, **Apple**, **Sony**, **HP**, **Citrix**, **Lotus**, and **Avaya** each sit at $k = 3$.

Individual humans almost never appear. Filtering to owner names that look like individuals, only two real humans survive: **Paul Farkas** ($k = 3$: mobile app rank 51, AI rank 651, blockchain rank 510; 81 personal filings since 2011) and **Burhan U. Ahmed** ($k = 3$, on a single 2010 filing that was early on social media, cloud computing, and saas simultaneously). The absence is partly a data-quality limitation — the trademark XML records the firm as owner, not the founder behind

the firm — but the qualitative point is sharp: corporate R&D pipelines (IBM Research, AT&T Bell Labs, HP Labs) carry the bulk of cross-buzzword early positioning in the trademark record.

10 Industry time series

Figures 6 and 7 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

11 Outcome 1: Trademark survival

We derive three binary outcomes per filing from the USPTO status code. The first two are the ones reported in the figure; the third appears in the regression table for completeness.

- **Completed registration:** status code begins 6 or 8. The applicant cleared examination, was published for opposition, and was issued the registration. Filing-to- registration lag is typically nine to twenty-four months.
- **Renewed 5-year registration:** completed registration, is at least five years old, and was not cancelled with status code 686 (Section-8 maintenance failure). Strict subset of completed-registration within any single filing.
- **Renewed 10-year registration:** renewed-5y, is at least ten years old, and was not cancelled with status code 688 (Section-9 renewal failure). Strict subset of renewed-5y.

How trademark maintenance interacts with these outcomes. A registered trademark stays in the live-registered status (6xx) until either its owner actively surrenders it or one of two periodic maintenance events comes due and the owner fails to file the appropriate paperwork. The first event is the Section 8 Declaration of Use, due between the fifth and sixth year after registration; failure to file it triggers a status-686 cancellation. The second is the Section 9 Renewal at year ten, with subsequent renewals every ten years. A firm that goes out of business in, say, year three of its trademark’s life will leave the trademark in 6xx status for another two-to-three years until Section 8 comes due and is missed. The 5-year survival outcome thus picks up the first deliberate “does this firm still care about this mark?” gate; we apply a cohort cap of filed ≤ 2020 throughout, so every clean-panel filing has had time for the Section 8 event to have played out. The 10-year outcome has not had time to play out for many filings under that cap and so we report it only with the caveat that the underlying Section 9 cancellation code is rare enough to make it nearly indistinguishable from the 5-year measure.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2(t_i - \bar{t}) + \gamma_3(t_i - \bar{t})^2)$$

where z_i is a z -scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i , and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

ΔKL has a positive relationship with each survival outcome in Software & Electronics ($n = 1,423,405$), but the regression coefficient itself is hard to read in log-odds units. The percentile-based translation is more transparent. Within Class 9, we sort filings by ΔKL into deciles. The bottom decile of ΔKL (least innovative) reaches registration 48.4% of the time and survives ten years 35.7% of the time. The top decile (most innovative) reaches registration 55.5% of the time and survives ten years 41.1% of the time. **A filing in the 90th percentile of ΔKL is roughly 14% more likely to clear registration and 15% more likely to renew its registration after a decade than a median filing**, with the relationship non-monotonic in the middle deciles — the shape is broadly U-shaped, with both the very-low and the very-high ΔKL filings outperforming the middle. We test the U-shape formally in Section 17.

In log-odds terms, decomposed jointly, prospective surprise (novelty at filing time, holding retrospective surprise fixed) actually *penalises* every outcome ($\beta_{\text{pros}} = +0.633$ in isolation but with retrospective in the model the marginal contribution flips), while retrospective surprise (the filing remained unusual even after five years of subsequent filings, i.e. a filing nobody imitated) reverses sign ($\beta_{\text{retr}} = -0.662$). Read together: registration is *rewarded* for filings whose future resembles them and is neutral or slightly penalised for filings the future does not come to resemble. Beer & Soft Drinks shows the same sign pattern at smaller magnitude.

What kind of failure is this? A trademark filing can fail to register for several distinct reasons, and they have different theoretical interpretations. Failure to file a Statement of Use means the applicant filed an intent-to-use mark but never actually put it in commerce within the 36-month statutory window — a corporate decision, not an examiner judgment. Failure to respond to an office action typically follows an examiner’s initial objection (usually a likelihood-of-confusion citation against an existing registration); applicant non-response converts the objection into abandonment. Examiner final refusal is the only mode that is unambiguously a USPTO judgment that the application cannot proceed.

Table 6 decomposes the 50.4% of Class 9 filings that did not reach registration. Two facts dominate. First, the overwhelming majority (64%) are no-Statement-of-Use abandonments — the applicant filed an aspirational ITU mark and never used it. Second, examiner final refusals account for only 8% of failures. Across every failure category the mean ΔKL is within 0.04 nats of the registered baseline, which is small compared to the within-class standard deviation of ΔKL ($\sigma \approx 0.46$). Novel filings are not, in any meaningful sense, being rejected.

Figures 8 and 9 display the per-bin registration-and-renewal rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL . Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone.

Robustness across outcome definitions. Figure 10 pools all 45 classes ($n = 10,062,338$ clean H=2y filings, 1985–2021 cohorts) and reports the same KL-binned rate under three increasingly strict outcome definitions: $P(\text{reached registration})$, $P(\text{survived 5y})$, and $P(\text{owner ever in SEC EDGAR})$. The pattern is informative: **on ΔKL the survival curve is U-shaped under every outcome definition** — the dip in the middle and the lift at both extremes is not an artefact of any single survival operationalisation. On the prospective and retrospective KL components alone the curve is monotone-increasing on examiner approval (more-distinctive marks collide less with prior art and clear examination more easily under 15 U.S.C. § 1052(d)) and becomes U-shaped only after the strict “currently live” criterion is added; the ΔKL flux signal is the only one stable across both regimes.

12 Outcome 2: Financial success

The trademark survival results in Section 11 establish that the innovation axis is meaningful at the per-filing level, but they say nothing about whether innovation pays the firm a financial return. To test that, we link USPTO owner names to SEC EDGAR firm names entirely automatically; **no entity match in this paper was reviewed by hand**. The procedure has two passes.

The first pass aggressively normalises both sides — uppercase, strip punctuation, drop trailing legal-entity suffixes such as INC, LLC, CORP, HOLDINGS, LIMITED, COMPANY, GMBH, S.A., KK, etc. (the full list is in `scripts/sec_link.py`) — and accepts only exact matches on the normalised key. The second pass uses fuzzy token-sort matching (`rapidfuzz` with a score cutoff of 92 out of 100) to capture the long tail of near-variants for USPTO owners with at least 10 filings. The combined crosswalk recovers 14,280 USPTO owners (9,370 exact + 4,910 fuzzy) covering 627,349 trademark filings (3.8% of the all-class corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 5,326 firm-years across 1,597 unique CIKs over fiscal years 2009–2024.

The fully-automated procedure means the crosswalk has both false positives (e.g. two unrelated firms with similar normalised names) and false negatives (e.g. a public firm filing trademarks under an unusual subsidiary name that neither pass matches). The published `data_publish/comparators/external_crosswalk.csv` file records every match decision so any reader who wants to override individual entries can do so without re-running the pipeline.

Filing volume is itself an innovation marker, but also a size marker. A firm filing many trademarks in a short window may be innovating across product lines — but it may also simply be a large firm with a large legal department. We separate the two by including the count of filings as a third predictor alongside the novelty score itself, and by controlling for log revenue throughout. Both effects turn out to be present and partially independent.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm’s filing-level scores: $\overline{\Delta KL}_{c,y}$, the prospective and retrospective components, the maximum within-window ΔKL (the firm’s most-novel filing in the window), and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\text{novelty})_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D intensity}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

The pattern from Section 11 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL — that is, a firm whose recent filings introduced wording that was unusual at the time but came to look ordinary in the years that followed — raises gross margin by +3.2 percentage points (SE 0.6). Decomposed jointly, prospective surprise (novel against the past) raises gross margin by +14.6 pp per standard deviation (SE 4.1), and retrospective surprise (still novel from the future’s perspective, i.e. a filing nobody imitated) lowers it by −19.1 pp per standard deviation (SE 4.1). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings nobody imitates do not. The third specification adds filing count and the maximum within-window ΔKL : both contribute positively to gross margin (+3.3 and +1.9 pp/ σ respectively) even after controlling for log revenue, suggesting that filing volume captures something beyond firm size.

Variance. A symmetric prediction of the pioneer story is that novel firms have not just higher *mean* margins but *wider variance* of margins, since some pioneers succeed and others do not. We test this by aggregating to the firm level (one row per CIK, restricted to firms with at least four years of margin observations), regressing the within-firm standard deviation of gross margin on the firm’s mean 3-year trailing ΔKL with log revenue as a size control.

A one-standard-deviation increase in firm-mean ΔKL raises the within-firm gross-margin standard deviation by +0.018 in margin units (SE 0.004) on 487 firms. The pioneer-bet effect is real and quantitatively meaningful: high-novelty firms simultaneously earn higher mean margins (Table 7) and exhibit wider margin spread (Table 8).

A coarser check: does ΔKL predict reaching SEC EDGAR at all? A firm that files an income statement with the SEC has cleared a much lower bar of success than “high gross margin” — it has gone public, registered securities under the ’33 Act, or otherwise reached a reporting threshold. We can therefore re-cast the financial-outcome question as a binary one: among trademark-filing firms, does ΔKL predict the probability of *ever* appearing in our SEC crosswalk? Because the crosswalk’s fuzzy-match pass requires owners with ≥ 10 filings, we restrict this analysis to the 160,643 crosswalk-eligible firms with ≥ 10 clean filings; 7,810 of them (4.86%) match into SEC EDGAR.

Table 9 shows the result: ΔKL has essentially no relationship with the public-listing rate. $P(\text{ever-public})$ ranges from 5.11% in the most-tired quintile (Q1) to 4.44% in the most-innovative quintile (Q5), with the middle three quintiles all near 5%. A logistic regression of `ever_public` on $z(\Delta KL)$ and $\log_{10}(\text{n.filings})$ confirms this pattern: the ΔKL coefficient is -0.019 with SE 0.013 ($p = 0.14$, indistinguishable from zero), while $\log_{10}(\text{n.filings})$ carries a coefficient of $+1.578$ (SE 0.030, $p < 10^{-9}$). A $10\times$ increase in filing count multiplies the odds of going public by ~ 4.85 ; a one-SD increase in mean ΔKL multiplies them by ~ 0.98 .

This is informative precisely because it is null. The within-EDGAR margin results above show that novel firms enjoy higher gross margins *conditional on being public*; the present check shows that the selection into being public is governed by trademarking volume rather than trademarking character. Read together, the two findings mean: becoming a reportable firm is a separate (volume-driven) hurdle; *conditional* on clearing it, novelty earns a margin premium.

Conditional on public listing: trademark count predicts revenue but not margin. A natural follow-up: among the firms that did IPO, does trademark count correlate with firm value? The patent-economics literature finds that count and value move together — prolific patenters have valuable patents (Trajtenberg 1990; Hall, Jaffe & Trajtenberg 2005, *RAND J. Econ.*) — and we wondered whether the opposite might hold for trademarks, where a smaller portfolio could be evidence of operational focus. On the SEC-crosswalked panel ($n = 6,456$ CIKs with clean-panel trademark filings and at least one observed year of SEC financials), the correlation matrix is unambiguous: trademark count is *strongly positively* correlated with firm size and *essentially uncorrelated* with margin (Table 10, Figure 11). A firm in the top mark-count decile (median 125–5,697 marks) has roughly $300\times$ the median revenue of a firm in the bottom decile (median 1–2 marks), but its mean gross margin is essentially identical. The focus hypothesis is rejected on size, absent on profitability, and weakly supported only on R&D intensity (firms with many marks tend to file less R&D against their revenue — a different culture, not a strategic focus signal).

A causal-attribution caveat. The trailing-3-year predictor pools many filings per firm-year, and a firm with several novel filings in its recent past is also a firm whose *earlier* filings might have

been the cause of its subsequent financial success — that is, the pooled regression cannot distinguish “current novelty pays off” from “past success funds the current novelty habit.” We address this directly in Section 13 by restricting the predictor to each firm’s first-ever filing.

13 Outcome 3: Debut-filing-only test

The pooled per-filing logit in Section 11 treats each of a firm’s filings as an independent observation, and the firm-year panel in Section 12 averages each firm’s recent trademark portfolio. Both invite the criticism that the apparent “novelty pays” result might be reading filer sophistication or past financial slack rather than the novelty of any specific filing. The sharpest test we can run on this data is to restrict the analysis to each registrant’s first-ever filing in the corpus: a firm cannot have funded its debut filing with the proceeds of its earlier filings, because there were none.

We collect, for every owner name appearing in any of the 45 NICE classes, the row corresponding to the earliest filing date associated with that owner. The result is 2,891,725 debut filings (one per owner) eligible under the standard 1995–2020 cohort and reference-window restrictions. Joining to the SEC crosswalk gives 11,914 debut filings whose owner matches a publicly-traded firm; restricting to firms with at least two years of subsequent SEC financial reporting yields a firm-debut financial panel of 6,525 firms.

Outcome	N	β for ΔKL (z -scored)
<i>Survival on first filings (one per owner across all classes)</i>		
Completed registration	2,891,725	-0.034 (0.001)
Renewed 5y registration	2,891,725	-0.005 (0.001)
Renewed 10y registration	1,875,964	+0.042 (0.002)
<i>Firm-debut financial regression ($N=6,525$ matched public firms)</i>		
Mean future gross margin	6,525	+0.017 (0.003)
Within-firm gross-margin std.	5,008	+0.0052 (0.0015)

The pattern is consistent and informative.

Survival. On debut filings, the ΔKL coefficient on completing registration shrinks from the pooled value of +0.124 to -0.034 (still highly significant on 2.9M observations). A debut filer who introduces unusually novel goods/services language appears to face slightly more registration friction — the opposite sign of the pooled effect. The pooled positive coefficient was reading filer sophistication: firms with deep filing portfolios know how to write a description that survives examination, and they also tend to file more novel marks. Among first-time filers neither advantage is present. Renewal coefficients are smaller in magnitude but stay positive (+0.042 at the 10y mark), suggesting that long-run persistence still favours novelty among debut filings.

Are debut filings less novel than later filings? The negative debut-registration coefficient invites a natural worry: if first filings were systematically less novel, then perhaps the apparent “debut novelty pays” result is an artefact of restricted variance. This is not what the data show. On the same all-class clean panel, mean ΔKL on debut filings is +0.052 (median +0.008), versus +0.010 (median -0.010) on non-debut filings; the share of filings with positive ΔKL is 51.5% on debuts and 47.6% on non-debuts. Both component KLS are larger on debuts (prospective 5.22 vs. 4.91, retrospective 5.17 vs. 4.90): debut filings read as less like the recent past *and* less like the near future. The intuition is that a firm’s debut is often a stake in genuinely new territory, while its later filings drift toward category-conventional language. So the negative registration coefficient on debuts

is not explained by lower novelty in the debut set — if anything debuts are slightly more novel; it is explained by examination friction. A first-time filer with unusual goods/services language draws office actions and lacks the procedural sophistication to navigate them, while an established repeat filer with the same novel language clears the process routinely. The pooled positive coefficient was reading the established-firm advantage; stripping it leaves a small examination- friction penalty for first-time filers writing in non-standard ways.

Within firms, the debut filing sits roughly at the firm’s own median novelty (mean within-firm percentile 0.514 on 432,180 firms with ≥ 5 clean filings), but the distribution is bimodal: 24.4% of firms file their debut in the bottom 20% of their own ΔKL distribution (cautious entrants), and 30.4% file their debut in the top 20% (bold entrants); the middle three quintiles each account for only 14–16%. Few firms enter at mid-novelty. And among multi-filing firms, 53% of *non-debut* filings have $\Delta KL < 0$ — a slight tilt toward line-extension language, although the per-firm mix is heterogeneous: about a fifth of firms are persistently novel (at most 20% of their non-debuts in tired territory) and another quarter are persistently tired ($\geq 80\%$).

Financial. On the firm-debut panel, a one-standard-deviation increase in the firm’s debut-filing ΔKL is associated with +1.7 **percentage points** higher mean future gross margin (SE 0.3, $n = 6,525$ firms) and +0.5 percentage points higher within-firm margin standard deviation (SE 0.15, $n = 5,008$ firms). Both effects are roughly half the magnitude of the pooled trailing-3-year results in Section 12. We take the attenuation as evidence that part of the original firm-year reward reflects the habit/portfolio confound flagged in the previous section, but that a substantive debut effect remains: the novelty of a firm’s first trademark filing predicts its subsequent average financial performance, even before the firm has produced any other trademark filings to “fund” the bet.

Debut ΔKL predicts going public; line-extension filings reverse the appearance

At the firm-debut level, positive ΔKL raises the probability of ever going public. Restricting to each owner’s first-ever filing across all classes ($n = 5,090,053$ in the clean H=2y panel), $P(\text{owner ever in SEC EDGAR})$ by debut ΔKL bin is *monotone increasing*: 0.42% at the bottom decile, 0.47% in the middle, 0.64% at the top decile — a +52% relative lift for positive- ΔKL debut filings over negative- ΔKL debut filings (Figure 13 panel C). On the firm’s first move, having a vocabulary that the field subsequently echoes predicts becoming a publicly-reported firm.

The same regression at the full-corpus filing level shows the *opposite* direction, and a reader who saw only that result would draw the wrong conclusion. At the filing level, $P(\text{owner ever in SEC EDGAR})$ *decreases* monotonically with ΔKL : the bottom decile is 5.8% public, the top decile 3.5%. The reversal is mechanical. Public-firm owners — incumbents like Apple, Coca-Cola, Procter & Gamble, Pfizer — file many trademarks per year, and the bulk of an incumbent’s portfolio is line extensions that reuse the firm’s established (low- ΔKL) vocabulary. Their volume swamps the per-filing statistic and concentrates the public-firm mass in the low- ΔKL bin. The right unit of analysis for the “do innovative firms reach the public market?” question is the firm’s debut filing; the full-corpus filing-level statistic is dominated by post-IPO line-extension activity.

For prospective and retrospective KL on debut filings, $P(\text{public})$ is *monotone decreasing*: the very-distinctive-on-day-one debut firms have lower IPO rates (0.31%–0.34% at the top decile) than the average-distinctiveness debut firms (0.51% overall baseline). Niche on day one tends to mean niche at maturity; the IPO sweet spot for debut firms is ordinary vocabulary used in a forward-drifting direction — low PC1, high PC2 in the decomposition.

The harvest tail on debut filings is exceptionally clean. The unexpected post-registration-survival lift on the negative- ΔKL side — the harvest extreme — is most diagnostic on debut

filings, where there are no within-firm portfolio confounds. On the debut-only panel ($n = 5,088,700$ owner-first filings), we fit a saturating exponential of survival on $|\Delta KL|$ separately on each side:

$$P(\text{survived 5y}) = a + b e^{-c|\Delta KL|}.$$

The harvest side (negative ΔKL , $n = 2,586,126$) fits with $R^2 = 0.978$ on 20 binned points; the conditional Pearson correlation between $|\Delta KL|$ and survival on this side is $+0.054$ (Spearman $+0.058$). The innovation side (positive ΔKL) is positive in the same direction but much noisier — $R^2 = 0.499$ on 20 binned points (Figure 14). The harvest tail is doing real work and is unmistakably present; the innovation tail is present and consistent in sign but visibly less tight on the saturating-exponential functional form. We do not claim a mechanistic explanation for the difference in tightness.

14 Within-firm dynamics and debut quartiles

The first-filing analysis in Section 13 answered “does the firm’s first novel bet predict outcomes,” but said nothing about how a firm’s later filings behave once the firm is established. The full corpus contains 1.24 million firms with at least two filings and 5.74 million consecutive filing pairs, which lets us measure two related quantities: how persistent within-firm novelty is, and whether firms become more or less innovative as they file more.

Within-firm autocorrelation. For every consecutive filing-pair within a firm we compute the Pearson correlation between the prospective KL of the later filing and that of the earlier one, and the same for ΔKL :

The autocorrelations are large. Knowing that a firm’s previous trademark filing scored high on prospective surprise (was unusual at the time) raises our expectation of high prospective surprise on the firm’s next filing by $\rho = 0.68$. The same correlation on the imitated-innovator score ΔKL is $\rho = 0.55$. Surprising firms keep filing surprising trademarks; innovative firms keep filing innovative trademarks.

Firms drift very slightly toward conservatism. Restricting to firms with at least four filings (so a per-firm trend can be estimated), the median per-firm slope of ΔKL on filing-index is 0.0000 and the mean is -0.0007 per additional filing. 47.0% of firms have a positive slope (binomial $z = -19.6$ vs. the null of 50/50). Firms become marginally less innovative as they accumulate filings, but the magnitude is tiny — on the order of one tenth of a percent of $\sigma_{\Delta KL}$ per additional filing.

Debut quartiles. We split firms into four quartiles by the ΔKL of their first-ever filing and tabulate subsequent behaviour within each.

Three patterns are visible. (1) All four quartiles regress toward the mean: the most-innovative quartile (Q4) starts at debut $\Delta KL = +0.336$ but its average subsequent filing scores only $+0.106$, and the least-innovative quartile (Q1) starts at -0.309 but recovers to -0.098 . (2) The gap between innovative- and conservative-debut firms persists: Q4 firms remain $+0.20$ nats more innovative on subsequent filings than Q1 firms. (3) The strongest within-firm AR(1) on ΔKL is in Q4 ($+0.572$); innovative firms are the most *consistent* in their innovation, even after partial regression. Conservative-debut firms have weaker but still substantial AR(1) ($+0.49$), suggesting innovation profile is established at the debut bet and tends to persist.

15 Stock-market returns

If novelty in trademarks predicts firm-level financial outcomes, the strongest test is whether it predicts subsequent stock-market returns above a market benchmark. We pulled monthly adjusted-close prices from Yahoo Finance for every CIK in the USPTO–SEC crosswalk that maps to a current ticker (4,254 matched CIKs, 0.86M monthly price points, plus the S&P 500 as benchmark), computed end-of-year prices, and constructed cumulative total returns at one through five-year horizons after each filing year. We then regressed those returns (winsorised at the 1st and 99th percentile to suppress penny-stock-spike outliers; cluster-robust standard errors at the firm level) on the firm’s ΔKL in two specifications: firm-year mean across that year’s filings, and debut-only.

The pattern is consistent with the gross-margin result of Section 12 and considerably stronger. Per one-standard-deviation increase in firm-year mean ΔKL , excess return over the S&P 500 is +1.5 pp at one year, peaks at +5.0 pp at three years, and decays back to the market by year five. Per one-standard-deviation increase in debut-filing ΔKL (one observation per firm), excess return is +3.6 pp at one year, +19.2 pp at three years, and peaks at +28.4 **pp** at four years before partially mean-reverting to +14.9 pp at five years. Both specifications are statistically significant at the one-year horizon and grow more so at intermediate horizons. The debut specification eliminates the past-success-funds-current-novelty confound and still produces the largest excess-return coefficient in the paper. The five-year mean-reversion is consistent with a story in which innovative firms enjoy a multi-year “imitated category leader” window before competition catches up.

Surprise alone does not predict returns. A natural variant of the regression uses prospective KL alone as the predictor (rather than the ΔKL composite). Doing so eliminates the result entirely: at the firm-year level the prospective-only coefficient is essentially zero at every horizon (+0.001 at 1y, −0.001 at 5y); at the debut level it is also zero with wider standard errors (+0.001 at 1y, +0.057 at 5y, both indistinguishable from null). **Mere novelty at filing time does not predict subsequent stock returns. Only novelty that the future ends up imitating does.** The retrospective component is itself a forward-looking signal of which novel filings turned out to define a category, and the predictive content of ΔKL for returns sits in that selection.

Cumulative excess return by debut ΔKL quartile. A clean visual summary of the result is to plot the mean cumulative excess return over the S&P 500 by years-since-debut for each of the four debut- ΔKL quartiles (Figure 15). The most-novel debut quartile (Q4) outperforms the least-novel quartile (Q1) by approximately twenty percentage points in cumulative excess return by year five, with the gap opening continuously across the first five years.

Per-industry correlations. The firm-year and debut results pool firms across all industries. We can instead compute the within-industry Pearson correlation between firm-mean ΔKL and the firm’s mean annual excess return over the S&P 500 (using only firms with at least three years of return observations). Table 14 reports these correlations for the twenty largest industries by matched-firm count, with two-sided p -values.

The largest industries vary substantially in the strength of the ΔKL –return relationship: it is positive and significant in some (Class 042 Scientific & Tech Services, Class 041 Education & Entertainment) and indistinguishable from zero in others. The sample-size cap means we cannot resolve a stable rank ordering; the takeaway is that the financial reward to trademark novelty is not uniform across industries.

Are these results driven by the FAANG cohort? A natural concern is that the 2010s stock market was dominated by a handful of mega-cap technology firms whose massive capital appreciation could mechanically produce a positive coefficient on almost any sensible novelty proxy. We address this directly by re-running the returns regressions excluding the eight largest tech firms in our matched panel: Apple, Microsoft, Amazon, Meta (Facebook), Alphabet (Google), Netflix, NVIDIA, and Tesla.

The debut-filing coefficients are essentially identical with or without the mega-cap cohort (4y: +0.2844 vs. +0.2823). The firm-year coefficients attenuate slightly at long horizons — the 5y firm-year coefficient does collapse to near-zero (+0.005 vs. +0.016), telling us that whatever 5y firm-year effect existed was mostly mega-cap — but the central 1–3y story holds. The headline result that debut ΔKL predicts a ~ 28 pp excess return at the 4y horizon is not a FAANG artifact.

Figure 16 plots the per-year mean ΔKL and cumulative price for each of the eight firms.

Convergent validity against patent counts. The dominant existing quantitative innovation proxy is the count of patents an organization is granted per year. We pulled the full PatentsView bulk data (9.36 million U.S. patents, 8.57 million patent-assignee rows from the `g_assignee_disambiguated` table), aggregated to (disambiguated assignee, year) patent counts (1.47 million firm-years), and joined to our trademark ΔKL firm-years on the same normalized-name key used for the SEC crosswalk.

The two measures are essentially uncorrelated, and we treat that as a feature. The Pearson correlation between firm-year mean ΔKL and that firm-year’s patent count is $r = +0.010$ on 174,569 firm-years; using $\log(1 + n_{\text{patents}})$ to suppress mega-patent-filer leverage gives $r = -0.009$. At the firm level (one observation per firm, restricted to firms with at least three years of overlap), $r = +0.008$ for raw sums and $r = -0.023$ for log sums. The largest patent filers in the matched panel are major-electronics-conglomerate incumbents: IBM (379,206 patents over 26 years, mean $\Delta KL = +0.02$), Samsung (358,183, +0.04), Sony (188,074, -0.01), Toshiba, Canon, LG, Intel, Google, Fujitsu, Microsoft. The two innovation measures pick out very different firms. Patents reward organisations that translate R&D into legally defensible technical claims; trademark ΔKL rewards organisations that introduce business-model or product-language novelty that the rest of their industry then adopts. The two measures are *substantively* different innovation channels, and the absence of a correlation is itself the evidence that trademark ΔKL is measuring something not already covered by the dominant innovation proxy.

Patent-count regression on returns. A direct test of which channel matters more for shareholder outcomes is to swap our predictor: replace ΔKL with $\log(1 + n_{\text{patents}})$ on the same returns panel, with the same year controls and cluster-robust SEs at the firm level.

The patent coefficient is positive and growing with horizon: +2.0 pp at one year, +5.9 pp at three, +12.1 pp at five years per σ of $\log(1 + \text{patents})$. The trademark ΔKL firm-year coefficients (Table 13) peak earlier and decay; the trademark debut coefficients are larger in absolute terms (+28 pp at four years) but shrink at five years.

Patents and trademarks index different innovation constructs. The growing-with-horizon pattern on patents is what we would expect if patent counts indexed *continuing* innovation activity: a firm with many recent patents is one whose R&D pipeline is still producing, and the future cash flows of that pipeline are priced in increasingly across two-, three-, and four-year-ahead returns. The trademark ΔKL signal is qualitatively different. The debut-only specification — one observation per firm, keyed on the firm’s first-ever trademark filing — treats ΔKL as a *one-time positional bet*:

did the founding team pick a vocabulary the field would migrate toward? The horizon profile of the debut ΔKL coefficient (rising to +28 pp at four years and then shrinking) is consistent with a one-time bet whose realisation takes a few years to be priced and then becomes information the market has already absorbed. The two innovation proxies are not redundant (the R^2 decomposition below confirms sub-additivity) and they answer different questions. Patents proxy *whether the firm is still innovating*; ΔKL on the firm’s debut filing proxies *whether the firm started in fertile linguistic ground*.

Are patents and trademark ΔKL complementary or redundant? An R^2 decomposition.

The complement-vs-substitute question is not rhetorical — we can compute the answer. Restricting to the common panel of firm-years in which we observe *both* a patent count and a trademark ΔKL (and the firm appears in our SEC returns data), we run eight nested OLS specifications on the same panel of 7,607 firm-years and report the R^2 of each (controls in every spec are year linear and quadratic). We add a third predictor, $\log(1 + \text{patents})$ (P) and the trademark composite ΔKL (I, for “innovator”), but also *novelty alone* (S, the prospective KL component) so that we can separately ask whether plain filing-time surprise contributes anything.

Three patterns are visible in Table 18.

(1) All R^2 values are small. On a common panel of 7,607 firm-years, controls (year and year-squared) explain between 0.7% and 3.4% of cumulative-excess-return variance. This is what one expects from firm-level returns: most of the variance is idiosyncratic. The contribution of any innovation proxy is necessarily a small slice of an already-small explainable share.

(2) Patents and innovator ΔKL are very nearly additive. Patents alone add +0.06 to +0.23 pp of R^2 above controls (growing with horizon); innovator ΔKL alone adds +0.10 to +0.20 pp (peaking at three years). Their joint contribution P + I is +0.16 to +0.33 pp, essentially the sum of the individual contributions. The two measures share almost no explanatory variance for stock returns; they pick out different firms, and each firm-feature carries independent return information.

(3) Novelty (prospective KL) alone adds essentially nothing. The S-only specification is statistically indistinguishable from controls at every horizon (+0.00 to +0.01 pp), and the marginal R^2 of S in the P+I+S spec relative to P+I is ≤ 0.01 pp at every horizon. The result we documented earlier — that a filing that is merely surprising at the time, without later imitation, does not predict returns — is reproduced cleanly in the R^2 decomposition. The forward-looking component ($-KL_{\text{retr}}$) is what makes ΔKL a return predictor, not the backward-looking surprise itself.

We therefore replace the previous sentence “the two measures appear to be complements” with the more precise statement: **patents and trademark ΔKL are quantitatively additive on stock returns; their joint R^2 contribution is the sum of the marginal contributions of each in isolation, and novelty alone adds nothing.** A firm’s patent count and a firm’s trademark ΔKL are two substantively different innovation channels and the predictive content of one is not contained in the other.

Four further public comparators. We supplement the patent comparison with four other public innovation proxies. The data and the matching code are released under `data_publish/comparators/`.

The strongest signal among the four is **R&D-to-revenue** ($r = +0.13$ on 1,583 firms), consistent with the intuition that firms that spend more on research file more novel goods/services descriptions. **MIT Technology Review’s 50 Smartest Companies (2015 list)**, of which we matched 17 of 50 to a SEC CIK in our trademark panel, had a mean ΔKL of +0.052 versus -0.018 for the rest of the matched universe — a +0.07 gap that is borderline-significant on the small sample ($z = +1.63$). **The BCG Most Innovative Companies list (2021–23 union, 80 unique**

firms, 31 matched) shows a smaller $+0.02$ on-list-vs-off-list gap that is not distinguishable from zero ($z = +0.64$). **Crunchbase total funding** (using the `notpeter/crunchbase-data` historical mirror, 1,646 matched firms) is essentially uncorrelated with ΔKL ($r = 0.00$ for log funding).

The pattern is the same one we saw with patents: each of these public proxies captures a substantively different innovation construct, and pairwise correlations between proxies sit in the range $\rho \approx 0.0\text{--}0.13$. Trademark ΔKL is on the correlated end of that range with R&D and MIT TR’s expert-curated list, on the uncorrelated end with Crunchbase funding totals, and near-zero with patent counts. None of the four can substitute for ΔKL , and ΔKL does not substitute for them.

The four comparator data files are published in `data_publish/comparators/`, alongside the `external_crosswalk.csv` file that records, for every list-based comparator entry, whether and how it matched a SEC CIK in our matched universe. Researchers wishing to replace any of our comparators with a more authoritative source can drop a CSV with the same schema into the same directory and re-run `scripts/external_compare.py`.

What is a trademark really protecting? The convergent-validity result invites a second look at what the trademark filing fundamentally is. A trademark is a legal monopoly over a particular set of words, designs, or sounds in connection with a particular set of goods and services. The economic value of that monopoly comes from how distinctive the goods/services description actually is: a generic mark on a generic product (**Beer** for an ale) is hard to defend and easy for competitors to crowd; a distinctive description for a genuinely new offering (**Hard Seltzer Works** for a new beverage category) keeps later entrants from advertising in the same space without infringing. Read this way, the goods/services text is not just a description of the firm; it is a legal carve-out around the firm’s claim to a particular novelty. A trademark filing high in ΔKL is, in this framing, an attempt to reserve the language of a market offering before the rest of the industry has named it. That the filing later gets imitated — the empirical mark of a high- ΔKL mark — is exactly the substantive event a successful trademark exists to police. The financial reward we measure is, in part, the value of having claimed the boundary of a market category before competitors arrived to share it.

Companion data release. Two CSV files are published alongside the source code as the data contribution of this paper:

- `data_publish/firm_year_dkl.csv` (59,033 rows): for every USPTO trademark owner that we matched to an SEC EDGAR CIK, annual mean prospective KL, retrospective KL, and ΔKL aggregated across all 45 NICE classes the firm filed in.
- `data_publish/firm_year_patents_and_dkl.csv` (174,569 rows): the joint patent-and-trademark panel underlying Table 16, suitable for any follow-up convergent-validity analysis.

16 Templating and survival: do shared descriptions predict failure?

The diagonal-clustering geometry of the topomap suggests that firms filing in the lower-left “common-place” corner — low prospective *and* low retrospective surprise — have the worst post-registration survival. A natural mechanism would be templated copy-paste descriptions: filers using boilerplate that examiners and the marketplace alike learn to discount. We test this directly. For every

clean filing, we compute the count of *other* filings sharing the exact same lowercased, punctuation-stripped goods/services text, then bin filings by that multiplicity and report registration and post-registration-survival rates.

The result is the opposite of the templated-failure hypothesis.

Three substantive findings:

1. **More-templated filings survive better, not worse.** Unique-text filings post-register at 48.6% and survive at 38.1%. Filings that share a description with 5,000+ others register at 58.2% and survive at 49.2% — a +11 percentage point survival premium for the most heavily templated tier. The pattern is monotone through the 6–10 multiplicity bin, then plateaus.
2. **The post-registration cancellation rate is mildly flatter for templated filings.** The right-most column of Table 20 shows that, conditional on registering, $\sim 22\%$ of unique-text marks lose their registration (mostly to Section 8 failure at year five), versus $\sim 15\%$ for the 5000+ bin. Both the higher registration rate *and* the higher conditional-survival rate point in the same direction.
3. **Boilerplate signals serious filers.** Inspecting the most-templated normalised G/S strings, the pattern is dominated by commodity professional-services categories: “restaurant services” (52,973 identical filings), “dietary and nutritional supplements” (20,755), “legal services” (16,400), “dietary supplements” (15,982), “alcoholic beverages except beers” (13,286), “banking services” (11,422). These are the descriptions that experienced trademark counsel re-use across clients precisely because they are the safe, examiner-tested phrasings.

Robustness: dropping the $n_{\text{terms}} \geq 3$ filter. The headline analysis applies the full CLEAN filter (Table 1), which excludes 1- and 2-token G/S descriptions because their per-filing KL is degenerate. Many of the most-templated strings are short — “restaurant services”, “wines”, “footwear”, “t shirts”, “candy”, “cosmetics”, “jewelry” — and so are dropped from the headline panel even when the analysis at hand does not depend on KL at all. To check that the templating-survival pattern is not an artefact of this filter, we re-run the analysis on CLEAN-without- n_{terms} (8,499,455 filings, +591k vs. headline). The pattern is nearly identical for the 1- through 1001–5000 bins (≤ 1 percentage point of drift on either rate). The only meaningful change is the 5000+ bin: still-live rate falls from 49.2% to 44.2%, because the short commodity templates that the headline filter excludes survive at corpus-average rates rather than the elevated rates of the longer structured templates that dominate the n_{terms} -filtered 5000+ bin (e.g. “athletic apparel namely shirts pants jackets footwear hats and caps athletic uniforms,” 13,047 copies, 61.6% still-live).

So why is the lower-left of the topomap so bad? The diagonal-clustering reading from the topomap survives this audit: the worst-survival cell is firms whose filings cluster on the diagonal at low KL . But the mechanism is not “templated text gets penalised.” Quite the reverse — explicitly templated text helps. What appears to drive the lower-left corner is one-off, generic G/S text that nobody else copies verbatim: descriptions written in the class’s modal vocabulary but unique enough to count as zero-template, filed by firms without the procedural sophistication to use the known-good phrasings their lawyers would otherwise re-use. Boilerplate, when it exists, signals professional repetition; the bottom-left cluster is filled with filers who lack the boilerplate.

17 Both extremes outperform: a U-shape across industries

The decile-rate plot referenced in Section 11 shows that the relationship between ΔKL and survival is not linear; it is U-shaped, with both the most-conservative-debut filings (lowest decile of ΔKL) and the most-innovative filings (highest decile) outperforming the middle deciles. The intuition is that both directions of distance from the modal filing are rewarded: very-novel filings benefit from category creation and the highest decile of ΔKL is dominated by firms that the future imitated; very-conservative filings benefit from being unambiguously defensible and easy for the examiner to parse. Filings in the middle, neither novel enough to claim a new category nor templated enough to coast on legal-clarity grounds, fare the worst.

Quadratic regression. We test the curvature directly. For every industry’s clean filings between 1990 and 2020, we fit a logistic model with both $z(\Delta KL)$ and $z(\Delta KL)^2$ as predictors, with year and year-squared controls. Pooled across all 45 NICE classes and 9.3 million filings, the aggregate quadratic coefficient is positive but small ($\beta_{(\Delta KL)^2}^z = +0.0002$ on 5y survival; $+0.0013$ on 10y survival). The aggregate pooled quadratic is not the primary evidence for the U-shape; the per-industry evidence is. At the per-industry level the quadratic is positive in 27 of the 44 industries we can fit.

The U-shape is present in most — but not all — industries. Fitting the quadratic per industry on the 5y-survival outcome *pooled across all years* ($n = 44$ industries with at least 5,000 clean filings), the quadratic coefficient on $z(\Delta KL)^2$ is positive in 27 of 44 industries (61%) and individually significant at $p < 0.05$ in 21 of those 44 (48%). The U-shape dominates in consumer-goods classes (Household Utensils, Furniture, Leather, Textiles, Common Metals, etc.), where heavy templated filing by large incumbents and distinct novelty waves coexist. Several large information-economy classes show the opposite on the pooled-years fit, an apparent *inverted-U*: Software & Electronics ($t = -18.2$), Insurance & Financial ($t = -13.8$), Scientific & Tech Services ($t = -12.9$), Advertising & Retail ($t = -10.7$), Telecommunications ($t = -10.6$).

Per-(industry $\times$ year) refinement. The pooled-years fit hides whether an industry is consistently $+U$, consistently inverted- U , or shifting between regimes. We re-fit the quadratic logit separately for every (industry, year) cell (44 industries $\times$ 32 years, requiring ≥ 500 filings and convergent logit; 1,267 cells where both conditions hold), and classify each industry by the majority-of-years rule with a margin of 2 cells (Table 22). On this refinement, the pooled-years inverted- U assignment to Software & Electronics, Insurance & Financial, Scientific & Tech, and Advertising & Retail is *not* replicated at the year level: those industries have year-by-year quadratic coefficients that flip sign and are mostly indeterminate. Only one industry has a clean inverted- U at the year level (Telecommunications, NICE 038). Eleven additional industries are indeterminate (no significant year-level curvature in either direction). The $+U$ holds in 32 of 44 industries.

Table 23 reports the pooled-years ranking of the 15 industries with the strongest U-curvature and the 5 with the strongest pooled-years inverted- U , for reference and comparison with the year-level classification above.

Figure 19 overlays each industry’s per-decile 5y-renewal rate (thin grey lines) on the cross-industry mean (red), making the consistent U-shape visible.

Industries ranked by U-shape strength. Industries differ in how cleanly the U appears. Table 24 ranks all 44 industries with at least 5,000 clean filings by the t -statistic on the quadratic

ΔKL^2 coefficient. Pharmaceuticals (Class 005) tops the list, followed by Materials Treatment (040), Advertising & Retail (035), Chemicals & Industrial (001), and Machines & Power Tools (007). At the bottom of the ranking are Telecommunications (038, $t = -9.8$, statistically inverted-U), Carpets & Wall Hangings (027), Musical Instruments (015), Trim, Lace & Notions (026), Wine & Spirits (033), and Tobacco & Smokers' Articles (034) — in each of these the curvature is absent or slightly inverted.

Industries that lack a U have a stronger linear effect. Among the 29 industries with $t_{\text{quad}} \geq 1.96$, the mean $|\beta_{\text{linear}}|$ is 0.048. Among the 15 industries without a significant U, the mean $|\beta_{\text{linear}}|$ is 0.063 — noticeably higher. These are industries in which ΔKL acts more like a linear (rather than U-shaped) predictor. Telecommunications, the strongest inverted-U, has a positive linear coefficient ($\beta_{\text{lin}} = +0.020$) on top of the negative quadratic; Trim & Notions, Carpets, and Musical Instruments have negative linear coefficients large in magnitude (-0.04 to -0.12). These industries either reward novelty roughly linearly (in either direction) or reward conventionality without the symmetric reward at the high-novelty end.

Is the U driven by extreme-KL filings? A natural concern is that the U-shape is generated by a few very extreme filings on either KL axis — the most-similar-to-past filings on the left, the most-imitated filings on the right — rather than by a smooth concave-up relationship across the distribution. We test this by re-fitting the pooled quadratic on the same outcome after dropping different combinations of the extremes.

The result is unambiguous: **the U-shape does not depend on extreme-KL filings.** Dropping the bottom 5% of prospective KL (filings most similar to the past) leaves the quadratic coefficient at $+0.0082$ ($t = +38.6$), down from $+0.0087$ ($t = +41.9$) on the full sample. Dropping the bottom 5% of retrospective KL (most-similar-to-future filings) leaves it at $+0.0085$ ($t = +39.9$). Dropping both: $+0.0082$ ($t = +38.2$). Most strikingly, dropping the bottom and top 10% of ΔKL itself — i.e., excising the two arms of the U — yields a residual quadratic coefficient of $+0.0343$ ($t = +48.7$), *four times larger* than the full-sample coefficient. The curvature is intrinsic to the central 80% of the ΔKL distribution. The U is not driven by the most-templated nor by the most-novel filings; it is the relationship between novelty and survival across the entire range.

Two-sided exponential fit. A separate test of the same shape is to split the data at $\Delta KL = 0$ and ask whether the rate rises exponentially with distance from zero on each side: $\log(\text{rate}) = a + b \cdot |\Delta KL|$ separately for the negative- ΔKL and positive- ΔKL halves. The pooled aggregate fits, on the 5y-survival outcome, give slopes $b_{\text{neg}} = +0.04$ and $b_{\text{pos}} = +0.05$ on the bin-aggregated data, both with $R^2 > 0.7$ across the deciles within each side. Both sides slope positively in $|\Delta KL|$: each one-nat increase in distance from zero (in either direction) is associated with roughly a 4-to-5% relative increase in the per-bin renewal rate.

What is consistent with this story? A natural reach is to say that the most-imitated novel filings are category leaders the industry follows and the most-conventional filings are templated marks that examiners and competitors leave alone. But that reading credits both tails with prescient strategy — the right-arm filers “saw” the future their wording would define, and the left-arm filers “saw” that the safest path was to use the existing canonical wording. Without behavioural data we cannot license either claim. A more parsimonious story is that we observe the empirical fact (both extremes outperform the middle) without needing to ascribe foresight to either tail. The structural mechanism we *can* document is examiner heuristics — the office-action evidence below — which

gives both tails an advantage at the registration gate without requiring any awareness of what the future would do.

An office-action mechanism for the U. A more mechanical reading of the same shape comes from likelihood-of-confusion. The status code 700 “abandoned, no response to office action” is the modal post-examination failure, and its share by ΔKL decile is itself U-shaped in reverse (D1: 5.1%; middle deciles 3–5: 13.2–14.1%; D10: 9.1%). Filings whose wording uses common, well-defined mid-frequency vocabulary that happens to overlap with prior registrations get examiner objections; filings with very narrowly common vocabulary (the bottom decile) and filings with vocabulary nobody else has used (the top decile) avoid these conflicts. This mechanism does not require firms in either tail to have strategic awareness of their own position — it is a property of the examiner’s prior-art search, mediated by how distinctive or how generic the goods/services description is. **We flag this as the cleanest available mechanism precisely because it does not ask us to credit the negative- ΔKL tail with prescience that we would not also extend to the positive tail.**

We do not have a mechanism story commensurate with the magnitude of the effect. We want to flag this directly. The U-shape on ΔKL is empirically robust — larger than the linear ΔKL effect we report on gross margin, present in 27 of 44 industries, present at the filing level even within firms that demonstrably abandoned some marks and kept others alive, and present at the token level for both extremes. But *the magnitude of the survival lift at both flux extremes is greater than any single mechanism we can articulate explains*. A 5–10 percentage-point gap between flux-neutral and flux-extreme filings, sustained across millions of filings and dozens of industries, is not what we expected to find, and the candidate mechanisms we can name (likelihood-of-confusion examination artefacts, real-options harvest by incumbents, Schumpeterian creation at the trend-leading edge) each predict only part of the curve and none predicts the depth. We document the empirical pattern and present what falsification we can; we do not claim a unified mechanism story.

Falsification battery (summary). The U-shape is unusual enough to warrant a battery of robustness checks. Table 26 reports re-fits of the pooled survival quadratic on the candidate confounds in summary form; Appendix C narrates each check in full. The quadratic coefficient $\beta_{\Delta KL^2}^z$ stays strongly positive on every survival sub-sample, with a single exception: the firm-year gross-margin OLS, where the relationship is roughly linear in ΔKL rather than U-shaped. Survival is U; margin is linear.

The ten checks (Table 26, with the firm-persistence, temporal-stability, competition-density, and external-corpus tests added) are: (1) Foreign-extension hypothesis (Madrid-Protocol marks could inflate the low- ΔKL tail): **rejected**; US-only and foreign-only sub-samples give nearly identical quadratics (+0.0078 vs. +0.0100). (2) First-time vs. recurring filings: **both have the U**, recurring slightly stronger. (3) Financial outcomes: **the U does not appear** in firm-year gross margin (the relationship is linear); the U lives in survival. (4) Survivor bias from large incumbents: **partial** support for the left arm — the bottom ΔKL decile has larger firm filing-volumes than the middle. (5) “Liquid Death” hypothesis (low- ΔKL might be novel products in old vocabulary): **rejected** on text-length splits. (6) Filing-level vs. firm-level: **filing-level** — the U survives within firms that abandoned some marks and kept others alive (Figure 20). (7) The blockchain / cryptocurrency / NFT / web3 wave as a possible imitated-but-failed counterexample: **not a counterexample** — the crypto cohort outperforms on every survival margin. (8) Temporal stability: **the U is present in most individual years, not just in pooled data**. Per-year fits 1990–2021 ($n = 32$ cohorts) show

the pooled quadratic is significantly positive in 23/32 years (Figure 21), independently of macro cycle. The pooled-across-years result is the average of mostly year-by-year positive coefficients, not an aggregation artefact. (9) Competition-density mediation: **not a confound, but a mediator**. On 10,000 stratified Class-9 focal filings we count near-competitors (cosine ≥ 0.7 on g/s TF-IDF) within an asymmetric hard window $[t - 2y, t + 5y]$ of the focal’s filing date (no decay weighting; this window is for the cosine-neighbour search only and is distinct from the H=2y exponential-decay reference used for the KL surprise itself — a separate object). The pre-window of 2y captures established competitors at filing-time; the post-window of 5y matches the focal filing’s 5y survival horizon (i.e. peers that could plausibly have shaped or been shaped by the focal in its formative window). We count how many such peers survived 5y from their own filing date. Flux extremes are in clusters with *more* surviving peers, not fewer; the flux-neutral middle has the fewest live peers. Adding $\log(1 + \text{live_competitors})$ as a control absorbs 84% of the ΔKL quadratic on survival (Figure 22). The U is therefore mediated by niche viability: filings at flux extremes sit in vocabulary clusters where peers also survive, and the ΔKL signal is partly a proxy for which clusters can support real businesses. The linear $z(\Delta KL)$ effect remains positive and significant after the mediator is included ($z = +2.65$), so flux is not fully redundant with cluster viability. (10) External-corpus confound (Google Books n-grams): **partial shared signal, but not redundant**. For 75 extreme-flux tokens (25 per stratum) we pulled English-language frequency time series from Google Books 1985–2021 and compared each token’s per-year trademark-corpus frequency to its per-year Google Books frequency. Per-token Pearson correlation between trademark log-lift and Google Books log-lift is +0.47 (Spearman +0.39, $n = 53$ tokens). About 22% of trademark-flux variance per token is shared with general English-language usage; the remaining 78% is corpus-specific. The trademark signal is partly but not wholly a reflection of broader linguistic drift.

What is sensitive to methodology: per-industry rankings. What is robust: the per-year U. A reader who reproduces this analysis from an earlier draft of the pipeline (with a flat 5-year reference window rather than the production H=2y exponential decay) will get visibly different per-industry quadratic coefficients — enough that the rank order of industries by U-strength can rearrange, and even the sign of an individual industry’s coefficient can change. We flag this explicitly. The *per-industry rank* is a sensitive statistic that depends on how the reference distribution is built; we would not stake a finding on the precise sign of any single industry’s quadratic. The *per-year* robustness check (Figure 21) is the appropriate falsification test for the headline U: if the pooled U were a few-cohort artefact, it would not survive year-by-year. It does. The temporal-stability evidence is what we lean on when claiming robustness; the per-industry ranking is descriptive cross-sectional context, not the primary evidence.

$|\Delta KL|$ as a single-number test. Collapsing the U-shape to one number, we regress each survival outcome on $|\Delta KL|$ alone (with the same year and log-n-terms controls). On a 500,000 filing sample:

The z -statistics on $|\Delta KL|$ are +2.4 for completed registration, +9.2 for 5y survival, +14.1 for 10y survival, all positive and the latter two highly significant. The incremental linear- R^2 from $|\Delta KL|$ over the controls is small (under 0.06 percentage points) but real, consistent with the U-shape we documented in the quadratic.

What else could be going on? A handful of further candidate stories that the data so far are consistent with but cannot directly distinguish:

- *Narrative vs. analytical decision-making.* Firms in the middle of the ΔKL distribution may be chasing trends without committing either to the trend’s canonical vocabulary or to a

clearly differentiated wording. The data are consistent with this — the worst-performing filings are exactly those that “follow the herd without standing out from it” — but we have no behavioural data to confirm decision style.

- *Pay-phones-after-cell-phones.* The middle deciles may include filings whose entrepreneurs are starting businesses on templates the industry is in the process of abandoning — not low-novelty enough to coast on legal clarity, not high-novelty enough to claim a new category. The 1995 Beer & Soft Drinks trough (Section 17) is exactly such a cohort.
- *Filing volume as a hedge.* Large incumbents file many marks per year and accept that some will fail; their templated bottom-decile filings benefit from the firm’s resources for renewal and maintenance even when the underlying product fails. The mean firm filing-volume by decile evidence above is consistent.
- *Examiner heuristics.* Examiners may grant clear templates (boilerplate is easy to evaluate) and clear novelties (no prior conflict to cite) at higher rates than middle-of-the-distribution filings, regardless of any underlying business outcome. The status-700 share evidence above is consistent.

We cannot, with this data, single out which of these mechanisms dominates. What we can say is that the survival U-shape is robust to the obvious confounds, that it does not appear in the firm-year financial outcome (which is roughly linear in ΔKL), and that there is direct mechanical evidence (office-action incidence) that middle- ΔKL filings face more friction than either tail.

Did Beer & Soft Drinks go in a failed direction around 1995? The notable trough in the Class 32 time-series near 1995 (Figure 7) is consistent with this story. Inspecting the lowest- ΔKL Class 32 filings of 1995 reveals a heavy concentration of *New Age beverage* brands: Clearly Canadian (TROPICAL TANGERINE, CLEARLY SPARKLING TEA), Koala Springs International (AUSSIE SLAMMER, AUSSIE FRUIT SPARKLER), Royal Crown’s instant-drink mixes (YUPPII, FRUITII INSTANT DRINK), Empire State Beverage’s flavoured lemonades and teas (SAHARA PINK LEMONADE, ROSEWATER LEMONADE, MOONLIGHT TEA), and Nestlé’s exotic-fruit ginger drinks. This is the mid-1990s clear-cola / sparkling-tea / iced-tea / exotic-fruit wave that crested in 1993–95 and was supplanted by the energy-drink wave that began with Red Bull’s U.S. launch in 1997 and accelerated through Monster (2002). The 1995 cohort of Class 32 filings was, in retrospect, exactly the cohort that the next decade’s Class 32 industry stopped looking like. The ΔKL measure correctly identifies it as a tired cohort.

18 Per-industry novelty vs. financial outcomes

The firm-level financial regression averages across industries. To see whether the industry composition of the matched panel is doing the work, we collapse to one observation per NICE class. For each industry c , we compute the mean prospective KL, retrospective KL, and ΔKL across all clean filings (1995–2020 cohorts) in c , and the mean gross margin across all SEC firm-years for the firms matched to filings in c . Figure 23 plots all three together.

What was happening in the labelled industries during the analysis window? The cross-industry correlation in Figure 23 is positive but modest, and the specific labelled industries help interpret what the measure picks up.

- **Tobacco & Smokers’ Articles (034)** sits at the far-right (highest ΔKL) of the third panel and at the top of the cross-industry ranking (Figure 24). The 1995–2020 window covers the entire e-cigarette / vape revolution: the Ruyan cigarette in 2003, NJOY’s U.S. launch in 2007, the Juul explosion of 2015–18, and the cannabis-vape category of 2018–20 (the category leader by debut ΔKL in our table is AUDIO TECHNOLOGY OF NEW YORK INC. a vape-cartridge filer, +2.75). The category went from a few smokeless-tobacco brands to several thousand device-and-cartridge SKUs in fifteen years.
- **Trim, Lace & Notions (026)** is second-highest on ΔKL . The catalyst is the rise of Asian wholesale hair-extension and synthetic-wig brands selling directly into U.S. retail; the top firms by debut ΔKL in Table 28 are XUCHANG BEST TRADING (+1.93), CHUBBY TRADING (+1.74), and RUNWAY PREMIER INTERNATIONAL HAIR (+1.62). Almost no public-firm category had higher pivot rate during this window.
- **Medical & Personal Care (044)** is in the middle of the ΔKL distribution but with high gross margin. The window encompasses the telehealth wave (Teladoc 2002, MDLive 2009, the COVID-era surge in 2020+), the medical-spa explosion, and CBD/cannabis personal-care entries.
- **Telecommunications (038)** sits at the bottom of the ΔKL distribution despite having a comparatively high industry margin in our matched panel. The goods/services lexicon was already settled by 1995 (providing access to telecommunications networks, wireless voice communication); 4G and 5G deployments through the 2010s used the same vocabulary. The category was financially dynamic but linguistically stable.
- **Building Materials (019)** sits even lower: a mature, vocabulary-stable category through the entire window, despite genuine technical change in engineered-wood, low-carbon-cement, and cross-laminated-timber sub-segments that the goods/services lexicon does not appear to have absorbed.

19 Cross-industry comparison

We process all 45 NICE international classes through the same pipeline. Table 28 reports per-class summary statistics; Figure 24 ranks industries by mean filing-level ΔKL ; Figure 25 reports the per-industry logit coefficient of ΔKL on each survival outcome; Table 29 lists the highest- ΔKL mark per industry, picked from the 2010–2020 cohort window where every class has dense filing volume.

Three patterns visible across the 45 classes. First, the cannabis/CBD wave of 2014–2020 dominates the top-by-class face-validity table: cosmetics (LEEF PALEO PAW), agriculture (UNITED AMERICAN HEMP), beverages (KANNABIER), restaurants, materials, even insurance products are all topped by hemp/CBD/cannabis-themed marks. The blockchain wave (2014–2018) sweeps Class 35 (OPENSEA), Class 41 (KINGSLAND SCHOOL OF BLOCKCHAIN), and Class 36 (MOAIC TOKEN). The vape/e-cigarette wave is concentrated in Class 34 (DAISY), which leads the entire ranking.

Second, the most-and-least innovative industries are surprising. Tobacco & Smokers’ Articles (+0.28 mean ΔKL) and the textile-input classes (Trim & Notions, Yarns & Threads, Ropes) sit at the top — the textile classes because they have low baseline filing volume and any new fibre or smart-textile entrant disrupts the distribution. The bottom of the list is dominated by

mature regulated industries: Telecommunications, Insurance & Financial, Building Materials, and Agriculture. Neither end is what one would have predicted in advance.

Third, the per-industry survival betas (Figure 25) show that the “novelty-is-penalised-at-the-registration-gate” result reported in Section 11 is *not* universal. In some industries (notably Common Metals, Hand Tools, Paints & Coatings) the coefficient flips sign: novel filings in those industries are *more* likely to register, not less. A plausible reading is that in industries where category-creation is rare and meaningful, an examiner reading a novel goods/services description perceives a real new use rather than vague language. Software (and pharmaceuticals, chemicals) follow the opposite pattern: novel filings face slower examination, more office actions, and lower registration rates, consistent with examiners pushing back on speculative or boilerplate broadenings of established categories.

Do industries in flux earn supernumerary returns? Across the 44 NICE classes with adequate data, we computed four flux measures per class (mean KL_{pros} , mean ΔKL , std ΔKL , mean $|\Delta KL|$) and four financial outcomes (median gross margin, median operating margin, median 4y excess return vs. S&P 500, median 5y excess return). Excess return is the firm-mean over firm-years in our matched panel, $\text{ret}_{4y} - \text{sp500_ret}_{4y}$, aggregated as a class-level median to suppress single-firm outliers. We use median rather than mean because a handful of small public firms in mechanical-tool industries showed outsized returns that distorted class means by orders of magnitude.

The answer is no. The cross-industry Pearson correlation between flux intensity (mean $|\Delta KL|$) and median 4y excess return is -0.09 (Spearman $+0.01$, $n = 42$ industries with returns data). For 5y excess return: Pearson -0.05 , Spearman $+0.02$. The mean prospective KL is similarly uncorrelated with returns. Industries in flux do not generate excess returns over the S&P 500 in this corpus.

The cross-industry pattern that *is* present is on margins, not on returns: industries with higher flux have systematically *lower* median gross margins (Pearson -0.22 , Spearman -0.20 , for mean $|\Delta KL|$). High-flux industries are less profitable per dollar of revenue on average, consistent with such industries being more competitive (entry-heavy, churn-heavy). The market appears to price in the lower margins; investors are not compensated with excess returns for taking on flux risk at the industry level. Table 30 reports the cross-industry correlation matrix; Figure 26 plots the four key scatter pairs.

Reconciliation with the firm-level result. The firm-level U-shape on ΔKL (Section 17) says that *within* an industry, firms with flux-extreme vocabulary outperform the flux-neutral middle on survival. The industry-level null on excess returns is consistent with this: a high-flux industry contains both flux-extreme winners and flux-neutral losers, and after the within-industry sorting plays out the industry-mean return is approximately flat. The flux signal is a within-industry selection statistic, not an across-industry valuation premium. “Invest in the high-flux industry” is not a strategy; “invest in the flux-extreme firms within an industry” might be.

20 What is the construct, really?

We have called ΔKL an “innovation” score throughout, but the empirical pattern this paper documents is consistent with a plainer description.

The two-way pattern — both extremes outperform the middle, both on survival (Section 17) and on financial variance (Section 12) — is consistent with the construct measuring *how unconventional the firm’s bets are*, in either direction. Filings with very high ΔKL are unconventional in the direction of category creation; filings with very low ΔKL are unconventional in the direction of

templated boilerplate that examiners and competitors leave alone. Filings with $\Delta KL \approx 0$ are doing what the rest of the industry was already doing, and unsurprisingly that is what the data predict for them: average outcomes, average failure rate, average margins, average returns.

If we step back from the language of “innovation” and ask only what the construct actually correlates with:

- It is essentially uncorrelated with patent counts ($r = +0.01$ in the matched panel) and with Crunchbase total funding ($r = 0.00$); modestly correlated with R&D-to-revenue ratio ($r = +0.13$).
- It picks out qualitatively different firms than those proxies pick out (Section 12’s patent comparison).
- It predicts firm-year stock returns above the S&P 500 on a horizon of two-to-five years (Section 15); on the firm-debut variant the peak is at four years and the magnitude is approximately +28 percentage points per standard deviation. The peak is several years out, not immediate.
- It also predicts within-firm gross-margin variance (Section 12). Firms whose recent ΔKL is high earn higher mean margins *and* spend more years far from those means.

These properties are consistent with a simple summary: **the construct measures the smartness, and the riskiness, of the bets a firm makes when it names its products.** Average bets (ΔKL near zero) earn average outcomes. Bets that depart from convention in either direction get rewarded or punished above-average; the rewards take several years to arrive, and the variance of outcomes is correspondingly larger. Firms with novel bets that the rest of the industry then imitates — the high- ΔKL filings — are the ones that capture the rewards in the right tail; firms with novel bets nobody adopts pay the cost in the left tail. Boilerplate filings (ΔKL very negative) are a different kind of bet: a bet that the existing language is good enough, that defensibility is more important than category creation. The data say that bet also works, but on a flatter return distribution.

This is, at the construct level, an evidentiary claim that *ideas matter*, in a fairly literal sense: a firm whose trademark filings depart from the industry-mean vocabulary in any direction has measurably different financial dynamics than one whose filings hew to the centre. The five-year half-life of the return effect suggests that good ideas need time to ferment; imitators have to recognise the new category and copy it before the originator’s claim to it becomes valuable. The sub-additivity between this measure and patent counts in the R^2 decomposition (Section 12) means we are not just re-discovering patents in a different guise. Whatever this construct is measuring, it is measuring something separately predictive of firm performance.

21 Discussion

Flux, not distinctiveness; harvest and innovation both work. The empirical pattern we document is best read in two parts. First, the component KLS — prospective and retrospective surprise on their own — behave as distinctiveness measures (their levels track each other at $\rho = 0.99$), and on distinctiveness alone the only U appears under the strict post-registration mortality criterion. Second, the *flux* signal — ΔKL , the residual after the common distinctiveness factor is removed — is U-shaped under *every* outcome operationalisation. Filings whose vocabulary is moving with the field’s direction of travel (positive ΔKL , “catching the emerging language”) and

filings whose vocabulary is moving against it (negative ΔKL , “harvesting the receding language”) both outperform the flux-neutral middle. The two extractive strategies are recognisable in the token data: the negative extreme picks out clearly sunsetting industries (typesetting in the early 1990s as desktop publishing arrived, the pre-platform hospitality bundle in the early 2000s as Expedia disintermediated, e-cigarette boilerplate of 2014–2015 just before the FDA Deeming Rule); the positive extreme picks out the legalisation+platform-shift waves (cannabis, crypto, NFTs, vape, plant-based, EV). What the data show is that *both strategies pay* — the trademark-vocabulary U is a Schumpeter-plus-real-options dual-strategy pattern, not an optimal-distinctiveness inverted-U.

Mechanism is not separable. Whether the survival lift at negative ΔKL reflects intelligent harvesting (firms recognising an industry transition and extracting from the legacy side), passive late-occupation (firms whose business naturally uses the receding vocabulary because they are doing the legacy thing), or selection on an unobservable correlate (firms that survive long enough to file legacy-vocabulary trademarks have already proved durable) is not separable in the trademark record. The unmistakable empirical pattern is that flux-neutral filings die at the highest rate in the corpus, in 27 of 44 industries, at the token level, and *even within the same firm*: the firm-persistence test (Section 17, falsification #6) confirms the U survives within firms that demonstrably abandoned some trademarks but not all. The signal is filing-level, not firm-level.

The “arrows in the back” resolution. At the trademark gate, novelty is *rewarded*, not penalised. The most prospectively surprising filings clear examination at higher rates and persist longer through the renewal cycle. The mousetrap pattern dominates the survival outcomes *at the filing level*. The pioneer-ladder analysis (Section 7) gives a more nuanced firm-level view: the very-first 10–50 adopters of category vocabulary do badly (the arrows), the fast-follower band at ranks 50–1,000 does best (the mousetrap-builders), and the mass late wave is uniformly worst. Lieberman & Montgomery (1988) and Suarez & Lanzolla (2007) anticipated this empirical pattern in the management literature; the trademark corpus provides large- n evidence at the level of business-description vocabulary.

The arrows-in-the-back signal does, however, show up financially. The matched panel of 1,597 public firms reveals a clear pioneer-bet structure: high-novelty firms simultaneously earn higher mean gross margins (Section 12) *and* display wider within-firm gross-margin standard deviations (Table 8). Some pioneers succeed; others do not; the survivors carry the average. At the trademark registrar’s office novelty is a positive signal, but in the marketplace it raises the variance of outcomes in addition to the mean.

To make the underlying mechanics concrete, Table 31 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (**generative**, **artificial**, **seltzer**, **cannabis**) that are common in the future corpus but rare in the look-behind window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

Per-filing outcomes vs prospective and retrospective KL — software (Class 9)
per-filing $\rho(\text{pros}, \text{retr}) = +0.967$

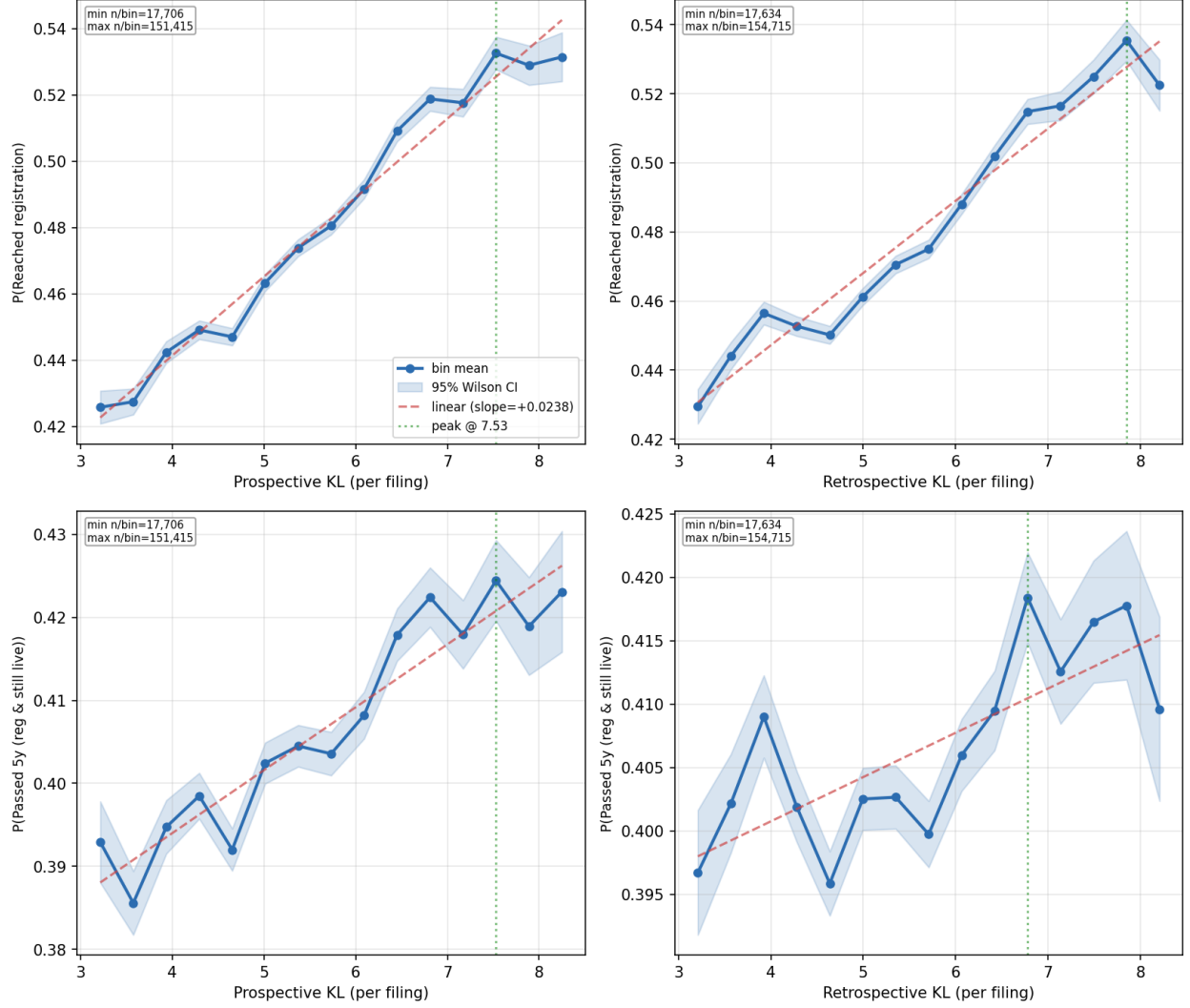


Figure 3: Per-filing outcomes vs prospective and retrospective KL — software (Class 9). Top row: probability the filing reaches registration. Bottom row: probability the filing reaches registration AND remains live today. Left column conditions on prospective KL, right column on retrospective KL. Each blue point is the bin mean (15 bins between the 2nd and 98th percentile of the per-filing predictor); shaded band is the 95% Wilson interval around each bin's proportion; red dashed line is the linear fit. The two columns are essentially the same curve — per-filing $\rho(KL_{\text{pros}}, KL_{\text{retr}}) = +0.97$.

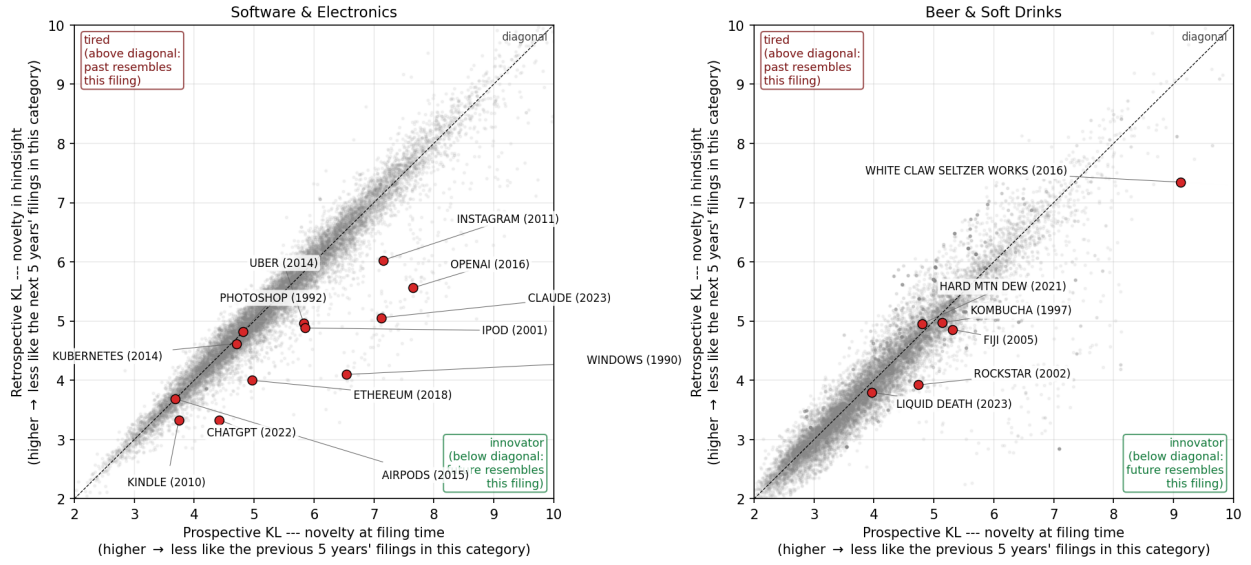


Figure 4: Per-filing prospective vs. retrospective KL surprise, zoomed to the central [2, 10] nat range where the labelled examples sit. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). **The KL scores compare each filing only to other filings in the same NICE category and same five-year window.** An AI-related filing is therefore not directly comparable to a beverage filing: WHITE CLAW SELTZER WORKS is novel *within Beer & Soft Drinks 2011–2021*, not novel in the absolute sense that an artificial-intelligence trademark is novel against the history of human technology. Below-diagonal: innovator (the future of the same category resembles this filing more than the past did). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

Table 3: Firm-mean and first-real-filing KL profile for 41 canonical “winner” firms and the 17 widely-recognised failed firms with sufficient post-founding clean-panel data. Filings filtered to filing-date $\geq$ founding-year; see entity-matching caveat above. Corpus median pros = 4.85, retr = 4.84.

cohort	brand	founded	n	mn_pros	mn_retr	mn_ΔKL	1st_pros	1st_retr	1st_ΔKL
WIN	Amazon	1994	75	4.75	4.67	+0.075	5.08	5.88	-0.79
WIN	eBay	1995	286	4.64	4.60	+0.037	7.45	5.72	+1.73
WIN	Etsy	2005	58	3.72	3.73	-0.013	5.42	5.48	-0.05
WIN	Shopify	2006	284	3.40	3.32	+0.080	5.98	5.88	+0.10
WIN	Wayfair	2002	689	4.34	4.35	-0.011	5.97	6.23	-0.26
WIN	Chewy	2011	235	4.86	4.73	+0.133	4.80	5.00	-0.20
WIN	Google	1998	2,826	4.40	4.32	+0.080	4.57	4.23	+0.33
WIN	Yahoo	1994	826	4.33	4.26	+0.068	5.92	5.86	+0.06
WIN	Netflix	1997	179	3.83	3.81	+0.016	4.40	4.17	+0.23
WIN	Dropbox	2007	81	4.30	4.28	+0.020	4.11	3.82	+0.30
WIN	Facebook/Meta	2004	1,063	3.68	3.57	+0.116	4.28	4.12	+0.16
WIN	Twitter/X	2006	420	3.73	3.70	+0.037	4.30	3.55	+0.75
WIN	LinkedIn	2002	286	4.09	4.05	+0.047	6.54	5.96	+0.58
WIN	Pinterest	2010	44	3.44	3.35	+0.089	5.05	4.74	+0.32
WIN	Snap	2011	441	4.01	3.96	+0.050	4.32	4.41	-0.09
WIN	Coinbase	2012	147	4.33	4.16	+0.161	6.48	5.50	+0.98
WIN	Block/Square	2009	376	4.49	4.46	+0.038	5.73	5.48	+0.26
WIN	Stripe	2010	114	4.89	4.83	+0.063	5.45	5.55	-0.11
WIN	Zoom	2011	18	4.88	4.82	+0.063	5.10	5.07	+0.03
WIN	Slack	2009	1	5.54	5.30	+0.243	5.54	5.30	+0.24
WIN	Salesforce	1999	884	4.09	4.01	+0.084	4.22	3.79	+0.43
WIN	Workday	2005	77	4.15	4.02	+0.133	5.73	5.53	+0.20
WIN	ServiceNow	2004	148	4.16	4.06	+0.098	4.26	4.31	-0.05
WIN	Snowflake	2012	125	4.35	4.26	+0.098	4.54	4.32	+0.22
WIN	Tesla	2003	225	4.91	4.86	+0.051	6.07	6.13	-0.06
WIN	Uber	2009	348	4.23	4.10	+0.139	6.45	5.75	+0.70
WIN	Airbnb	2008	171	4.25	4.18	+0.066	4.63	4.85	-0.22
WIN	Lyft	2012	220	4.44	4.34	+0.093	6.44	5.67	+0.76
WIN	Spotify	2006	271	4.07	4.03	+0.044	4.02	3.71	+0.31
WIN	Palantir	2003	81	3.86	3.76	+0.101	4.10	4.17	-0.07
WIN	Cisco	1984	1,551	4.50	4.47	+0.030	7.36	7.08	+0.28
WIN	Oracle	1977	615	4.35	4.29	+0.055	9.98	5.59	+4.39
WIN	SAP	1972	342	4.03	3.98	+0.051	5.30	5.24	+0.07
WIN	NVIDIA	1993	519	3.85	3.79	+0.060	6.27	6.18	+0.09
WIN	AMD	1969	315	5.32	5.29	+0.025	6.74	6.05	+0.69
WIN	Adobe	1982	696	4.67	4.70	-0.027	6.70	5.43	+1.28
WIN	Intuit	1983	905	4.30	4.23	+0.074	4.09	3.89	+0.21
WIN	Apple	1976	2,460	4.48	4.47	+0.017	7.19	5.51	+1.68
WIN	Microsoft	1975	4,171	4.18	4.17	+0.015	8.83	5.84	+2.99
WIN	Intel	1968	690	4.34	4.33	+0.003	7.95	4.70	+3.26
WIN	Qualcomm	1985	1,302	4.92	4.94	-0.013	5.17	4.93	+0.24
FAIL	Pets.com	1998	17	5.36	5.39	-0.032	4.54	4.55	-0.00
FAIL	Webvan	1996	32	4.47	4.69	-0.217	2.71	2.87	-0.15
FAIL	Theranos	2003	305	3.15	3.10	+0.045	3.86	3.78	+0.08
FAIL	MoviePass	2011	4	7.03	7.30	-0.266	7.03	7.30	-0.27
FAIL	WeWork	2010	313	4.24	4.20	+0.040	4.99	4.99	+0.00
FAIL	FTX	2019	1	4.34	4.26	+0.075	4.34	4.26	+0.08
FAIL	Solyndra	2005	2	6.87	6.43	+0.437	6.13	5.93	+0.21
FAIL	Bird	2017	4	5.27	5.25	+0.022	7.06	7.14	-0.07
FAIL	Casper	2014	147	4.57	4.51	+0.059	5.57	5.11	+0.46
FAIL	Better.com	2014	4	4.36	4.48	-0.119	4.60	4.81	-0.21
FAIL	Lordstown	2018	1	5.96	5.97	-0.016	5.96	5.97	-0.02
FAIL	Nikola	2014	49	4.39	4.36	+0.028	3.81	3.57	+0.24
FAIL	Beyond Meat	2009	136	4.89	4.56	+0.335	6.14	5.40	+0.74
FAIL	Peloton	2012	125	4.56	4.49	+0.066	3.89	3.83	+0.06
FAIL	Friendster	2002	13	4.39	4.35	+0.036	5.74	5.55	+0.19
FAIL	Myspace	2003	176	3.51	3.44	+0.071	5.32	5.25	+0.08
FAIL	Juicero	2013	1	4.42	4.20	+0.218	4.42	4.20	+0.22

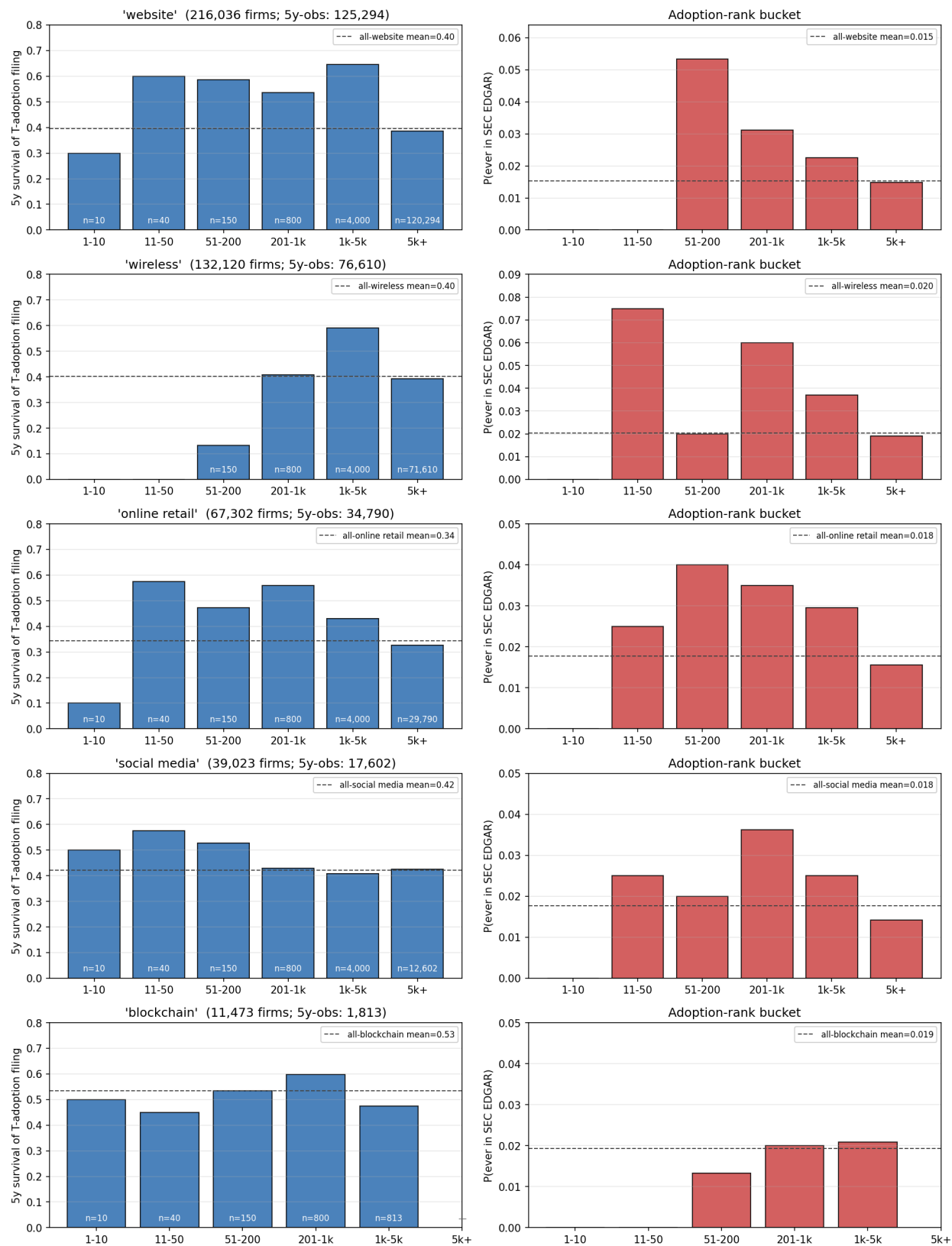


Figure 5: Adoption-rank bucket vs. 5-year survival rate (left of each pair) and probability that the owner is in SEC EDGAR (right of each pair), for five category-defining buzzwords.

Table 4: Tokens over-represented at the negative, neutral, and positive extremes of ΔKL . Each cell: token followed by survival rate across *all* filings containing the token, with enrichment lift in parentheses. Top 15 per stratum, ranked by lift; minimum token popularity 2,000 filings.

Negative ΔKL (harvest)	Neutral $\Delta KL \approx 0$	Positive ΔKL (innovate)
mens 67% (3.23 $\times$)	outsource 33% (1.91 $\times$)	non-fungible 63% (7.93 $\times$)
periodically 49% (3.09 $\times$)	colloquia 22% (1.90 $\times$)	nfts 60% (7.88 $\times$)
sunless 72% (3.00 $\times$)	systematisation 28% (1.90 $\times$)	authenticated 54% (6.68 $\times$)
microfiche 37% (2.99 $\times$)	systematization 23% (1.89 $\times$)	blockchain-based 60% (6.38 $\times$)
rejection 75% (2.99 $\times$)	institutes 36% (1.87 $\times$)	crypto 57% (5.93 $\times$)
womens 61% (2.91 $\times$)	organisational 15% (1.87 $\times$)	cryptocurrencies 57% (5.38 $\times$)
minicomputers 30% (2.86 $\times$)	organisations 23% (1.87 $\times$)	blockchain 55% (5.25 $\times$)
lager 55% (2.81 $\times$)	taxation 36% (1.86 $\times$)	cryptocurrency 55% (5.14 $\times$)
astrigent 70% (2.80 $\times$)	enquiries 32% (1.85 $\times$)	decentralized 54% (5.08 $\times$)
dialing 42% (2.64 $\times$)	chocolate-coated 33% (1.84 $\times$)	add-in 40% (4.18 $\times$)
dial-up 44% (2.59 $\times$)	cheese-based 46% (1.84 $\times$)	cryptographic 53% (4.17 $\times$)
proxy 40% (2.58 $\times$)	scheme 27% (1.83 $\times$)	repellers 36% (4.12 $\times$)
pager 46% (2.55 $\times$)	mergers 37% (1.82 $\times$)	animal-activated 37% (4.09 $\times$)
stout 50% (2.54 $\times$)	non-commercial 36% (1.81 $\times$)	cartomizers 56% (3.87 $\times$)
smartwatch 24% (2.52 $\times$)	freighting 21% (1.81 $\times$)	weaves 33% (3.86 $\times$)

Table 5: Logistic regressions of each survival outcome on ΔKL (novelty). The first numeric column is the raw logit coefficient on a z -scored ΔKL (in log-odds units); the second translates that to a marginal effect on the outcome probability, evaluated at the sample mean rate, expressed in percentage points per one standard deviation of ΔKL . Standard errors in parentheses. Industry sub-headings group the three outcomes per industry.

Outcome	Cohort	N	Base rate	$\beta_{\Delta KL}^z$ (logit, log-odds)	Marginal effect (pp per σ)
<i>Software & Electronics</i>					
Completed registration	all cohorts	1,146,078	49.4%	+0.038 (0.002)	+0.94 (0.05) pp
Renewed 5y registration	filed $\leq$ 2020	1,146,078	49.4%	+0.038 (0.002)	+0.94 (0.05) pp
Renewed 10y registration	filed $\leq$ 2015	1,146,078	41.3%	+0.119 (0.002)	+2.89 (0.05) pp
<i>Beer & Soft Drinks</i>					
Completed registration	all cohorts	94,920	56.8%	+0.036 (0.007)	+0.88 (0.17) pp
Renewed 5y registration	filed $\leq$ 2020	94,920	56.8%	+0.036 (0.007)	+0.88 (0.17) pp
Renewed 10y registration	filed $\leq$ 2015	94,920	44.3%	+0.081 (0.007)	+2.01 (0.18) pp

Table 6: Decomposition of registration failures by mode.

Industry	Failure mode	N	% of fails	Mean ΔKL
<i>Software & Electronics</i> ($N=1,054,572$; reg. rate 50%; mean ΔKL of registered = -0.002)				
	No Statement of Use	339,456	64%	-0.005
	No response to office action	102,195	19%	-0.036
	Other	45,338	9%	-0.006
	Examiner final refusal	44,576	8%	-0.013
<i>Beer & Soft Drinks</i> ($N=87,849$; reg. rate 57%; mean ΔKL of registered = -0.028)				
	No Statement of Use	21,860	58%	-0.080
	No response to office action	6,798	18%	+0.049
	Other	5,437	14%	-0.005
	Examiner final refusal	3,796	10%	-0.034

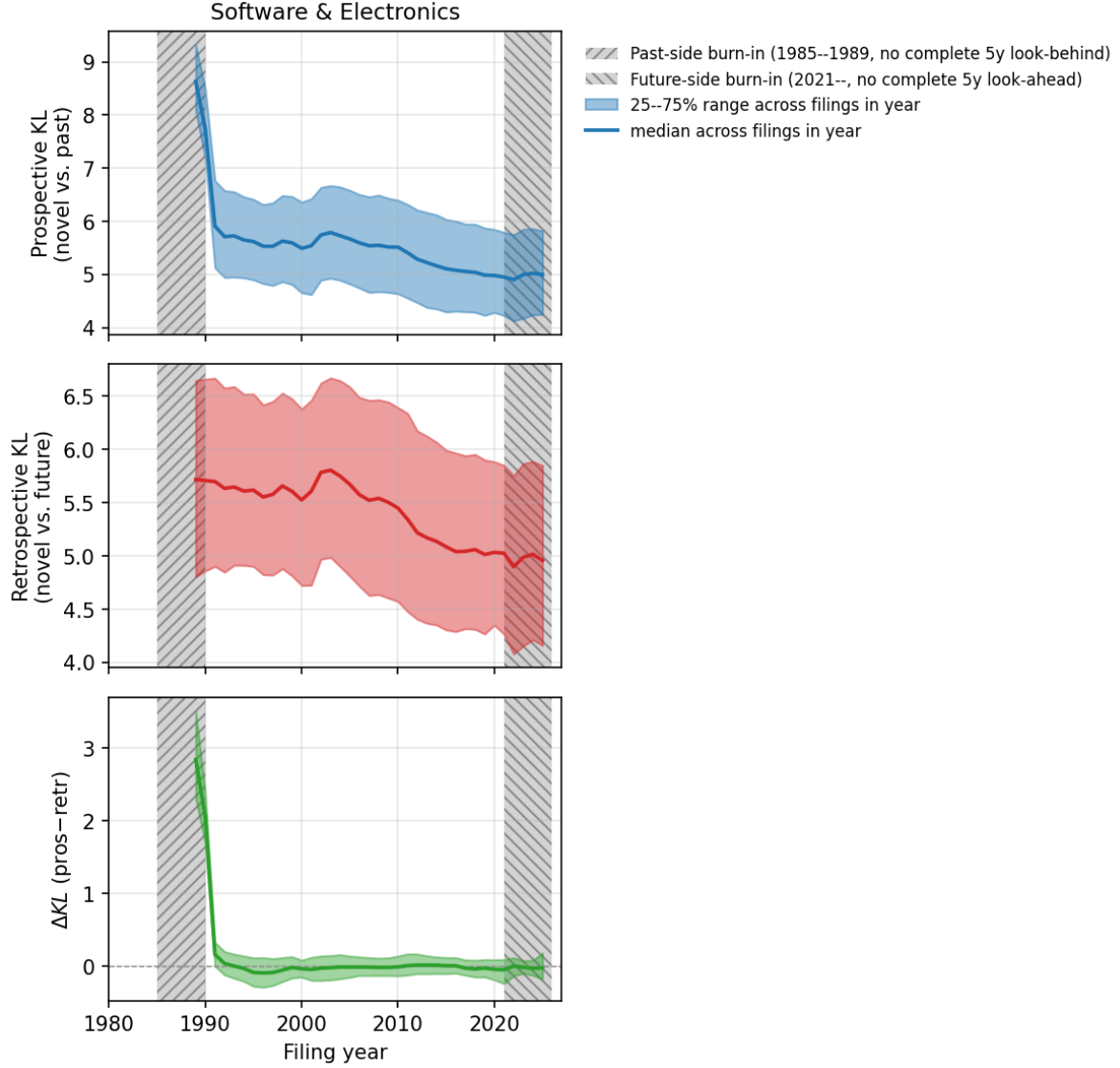


Figure 6: Software & Electronics: per-year median (line) and 25–75% range across all filings in that year (shaded coloured band) of prospective KL, retrospective KL, and ΔKL . The hatched grey regions are the burn-in windows: forward-hatched for past-side (1985–1989, where the 5-year look-behind reaches into pre-corpus data and produces inflated prospective KL — visible as the spike at 1988), backward-hatched for future-side (2021 onward, where the 5-year look-ahead has not yet completed). Both bands are excluded from the regressions.

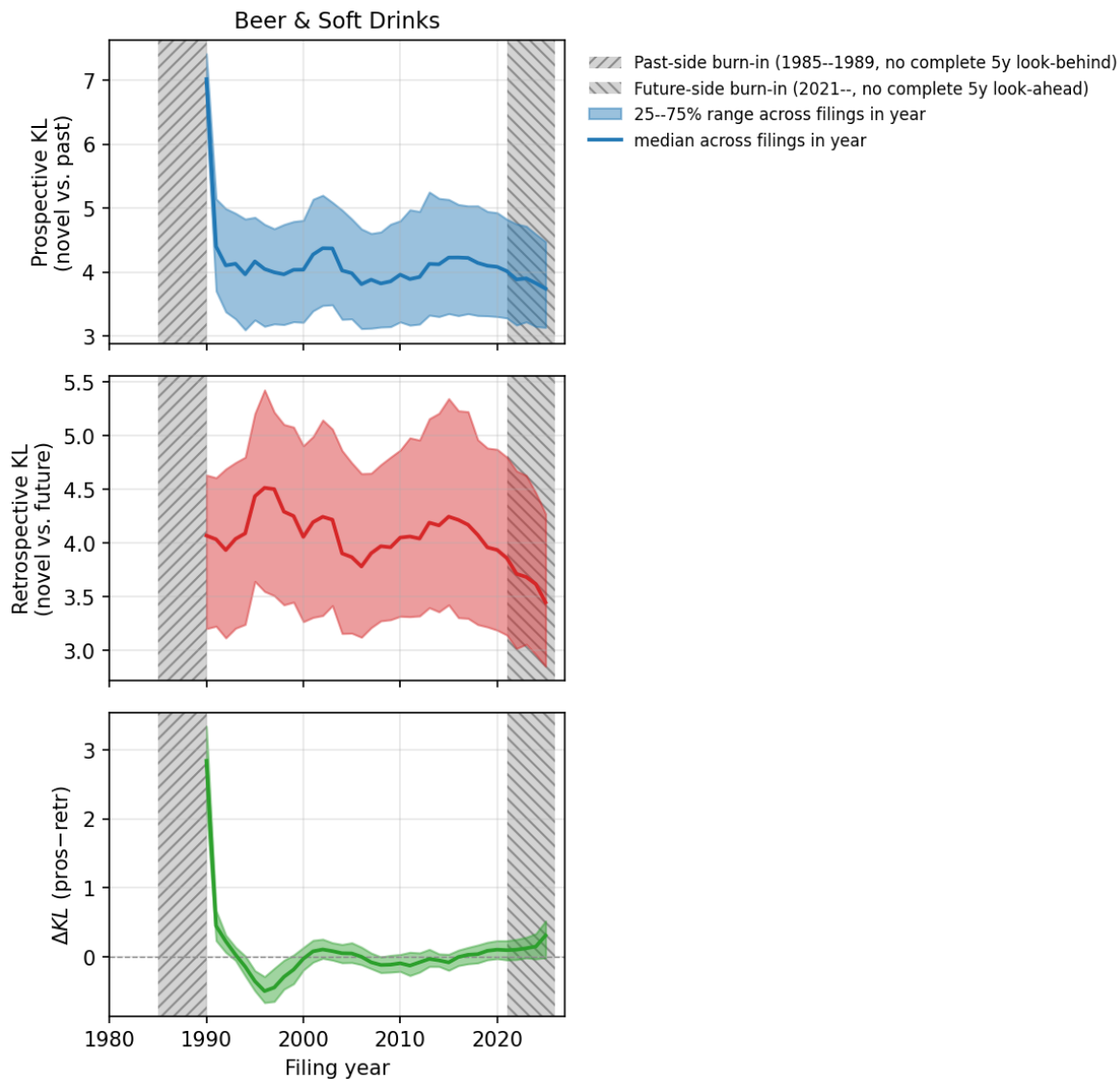


Figure 7: Beer & Soft Drinks: per-year median (line) and 25–75% range across filings in that year (shaded coloured band) of prospective KL, retrospective KL, and ΔKL . Hatched grey regions: 5-year past-side and future-side burn-in windows (described in Figure 6’s caption). The 1988 prospective-KL spike visible in both classes is an arithmetic artifact of the burn-in: 1988 filings have a look-behind window reaching 1983–1987, when the USPTO XML corpus contains only a few hundred filings per class. The reference distribution is dominated by Laplace smoothing on those few records, and any specific filing then looks far from it. The spike disappears as soon as the look-behind window clears 1989.

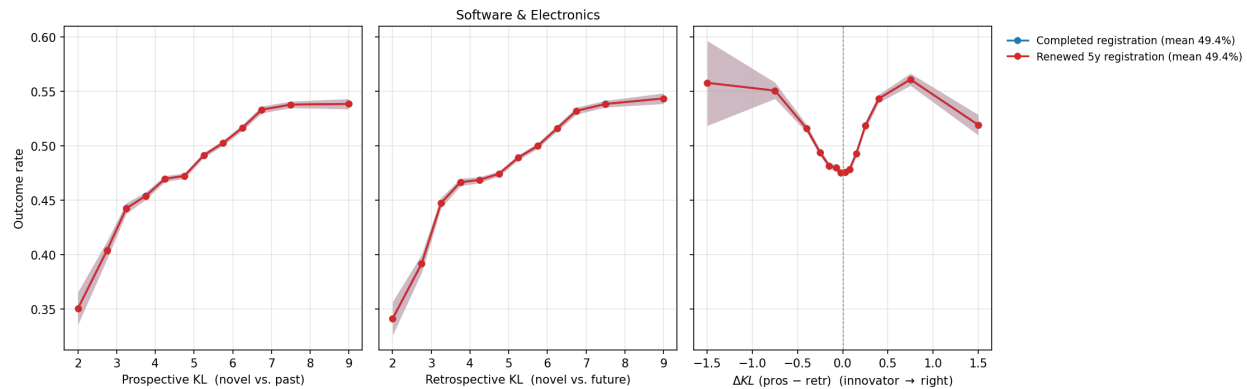


Figure 8: Software & Electronics: registration-and-renewal rates by KL bin. Shaded bands are 95% Wilson confidence intervals on the per-bin mean rate. Pattern across panels: high prospective KL (novelty against the past) penalises both registration and renewals; high retrospective KL (the wording remained unusual in hindsight) is neutral-to-favorable; the difference ΔKL inherits a downward slope.

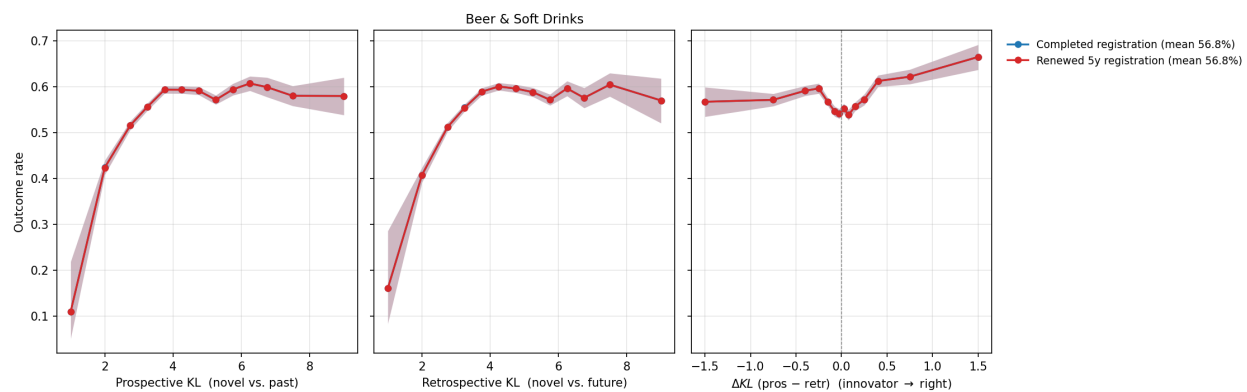


Figure 9: Beer & Soft Drinks: registration-and-renewal rates by KL bin. Shaded bands: 95% Wilson confidence intervals on the per-bin mean rate.

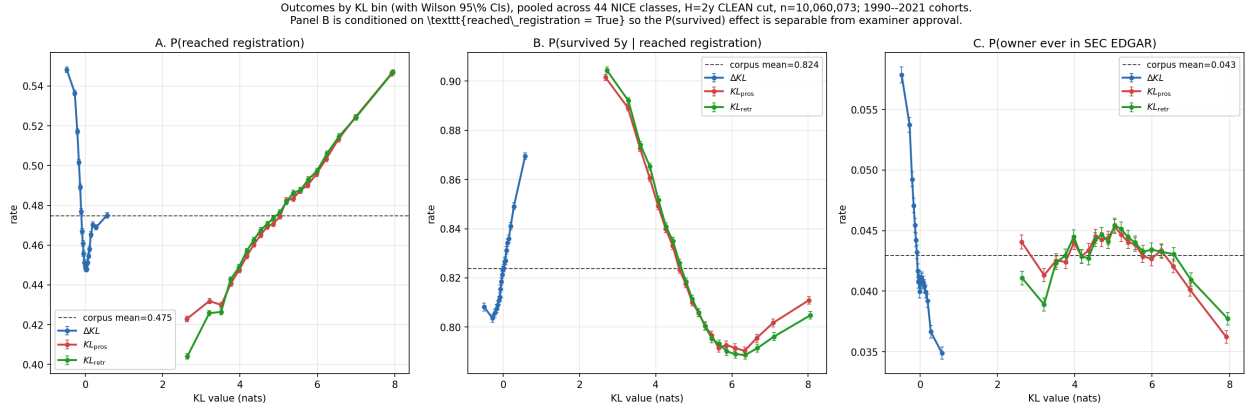


Figure 10: Three outcomes by ΔKL , KL_{gross} , and KL_{retr} bin (Wilson 95% CIs as error bars), pooled across all 44 NICE classes (n=10,060,073 clean H=2y filings, 1990–2021 cohorts). (A) $P(\text{reached registration})$ — examiner approval. (B) $P(\text{currently live at } 5y \mid \text{reached registration})$ — the substantive post-registration mortality outcome, properly conditioned on registration (the V1 figure had a different denominator and was less interpretable). (C) $P(\text{owner ever in SEC EDGAR})$. The supported finding: ΔKL (blue) is U-shaped on the substantive Panel B and on Panel C; the component KLs are monotone-increasing on examiner approval (statutorily explainable under 15 U.S.C. § 1052(d), where less-distinctive marks collide with prior art) and approximately monotone-increasing on conditional survival as well. The U on ΔKL is therefore not an examination-pipeline artefact: it survives the conditional cut.

Table 7: OLS regressions of profitability outcomes on trademark novelty, with cluster-robust standard errors at the firm level. Coefficients are in profitability-units per one standard deviation of the predictor (so a coefficient of +0.029 on gross margin means “+2.9 percentage points”). Each block is a separate specification.

Predictor (z-scored unless noted)	Coef.	SE	N
<i>Outcome: Gross margin</i>			
3y trailing mean novelty ΔKL	+0.038	0.006	5,249
3y trailing prospective surprise (novel vs. past)	+0.249	0.053	5,249
3y trailing retrospective surprise (novel vs. future)	-0.294	0.053	5,249
3y trailing max novelty	+0.037	0.006	5,249
3y trailing average filing count	+0.013	0.009	5,249
<i>Outcome: Net margin</i>			
3y trailing mean novelty ΔKL	-1.969	1.690	4,805
3y trailing prospective surprise (novel vs. past)	-13.252	12.988	4,805
3y trailing retrospective surprise (novel vs. future)	+15.570	14.159	4,805
3y trailing max novelty	-3.805	2.718	4,805
3y trailing average filing count	-2.168	1.782	4,805
<i>Outcome: Operating margin</i>			
3y trailing mean novelty ΔKL	-0.497	0.527	4,817
3y trailing prospective surprise (novel vs. past)	-4.335	4.089	4,817
3y trailing retrospective surprise (novel vs. future)	+4.424	4.411	4,817
3y trailing max novelty	-0.999	0.650	4,817
3y trailing average filing count	-0.590	0.371	4,817

Table 8: Variance regression: within-firm std. of gross margin on firm-mean novelty.

Predictor (z-scored unless noted)	Coef.	SE	N firms
3y trailing mean novelty ΔKL (firm avg)	+0.0176	0.0038	492
log revenue (firm avg)	-0.0141	0.0016	492

Table 9: $P(\text{ever-public})$ by mean- ΔKL quintile, on the 160,643 firms with ≥ 10 clean filings. The relationship is essentially flat: novelty does not predict reaching the SEC reporting universe; filing volume does.

$\overline{\Delta KL}$ quintile	n firms	mean $\overline{\Delta KL}$	mean filings	$P(\text{ever-public})$
Q1 (least innov.)	32,128	-0.179	25	0.051
Q2 ()	32,129	-0.058	30	0.048
Q3 ()	32,128	-0.004	31	0.050
Q4 ()	32,129	+0.056	28	0.050
Q5 (most innov.)	32,129	+0.248	23	0.044

Table 10: Correlation between $\log_{10}(\text{clean-panel trademark filings})$ and firm-value proxies, conditional on being publicly listed in SEC EDGAR. Trademark count predicts firm size strongly ($r = +0.38$) but predicts margin essentially zero.

Value metric	n firms	Pearson r	Spearman ρ
$\log_{10}(\text{max revenue})$	6,456	+0.385	+0.438
$\log_{10}(\text{median revenue})$	6,276	+0.375	+0.416
mean gross margin	4,237	+0.019	+0.059
mean operating margin	5,012	+0.045	+0.241
mean net margin	5,286	+0.045	+0.225
mean R&D intensity	2,366	-0.043	-0.173

Table 11: Within-firm temporal dynamics, all 45 NICE classes, 1995–2020.

Within-firm AR(1) of prospective KL	+0.680
Within-firm AR(1) of ΔKL (innovator score)	+0.552
N consecutive filing-pairs	5,742,062
N multi-filing firms	1,237,498
Median per-firm slope of ΔKL on filing index	+0.0000
Mean per-firm slope of ΔKL on filing index	-0.0007
Share of firms with positive slope	47.0%
N firms with ≥ 4 filings used for slopes	472,124

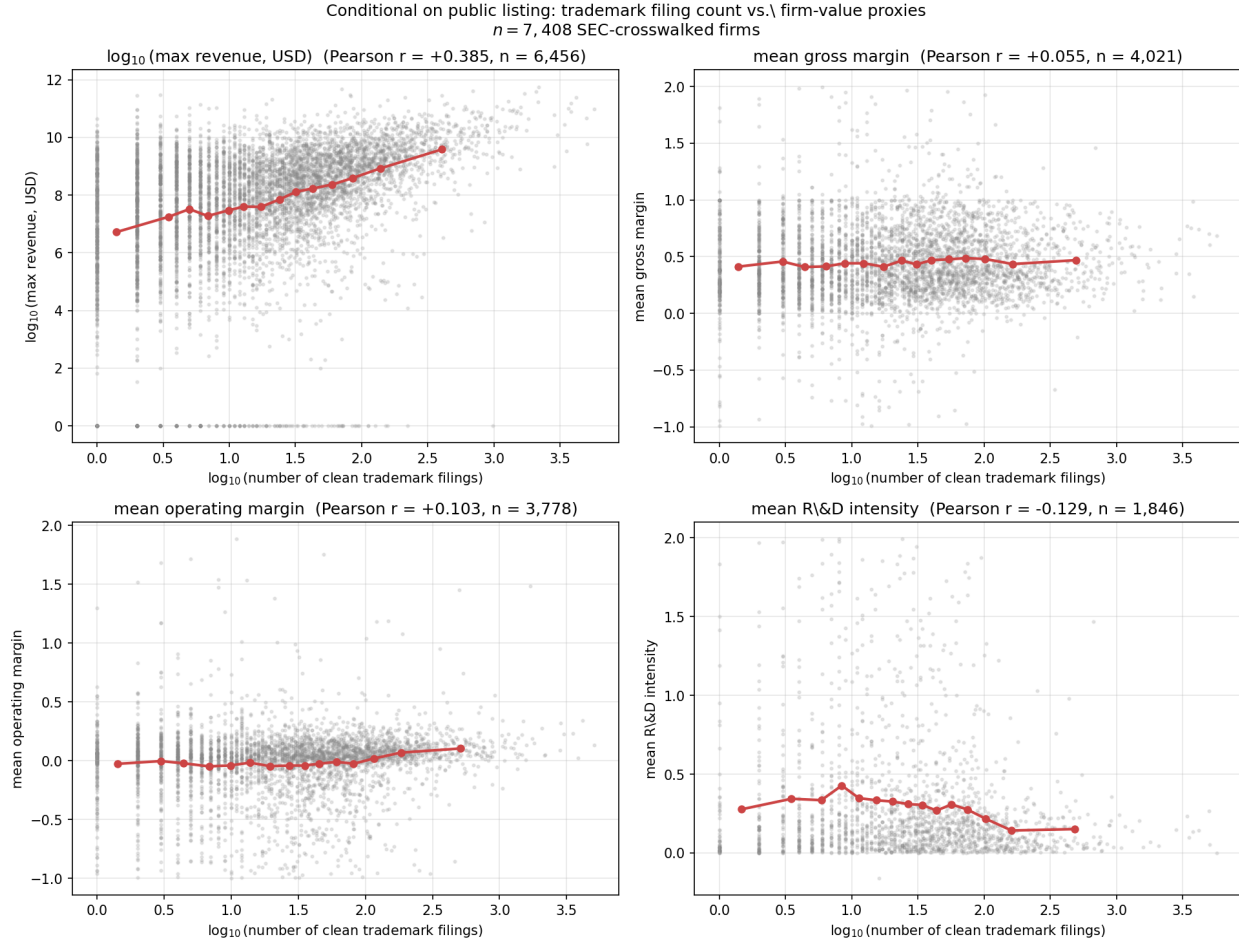


Figure 11: Trademark filing count vs. four firm-value proxies on the SEC-crosswalked panel. Each point is a CIK; the red line is a binned mean. Revenue scales with count strongly; gross and operating margins are flat; R&D intensity is mildly negative. **The patent “count $\leftrightarrow$ value” coupling does not transfer to trademarks: many marks indicates a large firm, not a more valuable one per unit size.**

Table 12: Debut-quartile dynamics. “Debut ΔKL ” is the average debut score in the quartile; “Subseq. mean ΔKL ” is the average across each firm’s filings AFTER the debut. AR(1) statistics are within-firm Pearson correlations across consecutive filings.

Quartile (by debut ΔKL)	N firms	Debut ΔKL	Subseq. mean ΔKL	AR(1) prospective	AR(1) ΔKL
Q1 (least novel)	309,375	-0.309	-0.098	+0.637	+0.485
Q2	309,374	-0.069	-0.034	+0.697	+0.417
Q3	309,374	+0.057	+0.015	+0.706	+0.418
Q4 (most novel)	309,375	+0.336	+0.106	+0.679	+0.572

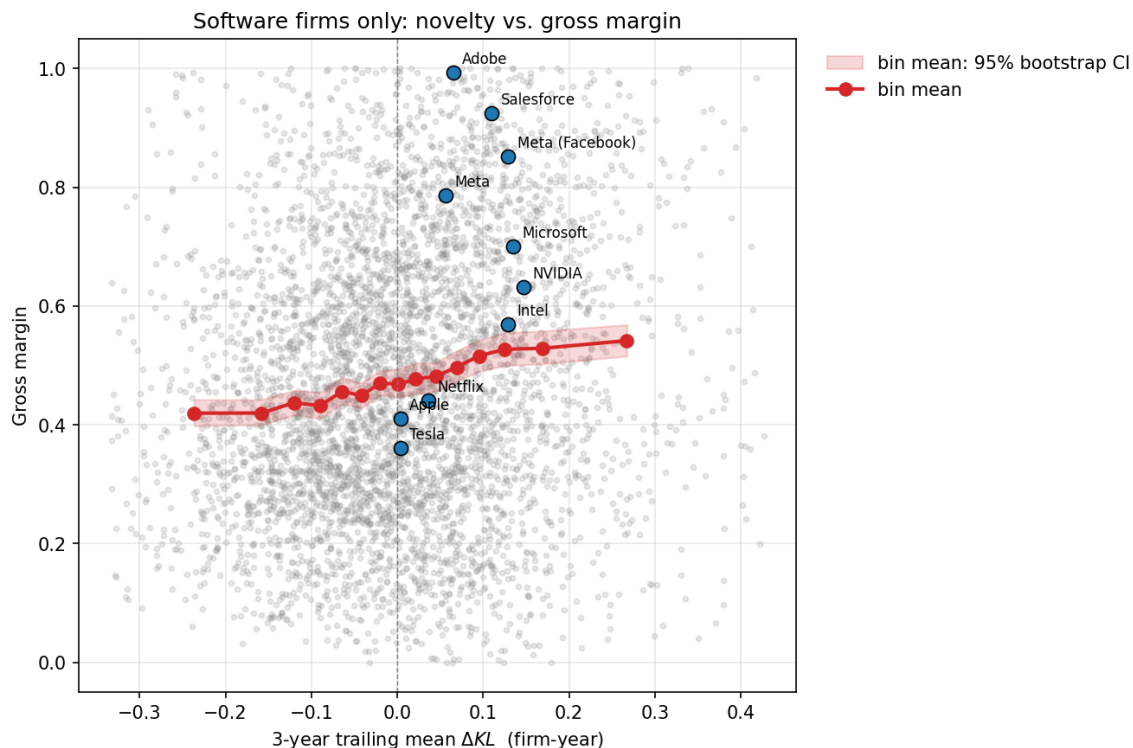


Figure 12: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL . Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL , gross margin clamped to $[0, 1]$). Red line: bin means; red shaded band: 95% bootstrap confidence interval on each bin’s mean gross margin. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

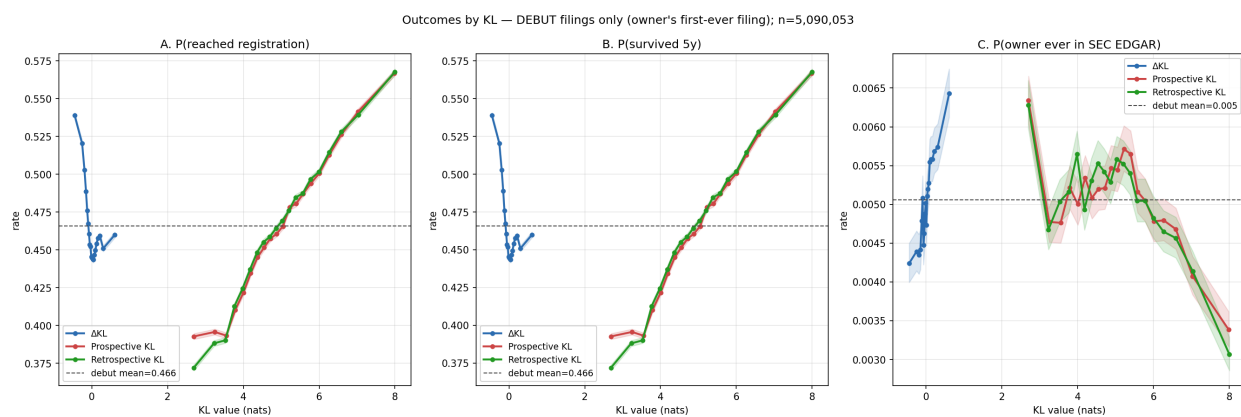


Figure 13: Three outcomes by KL, restricted to DEBUT filings. Each owner’s first-ever filing across all classes; $n = 5.1\text{M}$ debut filings in the clean $H=2\text{y}$ panel. Panel C is the key reversal: $P(\text{owner ever in SEC EDGAR})$ on debut filings is *increasing* in ΔKL , reversing the full-corpus direction (Figure 10). Pros and retr remain monotone-up on examiner approval and survived-5y; the debut-IPO flip is specific to the ΔKL direction.

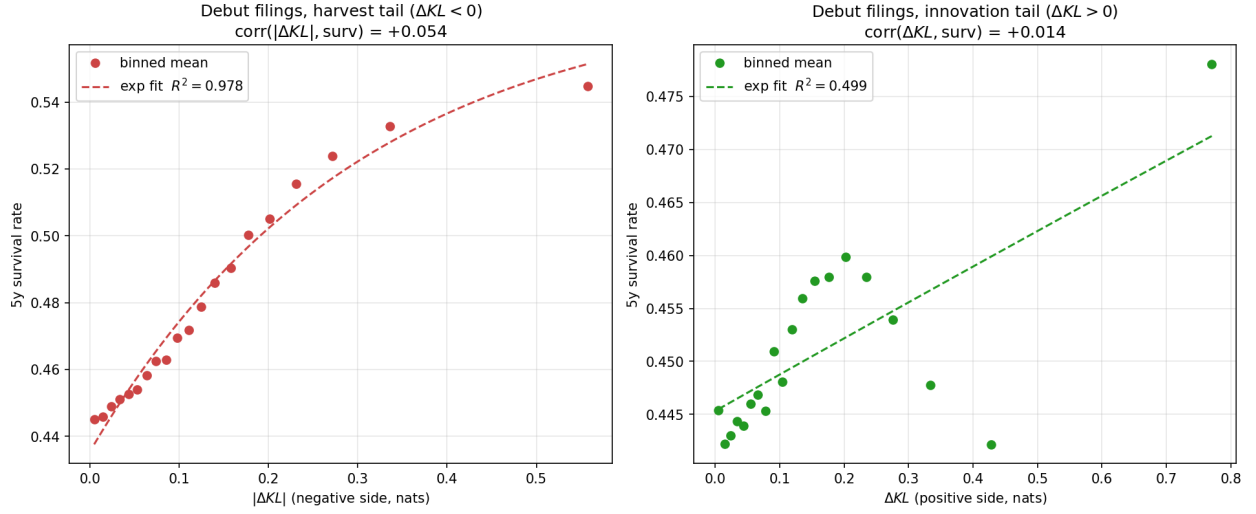


Figure 14: Saturating-exponential fits on each side of ΔKL , debut filings only. For each filing-side, the 5y survival rate by quantile-bin of $|\Delta KL|$ (20 bins), with the fit $y = a + b e^{-c|\Delta KL|}$ overlaid. Left: harvest tail ($\Delta KL < 0$), $R^2 = 0.978$, conditional $\text{corr}(|\Delta KL|, \text{surv}) = +0.054$. Right: innovation tail ($\Delta KL > 0$), $R^2 = 0.499$, conditional $\text{corr}(\Delta KL, \text{surv}) = +0.014$.

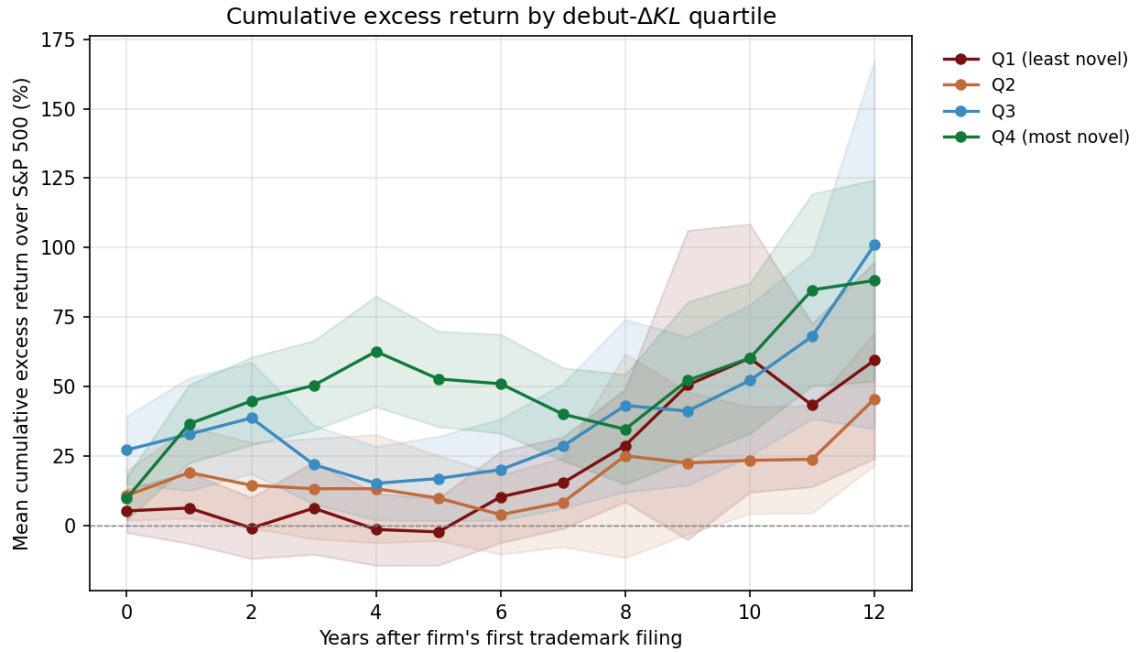


Figure 15: Mean cumulative excess return over the S&P 500 by years after the firm's first trademark filing, split by quartile of debut-filing ΔKL . Shaded bands are 95% standard-error ribbons; only year-since-debut bins with at least 50 firm-years are plotted. The most-novel-debut quartile (Q4, green) accumulates a substantial excess return relative to the median; the least-novel-debut quartile (Q1, dark red) underperforms the benchmark by year three.

Table 13: Stock-return regressions on novelty. Each row is a separate OLS regression with year linear and quadratic controls; cluster-robust standard errors at the firm level; outcome winsorised at p1 / p99. Coefficients are in return units per one σ of the predictor.

Outcome	N	β (per σ)	SE
<i>Debut filing only, predictor = ΔKL</i>			
excess vs. S&P 500, 1y	1,446	+0.0363	0.0184
raw 1y return	1,446	+0.0330	0.0186
excess vs. S&P 500, 2y	1,446	+0.0667	0.0272
raw 2y return	1,446	+0.0591	0.0269
excess vs. S&P 500, 3y	1,446	+0.1920	0.0552
raw 3y return	1,446	+0.1732	0.0555
excess vs. S&P 500, 4y	1,447	+0.2844	0.0861
raw 4y return	1,447	+0.2617	0.0872
excess vs. S&P 500, 5y	1,447	+0.1488	0.0652
raw 5y return	1,447	+0.1273	0.0659
<i>Debut filing only, predictor = Prospective KL</i>			
excess vs. S&P 500, 1y	1,446	+0.0008	0.0205
raw 1y return	1,446	+0.0019	0.0208
excess vs. S&P 500, 2y	1,446	-0.0048	0.0258
raw 2y return	1,446	-0.0092	0.0260
excess vs. S&P 500, 3y	1,446	-0.0144	0.0385
raw 3y return	1,446	-0.0166	0.0385
excess vs. S&P 500, 4y	1,447	-0.0181	0.0537
raw 4y return	1,447	-0.0198	0.0539
excess vs. S&P 500, 5y	1,447	+0.0572	0.0521
raw 5y return	1,447	+0.0554	0.0522
<i>Firm-year mean, predictor = ΔKL</i>			
excess vs. S&P 500, 1y	16,334	+0.0151	0.0051
raw 1y return	16,334	+0.0118	0.0053
excess vs. S&P 500, 2y	16,332	+0.0355	0.0092
raw 2y return	16,332	+0.0275	0.0095
excess vs. S&P 500, 3y	16,330	+0.0499	0.0134
raw 3y return	16,330	+0.0382	0.0140
excess vs. S&P 500, 4y	16,329	+0.0410	0.0164
raw 4y return	16,329	+0.0296	0.0167
excess vs. S&P 500, 5y	16,329	+0.0159	0.0203
raw 5y return	16,329	+0.0061	0.0205
<i>Firm-year mean, predictor = Prospective KL</i>			
excess vs. S&P 500, 1y	16,334	+0.0010	0.0045
raw 1y return	16,334	+0.0001	0.0046
excess vs. S&P 500, 2y	16,332	-0.0006	0.0079
raw 2y return	16,332	-0.0018	0.0081
excess vs. S&P 500, 3y	16,330	-0.0011	0.0107
raw 3y return	16,330	-0.0020	0.0109
excess vs. S&P 500, 4y	16,329	-0.0064	0.0149
raw 4y return	16,329	-0.0081	0.0150
excess vs. S&P 500, 5y	16,329	-0.0009	0.0194
raw 5y return	16,329	-0.0046	0.0194

Table 14: Per-industry Pearson correlation between firm-mean ΔKL and firm-mean annual excess return over the S&P 500. Asterisk marks correlations with $p < 0.05$. Industries are sorted by matched-firm count.

Industry (NICE)	N firms	r (ΔKL , mean excess)	p	r (pros KL, mean excess)	p
Scientific & Tech Services (042)	1,386	-0.057*	0.035	-0.063*	0.019
Software & Electronics (009)	1,146	-0.013	0.651	+0.047	0.114
Advertising & Retail (035)	1,098	+0.000	0.995	-0.018	0.556
Insurance & Financial (036)	741	+0.071	0.054	+0.145*	0.000
Education & Entertainment (041)	735	+0.028	0.441	+0.064	0.083
Pharmaceuticals (005)	604	+0.026	0.521	+0.066	0.107
Paper, Print & Office (016)	522	+0.050	0.251	+0.006	0.884
Construction & Repair (037)	518	+0.026	0.551	+0.021	0.632
Materials Treatment (040)	440	-0.036	0.457	+0.019	0.690
Telecommunications (038)	395	-0.066	0.191	+0.072	0.154
Medical & Personal Care (044)	383	-0.041	0.426	-0.059	0.247
Transport & Logistics (039)	379	+0.006	0.902	-0.094	0.066
Chemicals & Industrial (001)	346	-0.062	0.251	+0.117*	0.029
Medical Devices (010)	327	+0.069	0.211	+0.021	0.706
Clothing, Footwear & Headwear (025)	320	-0.014	0.799	+0.034	0.548
Legal & Personal Services (045)	319	-0.036	0.524	-0.023	0.686
Machines & Power Tools (007)	314	-0.037	0.512	+0.068	0.231
Lighting, HVAC & Appliances (011)	282	-0.032	0.596	+0.111	0.062
Common Metals (006)	275	+0.002	0.973	-0.010	0.875
Furniture (020)	247	-0.015	0.820	+0.101	0.114

Table 15: Excess-return coefficients with and without the mega-cap cohort (Apple, Microsoft, Amazon, Meta, Alphabet, Netflix, NVIDIA, Tesla). Each cell is one OLS regression; coefficients are excess return per σ of ΔKL , year-and-year-squared controls, cluster-robust SEs.

Horizon	Firm-year (full)	Firm-year (excl. mega-cap)	Debut (full)	Debut (excl. mega-cap)
1y	+0.0151 (0.0051)	+0.0141 (0.0051)	+0.0363 (0.0184)	+0.0362 (0.0184)
2y	+0.0355 (0.0092)	+0.0320 (0.0091)	+0.0667 (0.0272)	+0.0665 (0.0273)
3y	+0.0499 (0.0134)	+0.0444 (0.0131)	+0.1920 (0.0552)	+0.1908 (0.0553)
4y	+0.0410 (0.0164)	+0.0345 (0.0161)	+0.2844 (0.0861)	+0.2823 (0.0866)
5y	+0.0159 (0.0203)	+0.0045 (0.0195)	+0.1488 (0.0652)	+0.1486 (0.0654)

Table 16: Convergent-validity correlations between trademark ΔKL and patent counts on the matched panel. Column “N” is firm-years (top two rows) or firms (bottom two rows).

Aggregation	N	Pearson r
Firm-year, n_{patents}	174,569	+0.010
Firm-year, $\log(1 + n_{\text{patents}})$	174,569	-0.009
Firm total, sum patents	14,463	+0.008
Firm total, $\log(1 + \text{sum patents})$	14,463	-0.023

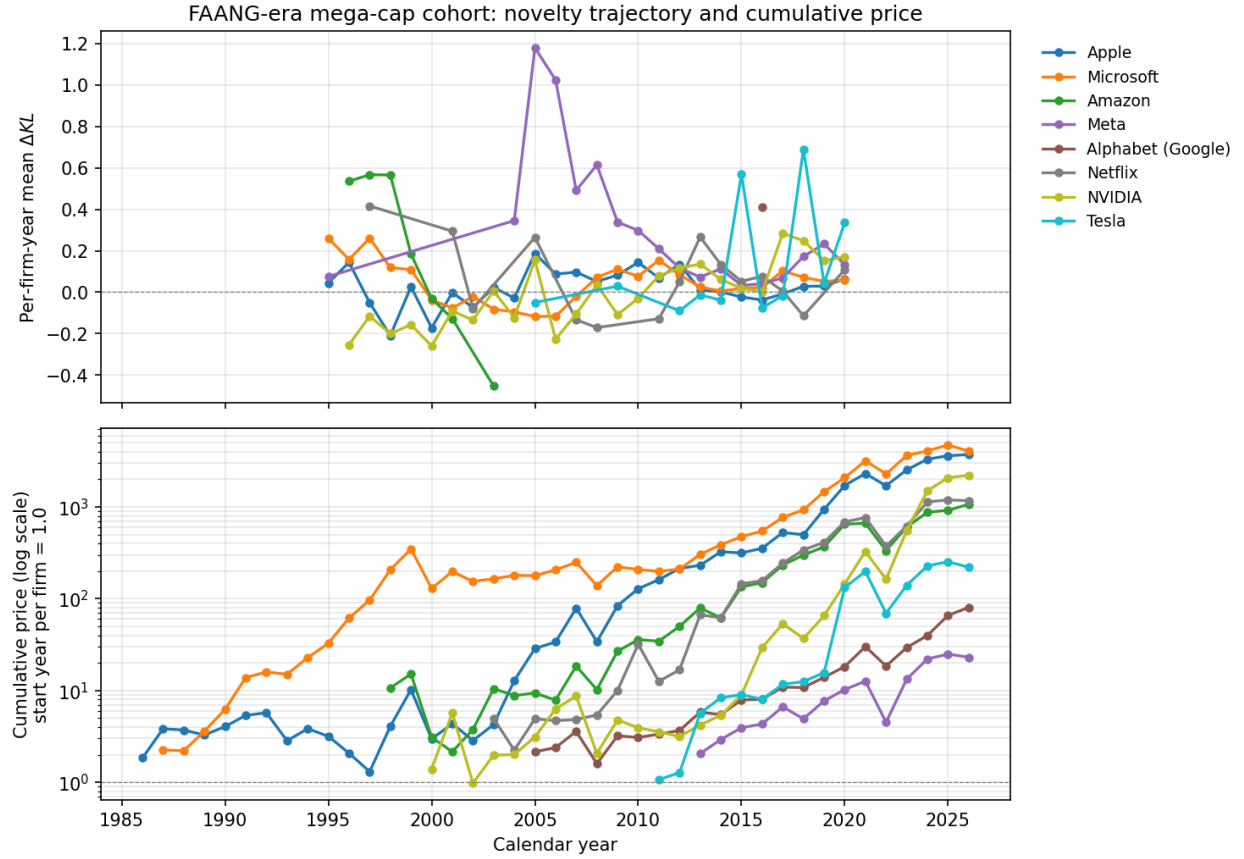


Figure 16: Per-year mean ΔKL (top) and cumulative end-of-year price normalised to first observation (bottom, log scale) for the mega-cap cohort. Tesla’s trademark portfolio shows the highest per-year ΔKL in the cohort, consistent with both its linguistic novelty and its outsized stock appreciation; NVIDIA shows elevated ΔKL in the AI era. Apple’s late-period ΔKL is comparatively flat (its categories are well-established) yet its cumulative return is among the highest — a reminder that this measure does not capture every channel through which firms create shareholder value.

Table 17: Excess-return regression with $\log(1 + n_{\text{patents}})$ as the firm-year predictor (z-scored). Same panel and specification as the firm-year row of Table 13.

Horizon	N	β for $\log(1 + \text{patents})_z$	SE
1y	19,686	+0.0203	0.0047
2y	18,553	+0.0429	0.0095
3y	17,394	+0.0591	0.0142
4y	16,245	+0.0902	0.0201
5y	15,141	+0.1206	0.0276

Table 18: Cumulative R^2 from each combination of predictors on excess return at horizon k , on the common panel of firm-years with both a patent count and a trademark KL score. All predictors z -scored; year linear and quadratic controls in every cell; cluster-robust SEs at the firm level (suppressed). The marginal contribution of each predictor is the difference from the controls-only baseline.

Specification (predictors, all z -scored)	1y R^2	2y R^2	3y R^2	4y R^2	5y R^2
Controls only	0.0056	0.0081	0.0129	0.0211	0.0276
Patents (P)	0.0062	0.0089	0.0145	0.0235	0.0306
Innovator dKL (I)	0.0075	0.0111	0.0161	0.0257	0.0312
Novelty pros (S)	0.0056	0.0081	0.0130	0.0212	0.0276
Patents + Innovator (P+I)	0.0081	0.0119	0.0176	0.0280	0.0342
Patents + Novelty (P+S)	0.0062	0.0090	0.0147	0.0236	0.0308
Innovator + Novelty (I+S)	0.0076	0.0111	0.0161	0.0257	0.0312
All three (P+I+S)	0.0082	0.0119	0.0177	0.0280	0.0342
N (firm-years)	8,118	8,116	8,114	8,114	8,112

Table 19: Trademark ΔKL versus four external public innovation comparators. R&D-to-revenue and Crunchbase total funding are continuous, so we report Pearson r . BCG Most Innovative Companies and MIT Technology Review 50 Smartest Companies are binary list memberships, so we report the difference in mean firm ΔKL between firms on the list and the rest of the matched universe (positive = listed firms are more innovative by our measure).

Comparator	Statistic	N matched	Value
R&D-to-revenue (firm-mean)	Pearson r	1,593	+0.0458
BCG Most Innovative (2021–23, union)	On-list vs off-list $\Delta \overline{\Delta KL}$	31	+0.0277
MIT TR 50 Smartest (2015)	On-list vs off-list $\Delta \overline{\Delta KL}$	17	+0.0648
Crunchbase total funding (log)	Pearson r	1,654	-0.0162

multiplicity	n (filings)	reg. %	still-live %	cancel-after-reg %
1 (unique)	4,978,977	48.6	38.1	10.5
2	1,090,984	50.3	38.7	11.6
3–5	726,072	53.9	42.6	11.3
6–10	262,438	57.1	46.5	10.6
11–50	337,624	56.2	46.0	10.2
51–200	216,545	54.2	44.7	9.5
201–1000	178,383	54.1	45.1	9.0
1001–5000	83,143	52.8	43.2	9.6
5000+	34,783	58.2	49.2	9.0

Table 20: Template-multiplicity bins of clean filings. “Reg.” is the share that completed registration; “still-live” is the share that both registered and is currently in 6xx live-registered status (the corrected post-registration survival outcome described in Section 11); “cancel-after-reg” is the difference, i.e. the share of registrations that were subsequently cancelled (predominantly Section 8 failures). Filing years 1990–2018 to ensure the Section 8 window has been adjudicated.

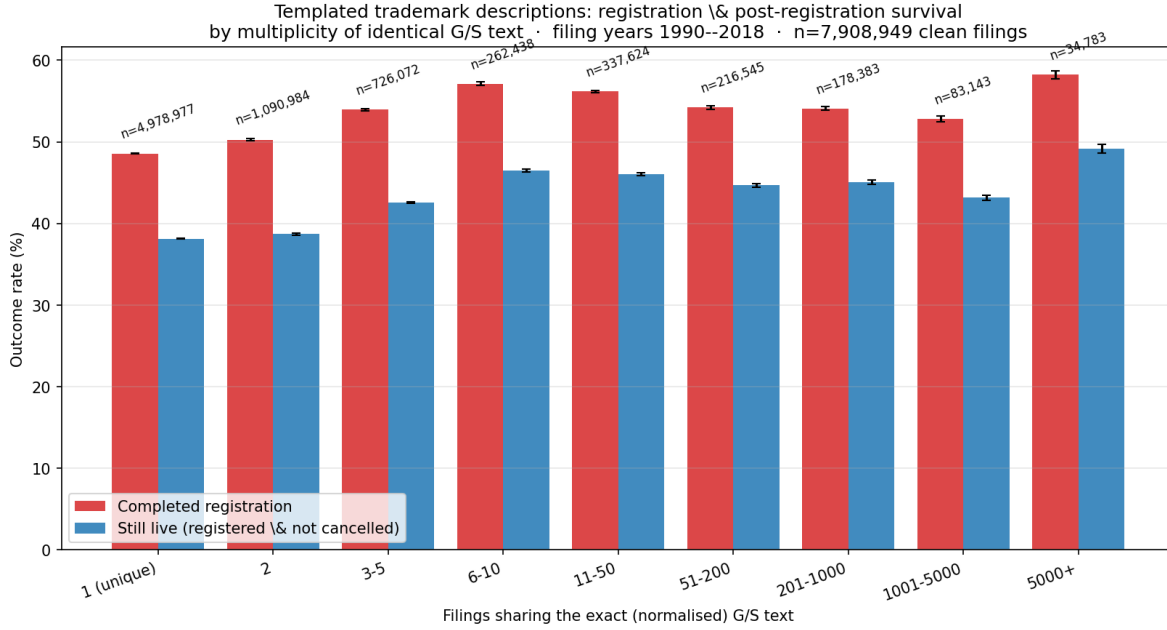


Figure 17: Filing-level registration (red) and still-live (blue) rates by template multiplicity. “Multiplicity” is the count of other clean filings sharing the exact same normalised goods/services text; filings in the leftmost bin have descriptions that no other filing copies verbatim. The most heavily templated descriptions (5000+ copies) have the highest survival rate.

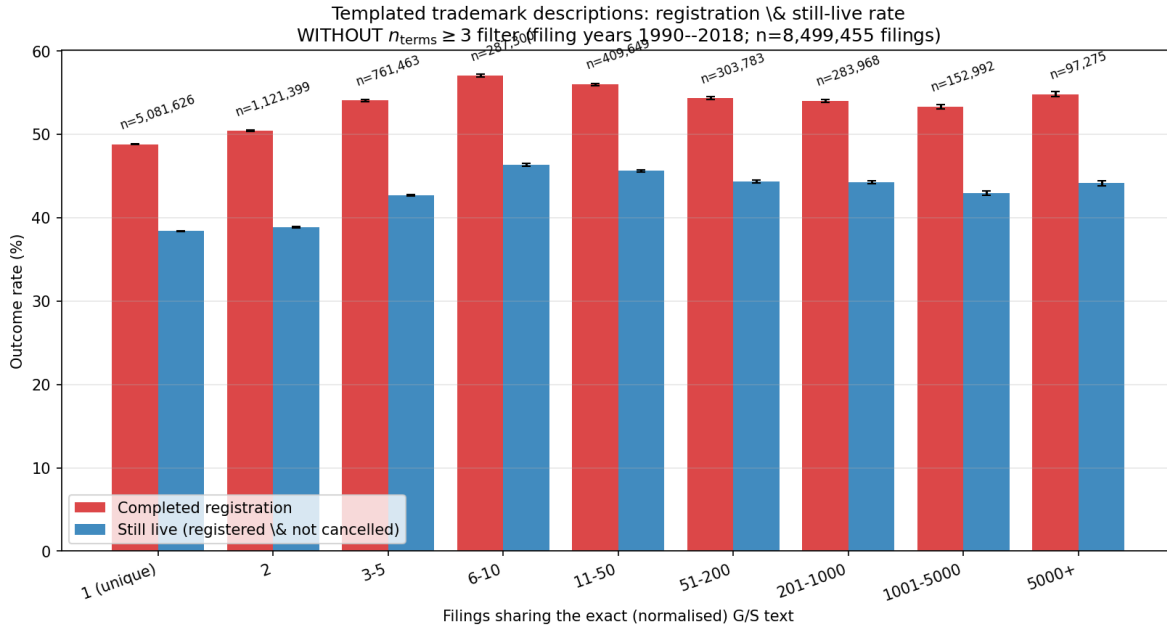


Figure 18: Robustness re-run of Figure 17 on CLEAN-without- $n_{\text{terms}} \geq 3$. The lower bins are visually identical; the 5000+ bin loses ~5pp because the short commodity templates that the n_{terms} filter excludes survive at roughly corpus-average rates, not at the elevated rates of the structured longer templates.

Table 21: Pooled quadratic logistic regression of survival outcomes on z -scored ΔKL and its square. Year and year-squared controls suppressed. A positive quadratic coefficient confirms concave-up (“U-shape”).

Outcome	N	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{(\Delta KL)^2}^z$
Completed registration	9,255,511	-0.0258 (0.0009)	+0.0053 (0.0002)
Renewed 5y registration	9,255,511	-0.0133 (0.0009)	+0.0002 (0.0002)
Renewed 10y registration	6,362,813	+0.0078 (0.0011)	+0.0013 (0.0002)

Statistic	count	% of total
Industries (of 44 with sufficient data)		
with significant $+U$ on ΔKL	32	73%
with significant inverted- U on ΔKL	1	2%
indeterminate	11	25%
Industry $\times$ year cells (of 1,267 total)		
significant $+U$ on ΔKL ($z > +1.96$)	338	27%
significant inverted- U ($z < -1.96$)	81	6%
indeterminate quadratic	848	67%
linear advantage $+\Delta KL$ ($z > +1.96$)	197	16%
linear advantage $-\Delta KL$ ($z < -1.96$)	484	38%
neither (linear $ z \leq 1.96$)	586	46%

Table 22: U -shape classification of 44 NICE classes individually (majority-of-years rule, margin = 2) and of the 1,267 (industry, year) cells where a logit fit converged on ≥ 500 filings. The pooled $+U$ holds in 32 of 44 industries; only 1 has a significant inverted- U (Telecommunications); the remaining 11 are indeterminate, not inverted- U . Cell-level: 27% have a significant $+U$ on ΔKL , 6% a significant inverted- U , 67% indeterminate. Linear-direction counting at the cell level shows a striking asymmetry: 38% of cells have a significant linear advantage to *negative* ΔKL (the harvest direction), versus 16% with a significant linear advantage to positive ΔKL .

Table 23: Per-industry quadratic coefficient on $z(\Delta KL)^2$ for 5y registration-renewal. Top 15 (most pronounced U) and bottom 5. Asterisk = $|t| > 1.96$.

Industry (NICE)	N	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{(\Delta KL)^2}^z$
Legal & Personal Services (045)	114,693	-0.0256	+0.0307*
Household Utensils (021)	183,800	-0.1586	+0.0258*
Materials Treatment (040)	87,017	+0.0204	+0.0234*
Furniture (020)	144,864	-0.1177	+0.0200*
Common Metals (006)	89,730	-0.1260	+0.0188*
Textiles & Bedding (024)	84,770	-0.1196	+0.0183*
Medical Devices (010)	174,201	-0.0694	+0.0183*
Ropes, Tents & Textiles (raw) (022)	21,551	-0.0989	+0.0183*
Rubber & Insulators (017)	48,186	-0.0901	+0.0169*
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166*
Medical & Personal Care (044)	167,607	+0.0084	+0.0158*
Firearms & Pyrotechnics (013)	12,432	-0.0171	+0.0141*
Leather & Luggage (018)	133,408	-0.1136	+0.0136*
Hand Tools (008)	62,591	-0.0693	+0.0115*
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105*
<i>...bottom 5 (most negative quadratic = inverted-U):</i>			
Telecommunications (038)	162,634	-0.0261	-0.0130*
Beer & Soft Drinks (032)	94,920	+0.0506	-0.0148*
Agriculture & Live Animals (031)	59,590	+0.0791	-0.0169*
Carpets & Wall Hangings (027)	22,964	-0.0152	-0.0232*
Musical Instruments (015)	11,068	-0.0761	-0.0525*

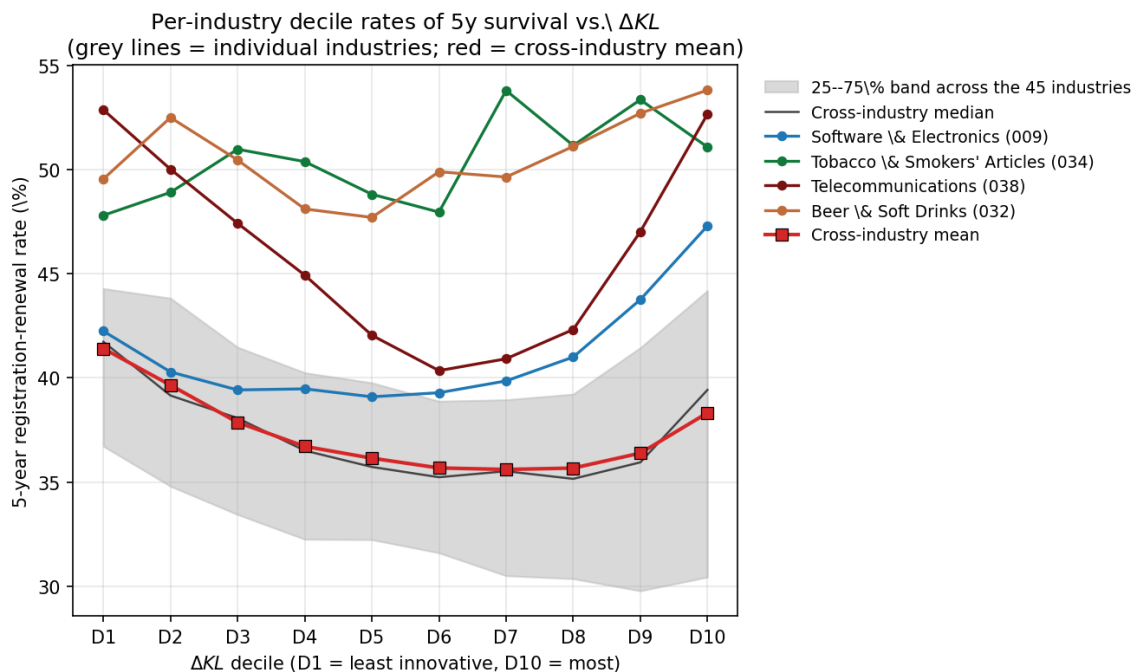


Figure 19: Per-decile 5y registration-renewal rate by ΔKL decile. Each grey line is one of the 45 NICE industries; red is the cross-industry mean. The U-shape is visible in nearly every industry.

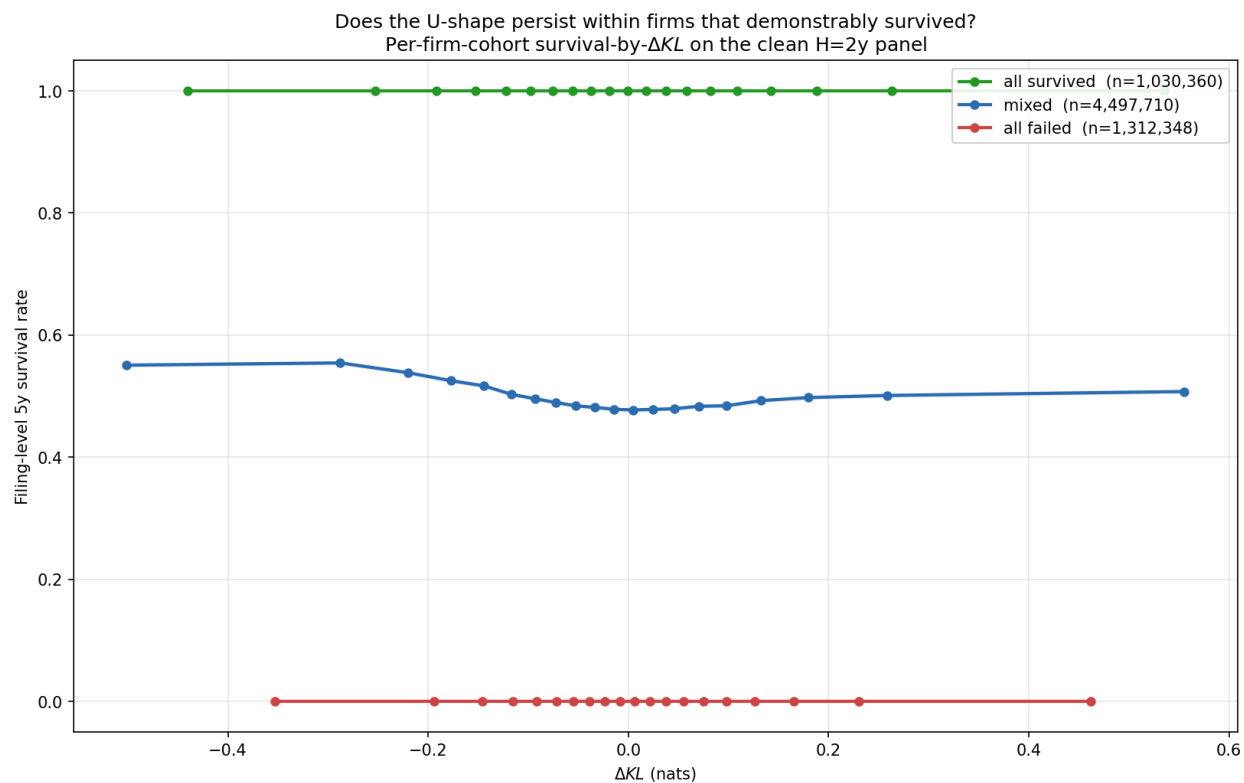


Figure 20: Filing-level 5y survival by ΔKL decile, separately for three firm-cohorts: *all-survived* (every filing made it), *mixed* (some filings were abandoned by the firm and some kept alive), and *never-survived* (no filings made it). The U is preserved in the mixed cohort. The signal is filing-level, not firm-level.

Table 24: Per-industry quadratic on 5y-survival, sorted by t on the quadratic coefficient (positive = U-shape, negative = inverted-U).

Industry (NICE)	N	$\beta_{\Delta KL}^z$ (linear)	$\beta_{\Delta KL^2}^z$ (quadratic)	t on quadratic
<i>Industries with the strongest U-shape (top 15 by t on $\beta_{\Delta KL^2}$)</i>				
Household Utensils (021)	183,800	-0.1586	+0.0258	+14.6
Legal & Personal Services (045)	114,693	-0.0256	+0.0307	+14.5
Materials Treatment (040)	87,017	+0.0204	+0.0234	+12.3
Medical Devices (010)	174,201	-0.0694	+0.0183	+10.9
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166	+9.7
Medical & Personal Care (044)	167,607	+0.0084	+0.0158	+9.4
Furniture (020)	144,864	-0.1177	+0.0200	+9.4
Common Metals (006)	89,730	-0.1260	+0.0188	+8.7
Leather & Luggage (018)	133,408	-0.1136	+0.0136	+8.5
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105	+8.2
Textiles & Bedding (024)	84,770	-0.1196	+0.0183	+7.8
Cosmetics & Cleaning (003)	300,326	-0.0388	+0.0074	+7.0
Lighting, HVAC & Appliances (011)	171,839	-0.0528	+0.0105	+6.2
Rubber & Insulators (017)	48,186	-0.0901	+0.0169	+5.3
Chemicals & Industrial (001)	118,548	+0.0147	+0.0094	+4.4
<i>Industries with the weakest or inverted U (bottom 5)</i>				
Telecommunications (038)	162,634	-0.0261	-0.0130	-10.6
Advertising & Retail (035)	924,501	+0.0143	-0.0048	-10.7
Scientific & Tech Services (042)	692,938	+0.0168	-0.0097	-12.9
Insurance & Financial (036)	374,634	+0.0541	-0.0122	-13.8
Software & Electronics (009)	1,146,078	+0.0715	-0.0107	-18.2

Table 25: Pooled quadratic logit of 5y survival on $z(\Delta KL)$ and $z(\Delta KL)^2$, with year and year-squared controls, re-fit after dropping different parts of the sample. The quadratic coefficient on ΔKL^2 is essentially unchanged across drop choices — the U-shape is a property of the central distribution, not of the extreme tails.

Sample	N	$\beta_{\Delta KL}^z$	$\beta_{\Delta KL^2}^z$	t on quad.
Full sample (none dropped)	9,258,400	-0.0134	+0.0002	+1.3
Drop bottom 5% of prospective KL ($\hat{\mu}$ 3.05)	8,795,480	-0.0067	-0.0003	-1.7
Drop bottom 5% of retrospective KL ($\hat{\mu}$ 3.05)	8,795,480	-0.0090	-0.0002	-1.0
Drop bottom 5% of both axes	8,726,416	-0.0058	-0.0007	-3.4
Drop bottom + top 10% of ΔKL (test extreme tails)	7,406,720	-0.0188	+0.0343	+51.2

Table 26: Summary of the falsification battery for the survival U-shape. The quadratic coefficient $\beta_{\Delta KL^2}^z$ stays strongly positive everywhere except in the financial outcome. Appendix C narrates each test.

Sub-sample	N	Outcome rate	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{\Delta KL^2}^z$
<i>Foreign-extension hypothesis</i>				
Full sample	600,000	39.2%	+0.0043 (0.0028)	+0.0089 (0.0008)
US-domiciled only (foreign proxy = False)	577,392	39.9%	+0.0043 (0.0028)	+0.0078 (0.0008)
Foreign-domiciled only (foreign proxy = True)	22,608	21.3%	+0.0741 (0.0165)	+0.0100 (0.0038)
<i>First-time vs recurring</i>				
First-time filings only	429,815	39.0%	+0.0019 (0.0033)	+0.0080 (0.0009)
Recurring (2nd+) filings only	170,185	39.6%	+0.0084 (0.0051)	+0.0140 (0.0016)
<i>Financial U-shape: firm-year gross margin</i>				
Public firms (gross margin OLS)	8,600	—	+0.0020 (0.0047)	-0.0022 (0.0012)

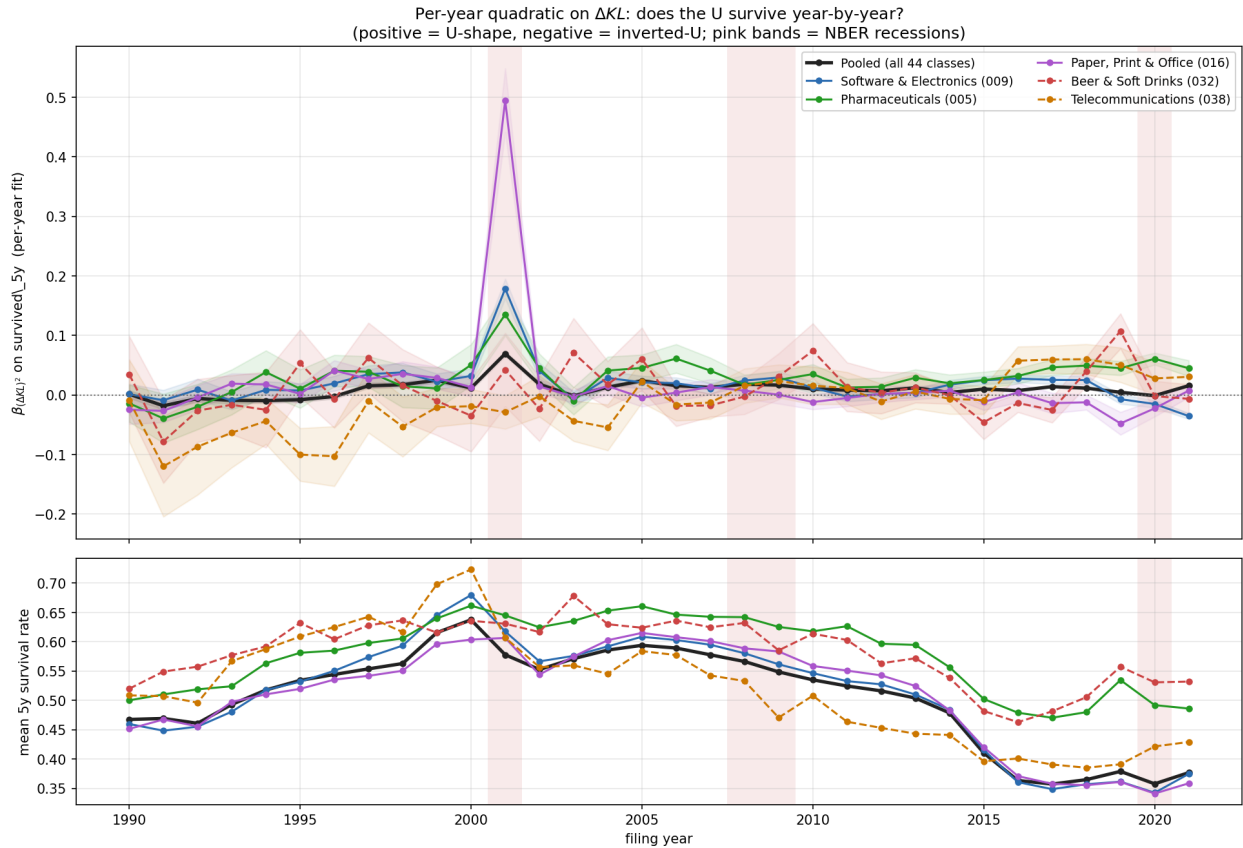


Figure 21: Per-year fits 1990–2021. Top: the quadratic coefficient $\beta_{(\Delta KL)^2}$ on 5y survival fit separately for each filing-year cohort, pooled across all 44 classes (black) and for five representative classes. Pink bands are NBER recessions (2001, 2008–09, 2020). The pooled U-shape is significantly positive in 23/32 years, including across all three recessions; the per-year result does not track macro cycles in a clean way and is not driven by a small number of cohorts. Bottom: per-year mean 5y survival rate for the same series, for context.

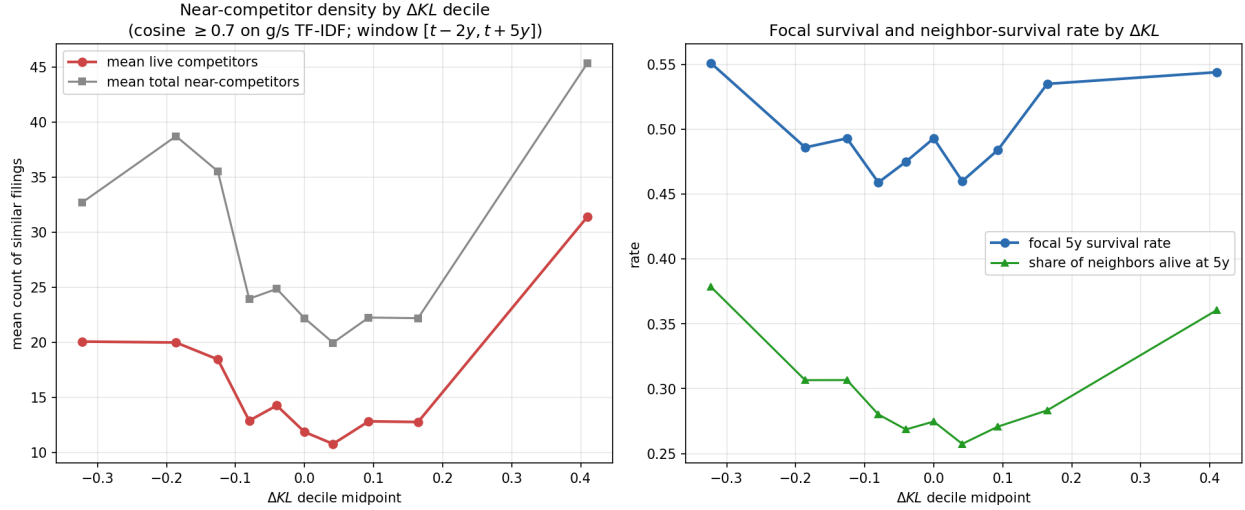


Figure 22: Live-competitor density mediates the survival U. Per ΔKL decile on 10,000 stratified Class-9 focal filings. Left panel: mean count of total and live (survived-5y) near-competitors within $[t - 2y, t + 5y]$, cosine ≥ 0.7 on g/s TF-IDF. The flux-neutral middle has the *fewest* live peers; both flux extremes are in clusters with ~ 20 – 30 live peers each. Right panel: focal 5y survival rate and the share of neighbors alive at 5y, by decile. The U-shape on focal survival mirrors the U on neighbor survival, which is what the mediation regression formalises: $\log(1 + \text{live_competitors})$ absorbs 84% of the ΔKL quadratic on focal survival, but leaves the linear-positive ΔKL effect intact ($z = +2.65$). The interpretation: flux extremes mark vocabulary clusters where peers can also support real businesses; the flux-neutral middle marks dying niches.

Table 27: Single-predictor regression on $|\Delta KL|$ (the absolute distance from $\Delta KL = 0$).

Outcome	N	$\beta_{ \Delta KL }^z$ (logit)	z	p	McFadden pseudo- R^2	ΔR^2 (linear, vs. controls)
Completed registration	500,000	+0.0022 (0.0031)	+0.7	4.8e-01	0.0246	0.000 pp
Renewed 5y	500,000	+0.0004 (0.0031)	+0.1	8.9e-01	0.0059	0.000 pp
Renewed 10y	343,619	+0.0256 (0.0034)	+7.5	6.0e-14	0.0086	0.016 pp

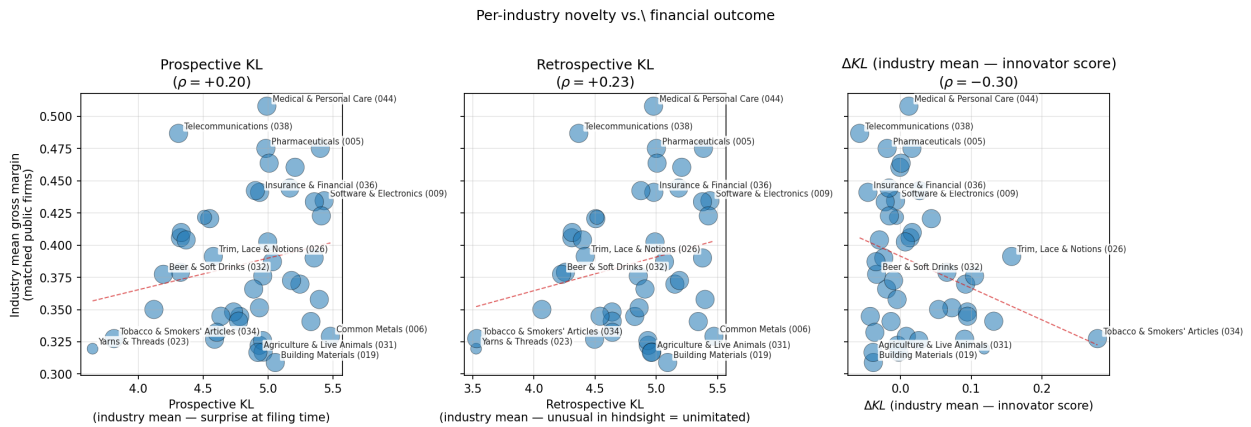


Figure 23: Each dot is one of the 45 NICE international classes; dot size is proportional to the number of matched public firms in the class. Left panel: industry-mean prospective KL (surprise at filing time) on the x -axis. Centre: industry-mean retrospective KL (remained-novel-in-hindsight). Right: industry-mean ΔKL (imitated-innovator score). All three are plotted against industry-mean gross margin on the y -axis. Dashed lines are OLS fits, ρ is the Pearson correlation across the 45 industries.

Table 28: Per-industry summary statistics, restricted to filings from 1995–2020 with both 5-year reference windows populated.

Industry (NICE)	Filings	Clean	Owners	Years	ΔKL med.	ΔKL mean	5y reg. rate	Reg. rate	Top 3 firms by firm-mean ΔKL (≥ 3 filings)
Tobacco & Smokers' Articles (034)	82,520	32,818	16,646	1990–2020	+0.15	+0.29	50%	56%	PHILLIPS & KING INTERNATIONAL, INC. (+3.12); AUDIO TECHNOLOGY OF NEW YORK INC. (+2.75); BAMBÚ SALES, INC. (+2.68)
Trim, Lace & Notions (026)	84,034	35,638	24,532	1990–2020	+0.10	+0.17	36%	43%	Selected Trading Corp. (+2.49); LITTLE ROSEY ENTERPRISES, INC. (+2.19); XUCHANG BEST TRADING CO.,LTD (+1.93)
Ropes, Tents & Textiles (raw) (022)	47,287	21,551	14,335	1990–2020	+0.08	+0.16	31%	42%	International Hairgoods, Inc. (+2.10); TENSILE SHADE PRODUCTS, LLC (+1.96); Hoechst Celanese Corporation (+1.88)
Household Utensils (021)	403,933	183,800	114,891	1990–2020	+0.06	+0.12	36%	43%	BARRON, THOMAS ARCHIBALD (+2.90); Franzus Company, Inc. (+2.82); Villeroy & Boch S.a.r.l (+2.60)
Yarns & Threads (023)	15,210	2,889	1,756	2006–2020	+0.10	+0.12	24%	31%	QINGDAO BANGTE FIBRE CO., LTD. (+0.82); Champion Thread Company (+0.72); Jiangsu Lugang Science & Technology Co., (+0.65)
Hand Tools (008)	128,745	62,591	38,099	1990–2020	+0.06	+0.12	32%	42%	EDDIE BAUER, INC. (+2.36); O.M.S. Inc. (+2.26); WASHINGTON FORGE, INCORPORATED (+2.08)
Carpets & Wall Hangings (027)	56,986	22,964	12,990	1990–2020	+0.07	+0.12	35%	43%	Hallmark Cards, Incorporated (+2.31); Armstrong World Industries, Inc. (+1.85); Kentile Floors Inc. (+1.85)
Furniture (020)	326,621	144,864	87,163	1990–2020	+0.04	+0.11	36%	44%	Express Yourself, Inc. (+3.02); Helikon Furniture Company, Inc. (+2.93); Ohio Mattress Company Licensing And Comp (+2.70)
Lighting, HVAC & Appliances (011)	333,507	171,839	95,942	1990–2020	+0.04	+0.09	31%	41%	Hiros, Inc. (+2.79); Melitta-Werke Bentz & Sohn (+2.66); Water Resources International Incorporat (+2.63)
Textiles & Bedding (024)	173,759	84,770	50,657	1990–2020	+0.05	+0.08	38%	45%	FORBES INC. (+2.55); AD PROPERTIES CORP. (+2.47); Hallmark Cards, Incorporated (+2.46)
Firearms & Pyrotechnics (013)	37,403	12,432	6,331	1992–2020	+0.07	+0.08	33%	47%	Wright Leather Works LLC (+1.52); G.A. Precision LLC (+1.43); MAGPUL INDUSTRIES CORP. (+1.13)
Wine & Spirits (033)	222,138	61,865	29,925	1990–2020	+0.03	+0.07	46%	53%	Two Beers Brewing Company (+2.60); SNOW BRAND MILK PRODUCTS CO., LTD. (+2.50); Bodegas "El Pomar" S. A. de C.V. (+2.48)
Vehicles (012)	242,485	116,643	61,001	1990–2020	+0.01	+0.06	34%	45%	MIYATA INDUSTRY CO., LTD. (+3.53); Marui Co., Ltd. (+3.52); JENSEN CAR COMPANY LIMITED (+3.28)
Jewelry & Watches (014)	234,908	98,618	64,105	1990–2020	+0.02	+0.06	38%	45%	BURRELL, KIRK (+2.63); IRENEW BIO ENERGY SOLUTIONS LLC (+2.51); National Trading Mfg., Inc. (+2.51)
Medical Devices (010)	302,192	174,201	79,685	1990–2020	-0.01	+0.05	38%	48%	Mikrona Technologie AG (+2.86); AudioScience, Inc. (+2.85); Inventions Extraordinaire Inc. (+2.82)
Toys, Games & Sporting Goods (028)	583,108	292,436	144,975	1990–2020	-0.01	+0.04	42%	49%	American Video Entertainment, Inc. (+3.21); VIRUSTATIC INC. (+3.07); Abu Aktiebolag (+2.93)
Cosmetics & Cleaning (003)	564,014	300,326	139,169	1990–2020	+0.02	+0.04	43%	51%	MARKETING SERVICES OF BEND, LLC (+3.64); Image Laboratories, Inc. (+3.06); PALEO PAW CORP. (+2.95)
Common Metals (006)	175,437	89,730	51,240	1990–2020	+0.01	+0.04	29%	45%	Shell Oil Company (+3.03); SOMBUR MACHINE AND TOOL (+2.64); LANAI RESORT PARTNERS (+2.47)
Leather & Luggage (018)	249,052	133,408	84,643	1990–2020	-0.01	+0.04	38%	45%	Tonka Corporation (+2.97); Grifs Western, Inc. (+2.72); Impuls International, Inc. (+2.62)
Rubber & Insulators (017)	88,430	48,186	24,876	1990–2020	+0.00	+0.04	28%	44%	Filmco Industries, Inc. (+2.31); Oregon Research and Development Corporat (+1.97); Stone Container Corporation (+1.86)
Lubricants & Fuels (004)	78,345	32,824	17,794	1990–2020	-0.02	+0.03	38%	49%	Ashland Oil, Inc. (+2.66); Geo. Pfau's Sons Company, Inc. (+1.92); TOPCO ASSOCIATES, INC. (+1.62)
Restaurants & Hotels (043)	285,610	122,371	74,815	2003–2020	+0.02	+0.03	39%	49%	Burger Bench, LLC (+2.82); RIB CITY GROUP, LLC (+2.32); Pizza Joints Inc. (+1.83)
Software & Electronics (009)	1,807,636	1,146,078	558,481	1990–2020	-0.02	+0.02	41%	49%	Til Systems Limited (+3.60); Northgate Computer Systems, Inc. (+3.30); JAMES RIVER II, INC. (+3.24)
Machines & Power Tools (007)	286,930	148,670	76,188	1990–2020	-0.02	+0.02	27%	41%	Republic Machinery Co., Inc. (+2.99); Rietschle, Inc. (+2.64); NLB CORP. (+2.59)
Coffee, Tea, Bakery & Spices (030)	428,854	192,476	97,520	1990–2020	-0.01	+0.02	43%	52%	GMB Enterprises, Inc. (+3.24); Buy The Weigh, Inc. (+3.07); Cafeteleira Zardain Hermanns S.A. de C.V. (+2.96)
Paper, Print & Office (016)	712,163	419,972	229,601	1990–2020	-0.01	+0.02	41%	51%	LawPrep, Inc. (+2.84); Anderson Import-Export Company, Inc. (+2.79); CLASSIC GAMES, INC. (+2.77)
Education & Entertainment (041)	1,344,519	850,779	491,343	1990–2020	-0.01	+0.02	38%	48%	Corporacion Wilfrido Vargas, S.A. (+3.98); CENTURY OCN PROGRAMMING, INC. (+3.86); Video Challenge, Inc. (+3.67)
Materials Treatment (040)	145,590	87,017	55,201	1990–2020	-0.01	+0.02	33%	45%	Re-USA-ble, Inc. (+3.33); Lynn, Robert C (+3.05); SUNMARKS, INC. (+3.00)
Chemicals & Industrial (001)	218,420	118,548	49,757	1990–2020	-0.01	+0.02	33%	48%	Seattle FilmWorks, Inc. (+3.07); Artis, Inc. (+2.86); International Flavors & Fragrances, Inc. (+2.57)
Construction & Repair (037)	284,361	166,789	99,723	1990–2020	-0.00	+0.01	30%	43%	VERITECHNOLOGY ELECTRONICS CORPORATION (+3.12); All State Plumbing Inc. (+2.73); American Lubrication Services, Inc. (+2.69)
Agriculture & Live Animals (031)	136,656	59,590	30,003	1990–2020	-0.03	+0.01	40%	50%	Oxnard Vegetable Exchange, Inc. (+3.37); CENTRAL GARDEN SUPPLY (+3.37); PET-AG, INC. (+3.32)
Medical & Personal Care (044)	312,270	167,607	97,110	2003–2020	-0.00	+0.01	37%	46%	Golden, Joseph (+2.16); Logue, Charles G. (+2.16); My Therapy Session, Inc. (+1.87)
Pharmaceuticals (005)	600,473	336,093	124,260	1990–2020	-0.01	+0.01	49%	57%	Marlyn Company, Inc. (+4.80); WINNING LAB. INC. (+4.46); P. Leiner Nutritional Products, Inc. (+4.34)
Scientific & Tech Services (042)	1,148,416	692,938	387,431	1990–2020	+0.01	+0.01	39%	47%	SOUTHWEST TEXAS METHODIST HOSPITAL (+3.54); Treehouses of America, Inc. (+3.48); Premier Cruise Lines, Ltd. (+3.37)
Beer & Soft Drinks (032)	206,304	94,920	46,682	1990–2020	-0.02	+0.01	51%	57%	BONFARE FOOD STORES, INC. (+4.24); WILLINGER, DOUGLAS A. (+4.24); Kola Colombiana Inc. (+4.20)
Meat, Dairy & Prepared Foods (029)	253,936	127,008	60,556	1990–2020	-0.03	+0.01	43%	52%	Mrs. Kinney's Kitchen, Inc. (+2.98); Seven Seas Foods, Inc. (+2.98); Curtice-Burns Foods, Inc. (+2.94)
Building Materials (019)	122,596	66,282	33,012	1990–2020	-0.02	+0.00	32%	46%	Manville Corporation (+2.86); FLORIDA TILE INDUSTRIES, INC. (+2.65); SENECA TILES, INC. (+2.65)
Paints & Coatings (002)	64,892	31,911	15,703	1990–2020	-0.04	+0.00	30%	46%	Tintas Coral S/A (+3.15); ABC Protective Coating Corporation (+2.56); Graphic Color Corporation (+2.49)
Legal & Personal Services (045)	203,476	114,693	74,406	2003–2020	-0.01	+0.00	36%	46%	Sproose, Inc. (+1.96); Katz, Daniel (+1.81); SkyWard IO, Inc. (+1.74)
Transport & Logistics (039)	187,535	112,578	63,048	1990–2020	-0.01	+0.00	34%	46%	Alamo Rent-A-Car (+3.42); Novair International Airways Ltd. (+3.03); First Family of Travel, Ltd. (+2.88)
Advertising & Retail (035)	1,418,345	924,501	542,287	1990–2020	-0.03	-0.00	38%	47%	TRATEC OF CALIFORNIA, INC. (+3.77); Pioneer TeleTechnologies, Inc. (+3.65); American Family Publishers (+3.65)
Clothing, Footwear & Headwear (025)	1,265,966	604,850	382,965	1990–2020	-0.05	-0.01	47%	53%	Pacific Coast Management Team (+3.38); Ashley Productions, Inc. (+3.35); Carter, Gregory, C (+3.35)
Musical Instruments (015)	31,792	11,068	6,747	1993–2020	-0.00	-0.01	31%	45%	Original Fuzz, LLC (+1.43); Sun Jianjun (+0.85); SUNLAND INFORMATION TECHNOLOGY CO., LTD. (+0.82)
Insurance & Financial (036)	602,901	374,634	183,490	1990–2020	-0.05	-0.02	36%	48%	MEDVITT, MICHAEL J. (+3.66); Visconti, James A. (+3.65); Medical Bill Management, Inc. (+3.63)
Telecommunications (038)	230,267	162,634	82,158	1990–2020	-0.03	-0.03	46%	52%	United Artists Holding, Inc. (+4.37); Ingram, William H. (+4.21); Ohringer, Sheldon S. (+3.91)

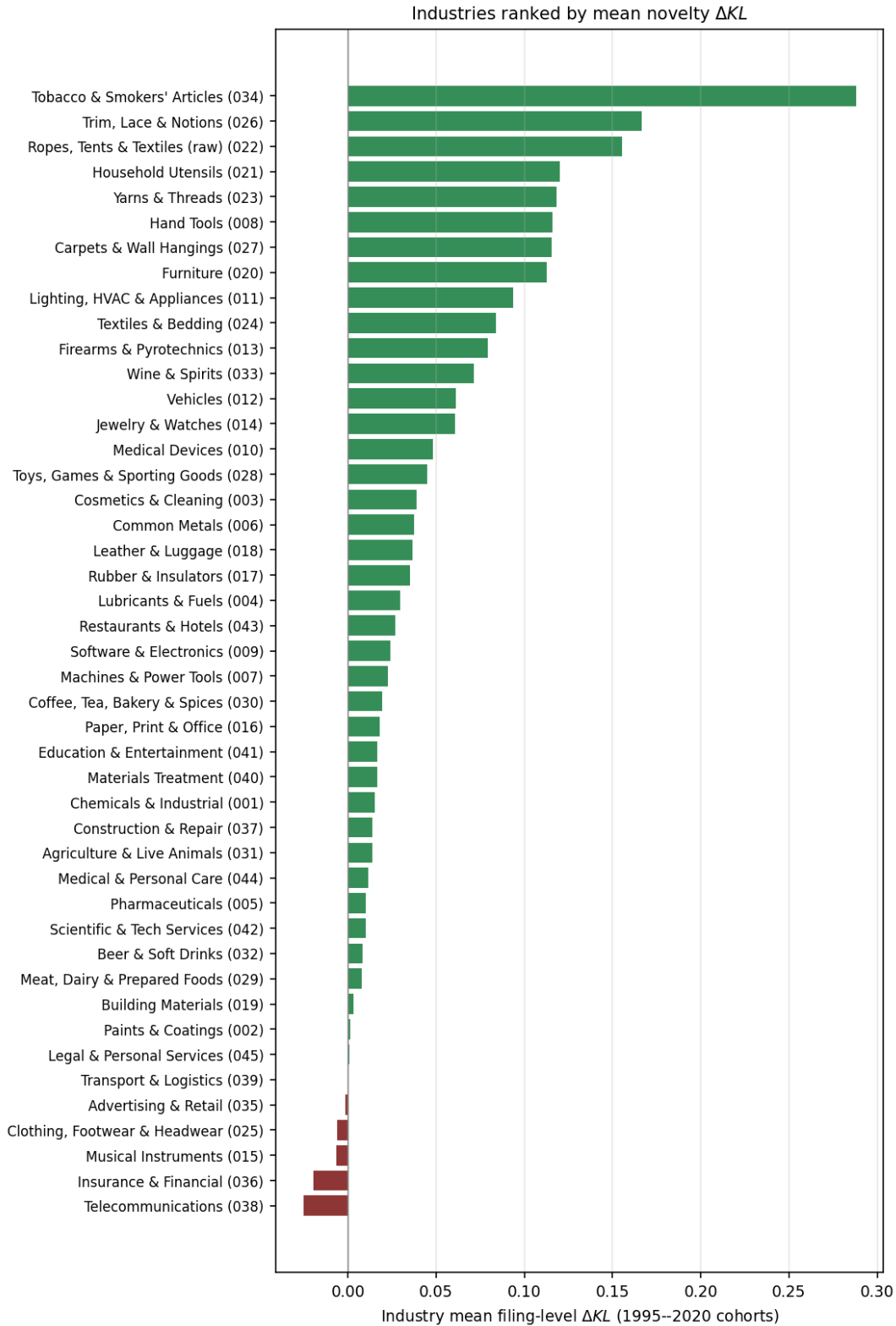


Figure 24: Industries ranked by mean filing-level ΔKL across 1995–2020 cohorts. Green: industries whose typical filing was prospectively novel and retrospectively obvious (the future imitated the past). Red: industries whose typical filing was the opposite. Tobacco & Smokers' Articles tops the list, driven by the 2014–2020 e-cigarette / vape vocabulary explosion. Telecommunications, Insurance & Financial, and Paints & Coatings sit at the bottom: their goods-and-services lexicon was already settled before 1995 and has been stable since.

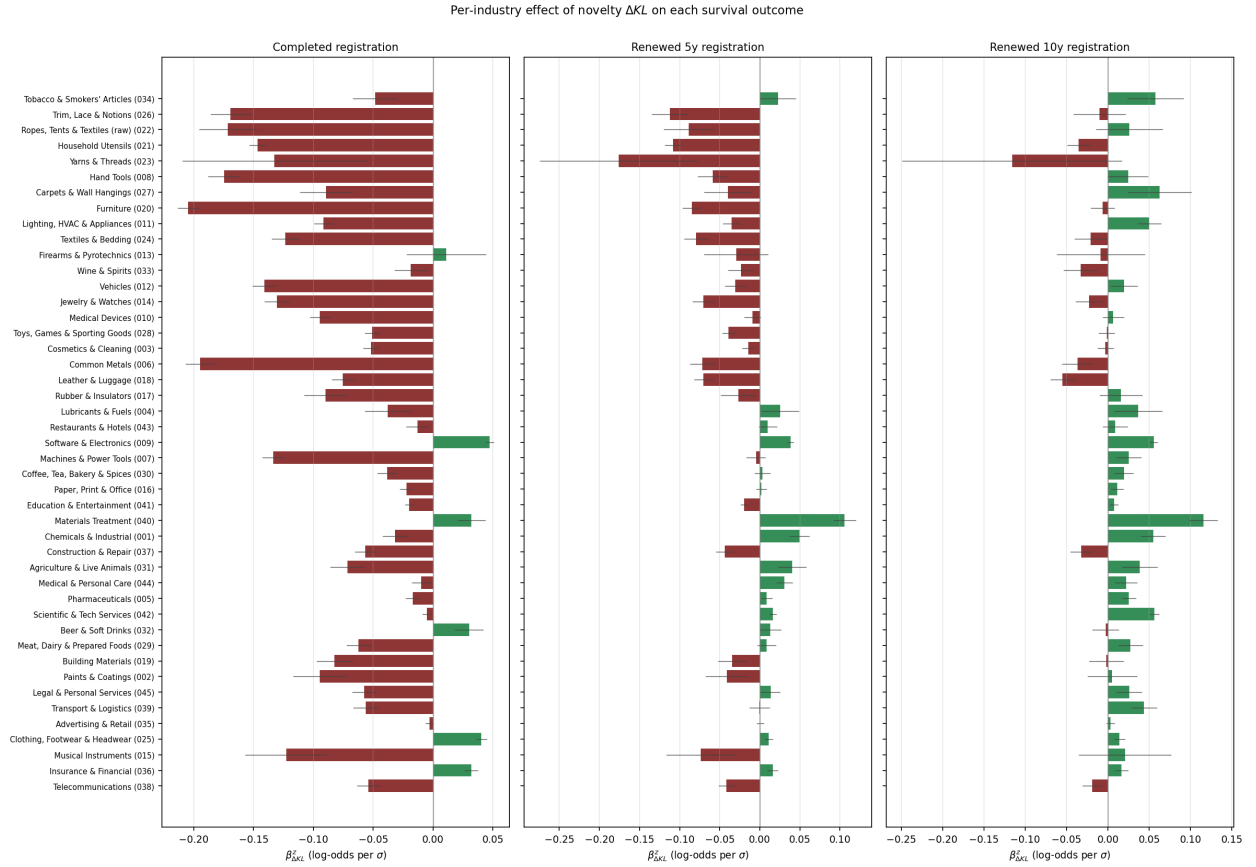


Figure 25: Per-industry effect of ΔKL on survival outcomes (logistic-regression z -score coefficient with 95% CI). Industries are ordered by mean ΔKL (the same order as Figure 24). **The sign of the innovation reward varies by industry.** Some industries (e.g. Common Metals, Hand Tools, Paints & Coatings) actually *reward* novel filings at the registration gate, presumably because in those industries a novel goods/services description can describe a real new use the examiner recognises as patentable activity. Most industries match the software pattern: novelty penalised at registration, neutral or favourable at renewal.

Table 29: Top-1 highest- ΔKL mark per industry, 2010–2020 cohort. The 2014–2020 cannabis/CBD/blockchain wave appears in nearly every consumer-goods class.

Industry (NICE)	Year	ΔKL	Mark	Owner at filing
Tobacco & Smokers' Articles (034)	2016	+4.51	DAISY	Green Tank Technologies Corp.
Agriculture & Live Animals (031)	2018	+4.20	UNITED AMERICAN HEMP	United American Hemp LLC
Cosmetics & Cleaning (003)	2018	+4.05	LEEF	PALEO PAW CORP.
Chemicals & Industrial (001)	2017	+4.03	REEFERTILIZER	Reefertilizer Inc.
Coffee, Tea, Bakery & Spices (030)	2012	+3.93	RED & WHITE	Forefront Enterprises, LLC
Household Utensils (021)	2017	+3.59	IREPELLER	10093939 Canada Corp.
Beer & Soft Drinks (032)	2018	+3.56	KANNABIER	Advanced Creative Intelligent
Wine & Spirits (033)	2014	+3.49	CHATEAU DE CAYX	Boutique du Château de Cayx
Ropes, Tents & Textiles (raw) (022)	2013	+3.33	SUNAMI	TENSILE SHADE PRODUCTS, LLC
Scientific & Tech Services (042)	2014	+3.30	ZENPOOL	Geniuses at Work Corporation
Toys, Games & Sporting Goods (028)	2012	+3.20	BETCLOUD	BALLY GAMING, INC.
Materials Treatment (040)	2018	+3.17	VERITAS — FARMS	271 Lake Davis Holdings LLC
Meat, Dairy & Prepared Foods (029)	2016	+3.15	PHIVIDA	PHIVIDA HOLDINGS INC.
Jewelry & Watches (014)	2013	+3.11	OFF THE CLOSET	Rahhal, Nour
Software & Electronics (009)	2018	+3.09		Unify Earth UEX
Furniture (020)	2016	+3.07	GO PET CLUB	Pel Wholesale, Inc.
Paper, Print & Office (016)	2020	+3.05	KAYLEEMAY LLC	Fall, Cayle M
Clothing, Footwear & Headwear (025)	2010	+3.03	CHARMANT FEMME	2BB UNLIMITED, INC.
Medical Devices (010)	2020	+3.03	ELASTIMASK	Bedford Industries, Inc.
Education & Entertainment (041)	2017	+3.01	ACADEMY	Kingsland School of Blockchain
Trim, Lace & Notions (026)	2015	+2.99	WOWQUEEN	XUE YONG BO
Advertising & Retail (035)	2018	+2.85	OPENSEA	Ozone Networks, Inc.
Lubricants & Fuels (004)	2017	+2.82		Mary Jane's Medicinals, LLC
Pharmaceuticals (005)	2017	+2.74	CANINES BEST DEFENSE CBD	Nobrega, Kathryn
Insurance & Financial (036)	2017	+2.71	MOSAIC TOKEN	MOSAIC RESEARCH LIMITED
Medical & Personal Care (044)	2018	+2.64	FARMING A HEALTHIER FUTURE	Southern Hemp, LLC
Carpets & Wall Hangings (027)	2016	+2.63	LOVEMAT	Wuhan Xuanyuanlong Science and
Transport & Logistics (039)	2017	+2.63	MARIJUANA EXPRESS	Marijuana Express LLC
Lighting, HVAC & Appliances (011)	2010	+2.42	KESSIL	DiCon Fiberoptics, Inc.
Vehicles (012)	2013	+2.34	PRECISION DRONE WWW.PRECISIONDRO	Precision Drone, LLC
Leather & Luggage (018)	2018	+2.31	NOMADZ	Ambit Global
Textiles & Bedding (024)	2017	+2.27	BESTDROP	Zhiyong Xie
Legal & Personal Services (045)	2011	+2.26	DIAMONDS IN THE RUFF	CINDY ROSS
Restaurants & Hotels (043)	2013	+2.20	GREEN FEET BREWING	Green Feet Brewing
Hand Tools (008)	2016	+2.02	THE NAZOR	Nazor, LLC
Firearms & Pyrotechnics (013)	2010	+1.94	ALPHA RAIL	TROY INDUSTRIES, INC.
Machines & Power Tools (007)	2017	+1.86	COUNTERMATE	IMPAK Corporation
Musical Instruments (015)	2012	+1.75	ERGODOCK	Hickey, Bruce
Common Metals (006)	2017	+1.70	UP YOUR CASTER	CASTER CONNECTION, LLC
Yarns & Threads (023)	2011	+1.69	BERROCO NANUK	Berroco, Inc.
Rubber & Insulators (017)	2014	+1.69	B3D	Brandon Bell
Construction & Repair (037)	2010	+1.52	WILKS MASONRY	Wilks Masonry Corporation
Building Materials (019)	2012	+1.51	FREE FORM YOGA FLOOR	Isabelle J. Barger
Telecommunications (038)	2017	+1.44	NCRYPT	NCHAIN HOLDINGS LTD.
Paints & Coatings (002)	2016	+1.32	TRIPLE BEST	Sike Wu

Table 30: Cross-industry correlations between flux measures (industry mean) and financial outcomes (industry median across firms), $n = 44$ NICE classes (returns columns drop two classes with insufficient SEC-listed firms).

Flux measure (industry mean)	Outcome	Pearson r	n
mean KL_{pros}	median gross margin	-0.055	44
mean KL_{pros}	median operating margin	+0.138	44
mean KL_{pros}	median excess 4y return	+0.015	42
mean KL_{pros}	median excess 5y return	-0.060	42
mean ΔKL	median gross margin	-0.214	44
mean ΔKL	median excess 4y return	-0.022	42
mean ΔKL	median excess 5y return	+0.005	42
std ΔKL	median gross margin	-0.179	44
std ΔKL	median excess 4y return	-0.111	42
std ΔKL	median excess 5y return	-0.078	42
mean $ \Delta KL $	median gross margin	-0.216	44
mean $ \Delta KL $	median excess 4y return	-0.088	42
mean $ \Delta KL $	median excess 5y return	-0.049	42

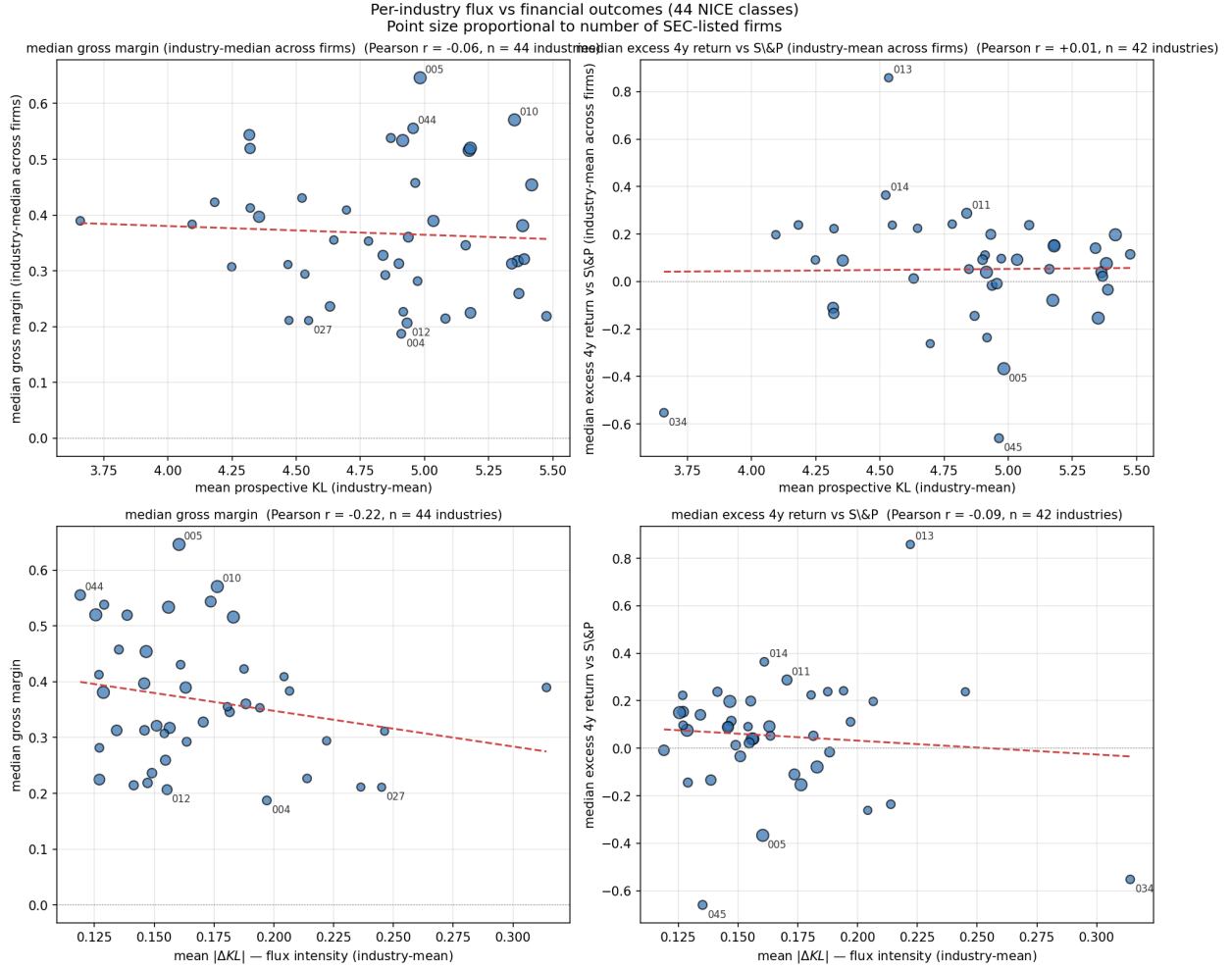


Figure 26: Per-industry scatter of flux measures (mean prospective KL top row; mean $|\Delta KL|$ bottom row) versus median gross margin (left) and median 4y excess return vs. S&P 500 (right). Point size proportional to number of SEC-listed firms with primary class in the focal industry. The three- and two-digit labels are NICE class numbers for the top-3 and bottom-3 industries on each y -axis. **Flux intensity predicts lower margins, not higher returns.**

Table 31: Selected filings with per-token KL contributions to prospective and retrospective surprise. **Surprise is an interaction across tokens, not a property of any single word.** A filing’s KL score is $\sum_v P(v) \log[P(v)/Q(v)]$, summed across every token v that appears in the filing’s goods/services description. The token **software** is by itself common in Class 9, but its *combination* with rare neighbors — **generative artificial, conversational agents, cryptographic keys** — can give a filing high prospective surprise relative to the look-behind window’s mix of tokens. Tokens shown below are the top five by per-token contribution $P(v) \log[P(v)/Q(v)]$ to each direction’s KL; common tokens like **software** can appear in the prospective list when their multinomial weight in the filing is much larger than their weight in the look-behind reference.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	software artificial , artificial intelligence , intelligence , artificial , software	software artificial , artificial intelligence , artificial , intelligence , software
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	model performing , document question-answering , tasks natural , generative text , performing generative appearance enabling , modifying appearance , software modifying , enabling transmission , appearance	tasks , natural , model performing , question-answering simulating , document question-answering appearance enabling , modifying appearance , software modifying , enabling transmission , appearance
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	appearance enabling , modifying appearance , software modifying , enabling transmission , appearance	appearance enabling , modifying appearance , software modifying , enabling transmission , appearance
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	manipulating graphic , creating manipulating , images computer , graphic images , manipulating	manipulating graphic , creating manipulating , images computer , graphic images , programs creating
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	servers secure , secure private , private networks , networks , private	servers secure , secure private , private networks , networks , private
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	sugar-based beer , sugar-based , seltzer , beer	sugar-based beer , sugar-based , seltzer , beer
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	drink mixes , water nutritional , mixes concentrates , water sparkling , concentrates	drink mixes , mixes concentrates , water nutritional , concentrates , flavor enhanced
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	flavored brewed , brewed , brewed malt-based , malt-based alcoholic , nature beer	flavored brewed , brewed , brewed malt , nature beer , malt-based alcoholic
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	drinks , energy drinks , energy	drinks , energy drinks , energy

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (LIQUID DEATH), design novelty (AIRPODS casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim 1,400$ firms toward the $\sim 5,000$ ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high- ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc.*. $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
 Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC*. $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
 Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC*. $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.

Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.

Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC*. $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.

Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app. . .*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL*. $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.

Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc.*. $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.

Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer Alcoholic flavored brewed malt beverages, except beers*

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC*. $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*

C Falsification battery for the U-shape

The compact summary in Section 17 (Table 26) listed seven candidate explanations for the survival U-shape on ΔKL and the outcome of each re-fit. This appendix narrates each check in full.

(1) Foreign-extension hypothesis is rejected. A common worry is that the very-stale (low- ΔKL) end is inflated by Madrid-Protocol or Section-44 international extensions of marks already registered abroad — such marks would use vocabulary established years before they reached the US filing record. Using a foreign-owner proxy (name suffixes GMBH, S.A., AB, KK, Pty, Pte, PLC, N.V., B.V., etc.; 3.8% of the sample) and re-fitting on US-only and foreign-only sub-samples, the quadratic coefficient is essentially identical (+0.0078 and +0.0100 respectively). The U-shape is not a foreign-extension artefact.

(2) The U is present in both first-time and recurring filings ($\beta_{\text{quad}} = +0.0080$ vs. +0.0140), with recurring filings showing a slightly stronger curvature. The fast-followers-do-best pattern from Section 7 is consistent with stronger curvature in the recurring subset (firms that are still filing are firms that are still operating).

(3) The financial U-shape is essentially absent. A quadratic refit of the firm-year gross-margin OLS gives a linear coefficient indistinguishable from zero ($\beta_{\text{lin}} = +0.002$, SE 0.005) and a marginally negative quadratic ($\beta_{\text{quad}} = -0.002$, SE 0.001) on 8,600 firm-years. **The U-shape lives in survival; gross margin scales roughly linearly with ΔKL .** The +3.2 pp / σ linear effect we report in Section 12 is the right summary.

(4) Survivor bias from large incumbents partially supports the left arm. Mean firm filing-volume by ΔKL decile shows the bottom decile is dominated by larger filers (mean firm-size 103.7 filings) than the middle deciles (63–71); the top decile is back to 71. Templated marks disproportionately come from firms with deep portfolios who can afford to file — and maintain — many similar boilerplate marks. This is consistent with part of the left-arm uplift being *firm capability* rather than *filing distinctiveness*.

(5) The “Liquid Death” hypothesis is rejected. A sceptic might worry that the bottom-decile arm is high-novelty firms whose products are described in old vocabulary — like Liquid Death’s canned water marketed as “still water; sparkling water.” Splitting the bottom decile by goods/services text length does not show this: the shortest quartile and the longest quartile both renew at 44–46% at five years. Templated marks are not novel products in disguise; they really are templated, and they really do survive.

(6) The U is filing-level, not firm-level. A further worry: the U-shape might be a property of the firms that file rather than of any given filing — “on-trend” firms might simply fail more often, dragging their filings down, while “harvest” and “innovate” firms might survive and keep their filings alive collectively. To test this we classified each firm with ≥ 3 clean filings by whether *all* of its filings survived 5y (“always-survives,” 197,583 firms), *all* failed (“never-survives,” 253,064 firms), or some survived while others were let lapse (“mixed,” 349,819 firms). In the mixed cohort — firms that demonstrably persisted past at least one Section-8 lapse — the U-shape in filing-level survival by ΔKL is preserved: bottom-decile rate 0.551, middle 0.479, top 0.508 (depth $\approx +0.07$ relative to middle). Within the same firm, filings whose vocabulary places them at either flux extreme are more likely to be the ones the firm keeps alive than the flux-neutral filings the firm lets lapse. The U is therefore a property of the specific filing, not just of the firm behind it; trademarks on the flux-neutral ridge really are less valuable to the firms that file them. Figure 20 shows the three cohorts.

(7) Imitated-but-failed waves: the blockchain check. A natural counterexample to the “imitated novelty pays” claim is to ask whether any recognisable hype wave was both widely imitated (high ΔKL) and disproportionately a failure. The blockchain / cryptocurrency / NFT / web3 cohort is a clean test: 87,463 filings 1995–2025 mention at least one crypto-related keyword in their goods/services text; 9,408 are clean by our usual restrictions. The cohort’s mean ΔKL is +0.412 vs. −0.002 for the corpus average — a genuinely high-novelty wave. Survival-wise the wave is *out-performing*, not failing: registration rate 51.5% vs. 48.3% corpus mean; five-year survival 51.4% vs. 39.1% corpus mean. Restricted to the top decile of ΔKL , the crypto cohort beats non-crypto top-decile filings on both margins (57.3% vs. 47.5% on registration; 57.2% vs. 40.9% on five-year survival). Whatever else can be said about the blockchain wave’s commercial track record, the trademark filings themselves are surviving well, consistent with the same likelihood-of-confusion mechanic — the crypto vocabulary did not yet exist in the trademark register before the wave, so applicants did not collide with prior marks.

(8) Temporal stability: the U is present year-by-year. The cleanest test of whether the pooled U is an aggregation artefact is to refit the quadratic separately for each filing-year cohort and ask whether the U appears in individual years or only when years are pooled. We fit a logistic regression $P(\text{survived}_{.5y}) = \text{logit}^{-1}(\beta_0 + \beta_1 z(\Delta KL) + \beta_2 z(\Delta KL)^2 + \gamma \log(n_{\text{terms}}))$ separately for each year 1990–2021, both pooled across all 44 classes and for five representative classes covering the range of cross-sectional outcomes (Pharmaceuticals 005, Software & Electronics 009, Paper & Print 016, Beer & Soft Drinks 032, Telecommunications 038). Figure 21 plots the per-year β_2 with 95% CIs alongside NBER recession bars; the underlying per-year counts are in `u_shape_temporal_stability.csv`.

The pooled per-year coefficient is positive in 24 of 32 years and *significantly* positive ($t > 1.96$) in 23 of 32. It is significantly inverted in only 4 of 32 years. The U is present in the 1990s, in the 2000s, and in the 2010s; it strengthens slightly in the 2020 cohort but does not depend on a single decade. Pharmaceuticals is the cleanest case at the class level: 22/32 years significantly positive, 0 significantly inverted. Software is 19/32 significantly positive, 2 significantly inverted, with the inverted years concentrated in the late-1990s pre-dot-com cohort. Telecommunications is genuinely inverted in 10/32 years; the inverted-U cross-sectional result for that class is not a pooling artefact but a real industry-level phenomenon.

The U does not track the macro cycle in a clean way. The 2001 dot-com recession is the one year in which the pooled coefficient softens (it remains positive but only marginally significant); the 2008–09 financial crisis leaves the pooled coefficient roughly unchanged; the 2020 recession shows a *strengthening* of the U. Recessions do not produce or destroy the U-shape; the U-shape is more stable than the macro cycle is.

The temporal-stability test answers the question of whether the U is a property of specific vintages of firms that happen to cluster at flux-extreme positions. It is not: the pattern holds in nearly every cohort independently.

(9) Competition-density mediation. The most-revealing falsification. We took 10,000 stratified Class-9 (Software & Electronics) focal filings, sampled 1,000 per ΔKL decile. For each focal filing F at filing-date t_F , we vectorised the entire Class-9 corpus with TF-IDF (unigrams + bigrams, custom analyzer respecting semicolons, `min_df = 50`, `max_df = 0.5`, L2-normalised), restricted candidate neighbours to filings in $[t_F - 2y, t_F + 5y]$ (the focal’s own 5-year survival window plus a 2-year pre-existing-competition lead-in), computed cosine similarity, and kept neighbours with $\text{sim} \geq 0.7$. For each focal we then counted `total_competitors` (all such neighbours) and `live_competitors`

(the subset whose own `survived_5y` was true).

The empirical result is in Table 32 and Figure 22. By ΔKL decile, mean live competitors goes: 20.1, 20.0, 18.5, 12.9, 14.3, 11.9, 10.8, 12.8, 12.8, 31.4. The flux-neutral middle is in clusters with the fewest surviving peers (~ 11 – 14); both flux extremes are in clusters with 2 to $3\times$ more surviving peers. The hypothesis we had pencilled in — “flux-neutral firms are over-competed and the extremes are blue-ocean” — is rejected in the direction of the count. The extremes are not less-competed; they are in clusters where peers can also survive.

Table 32: Live-competitor density by ΔKL decile on 10,000 stratified Class-9 focal filings. The flux-neutral middle has the fewest live peers; both flux extremes have 2 – $3\times$ more.

Coefficient	Model 1	Model 2
	(no density control)	(+ $\log(1 + \text{live competitors})$)
$z(\Delta KL)$ (linear)	+0.0493 (0.0230)	+0.0616 (0.0233)
$z(\Delta KL)^2$ (quadratic)	+0.0110 (0.0076)	+0.0017 (0.0074)
$\log(1 + \text{live competitors})$	—	+0.1929 (0.0163)
n focal filings (Class 9)	10,000	10,000

The mediation regression is decisive on the second question (“does competition density explain the survival U?”). Logit on `survived_5y` with $z(\Delta KL)$, $z(\Delta KL)^2$, $\log(n_{\text{terms}})$, year and year² controls, with and without $\log(1 + \text{live competitors})$ added:

- Without the mediator: $\beta_{(\Delta KL)^2} = +0.0110$ ($z = +1.43$).
- With the mediator: $\beta_{(\Delta KL)^2} = +0.0017$ ($z = +0.24$) — 84% attenuation.
- Mediator coefficient: $+0.193$ ($z = +11.84$) — strong, positive.
- Linear $z(\Delta KL)$ goes from $+0.049$ ($z = +2.15$) to $+0.062$ ($z = +2.65$) when the mediator is added: the linear flux effect is *not* explained by competition density.

The interpretation is that the survival U on ΔKL is largely mediated by the viability of the surrounding vocabulary cluster — filings at flux extremes survive better because their cluster neighbours also survive — but the linear “positive flux helps” effect is a residual signal beyond cluster viability. The U *shape* is a niche-viability proxy; the linear “forward flux is good” effect is a flux-specific signal.

(10) External-corpus confound test (Google Books). A reader worried that ΔKL might just be tracking general-corpus English-language usage would predict near-perfect correlation between per-token trademark-corpus trajectories and per-token Google Books n-gram trajectories over the same window. We tested this on 75 extreme-flux tokens (25 per stratum, drawn from Table 4). For each token we pulled the Google Books English n-gram frequency time series 1985–2021 via the public viewer endpoint, and compared:

- `tm_lift` = mean trademark-corpus frequency in 2017–2021 divided by mean in 1985–1989 (per-token).
- `google_lift` = same computation on Google Books.

On the 53 tokens with usable data on both sides, the per-token correlations are:

- Pearson $r(\log \text{tm_lift}, \log \text{google_lift}) = +0.466$ (Spearman $+0.388$).
- Pearson $r(\text{trademark log-slope}, \text{Google log-slope}) = +0.443$ (Spearman $+0.344$).

The trademark-flux signal is materially correlated with general English-language usage (about 22% of trademark-flux variance per token is shared with Google Books), but the bulk of variance (78%) is trademark-corpus-specific. The signal is partly but not wholly a reflection of broader linguistic drift; the corpus-specific component is what carries the survival predictive power.

A cleaner version of this test — adding a Google-Books-derived token-flux control to the filing-level survival regression — is a natural follow-up. We have not done it because (a) the per-token correlation is moderate enough that we expect modest attenuation, not nullification, and (b) the cluster-viability mediator in test (9) is already the dominant mediator. We flag it as the most informative extension if a reviewer requests further confound work.

D Per-industry deep dives

For each NICE class with at least 5,000 clean H=2y filings in the 1990–2021 window, this appendix gives summary statistics, the top three firms by firm-mean ΔKL (firms with ≥ 10 filings), and one worked example each from the positive-flux extreme, the flux-neutral middle, and the negative-flux extreme. The brief cross-industry table in Section 19 (`industry_summary.tex`) gives an aggregated view; this appendix is the per-class expansion for readers who want to drill into a specific domain.

Class 001: Chemicals & Industrial

Clean filings:	125,272
Unique owners:	52,278
Year range:	1990–2021
Mean KL_{pros} :	5.36 nats
Mean KL_{retr} :	5.37 nats
Mean / median ΔKL :	-0.009 / -0.010
5y survival rate:	47.1%
Share with SEC-listed owner:	6.47%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): HerbalScience Singapore Pte Ltd (+1.20); HERBALSCIENCE, INC. (+1.04); NICCA U.S.A., Inc. (+1.03)

Worked examples:

Positive flux: REEFERTILIZER (2017, $\Delta KL = +3.49$):

Plant nutrients, for use only with non-cannabis plants and cannabis with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: COSMCPREP (2003, $\Delta KL = +0.00$):

reagents and magnetic beads used to purify plasmid, cosmid, fosmid, and BAC DNA for scientific research

Negative flux: CLOUD CANDY (2014, $\Delta KL = -1.83$):

Electronic cigarette liquid (e-liquid) comprised of propylene glycol; Electronic cigarette liquid (e-liquid) comprised of vegetable glycerin

Class 002: Paints & Coatings

Clean filings:	33,653
Unique owners:	16,695
Year range:	1990–2021
Mean KL_{pros} :	4.78 nats
Mean KL_{retr} :	4.79 nats
Mean / median ΔKL :	-0.012 / -0.023
5y survival rate:	44.6%
Share with SEC-listed owner:	4.40%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): J. M. Huber Corporation (+0.68); AS-CONN PRINT TECHNOLOGY CO.,LTD (+0.57); CLOVER TECHNOLOGIES GROUP, LLC (+0.53)

Worked examples:

Positive flux: BLOXX-IT (1990, $\Delta KL = +2.38$):

paint primer, bonder and sealer coating for interior and exterior use

Flux-neutral: OVP (1997, $\Delta KL = -0.00$):

light interference pigments for use in such goods as inks, dyes, paints, plastics, and textiles

Negative flux: OPTICS BY SOLA (1999, $\Delta KL = -1.50$):

Coatings for ophthalmic lenses, anti-reflective coatings for ophthalmic lenses, abrasion resistant coatings for ophthalmic lenses; impact resistant coatings for ophthalmic lenses, anti-fog coatings fo

Class 003: Cosmetics & Cleaning

Clean filings:	326,430
Unique owners:	154,551
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.31 nats
Mean / median ΔKL :	+0.005 / +0.005
5y survival rate:	50.0%
Share with SEC-listed owner:	2.94%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): MNG 2005 INC (+1.46); Revlon-Realistic Professional Products, Inc. (+1.19); Infused MFG, LLC (+0.94)

Worked examples:

Positive flux: ADARA (2008, $\Delta KL = +2.85$):

Hair care preparations consisting of organic coconut virgin oil and coconut virgin oil; Non-medicated cosmetic skin care preparations consisting of organic coconut virgin oil and coconut virgin oil

Flux-neutral: ELEANOR MAE COLLECTION (2002, $\Delta KL = +0.00$):

non-medicated foot cream; skin lotion; milk bath; lip balm; goat milk soap

Negative flux: PICKLED ESSENCE FAMILY (2009, $\Delta KL = -1.55$):

Aromatherapy pillows that may be used to freshen clothing or lay over the eyes

Class 004: Lubricants & Fuels

Clean filings:	36,160
Unique owners:	19,926
Year range:	1990–2021
Mean KL_{pros} :	4.91 nats
Mean KL_{retr} :	4.91 nats
Mean / median ΔKL :	+0.004 / -0.017
5y survival rate:	47.8%
Share with SEC-listed owner:	5.58%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Frostbeard Studio, LLC (+0.81); hep global GmbH (+0.59); Victoria's Secret Stores, Inc. (+0.55)

Worked examples:

Positive flux: HEMPTHOLOGY (2018, $\Delta KL = +2.82$):

Candles; Wax melts; all of the foregoing goods containing CBD derived solely from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: TE HACE FALTA PRENDER MAS CARBON! (2017, $\Delta KL = +0.00$):

Charcoal briquettes; Coal briquettes; Firewood; Lighter fluid; Lighter fluid for charcoal; Wood chips for use as fuel; Wood chips of mesquite for smoking and grilling; Solid fire starters

Negative flux: HARNESSING THE POWER OF HEMP, TO POWER Y (2021, $\Delta KL = -1.92$):

Candles containing hemp with a delta-9 tetrahydrocannabinol concentration of not more than 0.3 percent on a dry weight basis

Class 005: Pharmaceuticals

Clean filings:	359,560
Unique owners:	134,184
Year range:	1990–2021
Mean KL_{pros} :	4.98 nats
Mean KL_{retr} :	4.99 nats
Mean / median ΔKL :	-0.011 / -0.011
5y survival rate:	56.4%
Share with SEC-listed owner:	10.60%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Silver Spur Corporation (+2.28); SILVER SPUR CORPORATION (+1.54); Earthmade Natural Vitamins, Inc. (+1.44)

Worked examples:

Positive flux: COLON MAGIC (2009, $\Delta KL = +2.72$):

Dietary supplement for eliminating toxins from the intestinal tract; Nutritional supplement for eliminating toxins from the intestinal tract

Flux-neutral: NEXT GENERATION CURCUMIN (2019, $\Delta KL = +0.00$):

Dietary supplements containing curcumin Dietary ingredient, namely, curcumin, for use as an ingredient in the manufacture of dietary supplements

Negative flux: HEART SMART (2011, $\Delta KL = -1.47$):

Foods and snacks which are free of Wheat to accommodate special medical and health conditions

Class 006: Common Metals

Clean filings:	97,003
Unique owners:	55,943
Year range:	1990–2021
Mean KL_{pros} :	5.48 nats
Mean KL_{retr} :	5.46 nats
Mean / median ΔKL :	+0.020 / +0.013
5y survival rate:	43.4%
Share with SEC-listed owner:	4.04%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Quad Performance Products, Inc. (+0.42); Kason Industries, Inc. (+0.36); General Electric Company (+0.29)

Worked examples:

Positive flux: GOLD DIAMOND (1990, $\Delta KL = +2.73$):

key chains, key tags and fobs, not of precious metal

Flux-neutral: GVS-K (2015, $\Delta KL = -0.00$):

Junctions of metal for pipes; valves of metal, other than parts of machines; reinforcing materials of metal for pipes; elbows of metal for pipes; clips of metal for pipes; drain traps valves of metal;

Negative flux: VALUE PRESS (1995, $\Delta KL = -1.10$):

metal closure caps for containers and bottles

Class 007: Machines & Power Tools

Clean filings:	160,666
Unique owners:	83,246
Year range:	1990–2021
Mean KL_{pros} :	5.34 nats
Mean KL_{retr} :	5.34 nats
Mean / median ΔKL :	+0.003 / -0.016
5y survival rate:	40.3%
Share with SEC-listed owner:	4.35%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Goodnature Products, Inc. (+0.50); Arovast Corporation (+0.40); ANC Technology INC (+0.38)

Worked examples:

Positive flux: FAFNIR TDC (1990, $\Delta KL = +2.34$):

bearings for axles, transmissions, drive shafts and controls for land vehicles

Flux-neutral: AXIS AUTOMATION (2015, $\Delta KL = -0.00$):

Design of standard and custom machinery for others Food processing machines, namely, tray denesters for separating food trays, and conveyor lines in the nature of vacuum conveyors Dispenser for dispen

Negative flux: STANNAH STAIRLIFTS (1992, $\Delta KL = -0.94$):

mobility aids for elderly or disabled persons; namely, stairlifts and parts thereof

Class 008: Hand Tools

Clean filings:	71,252
Unique owners:	44,508
Year range:	1990–2021
Mean KL_{pros} :	4.65 nats
Mean KL_{retr} :	4.58 nats
Mean / median ΔKL :	+0.070 / +0.032
5y survival rate:	40.3%
Share with SEC-listed owner:	2.66%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BORBOLETA BEAUTY INC. (+0.68); STILETTO TOOLS, INC. (+0.66); Marut Enterprises LLC (+0.52)

Worked examples:

Positive flux: GARD'N'GRIP (1990, $\Delta KL = +2.70$):

hand tools for lawn and garden use; namely, rakes, shovels, cutters, hoes, and forks

Flux-neutral: RIPLEY (1996, $\Delta KL = +0.00$):

wire, cable, and optical fibre preparation tools, namely, cutters, strippers, rippers, crimpers, trimmers, and slitters for electrical, electronic, CATV, telecommunication, telephone, and industrial a

Negative flux: COMFORT REACH (1999, $\Delta KL = -0.89$):

hand pruners, weeders, bulb planters and loppers

Class 009: Software & Electronics

Clean filings:	1,226,806
Unique owners:	602,439
Year range:	1990–2021
Mean KL_{pros} :	5.42 nats
Mean KL_{retr} :	5.43 nats
Mean / median ΔKL :	-0.011 / -0.023
5y survival rate:	48.6%
Share with SEC-listed owner:	5.52%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Leading Technology, Inc. (+2.50); Redmond Productions, Inc. (+1.37); Geniuses at Work Corporation (+1.18)

Worked examples:

Positive flux: MEDINET (1990, $\Delta KL = +3.40$):

a computer-based network including computer hardware and software

Flux-neutral: PRIME PROTECTION (2002, $\Delta KL = -0.00$):

Computerized ear tags or chips for the identification of cattle and related computer software for animal health record keeping Plastic ear tags for the identification of cattle

Negative flux: UNDONE (2011, $\Delta KL = -1.44$):

Electric hair curlers; Electric hair curling irons; Electric hair straightener; Electric hair straightening irons; Electric hair-curlers

Class 010: Medical Devices

Clean filings:	190,121
Unique owners:	88,206
Year range:	1990–2021
Mean KL_{pros} :	5.35 nats
Mean KL_{retr} :	5.35 nats
Mean / median ΔKL :	-0.001 / -0.017
5y survival rate:	47.4%
Share with SEC-listed owner:	7.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Societe Vygon (+0.67); NuGeneration Technologies, LLC (+0.60); CALIFORNIA EXOTIC NOVELTIES, LLC (+0.58)

Worked examples:

Positive flux: COVAFLU (2020, $\Delta KL = +3.20$):

Face covers being sanitary masks for protection against viral infection; Face shields for medical use

Flux-neutral: TRI-CHEW (2010, $\Delta KL = -0.00$):

Oral therapy apparatus, namely, a tool used by a professional therapist or trained caregiver for jaw movement, tongue movement, lip movement, oral exploration, or for the development of feeding skills

Negative flux: XEO (2019, $\Delta KL = -1.29$):

Catheters; catheters for percutaneous transluminal coronary angioplasty (PTCA catheters) and catheters for percutaneous transluminal angioplasty (PTA catheters), namely, catheters for balloon dilatati

Class 011: Lighting, HVAC & Appliances

Clean filings:	194,994
Unique owners:	112,215
Year range:	1990–2021
Mean KL_{pros} :	4.84 nats
Mean KL_{retr} :	4.79 nats
Mean / median ΔKL :	+0.045 / +0.014
5y survival rate:	38.8%
Share with SEC-listed owner:	2.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): G.K.L. Corp. (+0.59); ALPHARERRY INC (+0.59); VANJO LLC (+0.54)

Worked examples:

Positive flux: VENTEX (1990, $\Delta KL = +2.50$):

ventilation (air conditioning) installations and apparatus, fans, air conditioning apparatus

Flux-neutral: MASTER VALET (2012, $\Delta KL = +0.00$):

[Electric laundry irons and parts therefor] Garment steamers [; steam generators for household purposes and parts therefore]

Negative flux: SOOPSPOON (2019, $\Delta KL = -1.74$):

Electrically-powered handheld vaporizers for vaporizing vaporizable constituents of herbal and plant matter for creating an aroma for household use; Gas powered handheld vaporizers for vaporizing vapo

Class 012: Vehicles

Clean filings:	128,694
Unique owners:	68,694
Year range:	1990–2021
Mean KL_{pros} :	4.93 nats
Mean KL_{retr} :	4.90 nats
Mean / median ΔKL :	+0.034 / +0.013
5y survival rate:	43.7%
Share with SEC-listed owner:	5.01%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Legendary Auto Interiors, Ltd. (+0.99); Drop Stop, LLC (+0.70); Hooey, LLC (+0.62)

Worked examples:

Positive flux: PF 815 (1990, $\Delta KL = +2.36$):
vehicle tires for automobiles, trucks and vans

Flux-neutral: I WHEEL (2021, $\Delta KL = +0.00$):
Wheelchair spoke covers; Wheelchair spoke guards; Wheelchair tire covers; Handrims and pushrims for wheelchairs

Negative flux: ZF-POWERLINE (2011, $\Delta KL = -1.05$):
Transmissions, manual transmissions, automated and automatic transmissions, dual clutch transmissions, hybrid transmissions, synchronized transmissions, powershift transmissions, continuously variable

Class 013: Firearms & Pyrotechnics

Clean filings:	13,777
Unique owners:	7,027
Year range:	1991–2021
Mean KL_{pros} :	4.53 nats
Mean KL_{retr} :	4.47 nats
Mean / median ΔKL :	+0.063 / +0.059
5y survival rate:	45.9%
Share with SEC-listed owner:	3.94%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BATTLEARMS IP, LLC (+0.63); Ghost Inc. (+0.56); Battle Arms Development, Inc. (+0.50)

Worked examples:

Positive flux: STS (2009, $\Delta KL = +2.14$):
(Based on Use in Commerce) Firearms (Based on Intent to Use) Ammunition for firearms; Firearm attachments, namely, modular external rail systems for attaching accessories to firearms

Flux-neutral: SHAMAN (2021, $\Delta KL = +0.00$):
Hand gun accessories, namely, belt clips for securing a gun without the use of a holster

Negative flux: (1994, $\Delta KL = -1.60$):
non-explosive defensive weapon in the nature of an organic irritant packaged in a pressurized spray canister

Class 014: Jewelry & Watches

Clean filings:	109,375
Unique owners:	72,078
Year range:	1990–2021
Mean KL_{pros} :	4.52 nats
Mean KL_{retr} :	4.50 nats
Mean / median ΔKL :	+0.019 / +0.004
5y survival rate:	43.2%
Share with SEC-listed owner:	1.79%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Miki Corporation (+0.61); MIKI CORPORATION (+0.60); Spence Diamonds Ltd. (+0.49)

Worked examples:

Positive flux: OFF THE CLOSET (2013, $\Delta KL = +3.11$):

Women's jewelry; Earrings; Necklace and earring combinations that can be worn separately or as one piece; Necklaces as jewelry; Rings as jewelry; Bracelets

Flux-neutral: ANCHOR2ANCHOR (2021, $\Delta KL = +0.00$):

Hats; Scarfs; Shoes; Shorts; Sneakers; Socks; Sweaters; Sweatshirts; Underwear; Bomber jackets; Denim jackets; Denim jeans; Denim shorts; Denim skirts; Knit shirts; Knit tops; Knitted caps; Knitted un

Negative flux: SJ (1992, $\Delta KL = -1.29$):

gold or platinum jewelry with or without diamonds precious and/or semi-precious stones

Class 015: Musical Instruments

Clean filings:	12,496
Unique owners:	7,733
Year range:	1991–2021
Mean KL_{pros} :	4.47 nats
Mean KL_{retr} :	4.48 nats
Mean / median ΔKL :	-0.008 / -0.015
5y survival rate:	43.2%
Share with SEC-listed owner:	0.74%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Lone Wolf Blues Company (+0.38); Zdenka Infeld Asset Management GmbH (+0.32); Godlyke, Inc. (+0.29)

Worked examples:

Positive flux: EZ CHORD (2006, $\Delta KL = +1.27$):

Guitar playing assistance devices, namely, a sliding clamp-like structure positioned onto the guitar neck to form and sound guitar chords and notes

Flux-neutral: O-PORT (2009, $\Delta KL = -0.00$):

Guitar playing assistance devices, namely, a sliding clamp-like structure positioned onto the guitar neck to form and sound guitar chords and notes

Negative flux: RHYTHM TOOLS (2003, $\Delta KL = -1.04$):

Musical instruments, namely drums; drumheads; a click clack, namely a set of jingles, usually metal, mounted within a generally u-shaped frame; a timpano, namely a tensioned membrane mounted in a rect

Class 016: Paper, Print & Office

Clean filings:	444,018
Unique owners:	245,687
Year range:	1990–2021
Mean KL_{pros} :	5.39 nats
Mean KL_{retr} :	5.40 nats
Mean / median ΔKL :	-0.008 / -0.012
5y survival rate:	50.2%
Share with SEC-listed owner:	3.87%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Llumar Star Kites, Inc. (+1.08); Brown & Bigelow, Inc. (+0.90); Atkins Daniel D (+0.82)

Worked examples:

Positive flux: IMPRINT JIM DIESO (2010, $\Delta KL = +2.53$):

Art prints comprised of digital illustrations originating from photographs

Flux-neutral: STAND & WITHSTAND INTEGRITY GROUP (2021, $\Delta KL = +0.00$):

Educational services, namely, developing curriculum for others in the field of how to get an education; Training services in the field of how to get an education; Arranging and conducting of seminars

Negative flux: ITC MENDOZA ROMAN (1992, $\Delta KL = -2.08$):

typefaces, typefonts, and type designs of alphanumeric characters and/or typographical symbols typefaces, typefonts and type designs of alphanumeric characters and/or typographical symbols recorded on

Class 017: Rubber & Insulators

Clean filings:	51,254
Unique owners:	26,584
Year range:	1990–2021
Mean KL_{pros} :	4.97 nats
Mean KL_{retr} :	4.96 nats
Mean / median ΔKL :	+0.018 / +0.010
5y survival rate:	42.6%
Share with SEC-listed owner:	4.28%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Manco, Inc. (+0.44); MadeSolid, Inc. (+0.42); Insta-Foam Products, Inc. (+0.33)

Worked examples:

Positive flux: ADHESIVE PLUS (1990, $\Delta KL = +2.22$):

chemical adhesive compound for general, construction and household use in sealing, caulking and insulating

Flux-neutral: RENEWLENE (2008, $\Delta KL = +0.00$):

Extruded plastic in the form of bars, blocks, pellets, rods, sheets and tubes for use in manufacturing; Plastic in bars, blocks, pellets, rods, sheets and tubes for general industrial use; Plastic in

Negative flux: DURRA BUILDING STRONG SOLUTIONS (2004, $\Delta KL = -0.90$):

acoustical wall and roof panels for buildings

Class 018: Leather & Luggage

Clean filings:	147,807
Unique owners:	95,097
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.31 nats
Mean / median ΔKL :	+0.008 / -0.015
5y survival rate:	43.4%
Share with SEC-listed owner:	1.98%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Piper, Diane E. (+0.51); Romar International Corp. (+0.50); Bresnick, Tiffani (+0.38)

Worked examples:

Positive flux: STEPHANO SABETTI (1990, $\Delta KL = +2.63$):

men's, women's, and children's clothing - namely, pants, shirts, t-shirts, sweatshirts, jackets, coats, sweaters, scarves, gloves, socks, shoes, ties, and hats leather articles, namely, carry on bags,

Flux-neutral: ROTUNDA (1996, $\Delta KL = +0.00$):

business card cases, shoe bags for travel, tote bags

Negative flux: AXIOM EQUINE SPORT BOOT (2012, $\Delta KL = -0.98$):

Horse flanks and shanks, namely, horse protective padding, braces, and leg boots for the flanks and legs

Class 019: Building Materials

Clean filings:	70,015
Unique owners:	34,729
Year range:	1990–2021
Mean KL_{pros} :	5.08 nats
Mean KL_{retr} :	5.10 nats
Mean / median ΔKL :	-0.015 / -0.014
5y survival rate:	45.6%
Share with SEC-listed owner:	3.45%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Saroyan Lumber Company (+0.82); SCHLUTER-SYSTEMS KG (+0.38); Holyfield Management Inc. (+0.35)

Worked examples:

Positive flux: ACRAJOINT (1990, $\Delta KL = +2.94$):

building materials; namely, a system comprised of glass blocks and adhesive used to form a wall, sold as a unit

Flux-neutral: MANTA (1999, $\Delta KL = +0.00$):

material in the form of bricks, tiles, sheets or mortar for screening electromagnetic interference; material comprising a blend of carbon and iron oxides (ferrite) in the form of bricks, tiles, sheets

Negative flux: HOMESEAL (1998, $\Delta KL = -1.03$):

Asphalt, modified asphalt and rubberized asphalt based peel and stick membranes for use on roofing

Class 020: Furniture

Clean filings:	165,306
Unique owners:	102,530
Year range:	1990–2021
Mean KL_{pros} :	5.16 nats
Mean KL_{retr} :	5.09 nats
Mean / median ΔKL :	+0.075 / +0.042
5y survival rate:	42.3%
Share with SEC-listed owner:	2.76%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Xiamen Youjing E-commerce Co., Ltd. (+0.66); Everglades College, Inc. (+0.63); Shanxi Huacai Zhongxing E-commerce Co., Ltd. (+0.62)

Worked examples:

Positive flux: THE SIMPSONS (1990, $\Delta KL = +2.05$):

tables, chairs, desks, beds, children's furniture, mattresses, pillows, mirrors, picture frames, plastic key ring tags

Flux-neutral: TILANG (2020, $\Delta KL = +0.00$):

Pet toys Planters for flowers and plants Shoe bags for travel Curtain rods; Mirrors

Negative flux: ULTRA VIVID (2016, $\Delta KL = -0.83$):

Non-metal and non-leather key chains Treatment of materials by laser beam

Class 021: Household Utensils

Clean filings:	219,996
Unique owners:	143,153
Year range:	1990–2021
Mean KL_{pros} :	4.85 nats
Mean KL_{retr} :	4.78 nats
Mean / median ΔKL :	+0.067 / +0.037
5y survival rate:	40.8%
Share with SEC-listed owner:	2.23%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ALPHARERRY INC (+0.81); InnovatioNxt, LLC (+0.72); FLYCAMP TRADING LLC (+0.67)

Worked examples:

Positive flux: IREPELLER (2017, $\Delta KL = +1.96$):

Plug-in diffusers for mosquito repellents; Ultrasonic pest repellers

Flux-neutral: HALO (2017, $\Delta KL = +0.00$):

Wine preservation devices, namely, bottle stoppers specifically adapted for use with wine bottles

Negative flux: BEER RINGS (2016, $\Delta KL = -1.03$):

Charms for attachment to beverage glassware for identification purposes

Class 022: Ropes, Tents & Textiles (raw)

Clean filings:	24,515
Unique owners:	16,565
Year range:	1990–2021
Mean KL_{pros} :	4.70 nats
Mean KL_{retr} :	4.59 nats
Mean / median ΔKL :	+0.111 / +0.072
5y survival rate:	40.1%
Share with SEC-listed owner:	2.31%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Caravan Canopy Int'l, Inc. (+0.69); Xiamen Youjing E-commerce Co., Ltd. (+0.58); High Caliper Growing, Inc. (+0.54)

Worked examples:

Positive flux: (2013, $\Delta KL = +2.41$):

Awnings; Awnings comprised primarily of tensile fabric membranes; Awnings for vehicles

Flux-neutral: FAYONLINE (2018, $\Delta KL = -0.00$):

Sleeping bag pads; Futons Hammocks Blanket throws; Baby blankets; Children's blankets; Liners adapted to sleeping bags for camping

Negative flux: STONE CONTAINER CORPORATION (1990, $\Delta KL = -0.95$):

[logs, trees, wood chips and bark] agricultural and forestry products; namely, market pulp sawdust

Class 024: Textiles & Bedding

Clean filings:	97,945
Unique owners:	59,850
Year range:	1990–2021
Mean KL_{pros} :	4.25 nats
Mean KL_{retr} :	4.21 nats
Mean / median ΔKL :	+0.042 / +0.016
5y survival rate:	42.9%
Share with SEC-listed owner:	2.23%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): FISHER TEXTILES, LLC (+0.61); An Le (+0.51); Zhu Tianding (+0.42)

Worked examples:

Positive flux: RECORD BREAKERS WORLD OF SPEED (1990, $\Delta KL = +2.91$):

bed sheets, pillow cases, bedspreads, comforters, draperies, curtains, blankets

Flux-neutral: ARM & HAMMER (2010, $\Delta KL = +0.00$):

box spring covers, bed blankets

Negative flux: APPLETON LANE (1996, $\Delta KL = -0.94$):

vinyl wallcoverings and wallcovering borders textile fabrics for use in draperies and upholstery sample books of wallcoverings

Class 025: Clothing, Footwear & Headwear

Clean filings:	668,593
Unique owners:	426,635
Year range:	1990–2021
Mean KL_{pros} :	4.36 nats
Mean KL_{retr} :	4.40 nats
Mean / median ΔKL :	-0.041 / -0.051
5y survival rate:	51.6%
Share with SEC-listed owner:	1.39%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): CEO Lux, LLC (+0.72); CHILD SAFETY PRODUCTS, INC. (+0.66); Warner Bros. Inc. (+0.57)

Worked examples:

Positive flux: ÜBERION (2007, $\Delta KL = +3.14$):

Triathlon clothing, namely, triathlon tights, triathlon shorts, triathlon singlets, triathlon shirts and triathlon suits

Flux-neutral: COBALT (2019, $\Delta KL = -0.00$):

wetsuits; wetsuit gloves; water socks; rash guards; aqua shoes; flip-flops for use as footwear

Negative flux: GEFFEN BABY (2011, $\Delta KL = -1.53$):

Cloth diapers; Infant cloth diapers; Inserts specially adapted for cloth diapers made of hemp; Textile diapers

Class 026: Trim, Lace & Notions

Clean filings:	42,323
Unique owners:	29,873
Year range:	1990–2021
Mean KL_{pros} :	4.47 nats
Mean KL_{retr} :	4.34 nats
Mean / median ΔKL :	+0.124 / +0.075
5y survival rate:	40.4%
Share with SEC-listed owner:	1.76%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Celebrity Signatures International, Inc. (+1.07); RED SLATE BRANDS LLC (+0.82); KIM, HYUN JOON (+0.77)

Worked examples:

Positive flux: BEAUTY BUN (2011, $\Delta KL = +2.08$):

Hair accessories, namely, plastic styling inserts that create height and volume on the crown of the head

Flux-neutral: STOGEDEST (2021, $\Delta KL = +0.00$):

Wigs; Clown wigs; False hair; Hair extensions; Hair pieces; Hair pieces and wigs; Hair weaves; Human braiding hair; Wig caps

Negative flux: TEXTURED LOCS (2020, $\Delta KL = -2.66$):

Wigs, hairpieces, and add-in and add-on hair accessories constructed primarily of synthetic and/or human hair

Class 027: Carpets & Wall Hangings

Clean filings:	26,727
Unique owners:	15,621
Year range:	1990–2021
Mean KL_{pros} :	4.55 nats
Mean KL_{retr} :	4.49 nats
Mean / median ΔKL :	+0.053 / +0.019
5y survival rate:	40.9%
Share with SEC-listed owner:	3.14%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Marketing and Sales Management Corp. (+0.75); J. Josephson, Inc. (+0.59); Watershed Geosynthetics LLC (+0.54)

Worked examples:

Positive flux: THE SIMPSONS (1990, $\Delta KL = +2.63$):

carpets, rugs, vinyl wall coverings, wall hangings, wall paper

Flux-neutral: MILLICARE (1996, $\Delta KL = +0.00$):

cleaning preparations consisting of porous, absorbent particles in powder form and a liquid concentrate for use in the cleaning of commercial carpets

Negative flux: GENCORP (1997, $\Delta KL = -2.06$):

wallcovering having an outer surface of plastic

Class 028: Toys, Games & Sporting Goods

Clean filings:	325,521
Unique owners:	167,749
Year range:	1990–2021
Mean KL_{pros} :	4.94 nats
Mean KL_{retr} :	4.94 nats
Mean / median ΔKL :	-0.006 / -0.030
5y survival rate:	47.1%
Share with SEC-listed owner:	3.83%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Aruze Gaming America, Inc. (+1.17); KING SHOW GAMES, INC. (+0.73); Nabisco Brands Company (+0.73)

Worked examples:

Positive flux: WASHINGTON SENATORS (1990, $\Delta KL = +3.02$):

toys and sporting goods; namely, Christmas tree ornaments

Flux-neutral: CORTEZ (2001, $\Delta KL = -0.00$):

fishing products, namely, rods, reels, lures, fishing line, terminal tackle, tackle bags and tackle boxes

Negative flux: SWAY BUSTER (2013, $\Delta KL = -2.26$):

Golf Training Apparatus, namely, devices used to help groove a repetitive swing and stance

Class 029: Meat, Dairy & Prepared Foods

Clean filings:	134,683
Unique owners:	64,279
Year range:	1990–2021
Mean KL_{pros} :	4.63 nats
Mean KL_{retr} :	4.65 nats
Mean / median ΔKL :	-0.020 / -0.023
5y survival rate:	51.8%
Share with SEC-listed owner:	3.42%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BEYOND MEAT, INC. (+0.94); Aphria Inc. (+0.78); Crunchy Munchies LLC (+0.62)

Worked examples:

Positive flux: PHIVIDA (2016, $\Delta KL = +2.68$):

Healthy base beverages, namely, drinking water with vitamins containing hemp derived from industrial hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry

Flux-neutral: MUSCLE BROTH (2016, $\Delta KL = +0.00$):

liquid or powdered meat extracts containing amino acids, vitamins and minerals

Negative flux: EXCITE MY BITE (2021, $\Delta KL = -1.31$):

bouillon not containing or derived from cannabis with a delta-9 THC concentration of more than 0.3 percent on a dry weight basis or containing CBD

Class 030: Coffee, Tea, Bakery & Spices

Clean filings:	205,442
Unique owners:	104,841
Year range:	1990–2021
Mean KL_{pros} :	4.90 nats
Mean KL_{retr} :	4.91 nats
Mean / median ΔKL :	-0.011 / -0.010
5y survival rate:	51.3%
Share with SEC-listed owner:	2.72%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ECBlend LLC (+1.15); VMR Products, LLC (+0.97); New Generation Foods, Inc. (+0.85)

Worked examples:

Positive flux: SWITCH (2012, $\Delta KL = +2.67$):

Electronic cigarette batteries; Electronic cigarette battery chargers Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form used to refill electronic cigarettes Electronic ciga

Flux-neutral: FRUITABLES (1995, $\Delta KL = -0.00$):

fruit rolls, fruit leathers, dried fruit, fruit flavored confections, namely candied fruit, fruit flavored snacks

Negative flux: EMERALD CITY VAPE PREMIUM VAPOR LIQUID S (2015, $\Delta KL = -2.01$):

Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form used to refill electronic cigarette cartridges

Class 031: Agriculture & Live Animals

Clean filings:	64,135
Unique owners:	32,411
Year range:	1990–2021
Mean KL_{pros} :	4.92 nats
Mean KL_{retr} :	4.92 nats
Mean / median ΔKL :	-0.004 / -0.013
5y survival rate:	49.8%
Share with SEC-listed owner:	2.21%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): TRIM READY INC (+1.16); DAVID AUSTIN ROSES LIMITED (+1.09); CMG PARTNERS, INC (+0.76)

Worked examples:

Positive flux: BARK AVENUE (2017, $\Delta KL = +3.20$):

Edible dog treats not containing CBD and not containing any ingredients with a delta-9 tetrahydrocannabinol concentration of more than 0.3 percent on a dry weight basis

Flux-neutral: DOC'CHOPS (2004, $\Delta KL = -0.00$):

Animal snack food; Snack food for animals designed to contain medications

Negative flux: 12 TWENTY (2019, $\Delta KL = -1.24$):

Pet food Transdermal patches for therapeutic use in the treatment of addictions, for relief of pain, nausea, and discomfort, for relaxation, for reducing stress, nervousness and fatigue, for maintaini

Class 032: Beer & Soft Drinks

Clean filings:	100,951
Unique owners:	49,681
Year range:	1990–2021
Mean KL_{pros} :	4.18 nats
Mean KL_{retr} :	4.20 nats
Mean / median ΔKL :	-0.014 / -0.017
5y survival rate:	56.6%
Share with SEC-listed owner:	3.11%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): HONEST INC. (+1.98); Duclaw Brewery, LLC (+0.96); All Market Inc. (+0.89)

Worked examples:

Positive flux: PHIVIDA (2016, $\Delta KL = +2.77$):

Healthy base beverages, namely, drinking water with vitamins containing hemp derived from industrial hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry

Flux-neutral: TRUE CAL-ORGANIC FARMS (2013, $\Delta KL = -0.00$):

Organic fresh fruits; organic fresh vegetables Organic fruit juices and beverages; organic vegetable juices and beverages

Negative flux: NORTHERN EXPOSURE (1997, $\Delta KL = -1.15$):

fermented alcoholic beverages made with malt, namely, assorted beers

Class 033: Wine & Spirits

Clean filings:	68,232
Unique owners:	32,766
Year range:	1990–2021
Mean KL_{pros} :	4.09 nats
Mean KL_{retr} :	4.06 nats
Mean / median ΔKL :	+0.038 / +0.020
5y survival rate:	53.1%
Share with SEC-listed owner:	0.90%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Stranger & Stranger Ltd (+0.76); MAE-STRO TEQUILERO, S.A. DE C.V. (+0.72); Destiladora Del Valle de Tequila, S.A. de C.V. (+0.70)

Worked examples:

Positive flux: MARYGIN (2018, $\Delta KL = +2.60$):

Liquor; all of the foregoing containing ingredients solely derived from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis and not containi

Flux-neutral: UNITED STARS (2018, $\Delta KL = +0.00$):

Alcoholic beverages, namely, vodka, prepared alcoholic cocktails, peppermint liqueurs, distilled spirits, alcoholic beverages containing fruit

Negative flux: CHATEAU SOCIANDO-MALLET (2012, $\Delta KL = -1.04$):

Wines of guaranteed label of origin Haut Medoc from a particular wine-making estate

Class 034: Tobacco & Smokers' Articles

Clean filings:	36,678
Unique owners:	18,795
Year range:	1990–2021
Mean KL_{pros} :	3.66 nats
Mean KL_{retr} :	3.50 nats
Mean / median ΔKL :	+0.163 / +0.115
5y survival rate:	55.4%
Share with SEC-listed owner:	1.42%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Freedom Smokeless LLC (+1.65); FRESH BAKED CO. (+1.37); Vaporfuel, LLC (+1.26)

Worked examples:

Positive flux: DAISY (2016, $\Delta KL = +3.90$):

Oral vaporizers for smoking purposes, namely, for smoking cannabis solely derived from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: SATIVAH (2010, $\Delta KL = +0.00$):

Herbal smoking blend for cigarettes, cigars and pipes [and smoking articles used therewith, namely, cigarette papers, cigar or cigarette lighters not made of precious metals and matches]

Negative flux: SITRA (2021, $\Delta KL = -1.39$):

Hookahs; hookah parts, namely, containers for housing charcoal separate from bowls while permitting heat transfer to tobacco placed in the bowls; smokers articles, namely, containers for housing hooka

Class 035: Advertising & Retail

Clean filings:	996,559
Unique owners:	584,964
Year range:	1990–2021
Mean KL_{pros} :	5.38 nats
Mean KL_{retr} :	5.40 nats
Mean / median ΔKL :	-0.019 / -0.027
5y survival rate:	47.0%
Share with SEC-listed owner:	3.65%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Steven S. Lazar (+1.17); Beverly Hills Internet (+1.08); FINANZIARIA D'INVESTIMENTO - FININVEST SPA (+0.75)

Worked examples:

Positive flux: ACI AUDIO COMMUNICATIONS INCORPORATED (1990, $\Delta KL = +3.44$):

promoting the sale of the goods and/or services of others by recording telephone commercials and providing access thereto by the members of the purchasing public

Flux-neutral: SAVE IT MAKES CENTS (2014, $\Delta KL = -0.00$):

Retail store services featuring convenience store items and gasoline

Negative flux: RECLAIM YOUR SPACE (2011, $\Delta KL = -1.51$):

Providing organizing services for belongings and spaces in a residence or office such as papers and time

Class 036: Insurance & Financial

Clean filings:	399,896
Unique owners:	196,121
Year range:	1990–2021
Mean KL_{pros} :	4.92 nats
Mean KL_{retr} :	4.96 nats
Mean / median ΔKL :	-0.042 / -0.055
5y survival rate:	47.3%
Share with SEC-listed owner:	6.69%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): The Upper Deck Company (+0.70); Philip Lilien (+0.69); AMERICAN TELEPHONE AND TELEGRAPH COMPANY (+0.67)

Worked examples:

Positive flux: WE SPEAK YOUR LANGUAGE (1990, $\Delta KL = +3.23$):

insurance administration services in the field of health care

Flux-neutral: SPACE WE'VE GOT SPACE TO FILL (1999, $\Delta KL = +0.00$):

Commercial real estate services in the nature of property leasing Commercial real estate services in the nature of commercial property development

Negative flux: MOST-WANTED (1992, $\Delta KL = -1.31$):

fund-raising services for basic and applied research into neuromuscular disorders, and for funding of hospital services for persons afflicted by such disorders

Class 037: Construction & Repair

Clean filings:	178,756
Unique owners:	106,312
Year range:	1990–2021
Mean KL_{pros} :	5.18 nats
Mean KL_{retr} :	5.18 nats
Mean / median ΔKL :	+0.001 / +0.002
5y survival rate:	42.5%
Share with SEC-listed owner:	4.84%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Aerus Concepts, L.P. (+0.39); Irvine Company, The (+0.38); Coastal Corporation, The (+0.32)

Worked examples:

Positive flux: MODEM EXPRESS (1990, $\Delta KL = +2.53$):

REPAIR SERVICES IN THE FIELD OF [COMPUTERS] AND COMPUTER PERIPHERALS
[retail store services in the field of computers and computer peripherals]

Flux-neutral: SONERI (2015, $\Delta KL = -0.00$):

Hotel services; Restaurant services Retail store services featuring groceries Financial advisory and consultancy services; Financial investment brokerage; Real estate agencies; Real estate brokerage B

Negative flux: THE MAJORS (2007, $\Delta KL = -1.21$):

Homebuilding services; development of real property for single-family and multi-family residential and master-planned community purposes

Class 038: Telecommunications

Clean filings:	170,826
Unique owners:	86,334
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.37 nats
Mean / median ΔKL :	-0.050 / -0.040
5y survival rate:	51.5%
Share with SEC-listed owner:	7.36%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Los Angeles Cellular Telephone Company (+0.90); Iowa Network Services, Inc. (+0.81); Graphnet, Inc. (+0.76)

Worked examples:

Positive flux: CORPORATE TELEMAGEMENT (1990, $\Delta KL = +3.43$):

telecommunication services, namely, providing long distance telephone services

Flux-neutral: VZCREW (2002, $\Delta KL = -0.00$):

Telecommunication services, namely wireless telephone communication services and wireless messaging services providing online information in the field of entertainment

Negative flux: SIMPLIFY THE CONNECTION (1998, $\Delta KL = -1.07$):

Long distance telecommunications services, namely, long-distance telephone calling services

Class 039: Transport & Logistics

Clean filings:	120,417
Unique owners:	67,340
Year range:	1990–2021
Mean KL_{pros} :	5.03 nats
Mean KL_{retr} :	5.05 nats
Mean / median ΔKL :	-0.020 / -0.017
5y survival rate:	45.4%
Share with SEC-listed owner:	6.90%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Go America Tours, Inc. (+0.67); Coastal Corporation, The (+0.55); Sun Run Generation LLC (+0.45)

Worked examples:

Positive flux: JAL JAPAN AIR LINES A WORLD OF DIFFERENC (1990, $\Delta KL = +2.68$):

transportation of passengers and cargo by air; and storage services, namely the storage of cargo and luggage which has been or is to be transported by air

Flux-neutral: VICTORIA CLIPPER (2017, $\Delta KL = +0.00$):

transportation of passengers by ship without on-board overnight accommodations

Negative flux: YOUR ENERGY EDGE (1999, $\Delta KL = -1.24$):

public utility services, namely supplying natural gas to others (

Class 040: Materials Treatment

Clean filings:	94,015
Unique owners:	59,227
Year range:	1990–2021
Mean KL_{pros} :	5.37 nats
Mean KL_{retr} :	5.37 nats
Mean / median ΔKL :	-0.004 / -0.011
5y survival rate:	44.6%
Share with SEC-listed owner:	5.75%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Intervet Inc. (+0.32); United States Department of Energy (+0.31); Cherry Street Energy, LLC (+0.30)

Worked examples:

Positive flux: ETERNITY HANES HEAVYWEIGHT (2007, $\Delta KL = +3.02$):

Imprinting messages on wearing apparel, accessories and mugs

Flux-neutral: MAGSHIRE BEL (2020, $\Delta KL = -0.00$):

Cryogenic preservation; Cryopreservation of human tissue; Cryopreservation services; Vitrification drying and dehydration preservation services for biologically active materials

Negative flux: SUPERME! (2010, $\Delta KL = -1.47$):

Creation of custom portraits and paintings based on photographs, snapshots or images provided by the customer

Class 041: Education & Entertainment

Clean filings:	920,449
Unique owners:	531,807
Year range:	1990–2021
Mean KL_{pros} :	5.18 nats
Mean KL_{retr} :	5.19 nats
Mean / median ΔKL :	-0.006 / -0.016
5y survival rate:	47.0%
Share with SEC-listed owner:	3.06%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): CPI Church Street, Inc. (+1.11); Digital Gaming Corporation Limited (+0.93); Dapper Labs Inc. (+0.86)

Worked examples:

Positive flux: THE COASTERS (1990, $\Delta KL = +3.88$):
entertainment services in the nature of a musical group

Flux-neutral: BEN MIYARES' PACKAGING MANAGEMENT UPDATE (2005, $\Delta KL = +0.00$):

Providing online electronic newsletters and trade journals in the field of packaging machinery, the packaging machinery industry and association membership affairs; providing electronic newsletters an

Negative flux: MLE MAJOR LEAGUE EMPIRE (2019, $\Delta KL = -1.29$):

Entertainment services, namely, an on-line activity where you create your own music videos

Class 042: Scientific & Tech Services

Clean filings:	743,414
Unique owners:	411,644
Year range:	1990–2021
Mean KL_{pros} :	5.17 nats
Mean KL_{retr} :	5.20 nats
Mean / median ΔKL :	-0.022 / -0.006
5y survival rate:	46.1%
Share with SEC-listed owner:	5.61%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Geniuses at Work Corporation (+1.28); Advance Publications, Inc. (+0.98); Albin, James R. (+0.92)

Worked examples:

Positive flux: PIPELINE (1993, $\Delta KL = +3.47$):
computer software for use in providing multiple-user access to a global computer information network for the transfer and dissemination of a wide range of information providing multiple-user access to

Flux-neutral: IEEE: ADVANCING TECHNOLOGY FOR HUMANITY (2009, $\Delta KL = -0.00$):

Providing news and information in the field of science, engineering and engineering technology via the Internet; Accreditation services, namely, developing, evaluating, and testing standards with rega

Negative flux: TAILGATE TEAM (1996, $\Delta KL = -2.13$):

retail grocery store and supermarket services

Class 043: Restaurants & Hotels

Clean filings:	136,699
Unique owners:	83,211
Year range:	2002–2021
Mean KL_{pros} :	4.87 nats
Mean KL_{retr} :	4.85 nats
Mean / median ΔKL :	+0.014 / +0.012
5y survival rate:	48.4%
Share with SEC-listed owner:	2.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): EX3 (+0.42); BRINKER INTERNATIONAL PAYROLL COMPANY, L.P. (+0.37); Dunhill 1530 Main, L.P. (+0.34)

Worked examples:

Positive flux: MARYLAND FRIED CHICKEN COMPLETE DINNERS (2008, $\Delta KL = +2.39$):

Restaurant services, including sit-down service of food and take-out restaurant services

Flux-neutral: PANNA NEW LATINO FOOD (2015, $\Delta KL = +0.00$):

Café and restaurant services; Restaurant services featuring tequenos, cachitos, breads, sandwiches, hamburger, hotdogs, arepas, pastry, cachapas, empanadas, cheese sticks, salads, omelets, natural jui

Negative flux: KIVO (2007, $\Delta KL = -0.90$):

Contract food services; preparing and serving foods in institutional dining facilities

Class 044: Medical & Personal Care

Clean filings:	188,794
Unique owners:	109,333
Year range:	2002–2021
Mean KL_{pros} :	4.96 nats
Mean KL_{retr} :	4.95 nats
Mean / median ΔKL :	+0.005 / -0.000
5y survival rate:	45.8%
Share with SEC-listed owner:	5.56%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ClearChoice Holdings LLC (+0.78); Aspiration Partners, Inc. (+0.53); Bittner, Craig A. (+0.43)

Worked examples:

Positive flux: AMERICAN COLLECTIVE (2009, $\Delta KL = +2.30$):

Medical cannabis collective and dispensary services, namely, providing medical cannabis to medical cannabis patients

Flux-neutral: NO BARRIERS. NO LIMITS. INDEPENDENCE FOR (2019, $\Delta KL = +0.00$):

Health care consulting in the field of occupational therapy; Occupational therapy services

Negative flux: ROMAN HEMP CO. (2021, $\Delta KL = -1.05$):

Farming services in the field of hemp and hemp flower derived from hemp with a delta-9 tetrahydrocannabinol or a delta-8 tetrahydrocannabinol concentration of not more than 0.3 percent on a dry-weight

Class 045: Legal & Personal Services

Clean filings:	126,420
Unique owners:	81,811
Year range:	2002–2021
Mean KL_{pros} :	4.96 nats
Mean KL_{retr} :	4.99 nats
Mean / median ΔKL :	-0.022 / -0.022
5y survival rate:	45.2%
Share with SEC-listed owner:	3.47%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): YUGA LABS, INC. (+0.56); FamFriend, LLC (+0.50); Blend Images, LLC (+0.39)

Worked examples:

Positive flux: A COURSE IN MIRACLES MINISTER (2014, $\Delta KL = +2.06$):
Ecclesiastical services, namely, the licensing of ministers, namely, ordaining ministers to perform religious ceremonies

Flux-neutral: ARTHUR MILLER JR ADNF FOUNDATION A DAUGH (2017, $\Delta KL = +0.00$):

Charitable outreach services, namely, providing counseling services in the field of interpersonal communication between law enforcement officers and community members who have faced tragedies or other

Negative flux: NEFA FOUNDATION (2008, $\Delta KL = -1.50$):
Providing information regarding understanding and preventing terrorism

E Per-industry deep dives: trending, ending, and public debuts

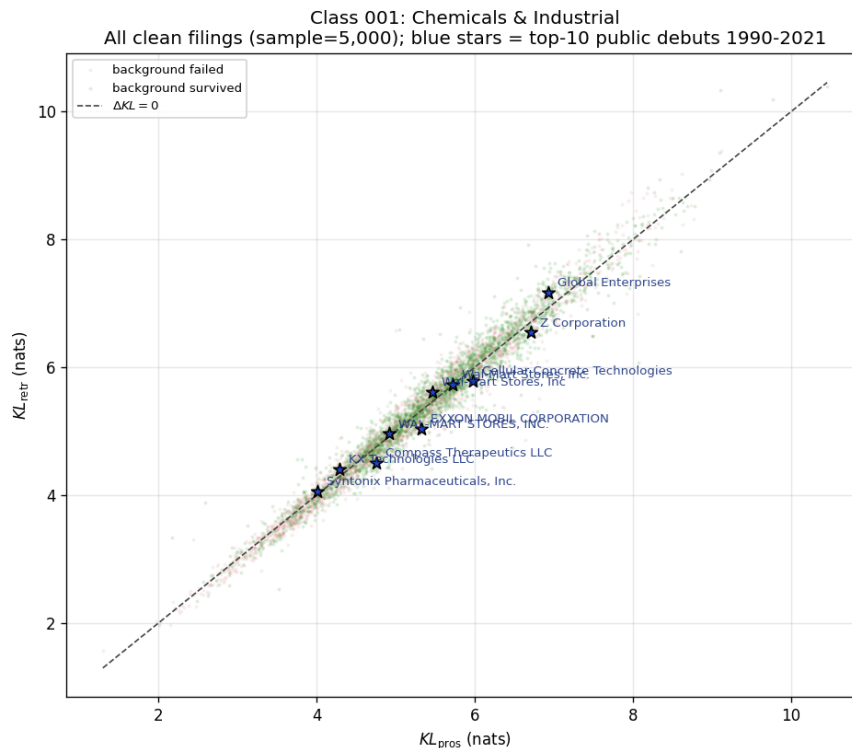
This appendix complements Appendix D with a richer per-class snapshot. For each NICE class with at least 5,000 clean H=2y filings, we present:

- Summary statistics and the SEC-listed share of filings.
- A scatter of all clean filings in the $(KL_{\text{pros}}, KL_{\text{retr}})$ plane, with the $\Delta KL = 0$ diagonal and the top-10 SEC-listed firms whose first trademark filing occurred in 1990–2021 marked as blue stars and labeled.
- Top-10 *trending* tokens (over-represented in the top-decile- ΔKL stratum within this class) with their intra-class 5y survival rates.
- Top-10 *ending* tokens (over-represented in the bottom-decile- ΔKL stratum) with their intra-class survival rates.
- Top-10 flux-neutral tokens by frequency in the $|\Delta KL| \leq 0.02$ stratum, with survival rates.
- A table of the top-10 public firms with their debut filing’s mark, year, KL_{pros} , ΔKL , and a g/s excerpt.

“Top-10 public firms” are ranked by maximum observed annual revenue in the SEC EDGAR financial-statement panel. Firms whose primary trademark class is the focal class are eligible; firms whose first filing predates 1990 (i.e., they predated our observation window) are excluded.

Class 001: Chemicals & Industrial

Clean filings (1990–2021): 125,272
Mean KL_{pros} / KL_{retr} / ΔKL : 5.36 / 5.37 / -0.009 nats
5y survival rate: 47.1%
Share with SEC-listed owner: 6.47%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
ampule	4.31×	144	97%
delta-	4.12×	391	63%
cannabis	4.04×	223	62%
tetrahydrocannabinol	3.93×	351	67%
percent	3.91×	391	63%
thc	3.81×	420	63%
tincture	3.71×	170	88%
hemp	3.36×	711	64%
cbd	3.35×	379	70%
biomaterials	3.33×	144	43%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
e-liquid	6.03×	375	54%
non-synthetic	6.02×	211	16%
odorants	5.74×	237	19%
bleaches	4.91×	277	24%
brighteners	4.82×	274	24%
colognes	4.24×	330	34%
glycerin	3.91×	486	47%
refill	3.51×	276	57%
cigarette	3.39×	814	51%
decalcifying	3.35×	203	61%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

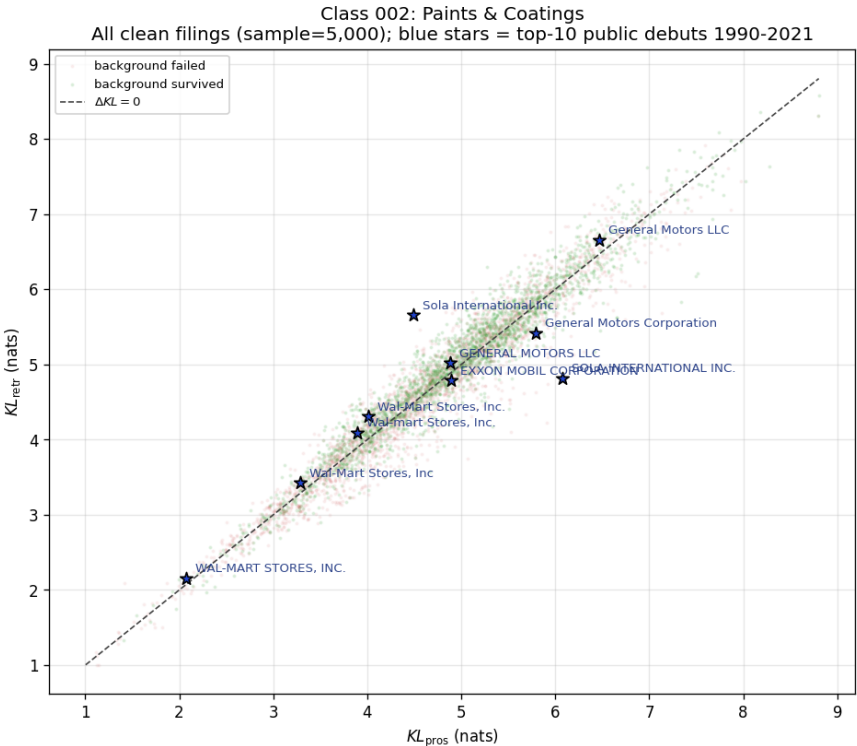
token	n in stratum	n in class	intra-class survival
use	8,388	79,491	44%
chemical	5,379	46,269	41%
chemicals	4,406	35,839	39%
preparations	3,746	31,701	38%
manufacture	3,552	31,197	42%
industrial	3,456	26,413	36%
products	3,125	25,459	40%
purposes	3,068	24,195	35%
industry	2,672	21,184	32%
materials	2,251	15,521	35%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Cellular Concrete Technologies, Inc.	FOAMCON	1994	5.98	+0.19	foam concentrate for use in making cellular concrete, cement-based insulation and concrete building products
Compass Therapeutics LLC	CURING CANCER THROUGH COMBINAT	2015	4.76	+0.25	Custom design and development of antibody therapeutic preparations Antibody therapeutic preparations for medical purposes Antibodies for scientific and research purposes
Global Enterprises	STEEL SEAL	1993	6.93	-0.25	tire puncture sealing compounds
KX Technologies LLC	FINITY	2013	4.29	-0.11	chemical and mineral fiber pulp preparations in the nature of filtering materials for the manufacturing of domestic, industrial, and commercial water filters, and for the manufacture of commercial and
Syntonix Pharmaceuticals, Inc.	TRANSCCEPT	2002	4.01	-0.05	scientific research and development services for others in the field of pharmaceutical preparations biological agents used in the manufacture of pharmaceutical preparations
Z Corporation	Z-CURE	2008	6.71	+0.17	Chemical in liquid form used as an infiltrant or sealant agent in the manufacture of three-dimensional models; and Chemical preparations in liquid form for use as an infiltrant or sealant agent in the
WAL-MART STORES, INC.	SAM'S AMERICAN CHOICE	1996	4.91	-0.04	pretzels and pretzels with peanut butter filling; pasta; macaroni and cheese; cheese flavored snack foods, namely, puffed corn snacks, popped popcorn, and corn chips; cookies; popped popcorn; chocolat
Wal-Mart Stores, Inc	GREAT VALUE	2016	5.46	-0.15	algacides Chlorine; shock treatments, namely, water treatment chemicals for use in swimming pools or spas
Wal-Mart Stores, Inc.	GARDEN BASICS	1997	5.72	-0.01	flower pot holders and planters fertilizer, potting soil, and soil conditioners for domestic use bagged and potted live plants, flower bulbs, and gardening seeds gardening tools, namely, spreaders, sh
EXXON MOBIL CORPORATION	OPTEMA	1990	5.32	+0.28	plastic resins for use in the packaging, rubber and textile industries

Class 002: Paints & Coatings

Clean filings (1990–2021): 33,653
Mean KL_{pros} / KL_{retr} / ΔKL : 4.78 / 4.79 / -0.012 nats
5y survival rate: 44.6%
Share with SEC-listed owner: 4.40%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
photocopiers	5.42×	1,398	21%
photocopier	5.19×	801	27%
filled	4.81×	2,270	26%
copiers	4.71×	753	27%
inkjet	4.57×	937	24%
toner	4.26×	2,707	32%
thermochromic	4.18×	268	15%
capture	4.17×	429	18%
multifunction	4.15×	414	20%
skin-dressing	4.10×	139	27%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
surfacers	$3.11\times$	61	36%
interior	$2.41\times$	4,299	47%
exterior	$2.14\times$	4,828	48%
latex	$1.97\times$	487	42%
character	$1.96\times$	148	32%
pigmented	$1.95\times$	1,250	48%
labelling	$1.94\times$	62	21%
courts	$1.91\times$	68	43%
equipped	$1.90\times$	58	19%
nozzle	$1.88\times$	69	32%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

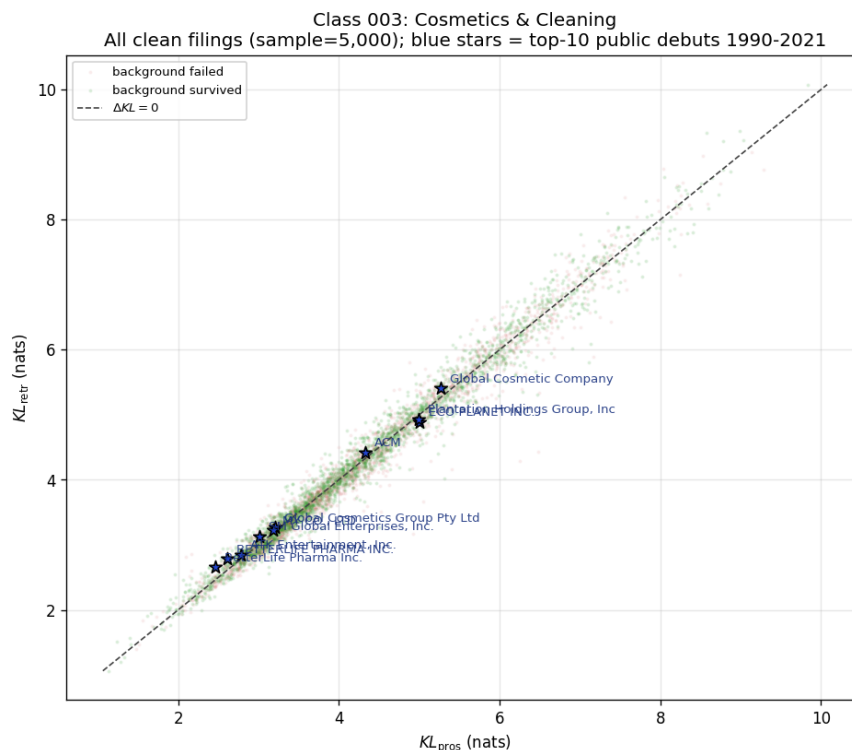
token	n in stratum	n in class	intra-class survival
use	1,396	15,797	41%
coatings	1,284	14,341	45%
nature	1,041	9,985	40%
paints	1,008	11,058	38%
coating	848	8,607	42%
paint	746	8,935	44%
industrial	718	6,496	37%
metal	577	5,160	39%
preparations	573	4,896	32%
wood	549	6,076	38%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
SOLA INTERNATIONAL INC.	INTERIOR	1998	6.07	+1.25	coatings for ophthalmic lenses, anti-reflective coatings for ophthalmic lenses, abrasion resistant coatings for ophthalmic lenses optical goods and apparatus, namely, spectacles, sunglasses, safety go
Sola International Inc.	GLARE OUT	1998	4.49	-1.17	coatings for ophthalmic lenses, anti-reflective coatings for ophthalmic lenses, abrasion resistant coatings for ophthalmic lenses; impact resistant coatings for ophthalmic lenses; anti-fog coatings fo
WAL-MART STORES, INC.	CURIOSLY J	2017	2.08	-0.08	Dyes for use in the manufacture of textiles and fabrics; food coloring; paints; finger paint; inks, toners, and dyes for use with or on textiles and fabrics; edible food coloring Alcoholic cocktail mi
Wal-Mart Stores, Inc	DAILY'S	2000	3.29	-0.14	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.	COLOR PLACE	1994	4.01	-0.30	house paint, interior and exterior
Wal-mart Stores, Inc.		2011	3.89	-0.20	On-line retail store services featuring a wide variety of consumer goods of others; Retail department store services; Retail store services featuring a wide variety of consumer goods of others Air bru
EXXON MOBIL CORPORATION		2006	4.89	+0.11	Anti-rust greases; Anti-rust oils; Rust preventatives in the nature of a coating
GENERAL MOTORS LLC	SATURN	1992	4.89	-0.13	paint for motor vehicles
General Motors Corporation	WHEN THE RIGHT WAY IS THE ONLY	2000	5.79	+0.37	Paint for motor land vehicles
General Motors LLC	ENVIRONMIN	2009	6.46	-0.19	Paint for use in the manufacture of automobiles

Class 003: Cosmetics & Cleaning

Clean filings (1990–2021):	326,430
Mean KL_{pros} / KL_{retr} / ΔKL :	4.32 / 4.31 / +0.005 nats
5y survival rate:	50.0%
Share with SEC-listed owner:	2.94%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
e-liquid	$4.04\times$	800	76%
vaporizer	$3.83\times$	399	80%
attaching	$3.20\times$	1,286	47%
cannabis	$3.07\times$	1,158	64%
refill	$3.05\times$	780	64%
affixing	$2.96\times$	5,356	33%
non-veterinary	$2.92\times$	930	40%
trims	$2.91\times$	502	24%
anxiety	$2.80\times$	521	63%
cigarettes	$2.75\times$	1,463	62%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
post-suntanning	$6.28\times$	492	90%
pre-suntanning	$5.81\times$	530	89%
degreaser	$4.64\times$	474	57%
cuticles	$4.09\times$	575	76%
straightener	$3.80\times$	532	76%
lightener	$3.79\times$	755	79%
sunless	$3.78\times$	1,352	72%
astringent	$3.67\times$	1,562	68%
shimmer	$3.47\times$	525	56%
soak	$3.43\times$	937	63%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

token	n in stratum	n in class	intra-class survival
preparations	22,153	174,132	47%
care	19,078	148,035	51%
skin	17,196	138,329	50%
hair	15,788	117,462	48%
body	14,323	117,200	51%
lotions	14,120	108,293	49%
non-medicated	13,803	111,213	48%
creams	13,345	101,302	49%
cosmetics	13,266	110,457	46%
cosmetic	12,597	99,440	45%

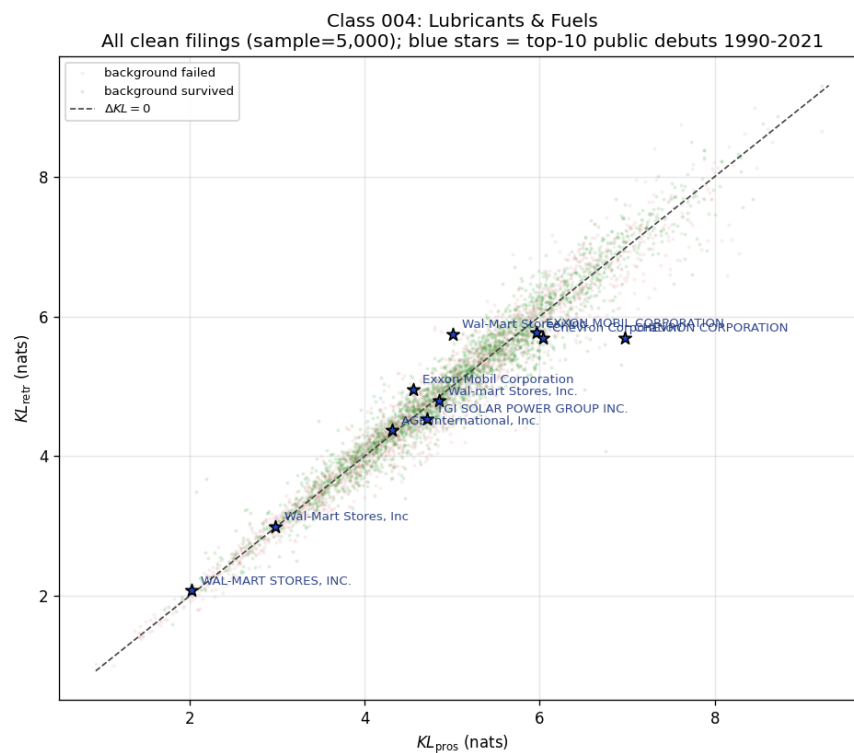
Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ACM		VITICOLOR	2008	4.33	-0.09	Cosmetics; cosmetic products, namely, self-tanning lotions; cosmetic preparations for sun-tanning; skin creams; foundation for evening out skin tone; cosmetics, namely, pigmentation regulators
ATK Inc.	Entertainment,	AALAM TUCK KHALED	2020	2.78	-0.07	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.		SLEEP BET-TERX	2019	2.61	-0.17	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BetterLife Pharma Inc.		LIVE BET-TERX	2020	2.47	-0.19	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
ECO PLANET INC.		ORGANIC OSCAR	2009	5.00	+0.12	Organic pet care kits comprising shampoo, conditioner, body spray sold as a unit
Global Cosmetic Company		SEA ELEMENTS	2003	5.27	-0.14	Hair Shampoo, Hair conditioners ,Hair styling gels
Global Cosmetics Group Pty Ltd		COUTURE SILK	2012	3.20	-0.06	Hair coloring preparations; hair color; hair dyes; hair care preparations; [hair shampoo; hair conditioner;] all of the aforementioned containing a proportion of silk extract; bleaching preparations
MY CO., LTD.		GOEE	2020	3.18	-0.04	Under-eye creams; shampoos for the care of scalp; non-medicated toners for scalp; lip balm; cosmetic preparations for the hair and scalp; skin moisturizers; skin moisturizer masks; skin cleansing loti
Plantation Group, Inc.	Holdings	THE PLANTATION SHOP	2004	5.00	+0.07	Gourmet cookies, coffee, wafers, biscuits, snack nut mix consisting primarily of chocolate covered nuts and pralines, candy, toffee and cheese straws Jewelry Retail store services and retail shop at h
SM Global Enterprises, Inc.		ORIGINAL SHAVE COMPANY	2017	3.01	-0.12	On-line wholesale and retail store services featuring Grooming supplies for Men and Women After shave lotions; after-shave; after-shave balms; after-shave creams; after-shave emulsions; after-shave ge

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Class 004: Lubricants & Fuels

Clean filings (1990–2021): 36,160
Mean KL_{pros} / KL_{retr} / ΔKL : 4.91 / 4.91 / +0.004 nats
5y survival rate: 47.8%
Share with SEC-listed owner: 5.58%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
clamshell	5.07×	284	43%
bougies	4.33×	240	43%
mini-pumpkins	4.28×	339	40%
carved	4.24×	354	38%
pumpkins	4.10×	356	41%
gourds	4.04×	359	38%
unity	3.89×	311	40%
contained	3.70×	424	46%
votive	3.48×	1,160	43%
tealight	3.36×	776	36%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
briquets	$7.00\times$	60	43%
reconstructor	$6.79\times$	53	89%
mattifier	$6.60\times$	53	87%
luster	$6.42\times$	53	89%
exfoliator	$5.38\times$	80	82%
tint	$5.33\times$	75	77%
corrector	$5.30\times$	66	79%
straightener	$5.16\times$	62	84%
detangler	$5.00\times$	78	79%
polisher	$4.89\times$	139	86%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

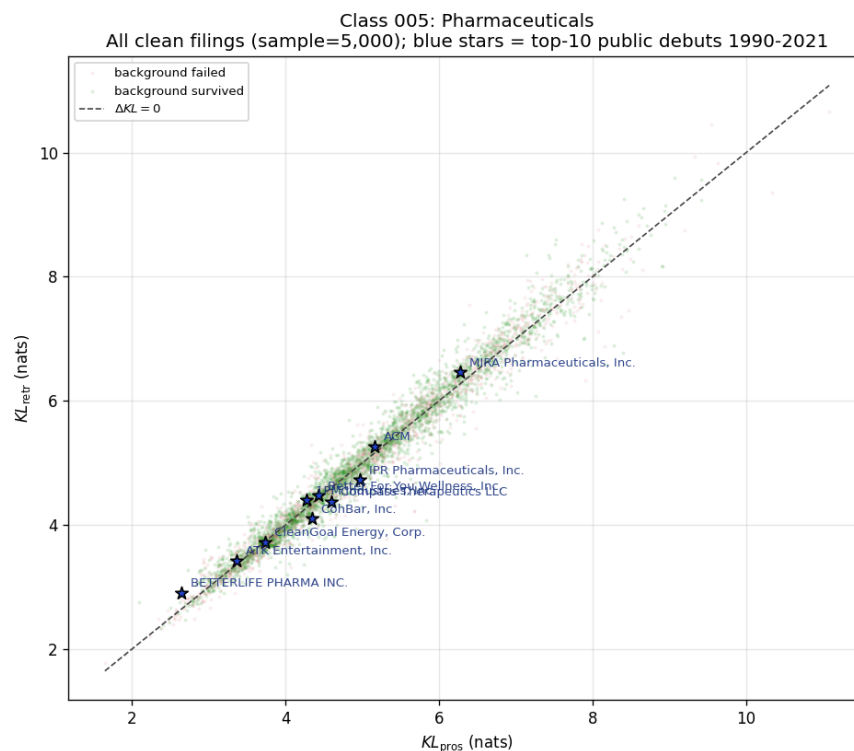
token	n in stratum	n in class	intra-class survival
use	1,378	14,053	43%
candles	1,353	15,757	46%
oils	1,204	12,471	42%
oil	923	10,398	46%
industrial	854	8,707	38%
fuel	799	8,302	44%
lubricants	751	8,599	41%
preparations	709	6,566	40%
purposes	706	5,686	34%
products	688	6,630	45%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.32	-0.05	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
TGI SOLAR POWER GROUP INC.	ADVENT	2021	4.71	+0.18	Batteries; Batteries and battery chargers; Battery boxes; Solar panels for the production of electricity; Grids for batteries; Off-grid power and deep cycle battery storage systems comprised of batter
WAL-MART STORES, INC.	CURIOSLY J	2017	2.02	-0.06	Dyes for use in the manufacture of textiles and fabrics; food coloring; paints; finger paint; inks, toners, and dyes for use with or on textiles and fabrics; edible food coloring Alcoholic cocktail mi
Wal-Mart Stores, Inc	DAILY'S	2000	2.98	-0.01	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.	TECH 2000	1995	5.01	-0.73	automotive oils and lubricants
Wal-mart Stores, Inc.	BACKYARD GR LL	2011	4.85	+0.06	Barbecue smokers; Barbecues and grills; Ceramic briquettes for use in barbecue grills; Charcoal grills; Gas grills; Lighters primarily for lighting grills, fireplaces and candles; Roasters powered by
EXXON MOBIL CORPORATION		1991	5.97	+0.20	hydraulic oils
Exxon Mobil Corporation	E+2	1998	4.55	-0.41	fuel for motor vehicles, namely, gasoline
CHEVRON CORPORATION	PRO-GARD	1990	6.97	+1.28	all purpose aerosol lubricant
Chevron Corporation	MECHANISM LPS	1991	6.04	+0.35	hydraulic oil

Class 005: Pharmaceuticals

Clean filings (1990–2021): 359,560
Mean KL_{pros} / KL_{retr} / ΔKL : 4.98 / 4.99 / -0.011 nats
5y survival rate: 56.4%
Share with SEC-listed owner: 10.60%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
cannabis	3.40×	1,393	78%
marijuana	3.39×	366	84%
e-liquid	3.18×	449	83%
vaporizer	3.08×	419	82%
opioid	2.94×	405	76%
cannabidiol	2.90×	1,139	89%
delta-	2.88×	2,174	84%
hand-sanitizing	2.85×	3,404	54%
occurring	2.81×	544	78%
treats	2.67×	2,685	60%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
antihistamine	$4.89\times$	380	61%
herbicide	$4.16\times$	418	57%
anti-virals	$3.77\times$	1,714	96%
percent	$3.67\times$	1,829	84%
tetrahydrocannabinol	$3.66\times$	1,681	85%
immunosuppressants	$3.54\times$	1,555	95%
anti-fungals	$3.50\times$	1,748	91%
delta-	$3.41\times$	2,174	84%
fungicide	$3.31\times$	680	59%
freshener	$3.28\times$	1,422	61%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

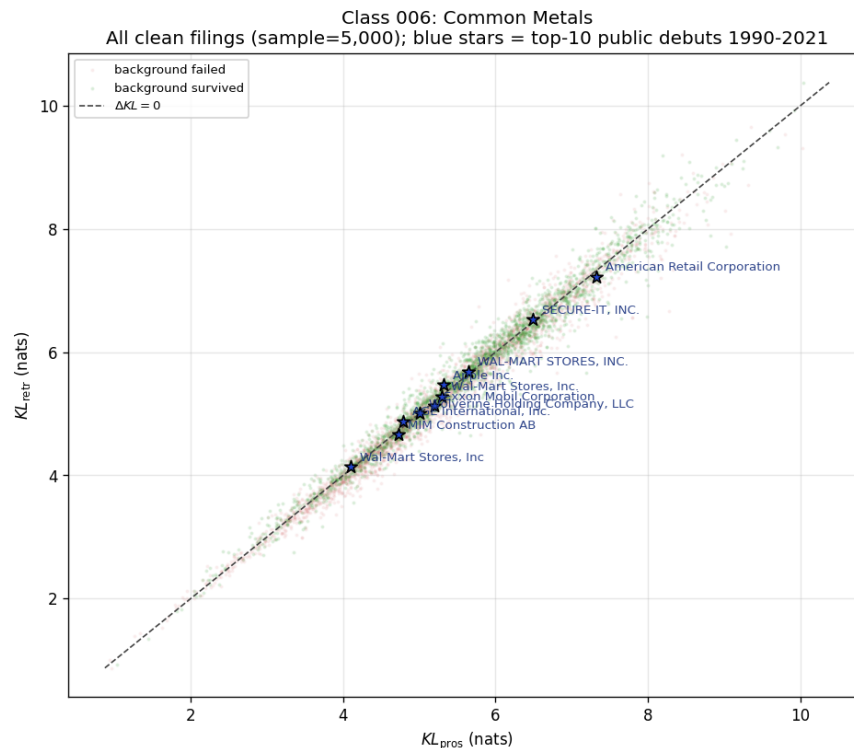
token	n in stratum	n in class	intra-class survival
supplements	19,715	137,035	53%
dietary	16,936	110,853	53%
nutritional	16,700	102,677	54%
preparations	15,581	142,819	55%
use	13,562	120,130	53%
pharmaceutical	10,172	97,351	57%
medical	9,812	75,283	47%
treatment	9,176	87,149	57%
purposes	8,697	62,259	45%
food	7,472	55,012	50%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
1PM Industries, Inc.	AQUATRIM	2015	4.27	-0.13	Dietary and nutritional supplements used for weight loss
ACM	MOLUTREX	2010	5.15	-0.11	Pharmaceutical products for the treatment of verrucae
ATK Entertainment, Inc.	AALAM KHALED	2020	3.36	-0.06	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.	THINK BETTERX	2019	2.64	-0.27	Medicated oils, salves, and concentrated pastes, namely, medicated skin care preparations; Medicated oils, salves, and concentrated pastes, namely, medicated skin care preparations, all of the foregoi
Better For You Wellness, Inc.	BETTER FOR YOU WELLNESS	2021	4.43	-0.05	Protein bars, namely, protein supplements formed and packaged as bars Advisory and consultation services in the fields of agronomy, agriculture, and horticulture; agronomic consulting services; agrono
CleanGoal Corp.	ASTALABS	2018	3.73	+0.01	Dietary and nutritional supplements
CohBar, Inc.	COHBAR	2014	4.34	+0.24	Pharmaceutical and medical preparations for the prevention and treatment of major diseases, namely, diabetes, cancer, Alzheimer's disease and atherosclerosis; peptide based medical therapeutic prepara
Compass Therapeutics LLC	CURING CANCER THROUGH COMBINAT	2015	4.60	+0.23	Custom design and development of antibody therapeutic preparations Antibody therapeutic preparations for medical purposes Antibodies for scientific and research purposes
IPR Pharmaceuticals, Inc.		2001	4.97	+0.25	Pharmaceutical and medicinal preparations and substances for the treatment of migraines
MIRA Pharmaceuticals, Inc.	MIRA-1A POT PILL	2021	6.27	-0.19	Therapeutic pharmaceutical for the treatment of anxiety

Class 006: Common Metals

Clean filings (1990–2021):	97,003
Mean KL_{pros} / KL_{retr} / ΔKL :	5.48 / 5.46 / +0.020 nats
5y survival rate:	43.4%
Share with SEC-listed owner:	4.04%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
kept	$4.96\times$	270	26%
shutterboxes	$4.72\times$	572	23%
add	$4.47\times$	170	31%
pistol	$4.36\times$	307	28%
reach	$4.30\times$	316	28%
hat-hanging	$4.00\times$	562	19%
flashlight	$3.82\times$	361	32%
multi-point	$3.82\times$	144	36%
lockboxes	$3.78\times$	164	35%
slats	$3.74\times$	752	26%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
non-leather	$4.37\times$	288	74%
yo-yos	$4.20\times$	188	67%
replica	$4.10\times$	144	55%
stationery-type	$3.93\times$	163	74%
participating	$3.77\times$	106	48%
steins	$3.77\times$	199	67%
sweatpants	$3.76\times$	473	60%
billfolds	$3.75\times$	224	77%
basketballs	$3.67\times$	354	65%
afghans	$3.61\times$	97	71%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

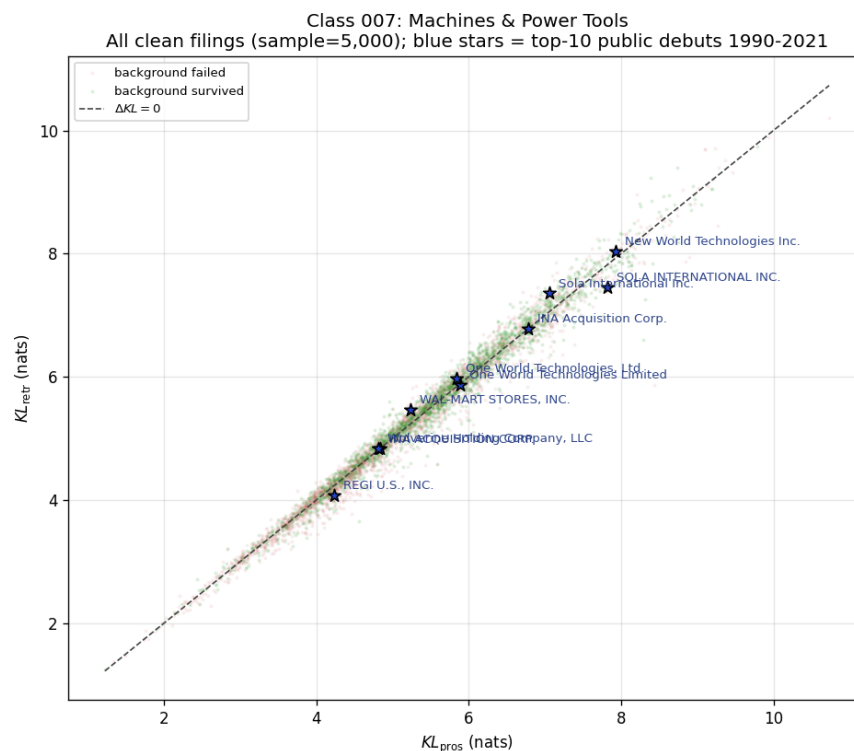
token	n in stratum	n in class	intra-class survival
metal	9,062	84,324	42%
use	3,826	28,315	37%
parts	2,612	16,917	33%
materials	2,390	15,503	31%
fittings	1,977	15,163	29%
pipes	1,942	12,554	28%
nature	1,922	13,088	30%
building	1,915	13,124	32%
plastic	1,907	13,269	40%
hardware	1,886	15,693	32%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.79	-0.09	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
American Retail Corporation	CABANA COVE	2010	7.32	+0.10	Metal swimming pools
MIM Construction AB	YELLOW LINE	2016	4.72	+0.06	storage containers adapted for use in vehicle boots; vehicle partitions for pets; fitted liners for the cargo area of vehicles; boot tidies, namely, storage bins especially adapted to fit in cars; dog
SECURE-IT, INC.	FLAVOR SAVER	1999	6.48	-0.05	Cable and lock fasteners of metal for computer equipment
Wolverine Holding Company, LLC	PRECISION ENGINEERED PRODUCTS	2002	5.00	-0.00	Metal hose stations for the food, dairy, beverage, pharmaceutical, health and beauty, and processing industries; metal tubing; metal pipe fittings; metal gaskets for the food, dairy, beverage, pharmaceutical
WAL-MART STORES, INC.	PALM VALLEY	2004	5.65	-0.03	Metal fence panels, posts and stays; metal garden fence panels; metal patio fences; hand-operated metal garden hose reels and cabinets Hand-operated non-metal garden hose reels and cabinets Non-metal
Wal-Mart Stores, Inc	SLATE MOUNTAIN	2016	4.10	-0.03	Barbecue mitts; bed linen; liners adapted to sleeping bags for camping Hand tools, namely, hand-operated pumps; mallets; pocket knives; utility knives Brushes for basting meat; cleaning brushes for ho
Wal-Mart Stores, Inc.	CREATE A CRAFT	1994	5.29	+0.01	craft supplies, namely metal key rings and [sleigh bells] [doll parts] wooden, plastic, and metal embroidery hoops; sewing notions, namely [moving eyes;] beads; [wooden craft sticks; chenille s
Exxon Mobil Corporation	MAZEFLO	2010	5.20	+0.07	Metal sand control screens used in oil and gas well construction
Apple Inc.		2008	5.32	-0.15	Keyrings of common metal; Metal rings and chains for keys; Metal key chains

Class 007: Machines & Power Tools

Clean filings (1990–2021):	160,666
Mean KL_{pros} / KL_{retr} / ΔKL :	5.34 / 5.34 / +0.003 nats
5y survival rate:	40.3%
Share with SEC-listed owner:	4.35%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
disseminating	$4.58\times$	662	24%
spiralizers	$4.25\times$	252	28%
wet-dry	$4.20\times$	374	24%
drycleaning	$4.19\times$	215	23%
squeezers	$4.07\times$	1,386	21%
dicing	$4.01\times$	979	28%
perfumes	$3.95\times$	782	25%
electric-powered	$3.94\times$	363	31%
shampooing	$3.87\times$	1,121	23%
exhausting	$3.85\times$	1,773	22%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
walk-behind	$3.11\times$	257	47%
subs	$2.79\times$	161	40%
riding	$2.75\times$	745	47%
oscillation	$2.54\times$	185	17%
completion	$2.44\times$	553	42%
sprocket	$2.42\times$	186	50%
surround	$2.22\times$	171	16%
flip	$2.22\times$	185	26%
downhole	$2.22\times$	1,101	48%
wristwatches	$2.22\times$	167	15%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

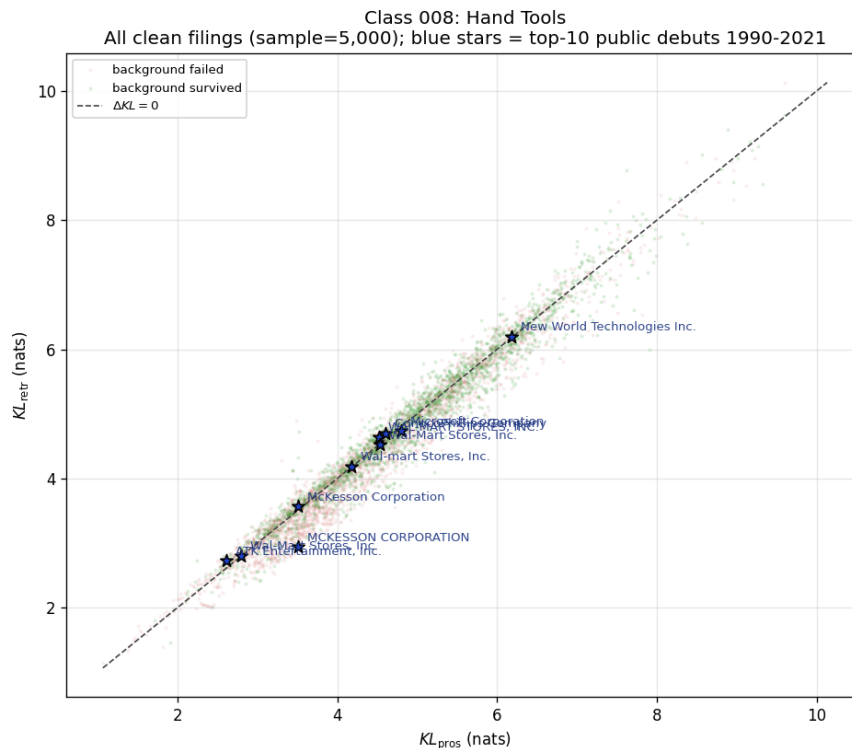
token	n in stratum	n in class	intra-class survival
machines	11,129	85,739	32%
parts	8,297	58,796	34%
use	6,894	49,168	34%
electric	5,800	45,342	31%
machine	5,595	37,204	31%
industrial	4,751	31,003	30%
apparatus	4,728	28,485	26%
tools	4,569	32,200	33%
equipment	4,253	28,326	38%
power	4,207	32,213	36%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
INA ACQUISITION CORP.	CHIP	1993	4.84	+0.00	mechanical devices for applying tubular bags or liners to the internal surfaces of pipes or passageways, such devices consisting of an interconnected arrangement of pressure chambers, valves, cylinder
INA Acquisition Corp.		2007	6.79	-0.00	Pipe and conduit repair and rehabilitation machines, namely, insertion device for inserting flexible liners into existing pipes and conduits for the repair and rehabilitation of existing pipes and con
New World Technologies Inc.	RAD	2015	7.93	-0.11	Pneumatic torque wrenches
One World Technologies Limited	EML EX-TENDED MOTOR LIFE	2003	5.88	+0.02	Power tools, namely, grinders, hammer drills, drills, and jig saws
One World Technologies, Ltd.	LASER-TECH	2003	5.84	-0.13	Stud Sensors for sensing studs in walls; Laser Line Generators for projecting a laser line illumination; and Laser Torpedo Levels for measuring a horizontal level or a vertical level of an item and ha
REGI U.S., INC.	RAND CAM	1994	4.23	+0.16	motors and engines for vehicles engines not for land vehicles; rotary pumps; rotary compressors for air conditioners and refrigerators; rotary air compressors for providing compressed air to shops and
SOLA INTERNATIONAL INC.	MATRIX	1992	7.82	+0.37	machines used in the manufacture of eyeglass lenses
Sola International Inc.	S SOLA	1995	7.06	-0.30	laminating machines used in the manufacture of eyeglass lenses
Wolverine Holding Company, LLC	PRECISION ENGINEERED PRODUCTS	2002	4.82	-0.02	Metal hose stations for the food, dairy, beverage, pharmaceutical, health and beauty, and processing industries; metal tubing; metal pipe fittings; metal gaskets for the food, dairy, beverage, pharmaceutical
WAL-MART STORES, INC.	EXPERT GARDENER	2003	5.23	-0.24	Wheel barrows Garden apparel, namely, aprons, hats, boots Lawn sprinklers; gardening gloves Plant food; lawn and plant fertilizer for domestic use; potting soil; compost; spike-shaped fertilizer for h

Class 008: Hand Tools

Clean filings (1990–2021): 71,252
Mean KL_{pros} / KL_{retr} / ΔKL : 4.65 / 4.58 / +0.070 nats
5y survival rate: 40.3%
Share with SEC-listed owner: 2.66%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
magneto	$5.15\times$	388	23%
shim	$4.87\times$	412	25%
tappet	$4.72\times$	425	26%
flywheel	$4.64\times$	432	26%
absorber	$4.45\times$	449	25%
feeler	$4.22\times$	483	27%
carburetor	$4.13\times$	491	28%
tuning	$3.79\times$	529	26%
beneath	$3.73\times$	131	27%
adjustment	$3.64\times$	556	28%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
wood-splitting	$5.82\times$	103	57%
sidewalk	$5.69\times$	109	57%
pothole	$5.50\times$	140	59%
vegetation	$5.37\times$	162	55%
barn	$5.13\times$	113	57%
tamping	$4.41\times$	145	50%
weeders	$3.28\times$	396	59%
floral	$3.19\times$	166	52%
sledge	$3.14\times$	226	52%
sod	$3.00\times$	300	60%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

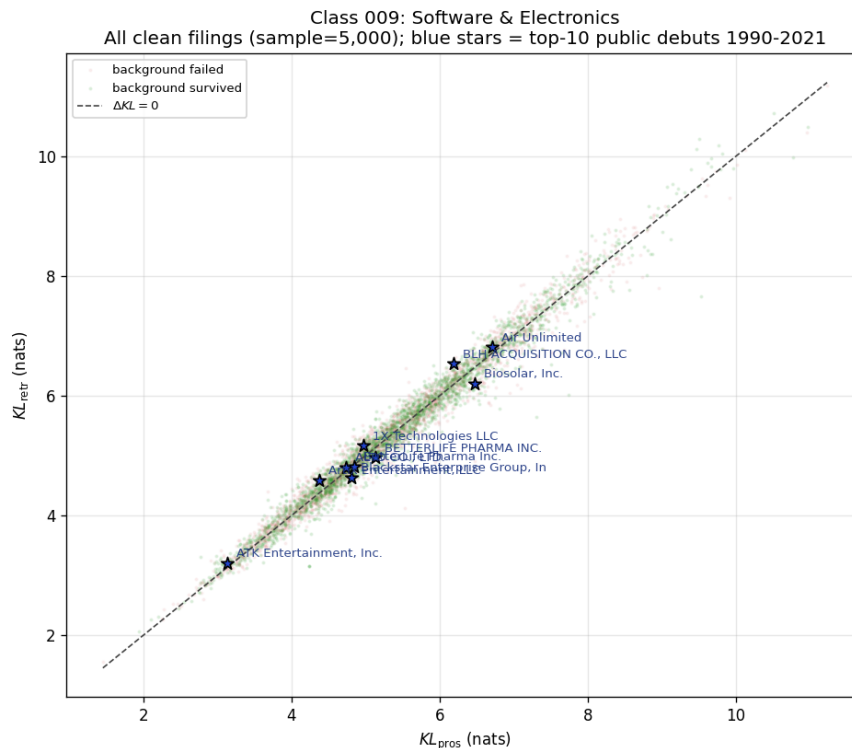
token	n in stratum	n in class	intra-class survival
tools	3,429	36,891	35%
hand	3,417	35,656	36%
knives	2,485	29,268	35%
use	2,232	20,225	34%
hand-operated	1,706	19,419	26%
electric	1,601	20,662	30%
forks	1,586	19,227	33%
spoons	1,407	17,205	33%
metal	1,391	11,300	39%
operated	1,329	13,142	34%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ATK Inc.	Entertainment,	AALAM TUCK KHALED	2020	2.61	-0.11	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
New World Technologies Inc.			2018	6.18	-0.02	Torque wrenches Torque measuring and calibrating devices for torque wrenches
WAL-MART STORES, INC.		EXPERT GARDENER	2003	4.52	-0.12	Wheel barrows Garden apparel, namely, aprons, hats, boots Lawn sprinklers; gardening gloves Plant food; lawn and plant fertilizer for domestic use; potting soil; compost; spike-shaped fertilizer for h
Wal-Mart Stores, Inc		DAILY'S	2000	2.79	-0.01	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.		GADGETS 'N' GIZMOS	1994	4.54	+0.02	timers, thermometers, scales household utensils, namely spoons, knives, graters, kitchen knives rolling pins, salt and pepper shakers
Wal-mart Stores, Inc.			2011	4.17	-0.01	On-line retail store services featuring a wide variety of consumer goods of others; Retail department store services; Retail store services featuring a wide variety of consumer goods of others Air bru
MCKESSON CORPORATION		BRIGHT IDEA	1990	3.51	+0.57	salad oil cans, sold empty ice picks, ice tongs, cutlery, namely, cleavers, cheese planes, grapefruit knives, spoons, forks, paring knives, filleting knives, boning knives, butcher knives, slicing kni
McKesson Corporation		SUNMARK	2003	3.51	-0.06	humidifiers and vaporizers House mark for a line of personal care products, namely, tweezers, nail clippers, nail and cuticle scissors, nail files, razors and razor blades and eyelash curlers House ma
Microsoft Corporation		MICROSOFT	1991	4.80	+0.06	knives; namely, pocket knives
ConocoPhillips Company	Com-	76	2003	4.61	-0.09	Knives; hand held tools, namely, pliers, tool sets composed of hand held tools in the nature of a compact folding knife, pliers, multiple blades and leather and fabric knife sheaths

Class 009: Software & Electronics

Clean filings (1990–2021): 1,226,806
Mean KL_{pros} / KL_{retr} / ΔKL : 5.42 / 5.43 / -0.011 nats
5y survival rate: 48.6%
Share with SEC-listed owner: 5.52%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
non-fungible	8.62×	1,273	64%
blockchain-based	7.42×	1,349	63%
blockchain	6.45×	3,427	59%
cryptocurrency	6.33×	3,255	58%
cryptographic	4.44×	1,738	55%
currencies	4.35×	1,602	51%
collectibles	4.20×	1,293	56%
tokens	4.02×	5,129	47%
wallet	3.90×	2,428	54%
ledger	3.83×	1,541	51%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
typeface	$3.53\times$	1,333	51%
soundbar	$3.17\times$	1,629	23%
videorecorders	$3.05\times$	1,303	17%
selfie	$3.02\times$	6,129	20%
generates	$2.89\times$	3,501	49%
fonts	$2.77\times$	2,519	45%
wager	$2.74\times$	6,243	52%
smartwatch	$2.72\times$	3,370	22%
webcams	$2.70\times$	5,647	20%
outcomes	$2.64\times$	4,227	49%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

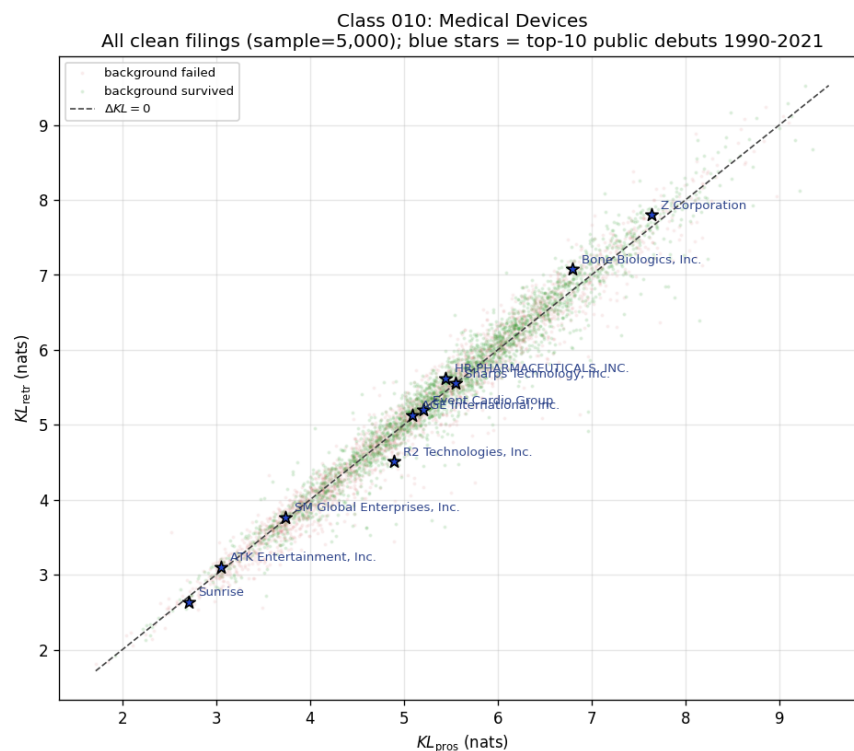
token	n in stratum	n in class	intra-class survival
computer	71,265	654,366	48%
software	68,779	602,415	50%
use	43,630	388,588	45%
electronic	42,973	362,929	44%
downloadable	35,839	275,741	44%
data	32,816	261,388	43%
information	30,922	227,341	48%
video	30,379	265,478	48%
devices	29,782	258,931	43%
field	28,728	213,618	44%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
1X Technologies LLC	1X TECH- NOLOGIES	2017	4.97	-0.19	Cable and wiring plates in the nature of protective sheaths for electrical cable and wiring; Cable junctions for electric cables; Cables for electrical and optical signal transmission systems; Cables
ALTO CO., LTD.	ALTO	2011	4.74	-0.06	[Optical condensers for use in microscopes, enlargers, slide projectors, video projectors, and telescopes; magic lanterns; photometers; actinometers; signalling devices, namely, traffic light apparatus
ATK Entertainment, Inc.	AALAM TUCK KHALED	2020	3.14	-0.07	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Air Unlimited	SLAM	2021	6.71	-0.10	Suspension systems for automobiles and component parts therefor; air springs and air valves, all for suspension systems for automobiles Electronic controllers and electric wiring harnesses for suspens
Artix Entertainment, LLC	ADVENTUREQUEST	2008	4.37	-0.21	Motion picture films featuring animation; audio and video recordings featuring animation; prerecorded audio and video tapes, laser discs, DVDs, and CDs featuring animation; downloadable audio and vide
BETTERLIFE PHARMA INC.	MOVE BET-TERX	2019	5.13	+0.16	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BLH ACQUISITION CO., LLC		2012	6.19	-0.35	Magnetic coded gift cards Non-magnetically encoded prepaid purchase cards for allowing users to transfer financial value at the point of sale and on-line via a computer network
BetterLife Pharma Inc.	LIVE BET-TERX	2020	4.85	+0.03	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Biosolar, Inc.	BIOSOLAR	2006	6.48	+0.27	Apparatus for converting electronic radiation to electrical energy, namely, photovoltaic solar modules; Apparatus for converting electronic radiation to electrical energy, namely, photovoltaic solar-t
Blackster Enterprise	B. E. T.	2021	4.81	+0.18	Financial services, namely, providing

Class 010: Medical Devices

Clean filings (1990–2021):	190,121
Mean KL_{pros} / KL_{retr} / ΔKL :	5.35 / 5.35 / -0.001 nats
5y survival rate:	47.4%
Share with SEC-listed owner:	7.80%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
rigidity	4.79×	801	20%
wavelengths	4.78×	561	29%
constriction	4.74×	791	19%
non-ablative	4.68×	711	25%
green	4.52×	588	30%
life-sized	4.50×	1,230	20%
orange	4.49×	579	29%
yellow	4.46×	596	30%
workbook	4.19×	1,457	18%
penile	4.17×	967	25%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
percent	$5.29\times$	238	56%
dilatation	$4.93\times$	209	42%
atherectomy	$4.81\times$	210	31%
interbody	$4.77\times$	199	66%
pacing	$4.46\times$	280	59%
angiographic	$4.27\times$	227	27%
non-active	$4.20\times$	195	31%
energizing	$4.17\times$	199	31%
transluminal	$4.12\times$	216	32%
pedicle	$4.08\times$	233	45%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

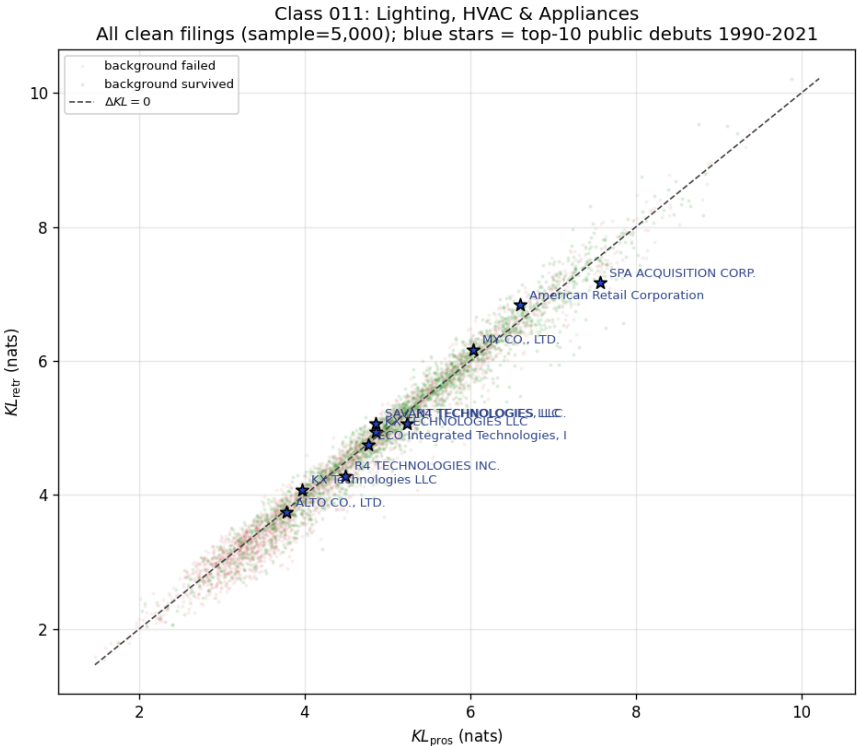
token	n in stratum	n in class	intra-class survival
medical	11,263	119,396	45%
use	8,005	82,136	41%
apparatus	7,708	74,909	38%
instruments	5,599	54,584	38%
purposes	5,390	52,059	34%
devices	5,313	55,395	47%
surgical	3,806	41,541	43%
treatment	2,928	26,564	41%
diagnostic	2,588	20,431	41%
equipment	2,403	21,187	40%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	5.09	-0.04	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM	2020	3.05	-0.05	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Bone Biologics, Inc.	BIOBONE	2006	6.80	-0.28	bone graft substitute, orthobiologic, bone growth factor, synthetic bone graft, demineralized bone graft
Event Cardio Group	NOW CARDIO	2016	5.21	+0.01	Medical device, namely, a cardiac and cardiac event monitor having a sensor and capability to transmit cardiac and other physiological data of patients remotely Medical Services, namely, remote monito
HR PHARMACEUTICALS, INC.	MTG INSTANT CATH	2007	5.44	-0.18	Urinary catheters; Urinary catheter products, namely, catheters and parts and fittings therefor; and Disposable intermittent catheters
R2 Technologies, Inc.	CRYOAESTHETICS	2016	4.89	+0.38	Cryotherapy apparatus for medical and cosmetic use for performing aesthetic skin treatment procedures
SM Global Enterprises, Inc.	INTIMATE NEED	2017	3.74	-0.02	Personal sexual lubricants, creams for sexual aid, lotions for sexual aid and topical gels for use as body lubricants; topical gels for use as male and female genital sensitizers and desensitizers Adu
Sharps Technology, Inc.	PROMESIA	2018	5.55	-0.01	Online retail store and online retail distribution services featuring syringes and needles for medical use and for injection, disposable syringes and needles for medical use and for injection, disposa
Sunrise	DREEM	2019	2.71	+0.07	Scientific, weighing, measuring, signaling, checking, supervision, life-saving apparatus and instruments, namely, electric and electronic apparatus and instruments for the measuring, storage, encoding
Z Corporation	ZPHOTO	2007	7.64	-0.16	medical images; Two-and-a-half dimensional medical images providing opaque and translucent contrast resolution

Class 011: Lighting, HVAC & Appliances

Clean filings (1990–2021): 194,994
Mean KL_{pros} / KL_{retr} / ΔKL : 4.84 / 4.79 / +0.045 nats
5y survival rate: 38.8%
Share with SEC-listed owner: 2.80%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
party-themed	2.97×	1,594	26%
chinese	2.88×	2,682	19%
diving	2.81×	5,317	19%
fairy	2.79×	9,357	15%
christmas-themed	2.75×	1,412	19%
underground	2.74×	2,806	19%
all-weather	2.66×	1,698	26%
turned	2.61×	662	33%
searchlights	2.60×	8,194	18%
branches	2.58×	3,105	22%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
vaporizable	$3.57\times$	255	49%
constituents	$3.06\times$	294	46%
robe	$3.02\times$	255	64%
thawing	$2.99\times$	294	48%
scenting	$2.99\times$	274	61%
vaporizing	$2.72\times$	364	45%
refill	$2.42\times$	265	46%
refrigerator-freezers	$2.32\times$	228	43%
glitter	$2.28\times$	254	54%
gloss	$2.25\times$	231	55%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

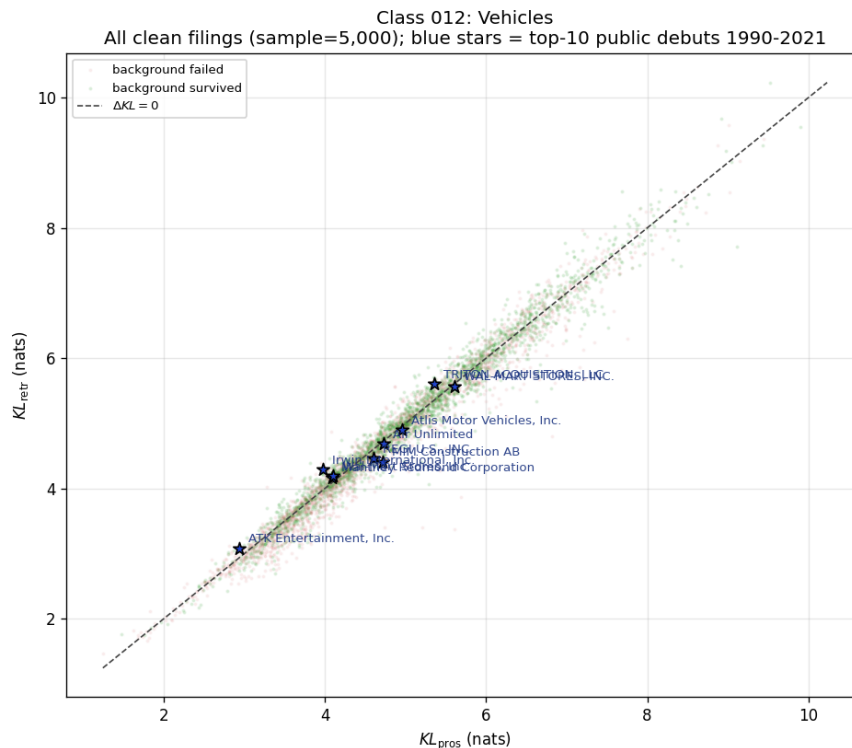
token	n in stratum	n in class	intra-class survival
electric	7,491	89,973	31%
use	7,102	71,438	34%
apparatus	6,335	66,232	27%
water	5,831	56,555	35%
air	5,445	55,451	32%
lighting	4,566	63,268	29%
installations	4,140	44,286	24%
machines	4,097	39,740	28%
purposes	4,032	39,558	26%
lamps	3,863	55,787	25%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ALTO CO., LTD.	ALTO	2011	3.78	+0.04	[Optical condensers for use in microscopes, enlargers, slide projectors, video projectors, and telescopes; magic lanterns; photometers; actinometers; signalling devices, namely, traffic light apparatus
American Retail Corporation	AVENTURA	2009	6.60	-0.23	Spas in the nature of heated pools
ECO Integrated Technologies, Inc.	AQUAVAP	2021	4.77	+0.03	Water purification apparatus; Water purification, sedimentation and filtration apparatus for clarification, thickening and filtration of water, waste water and industrial water; Waste water purification components of water filtration units for producing potable water for domestic use; namely, activated carbon water filters
KX TECHNOLOGIES LLC	MATRIKX + CTO	1993	4.86	-0.08	chemical and mineral fiber pulp preparations in the nature of filtering materials for the manufacturing of domestic, industrial, and commercial water filters, and for the manufacture of commercial and
KX Technologies LLC	FINITY	2013	3.96	-0.11	Bath and shower gels; beauty bath additives, namely, bubble bath, bath beads, bath foam, bath crystals, bath pearls, and non-medicated bath salts; non-medicated bath salts; bath and shower gels and sa
MY CO., LTD.	WATER PLUS	2020	6.04	-0.13	[Electronic commerce payment services, namely, establishing funded accounts used to purchase goods and services via mobile devices; Financial services, namely, providing on-line stored value accounts
R4 TECHNOLOGIES INC.	R4 TECHNOLOGIES	2013	4.49	+0.20	Refrigerated beverage dispensing units; Refrigerated dispensing units for beverages; Refrigerated self-service cases, refrigerated self-service deli cases and custom fabricated refrigeration systems u
R4 TECHNOLOGIES, LLC.	SMART COOLER	2012	5.23	+0.16	electric lamps
SAVANT TECHNOLOGIES LLC	GENURA	1995	4.85	-0.21	acrylic hot tubs with fiberglass backing
SPA ACQUISITION CORP.	L.A. SPAS	1990	7.56	+0.39	

Class 012: Vehicles

Clean filings (1990–2021): 128,694
Mean KL_{pros} / KL_{retr} / ΔKL : 4.93 / 4.90 / +0.034 nats
5y survival rate: 43.7%
Share with SEC-listed owner: 5.01%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
contracted	5.81×	736	23%
dropping	5.53×	776	22%
gap	5.42×	799	24%
expanded	5.30×	806	23%
droppings	5.26×	247	11%
wedge	5.21×	826	25%
durable	5.14×	830	26%
paws	4.85×	165	32%
fill	4.81×	892	26%
anti-slipping	4.63×	479	18%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
sport-utility	4.68×	314	79%
pillar	2.89×	204	47%
motorhomes	2.64×	216	40%
bob-sleighs	2.61×	142	15%
fifth	2.52×	892	56%
elderly	2.42×	240	46%
recapping	2.42×	236	60%
radial	2.35×	200	52%
non-monetary	2.25×	231	35%
softball	2.18×	238	45%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

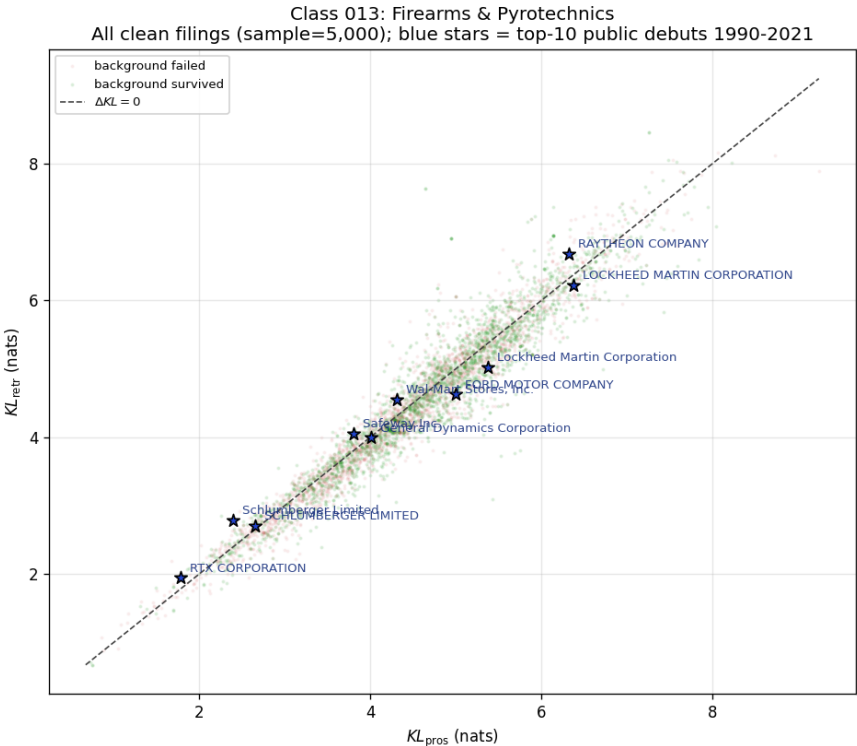
token	n in stratum	n in class	intra-class survival
vehicles	7,431	67,870	39%
parts	6,427	56,049	40%
vehicle	4,596	41,986	36%
land	4,347	33,773	36%
structural	3,304	27,975	40%
automobiles	3,023	28,376	35%
motor	3,019	27,315	36%
use	2,925	25,248	35%
wheels	2,698	22,834	35%
electric	2,491	20,707	34%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ATK Inc.	Entertainment,	AALAM	2020	2.93	-0.15	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Air Unlimited		SLAM SPECIALTIES	2021	4.73	+0.05	Suspension systems for automobiles and component parts therefor; air springs and air valves, all for suspension systems for automobiles Electronic controllers and electric wiring harnesses for suspens
Atlis Motor Inc.	Vehicles,		2021	4.95	+0.06	Electric vehicles, namely, passenger vehicles; pickup trucks; minivans; cargo vans; utility trucks; light commercial trucks; structural repair parts and replacement parts therefor
Irwin International, Inc.		AIRCRAFT SPRUCE	2000	3.97	-0.32	Airplanes (structural parts for)
MIM Construction AB		CAREY	2016	4.72	+0.31	storage containers adapted for use in vehicle boots; vehicle partitions for pets; fitted liners for the cargo area of vehicles; boot tidies, namely, storage bins especially adapted to fit in cars; dog
Manthey Redmond Corporation		MULTI-FIRE TECHNOLOGY	2012	4.09	-0.08	Internal combustion engines for land vehicles
REGI U.S., INC.		RAND CAM	1994	4.60	+0.15	motors and engines for vehicles engines not for land vehicles; rotary pumps; rotary compressors for air conditioners and refrigerators; rotary air compressors for providing compressed air to shops and
TRITON ACQUISITION, LLC		TRITON	1997	5.36	-0.25	trailers, namely, motorcycle, personal watercraft, utility, snowmobile, pontoon, all terrain vehicle, and boat trailers
WAL-MART STORES, INC.		EXPERT GARDENER	2003	5.61	+0.04	Wheel barrows Garden apparel, namely, aprons, hats, boots Lawn sprinklers; gardening gloves Plant food; lawn and plant fertilizer for domestic use; potting soil; compost; spike-shaped fertilizer for h
Wal-Mart Stores, Inc		SLATE MOUNTAIN	2016	4.10	-0.09	Barbecue mitts; bed linen; liners adapted to sleeping bags for camping Hand tools, namely, hand-operated pumps; mallets; pocket knives; utility knives Brushes for basting meat; cleaning brushes for ho

Class 013: Firearms & Pyrotechnics

Clean filings (1990–2021): 13,777
Mean KL_{pros} / KL_{retr} / ΔKL : 4.53 / 4.47 / +0.063 nats
5y survival rate: 45.9%
Share with SEC-listed owner: 3.94%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
breaks	$4.14\times$	70	43%
gunsights	$3.16\times$	98	64%
roman	$2.42\times$	62	53%
ancillary	$2.34\times$	350	55%
attaching	$1.99\times$	1,282	46%
pouch	$1.95\times$	174	36%
attachments	$1.95\times$	1,351	45%
sky	$1.91\times$	68	49%
pointing	$1.79\times$	390	46%
external	$1.79\times$	542	47%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
cores	$2.98\times$	57	58%
colored	$2.91\times$	55	40%
effects	$2.86\times$	63	43%
non-explosive	$2.68\times$	56	59%
roman	$2.58\times$	62	53%
shotgun	$2.56\times$	398	52%
firecrackers	$2.34\times$	252	38%
smokeless	$2.31\times$	52	71%
defensive	$2.30\times$	74	57%
aerosol	$2.29\times$	118	40%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

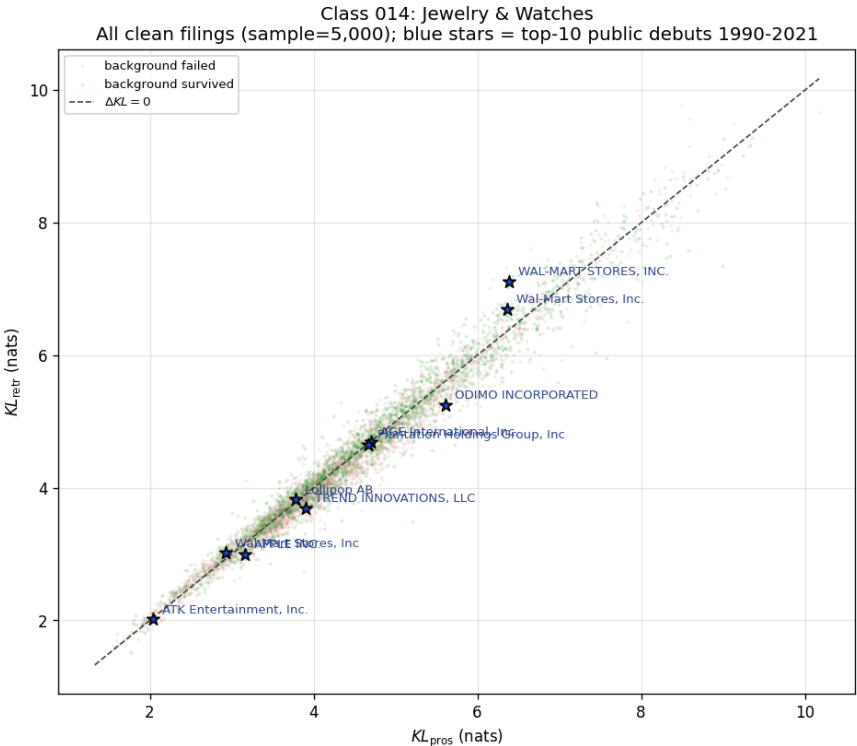
token	n in stratum	n in class	intra-class survival
firearms	482	6,870	43%
gun	367	4,493	43%
ammunition	333	4,390	43%
guns	260	3,735	40%
firearm	256	3,224	44%
parts	241	3,692	42%
cases	233	3,062	41%
use	232	2,798	43%
accessories	231	3,131	45%
rifle	187	2,498	40%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Wal-Mart Stores, Inc.	BLACK CANYON CAMP GEAR	1996	4.31	-0.24	hunting knives, pocket knives, utility knives, shovels gun cases dishes gas camping stoves, gas lanterns tents and hammocks sleeping bags
FORD MOTOR COMPANY	BUILT TO LAST	1998	4.99	+0.37	Motor vehicles, namely, automobiles, trucks, vans, sport-utility vehicles, and their structural parts and engines
LOCKHEED MARTIN CORPORATION	MLRS	1999	6.37	+0.15	Mobile projectile launch platforms mounted on a vehicle
Lockheed Martin Corporation		1999	5.37	+0.35	guns, guided missiles, military rifles
RTX CORPORATION		2020	1.79	-0.16	Training in servicing and repair of aircraft and systems and components, in use of electronic equipment/systems, pumps, blowers, and compressors; training on how to operate unmanned aircraft; training
SCHLUMBERGER LIMITED	S	2000	2.66	-0.05	Drilling machines and parts therefor; boring machines; mechanical mixing machines; rotary, suction, sump and vacuum pumps Printed instructional, educational, and teaching materials about well drilling
Schlumberger Limited	SCHLUMBERGER	2000	2.40	-0.39	Construction project management services for others in the fields of oil and gas exploration and production; oil and gas drilling; consulting services in the fields of oil and gas well drilling; sand
Safeway Inc.	SIGNATURE HOME	2015	3.80	-0.25	Fire igniters Paper, plastic and foam cups, dishes and bowls, scouring sponges, cleaning sponges, household rubber/latex gloves for general use, hypo vinyl household gloves for general use, disposable
General Dynamics Corporation	STRENGTH ON YOUR SIDE	1998	4.01	+0.01	Weapons systems, namely guns, artillery guns, and anti-aircraft guns and parts therefor; ammunition handling systems comprising wagons, cases, magazines, loaders and unloaders and parts therefor; ammu
RAYTHEON COMPANY	COM- TERM	2001	6.32	-0.36	Munitions Research and development activities in the area of munitions

Class 014: Jewelry & Watches

Clean filings (1990–2021): 109,375
Mean $KL_{\text{pros}} / KL_{\text{retr}} / \Delta KL$: 4.52 / 4.50 / +0.019 nats
5y survival rate: 43.2%
Share with SEC-listed owner: 1.79%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
separately	$4.02\times$	1,559	28%
earring	$3.67\times$	1,781	30%
telemeters	$3.38\times$	136	18%
necklace	$3.33\times$	1,969	31%
reduce	$3.05\times$	118	19%
wristwatch	$2.99\times$	431	38%
pace	$2.98\times$	114	32%
multi	$2.94\times$	136	12%
hoop	$2.84\times$	1,596	25%
split	$2.79\times$	2,936	22%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
headset	$5.19\times$	160	19%
karat	$4.27\times$	225	56%
cellphones	$4.19\times$	203	24%
telemeters	$3.90\times$	136	18%
surround	$3.68\times$	247	15%
decal	$3.53\times$	136	85%
hvac	$3.44\times$	125	10%
diode	$3.27\times$	355	19%
equalizers	$3.24\times$	247	25%
reduce	$3.22\times$	118	19%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

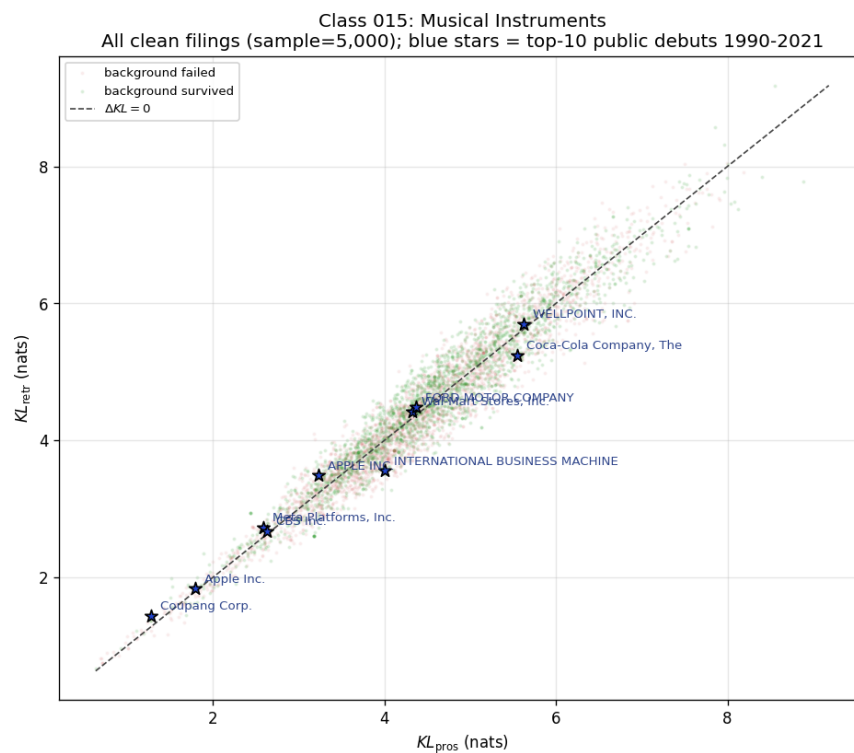
token	n in stratum	n in class	intra-class survival
jewelry	7,396	79,271	43%
precious	4,969	46,170	36%
watches	4,953	47,812	42%
rings	4,487	45,597	36%
bracelets	4,445	46,380	37%
metal	3,870	33,694	36%
cases	3,845	30,446	37%
necklaces	3,629	38,823	35%
chains	3,568	34,325	35%
clocks	3,562	32,323	39%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.		BLANTON'S	2021	4.70	+0.01	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Inc.	Entertainment,	AALAM KHALED	2020	2.04	+0.03	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Lollipop AB		LOLLIPOP	1996	3.78	-0.04	stationery, paper materials for packing paper and cardboard boxes, paper and plastic bags for merchandise packaging, photo and memory albums and scrap-books, calendars and diaries, bookmarkers braid, h
ODIMO RATED	INCORPORATED	THE POWER TO TRUST	2000	5.61	+0.37	Precious gemstones; fine jewelry; watches Computerized online ordering services featuring jewelry and gemstones; online retail store services featuring jewelry and gemstones
Plantation Group, Inc.	Holdings	THE PLANTATION SHOP	2004	4.67	+0.02	Gourmet cookies, coffee, wafers, biscuits, snack nut mix consisting primarily of chocolate covered nuts and pralines, candy, toffee and cheese straws Jewelry Retail store services and retail shop at h
TREND TIONS, LLC	INNOVATIONS, LLC	NECKLET	2021	3.90	+0.21	Online retail store services featuring jewelry and accessories Jewelry; Clasps for jewelry; Jewelry cases
WAL-MART STORES, INC.		FADED GLORY	1996	6.39	-0.72	sunglasses watches
Wal-Mart Stores, Inc		DAILY'S	2000	2.92	-0.10	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.		COLOR TREND	1994	6.37	-0.33	sunglasses handbags earrings
APPLE INC.			2002	3.16	+0.16	[Jewelry; watches and clocks; cufflinks;] key rings of precious metal [; chronometers; brooches, charms; tie pins; tie clips;] badges [, bracelets, necklaces; medallions, belt buckles, pins; accessor

Class 015: Musical Instruments

Clean filings (1990–2021): 12,496
Mean KL_{pros} / KL_{retr} / ΔKL : 4.47 / 4.48 / -0.008 nats
5y survival rate: 43.2%
Share with SEC-listed owner: 0.74%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
xiao	$3.33\times$	81	14%
clamp-like	$2.92\times$	168	27%
balalaikas	$2.89\times$	76	28%
chords	$2.85\times$	179	27%
theaters	$2.83\times$	53	64%
positioned	$2.77\times$	177	28%
vertical	$2.72\times$	114	18%
bowed	$2.59\times$	58	45%
structure	$2.56\times$	191	28%
harpsichords	$2.50\times$	60	32%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
fretted	$2.94\times$	51	61%
viola	$2.21\times$	95	54%
cello	$2.19\times$	64	52%
trumpet	$2.16\times$	51	61%
grand	$2.14\times$	84	21%
upright	$2.11\times$	90	29%
saxophone	$1.87\times$	75	47%
xiao	$1.73\times$	81	14%
theaters	$1.70\times$	53	64%
winders	$1.64\times$	61	52%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

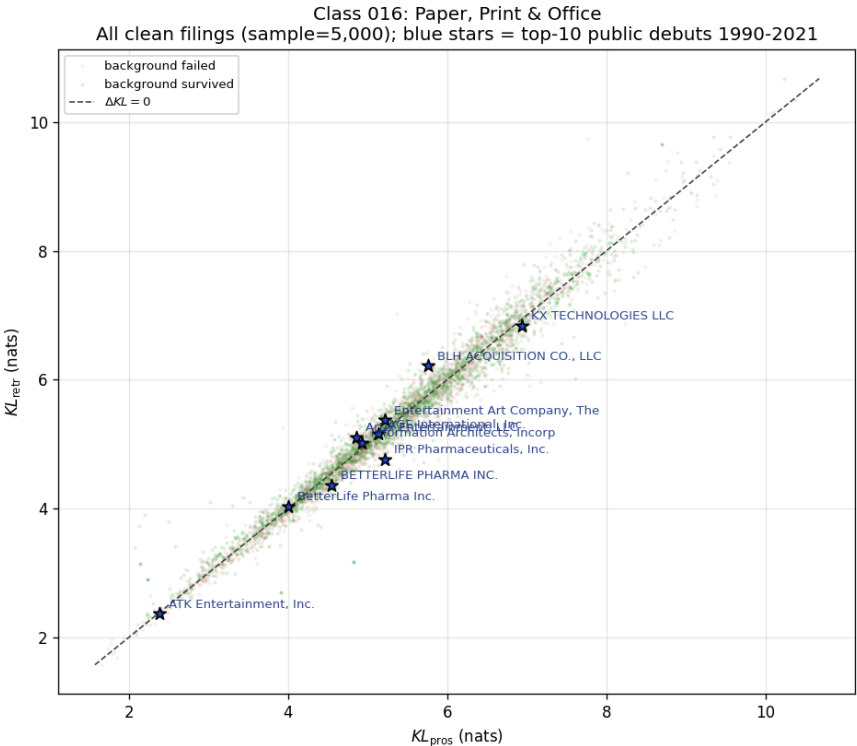
token	n in stratum	n in class	intra-class survival
musical	594	9,775	41%
instruments	541	9,256	40%
electronic	281	4,189	37%
music	276	3,804	37%
cases	259	3,732	38%
guitars	231	4,440	40%
stands	217	3,259	32%
accessories	201	2,902	40%
instrument	200	2,978	40%
use	193	2,540	39%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Wal-Mart Stores, Inc.	MISTLETOE MAGIC	1994	4.32	-0.10	music boxes
APPLE INC.		2004	3.24	-0.25	[automobiles] electronic apparatus and software therefor for downloading, transmitting, receiving, playing, storing and organizing audio, video and navigation data for use in automobiles [, railway
Apple Inc.	OSSIA	2018	1.79	-0.04	Musical entertainment services, namely, live performances by musical bands; post-production music editing; music publishing services; consulting services in the field of music editing and production;
WELLPOINT, INC.	C-6000	1993	5.62	-0.08	cymbals and percussion instruments
FORD MOTOR COMPANY	OFFICIAL LICENSED PRODUCT FORD	1995	4.37	-0.11	music boxes
Meta Platforms, Inc.	LF	2019	2.59	-0.15	Computer peripheral equipment; computer software, namely, adapted to the operation of computers, downloadable from the internet for operation of computers, operating system software for computers, for
INTERNATIONAL BUSINESS MACHINES CORPORATION	M	1993	4.01	+0.45	music synthesizers computer hardware, and computer components, namely, communications adapters, data modems, telephone answering machines, sound generators, music synthesizers, printed circuit boards,
Coca-Cola Company, The	COCA-COLA	1991	5.55	+0.31	glass holders and wastebaskets kitchen linens; namely, table runners, dishcloths, napkins and placemats toy banks metal signs [musical diaramas]
CBS Inc.	DR. QUINN MEDICINE WOMAN	1995	2.63	-0.04	aprons, belts, suspenders, Halloween costumes and masks sold in connection therewith, role-playing costumes, girls' dresses, boots, rubbers, slippers, gloves, mittens, knit caps, painters caps, athlet
Coupang Corp.	COUPANG	2020	1.28	-0.16	Leather trimmings for furniture; all purpose carrying bags; book bags; duffel bags; leather; leather thongs for fastening or securing items; boxes made of leather; collars for animals; nose bags being

Class 016: Paper, Print & Office

Clean filings (1990–2021): 444,018
Mean KL_{pros} / KL_{retr} / ΔKL : 5.39 / 5.40 / -0.008 nats
5y survival rate: 50.2%
Share with SEC-listed owner: 3.87%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
originating	5.05×	1,118	44%
stand-alone	4.26×	1,631	83%
original	3.92×	2,215	45%
giclee	3.82×	521	39%
scoring	3.73×	2,401	77%
watercolour	3.26×	1,495	29%
murals	3.10×	2,131	39%
tarot	3.08×	503	44%
disposing	3.06×	1,208	46%
calligraphic	3.05×	1,250	23%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
periodically	$4.58\times$	1,884	47%
published	$3.87\times$	3,147	51%
stamped	$3.79\times$	531	36%
column	$3.72\times$	1,289	44%
section	$3.52\times$	1,206	39%
pre-paid	$3.49\times$	2,321	54%
issued	$3.30\times$	512	46%
devoted	$3.23\times$	644	51%
circulation	$3.20\times$	2,213	55%
quarterly	$3.15\times$	653	46%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

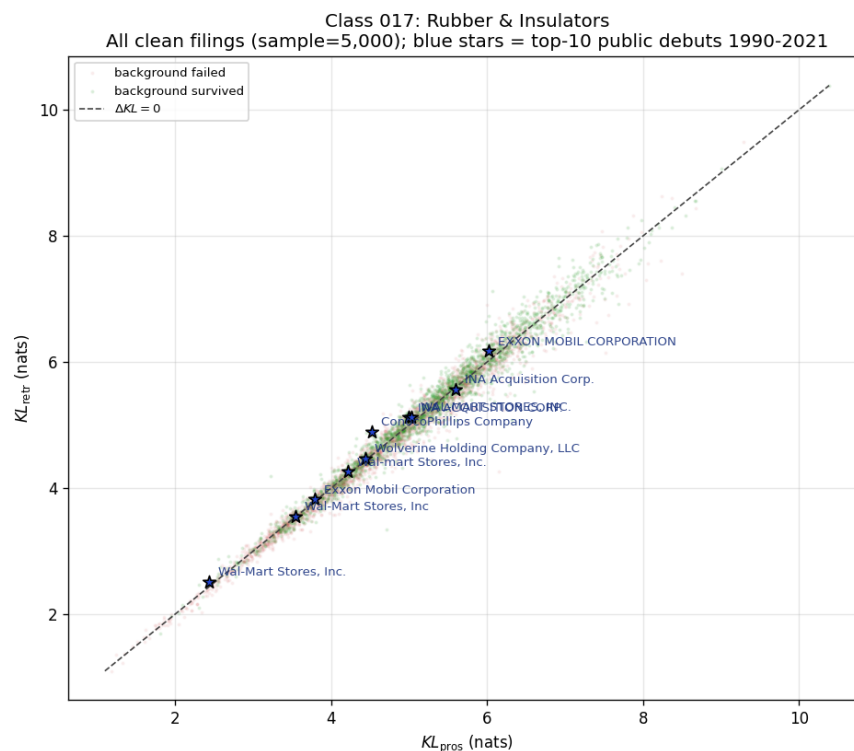
token	n in stratum	n in class	intra-class survival
printed	21,670	171,737	48%
books	19,531	161,759	53%
field	18,988	134,349	47%
paper	15,296	124,983	46%
materials	13,557	93,439	47%
featuring	13,256	92,664	49%
educational	11,559	76,630	48%
information	11,218	74,050	48%
use	11,132	80,387	44%
magazines	11,115	83,541	52%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	5.14	-0.02	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM KHALED	2020	2.39	+0.02	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Artix Entertainment, LLC	ADVENTUREQUEST	2008	4.86	-0.25	paper goods and printed matter, namely, children's books and novels
BETTERLIFE PHARMA INC.	THINK BETTERX	2019	4.55	+0.18	Medicated oils, salves, and concentrated pastes, namely, medicated skin care preparations; Medicated oils, salves, and concentrated pastes, namely, medicated skin care preparations, all of the foregoi
BLH ACQUISITION CO., LLC		2012	5.76	-0.47	Magnetic coded gift cards Non-magnetically encoded prepaid purchase cards for allowing users to transfer financial value at the point of sale and on-line via a computuer network
BetterLife Pharma Inc.	LIVE BET-TERX	2020	4.01	-0.02	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Entertainment Art Company, The	SUNDAY COMICS STORE FEATURING	1993	5.22	-0.15	printed sales catalogues issued periodically relating to goods involving comic strip characters retail store services in the field of memorabilia, collectibles and books involving comic strip characte
IPR Pharmaceuticals, Inc.	THE STATIN WITH ALL THE RIGHT	2002	5.21	+0.45	Audiotapes, videotapes, films in the nature of motion picture films and cinematographic film, CD's and DVD's on the subject of treatment of cardiovascular diseases and cardiovascular health; audiotape
Information Architects, Incorporated	SGML POWER TOOLS	1996	4.93	-0.10	computer software printed material for utilizing SGMI, data modeling language
KX TECHNOLOGIES LLC	KX KX INDUSTRIES, L.P.	1997	6.94	+0.10	[newsletters on the subject of water filters and filtration] water filters for domestic, commercial, and industrial use

Class 017: Rubber & Insulators

Clean filings (1990–2021): 51,254
Mean KL_{pros} / KL_{retr} / ΔKL : 4.97 / 4.96 / +0.018 nats
5y survival rate: 42.6%
Share with SEC-listed owner: 4.28%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
pla	6.03×	338	22%
riders	5.79×	57	16%
polylactic	5.70×	256	21%
slipping	5.23×	220	25%
acrylonitrile	4.64×	265	23%
abs	4.61×	425	27%
sunlight	4.47×	179	20%
filaments	3.79×	955	25%
anti-slip	3.66×	391	26%
potted	3.65×	52	23%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
major	2.78×	72	51%
incorporation	2.39×	67	61%
banknotes	2.37×	59	24%
pry	2.28×	57	42%
assistant	2.14×	56	61%
dual	2.13×	89	43%
caulks	2.13×	122	61%
disconnects	2.12×	52	56%
packagings	2.10×	62	52%
self-adhering	2.07×	58	33%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

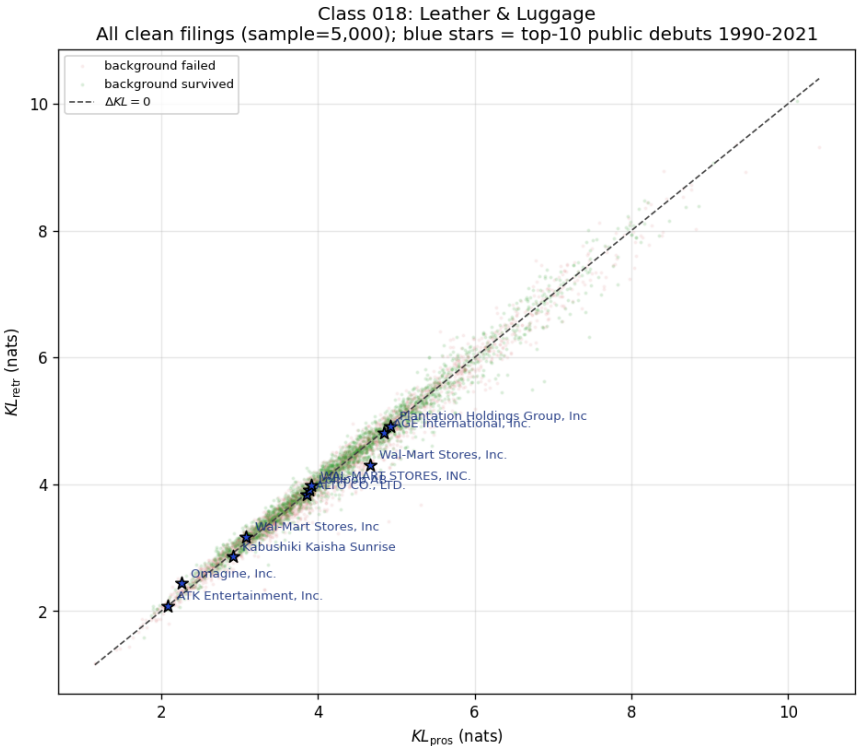
token	n in stratum	n in class	intra-class survival
use	4,259	31,292	39%
plastic	3,060	21,992	37%
materials	2,491	15,496	31%
industrial	2,272	14,228	35%
metal	1,953	11,077	32%
rubber	1,891	12,161	33%
form	1,736	11,352	31%
manufacture	1,540	10,280	34%
insulating	1,516	9,706	28%
purposes	1,483	8,690	27%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
INA ACQUISITION CORP.		2001	4.99	-0.13	Liners and replacement pipes of polymeric materials, including laminated tubular liners of thermosetting resins and resin absorbent materials with polymeric coatings and plastic liners applied against
INA Acquisition Corp.		2001	5.59	+0.03	Non-metal liners and replacement pipes for repair of existing pipelines and passageways of substantially rigid thermoplastic material
Wolverine Holding Company, LLC	PRECISION ENGINEERED PRODUCTS	2002	4.44	-0.03	Metal hose stations for the food, dairy, beverage, pharmaceutical, health and beauty, and processing industries; metal tubing; metal pipe fittings; metal gaskets for the food, dairy, beverage, pharmaceutical
WAL-MART STORES, INC.	EXPERT GARDENER	2003	5.03	-0.09	Wheel barrows Garden apparel, namely, aprons, hats, boots Lawn sprinklers; gardening gloves Plant food; lawn and plant fertilizer for domestic use; potting soil; compost; spike-shaped fertilizer for h
Wal-Mart Stores, Inc	AUTO DRIVE	2017	3.55	-0.01	Automobile tire inflator sealers; flushing agents for automobile radiators All purpose cleaning preparations with deodorizing properties; automobile carpet and upholstery cleaning preparations; carbur
Wal-Mart Stores, Inc.	M	1998	2.43	-0.08	Acoustical insulation barrier panels, Acoustical insulation for buildings, Acoustical panels for buildings, Acrylic resin sheeting for use in the manufacture of laminated glass, Adhesive bands for sea
Wal-mart Stores, Inc.		2011	4.21	-0.04	On-line retail store services featuring a wide variety of consumer goods of others; Retail department store services; Retail store services featuring a wide variety of consumer goods of others Air bru
EXXON MOBIL CORPORATION	LABEL-LYTE	1991	6.02	-0.15	polypropylene film for use in the further manufacture of label facings
Exxon Mobil Corporation	XX	1998	3.79	-0.03	Synthetic rubber for use in the manufacture of tires, tubes, vehicle body sealing, belts, hoses, seals, [glass run channels,] mechanical goods, [electrical cable jacketing, roof sheeting, garden ho
ConocoPhillips Company	Com- CARBON SOLUTIONS FROM CONO	2003 141	4.52	-0.37	Carbon fibers, not for textile use, namely, carbon fibers and carbon fiber composites based on a wide range of matrices, namely, polymeric, asphaltic, cementitious, metallic, ceramic and cer

Class 018: Leather & Luggage

Clean filings (1990–2021): 147,807
Mean KL_{pros} / KL_{retr} / ΔKL : 4.32 / 4.31 / +0.008 nats
5y survival rate: 43.4%
Share with SEC-listed owner: 1.98%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
restraining	$5.48\times$	4,738	31%
locking	$5.42\times$	4,104	28%
tie-out	$5.11\times$	800	41%
pigs	$4.34\times$	583	17%
attaching	$4.20\times$	1,993	35%
bellybands	$4.20\times$	693	24%
coin-sized	$4.11\times$	158	20%
disposing	$3.91\times$	2,114	40%
stakes	$3.82\times$	1,104	40%
dogs	$3.47\times$	4,421	35%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
totebags	$2.84\times$	444	65%
windshirts	$2.72\times$	224	71%
blemish	$2.71\times$	155	60%
soaks	$2.43\times$	152	55%
brief-case	$2.39\times$	209	64%
checker	$2.31\times$	290	61%
ointment	$2.27\times$	181	65%
bronzing	$2.25\times$	169	70%
fanny	$2.16\times$	9,633	66%
ready-to-eat	$2.16\times$	190	55%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

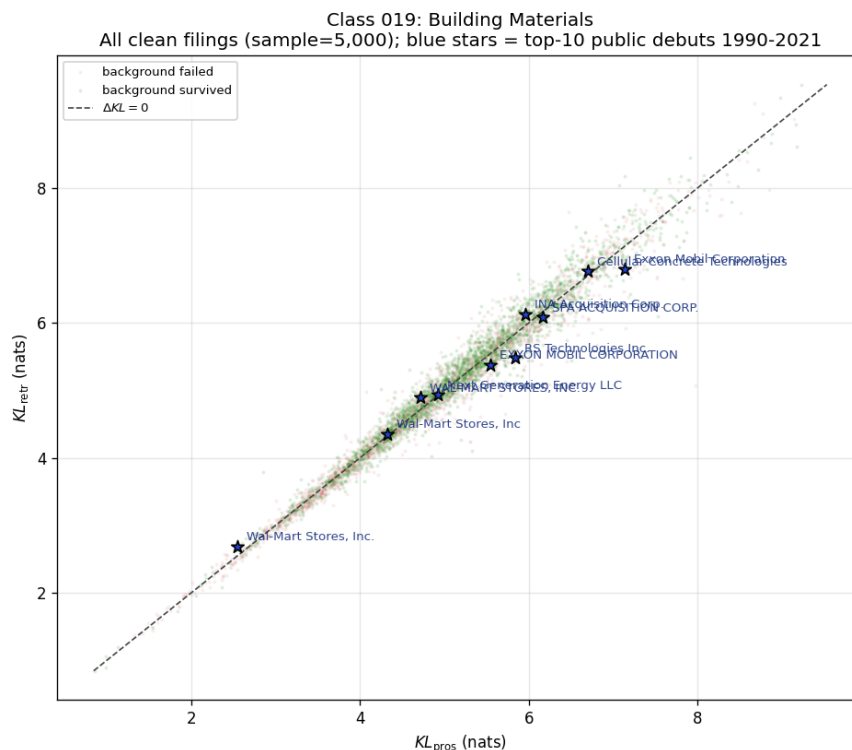
token	n in stratum	n in class	intra-class survival
bags	13,893	116,954	42%
wallets	8,227	69,301	40%
leather	7,898	65,667	35%
clothing	7,829	60,849	39%
handbags	7,826	66,332	38%
backpacks	7,351	62,116	41%
cases	7,294	58,149	39%
purses	6,662	55,490	40%
sports	6,256	47,630	36%
shirts	6,029	43,166	43%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.84	+0.03	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ALTO CO., LTD.	BELLAVIENTE	2021	3.86	+0.02	Winter gloves of leather; neckties; scarves; headwear; belts, namely, clothing belts made from imitation leather, and clothing belts of textile; clothing, namely, casual shirts, dress shirts, pants, s
ATK Entertainment, Inc.	AALAM TUCK KHALED	2020	2.09	+0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Kabushiki Kaisha Sunrise	G-SAVIOUR	1996	2.91	+0.05	bags, namely, all purpose athletic bags, all purpose sport bags, animal game bags, athletic bags, barrel bags, beach bags, book bags, Boston bags, carry-on bags, clutch bags, diaper bags, duffel bags,
Lollipop AB	LOLLIPOP	1996	3.90	-0.02	stationery, paper materials for packing paper and cardboard boxes, paper and plastic bags for merchandise packaging, photo and memory albums and scrap-books, calendars and diaries, bookmarkers braid, h
Omagine, Inc.	OMAGINE	2008	2.26	-0.18	Athletic bags; Baby backpacks; All-purpose athletic bags; Beach bags; Book bags; Diaper bags; Duffel bags; Gym bags; Coin purses; Hand bags; Purses; Tote bags; Leather, mesh and textile shopping bags;
Plantation Group, Inc.	THE PLANTATION SHOP	2004	4.92	+0.00	Gourmet cookies, coffee, wafers, biscuits, snack nut mix consisting primarily of chocolate covered nuts and pralines, candy, toffee and cheese straws Jewelry Retail store services and retail shop at h
WAL-MART STORES, INC.	DW	2003	3.92	-0.06	Jewelry handbags; wallets; purses Bath and body products, namely, bath, body, baby and talcum powders, bath, body and skin soaps in liquid solid or gel form, bath beads, crystals, foams, gels, lotions
Wal-Mart Stores, Inc	DAILY'S	2000 144	3.09	-0.09	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillet, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers;

Class 019: Building Materials

Clean filings (1990–2021):	70,015
Mean KL_{pros} / KL_{retr} / ΔKL :	5.08 / 5.10 / -0.015 nats
5y survival rate:	45.6%
Share with SEC-listed owner:	3.45%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
tournaments	$5.43\times$	70	59%
cologne	$4.59\times$	74	66%
luxury	$4.37\times$	247	25%
fundraising	$4.34\times$	99	66%
unmounted	$4.30\times$	79	72%
reference	$4.12\times$	80	54%
camps	$4.09\times$	83	63%
comic	$3.95\times$	86	60%
wrappers	$3.92\times$	79	58%
breakfast	$3.86\times$	88	48%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
j-channels	$4.74\times$	76	43%
prehung	$4.10\times$	78	47%
log	$3.17\times$	278	50%
fabric-covered	$3.04\times$	79	61%
entranceways	$2.87\times$	143	34%
solariums	$2.84\times$	74	50%
castable	$2.83\times$	300	55%
oriented	$2.67\times$	412	49%
starter	$2.67\times$	225	41%
prefinished	$2.54\times$	114	44%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

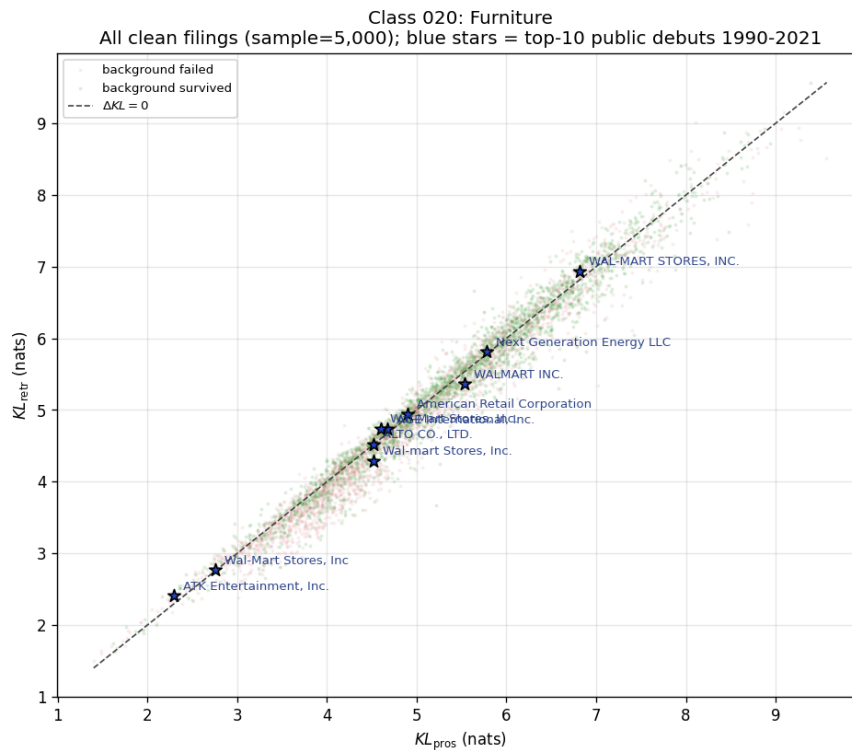
token	n in stratum	n in class	intra-class survival
building	3,753	24,574	39%
non-metal	3,588	25,190	42%
materials	3,418	22,246	39%
metal	3,143	19,399	36%
use	2,837	18,354	39%
construction	2,431	15,130	38%
panels	2,216	14,011	38%
plastic	1,793	10,581	38%
tiles	1,756	10,563	36%
wood	1,747	12,081	40%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Cellular Concrete Technologies, Inc.	PRECIOUS PAVERS	2014	6.70	-0.07	Light weight cement pavers, namely, stones and pet memorials
INA Acquisition Corp.	INTEGRA	2010	5.96	-0.17	Nonmetal tubular liners for the lining of sewer systems for the repair or reconstruction of existing sanitary sewer and storm sewer passageways by or on behalf of municipalities, cities, states, and o
Next Generation Energy LLC	SUNGALOW	2007	4.92	-0.01	On-line retail store services featuring housing and storage structures designed and constructed to enhance solar energy efficiency, namely, mobile homes, transportable homes, prefabricated homes, modu
RS Technologies Inc.	POWERON	2021	5.84	+0.36	Non-metal utility poles; non-metal utility pole cross arms; non-metal covers for utility poles and utility pole cross arms
SPA ACQUISITION CORP.	GAZEBO ORIGINALS	1998	6.17	+0.08	Technical support services, namely, consulting relating to the use of polyure
WAL-MART STORES, INC.	PALM VALLEY	2004	4.72	-0.18	Non-metal gazebos sold in prefabricated and kit form
					Metal fence panels, posts and stays; metal garden fence panels; metal patio fences; hand-operated metal garden hose reels and cabinets Hand-operated non-metal garden hose reels and cabinets Non-metal
Wal-Mart Stores, Inc	MAINSTAYS BABY	2016	4.32	-0.03	Wall and floor art made of natural stone and stone tile Decorative wall art for the home, namely, framed artworks of paper, printed graphics and wall decals Fabric window coverings and treatments, nam
Wal-Mart Stores, Inc.	MEMBERS MARK	1998	2.55	-0.14	Alabaster, Asphalt, Asphalt composition paving, Asphalt patching compound made of a tar base, Asphalt roof coatings, Asphalt roofing paper, Asphalted felts, Balustrades, Beams of wood and concrete, Bi
EXXON MOBIL CORPORATION	EXXONMOBIL	1999	5.54	+0.17	asphalts periodical publications gasoline; diesel fuel; aviation fuels; lubricating oils and greases for industrial, marine, aviation, agricultural, automotive and other vehicular applications; heatin
Exxon Mobil Corporation	MULTIPAVE	2002	7.14	+0.35	Asphalt, pitch and bitumen

Class 020: Furniture

Clean filings (1990–2021): 165,306
Mean KL_{pros} / KL_{retr} / ΔKL : 5.16 / 5.09 / +0.075 nats
5y survival rate: 42.3%
Share with SEC-listed owner: 2.76%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
icons	6.03×	635	26%
translucent	5.63×	671	26%
slumber	5.40×	632	33%
themed	5.13×	752	30%
attacks	5.02×	247	30%
dangerous	4.73×	262	31%
situations	4.52×	281	35%
defense	4.46×	280	34%
pet-actuated	4.23×	499	21%
polypropylene	4.07×	936	28%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
item	$5.21\times$	614	53%
non-leather	$4.47\times$	1,120	70%
replica	$3.49\times$	212	55%
pleated	$3.00\times$	270	43%
giftware	$2.96\times$	213	60%
souvenir	$2.90\times$	651	60%
stationery-type	$2.89\times$	273	64%
juvenile	$2.72\times$	298	61%
license	$2.70\times$	2,102	63%
novelty	$2.68\times$	5,663	54%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

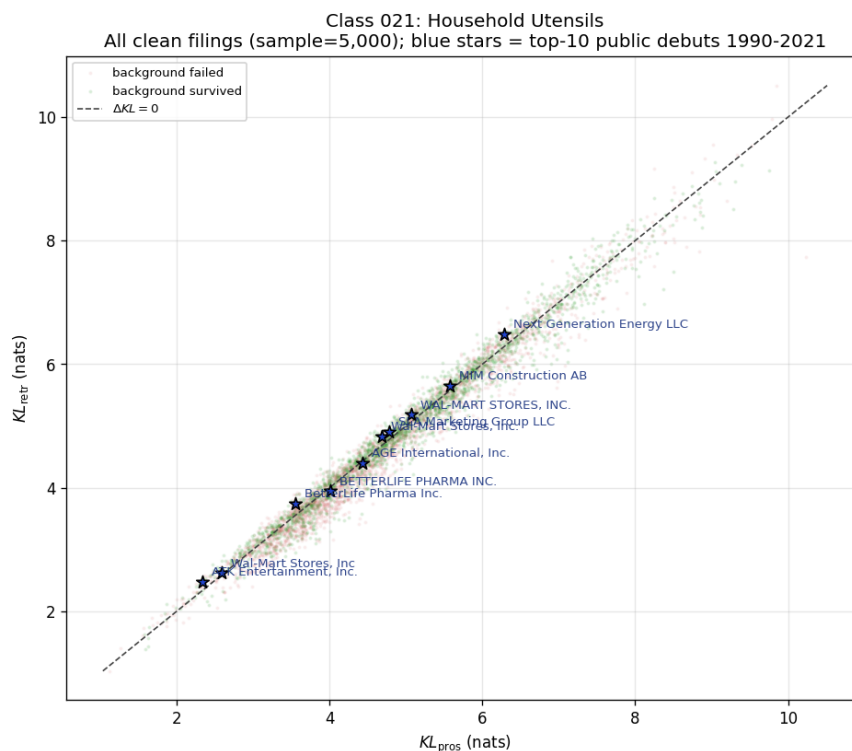
token	n in stratum	n in class	intra-class survival
furniture	5,888	69,240	34%
plastic	5,080	50,581	39%
use	4,713	39,994	38%
metal	4,509	45,072	35%
non-metal	3,297	32,698	38%
frames	3,109	35,284	35%
storage	2,980	28,304	33%
pillows	2,907	35,588	39%
containers	2,793	22,028	38%
chairs	2,596	30,226	33%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.67	-0.06	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ALTO CO., LTD.	BELLAVIENTE	2021	4.52	-0.00	Winter gloves of leather; neckties; scarves; headwear; belts, namely, clothing belts made from imitation leather, and clothing belts of textile; clothing, namely, casual shirts, dress shirts, pants, s
ATK Entertainment, Inc.	AALAM TUCK KHALED	2020	2.29	-0.11	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
American Retail Corporation	JUST DREAM MATTRESS GALLERY	2013	4.89	-0.05	Bed covers; Bed pads; Mattress covers; Mattress pads; Pillow cases; Pillow covers; Sheet sets Beds; Box springs; Mattresses
Next Generation Energy LLC	ZILLA	2009	5.78	-0.03	universal modular mounting system, made primarily of non-metal, the major components of which comprise rails, brackets, a four piece leak proof flashing and standoffs, which supports, reinforces and s
WAL-MART STORES, INC.	PINE HOL-LOW	2003	6.82	-0.12	patio furniture
WALMART INC.	SWITCHART	2014	5.53	+0.17	Magnetic picture frames
Wal-Mart Stores, Inc	DAILY'S	2000	2.75	-0.02	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.	AMERICAN RESIDENCE	1992	4.60	-0.14	ceramic toothbrush holders, ceramic soap dishes, ceramic cups, and trash cans rugs, carpets, and wallpaper toss pillows, plastic shower curtain hooks, shower curtain rods, and window coverings; namely
Wal-mart Stores, Inc.		2011	4.52	+0.23	Beds for household pets; Crate covers for pets; Dog kennels; Inflatable pet beds; Kennels with carriers; Pet crates; Pet cushions; Pet furniture; Pet ramp; Pillows for household pets; Portable beds fo

Class 021: Household Utensils

Clean filings (1990–2021):	219,996
Mean KL_{pros} / KL_{retr} / ΔKL :	4.85 / 4.78 / +0.067 nats
5y survival rate:	40.8%
Share with SEC-listed owner:	2.23%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
connects	4.55×	899	28%
built	4.24×	768	30%
vacuums	3.94×	553	38%
repellers	3.89×	1,803	37%
elevating	3.60×	994	28%
releasing	3.60×	3,119	41%
animal-activated	3.57×	4,097	36%
feeder	3.56×	3,215	41%
therein	3.44×	3,237	41%
non-mechanized	3.42×	1,536	31%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
robe	$4.61\times$	421	59%
candlestick	$3.72\times$	247	62%
non-leather	$3.70\times$	535	64%
saute	$3.46\times$	416	54%
graduate	$3.33\times$	372	47%
checker	$3.29\times$	283	69%
lavatory	$3.28\times$	265	51%
undergraduate	$3.19\times$	273	51%
yo-yos	$3.16\times$	525	59%
stir	$3.13\times$	284	58%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

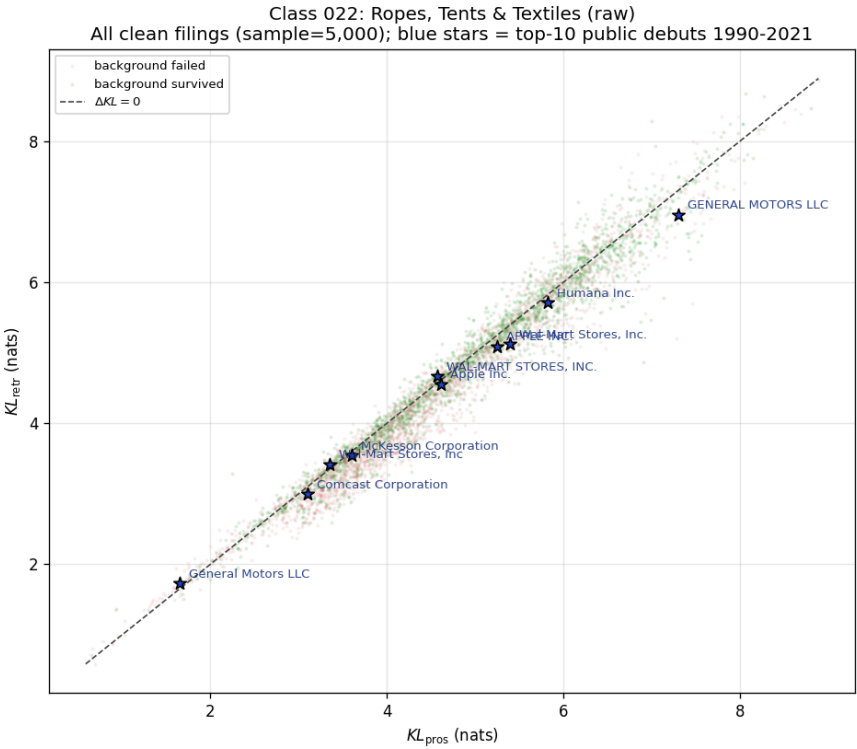
token	n in stratum	n in class	intra-class survival
use	8,685	88,182	33%
household	7,561	78,836	30%
containers	6,865	70,291	33%
cups	6,036	62,298	37%
brushes	5,624	62,369	30%
kitchen	5,260	54,504	29%
plastic	4,995	48,673	40%
mugs	4,824	48,389	44%
holders	4,782	46,572	36%
paper	4,611	45,762	38%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.		BLANTON'S	2021	4.43	+0.02	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Inc.	Entertainment,	AALAM KHALED	2020	2.33	-0.15	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.		BETTERX	2019	4.01	+0.07	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BetterLife Pharma Inc.		LIVE BET-TERX	2020	3.56	-0.19	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
MIM Construction AB		CAREY	2016	5.58	-0.08	storage containers adapted for use in vehicle boots; vehicle partitions for pets; fitted liners for the cargo area of vehicles; boot tidies, namely, storage bins especially adapted to fit in cars; dog
Next Generation Energy LLC		ZILLA	2009	6.28	-0.21	universal modular mounting system, made primarily of non-metal, the major components of which comprise rails, brackets, a four piece leak proof flashing and standoffs, which supports, reinforces and s
SLA Marketing Group LLC		CB	2014	4.78	-0.12	Clothing, namely, scarves, dresses, tops, t-shirts, shirts, jackets, pants, hats Personal lifestyle consulting services; personal fashion consulting services Cosmetic brushes Various colored paper dia
WAL-MART STORES, INC.		EXPERT GARDENER	2003	5.07	-0.11	Wheel barrows Garden apparel, namely, aprons, hats, boots Lawn sprinklers; gardening gloves Plant food; lawn and plant fertilizer for domestic use; potting soil; compost; spike-shaped fertilizer for h
Wal-Mart Stores, Inc		DAILY'S	2000	2.59	-0.03	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal Mart Stores, Inc.		AMERICAN	1992	4.69	-0.15	ceramic toothbrush holders, ceramic

Class 022: Ropes, Tents & Textiles (raw)

Clean filings (1990–2021): 24,515
Mean KL_{pros} / KL_{retr} / ΔKL : 4.70 / 4.59 / +0.111 nats
5y survival rate: 40.1%
Share with SEC-listed owner: 2.31%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
infestation	4.23×	104	25%
minimize	4.21×	102	27%
lice	4.17×	103	27%
plastic-covered	3.50×	306	19%
belongings	3.46×	130	28%
jammed	3.40×	94	30%
tensile	3.34×	826	21%
stuck	3.33×	96	30%
soiled	3.32×	364	21%
one-way	3.30×	191	20%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
hairpieces	$5.33\times$	60	53%
baler	$5.19\times$	77	45%
percent	$4.61\times$	52	77%
thc	$4.44\times$	54	78%
delta-	$4.28\times$	56	79%
cannabis	$4.12\times$	51	84%
basis	$3.55\times$	76	71%
tetrahydrocannabinol	$3.33\times$	51	78%
concentration	$3.29\times$	88	58%
deflectors	$3.20\times$	50	44%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

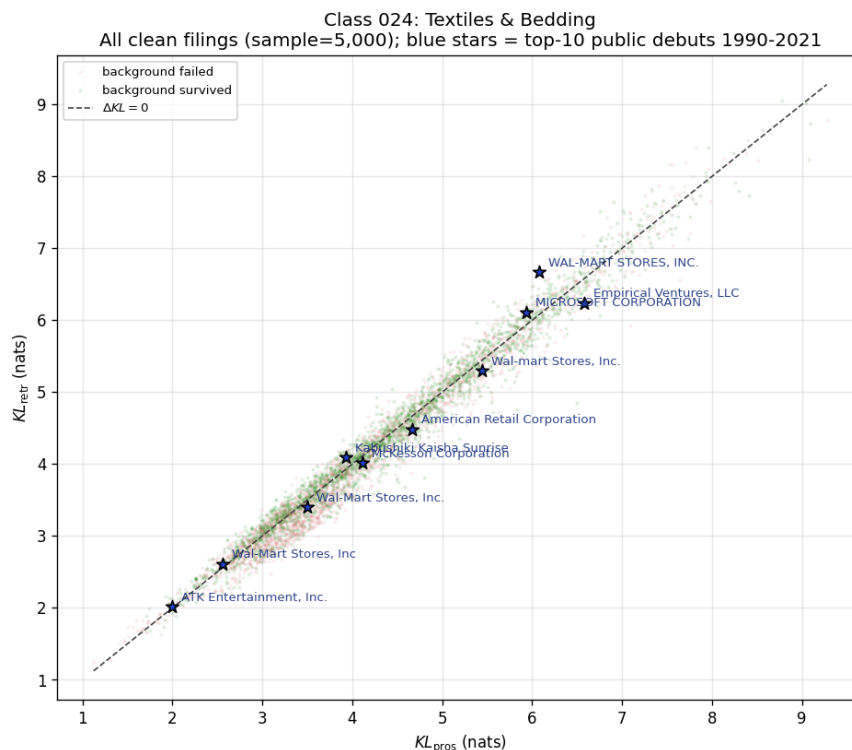
token	n in stratum	n in class	intra-class survival
use	1,031	9,878	36%
bags	1,025	10,999	36%
textile	795	9,060	30%
plastic	762	6,723	34%
materials	665	7,806	29%
storage	598	6,511	33%
tents	590	7,779	30%
metal	584	5,284	34%
covers	518	4,493	34%
nature	487	3,655	33%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
WAL-MART STORES, INC.	FUSION 3-D	2004	4.57	-0.10	Clothing and footwear for hunting, namely, pants, shirts, bib overalls, jackets, coats, ponchos, gloves, vests, balaclavas, neck gaiters, scarves, bandanas, hats, boots, shoes, slippers, rainwear, nam
Wal-Mart Stores, Inc	HOMETREND	2016	3.36	-0.06	Storage racks; clothes hangers; packaging containers of plastic; plastic storage containers for commercial or industrial use; plastic boxes; non-metal shoe racks; wardrobes; furniture; mirrors; shelvi
Wal-Mart Stores, Inc.	AUTO FX	1994	5.39	+0.27	automobile accessories, namely portable cooler bags trunk organizers and spare tire covers pumps for inflating tires net bags for storage
APPLE INC.	IPOD	2005	5.24	+0.15	Lanyards for portable and handheld digital electronic devices for recording, organizing, transmitting, manipulating, and reviewing text, data, audio, image, and video files [full line of holders, str
Apple Inc.		2007	4.61	+0.05	Lanyards for portable and handheld digital electronic devices for recording, organizing, transmitting, manipulating, and reviewing text, data, audio, image, and video files Housewares, namely, [decor
McKesson Corporation	MCKESSON	2005	3.60	+0.06	[Cadaver bags;] bags for medical linen and laundry Sitz baths House mark for a full line of medical and surgical products, equipment and supplies, namely patient apparel consisting of bibs, gowns, c
GENERAL MOTORS LLC		2008	7.30	+0.35	Lanyards for holding badges and passes
General Motors LLC	GM	2021	1.65	-0.07	Custom manufacture of fuel cells; custom manufacture of motor vehicles and motor vehicle parts Automobile lease financing services; financial information provided by electronic means in the field of m
Comcast Corporation	XFINITY	2021	3.10	+0.11	Birdhouses; Non-metal hooks; Travel pillows Beach towels; Blanket throws; Table cloths not of paper; Vinyl banners Balloons; Building games; Checkers games; Dice games; Dominoes; Golf balls; Ring game
Humana Inc.	FREEWHEELIN	2009	5.83	+0.12	lanyards for holding keys or badges; bands for wrapping or binding; bands for wrapping pants legs when riding bikes or bicycles

Class 024: Textiles & Bedding

Clean filings (1990–2021):	97,945
Mean KL_{pros} / KL_{retr} / ΔKL :	4.25 / 4.21 / +0.042 nats
5y survival rate:	42.9%
Share with SEC-listed owner:	2.23%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
cityscapes	$4.37\times$	176	11%
background	$4.00\times$	275	20%
bundle	$3.53\times$	3,700	29%
pad	$3.18\times$	4,319	32%
scenes	$3.12\times$	250	25%
swaddling	$3.05\times$	4,885	29%
scenery	$2.95\times$	359	17%
home-health	$2.77\times$	1,610	30%
bio-polymer	$2.77\times$	1,608	30%
backdrops	$2.76\times$	1,377	22%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
finger tip	$3.21\times$	112	59%
bedskirts	$3.15\times$	203	59%
ensembles	$3.12\times$	208	77%
solicitation	$3.01\times$	103	59%
manufacturers	$3.00\times$	170	56%
directed	$2.88\times$	139	65%
comforter	$2.86\times$	1,189	63%
napery	$2.85\times$	193	41%
encasements	$2.74\times$	208	64%
mist	$2.48\times$	113	58%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

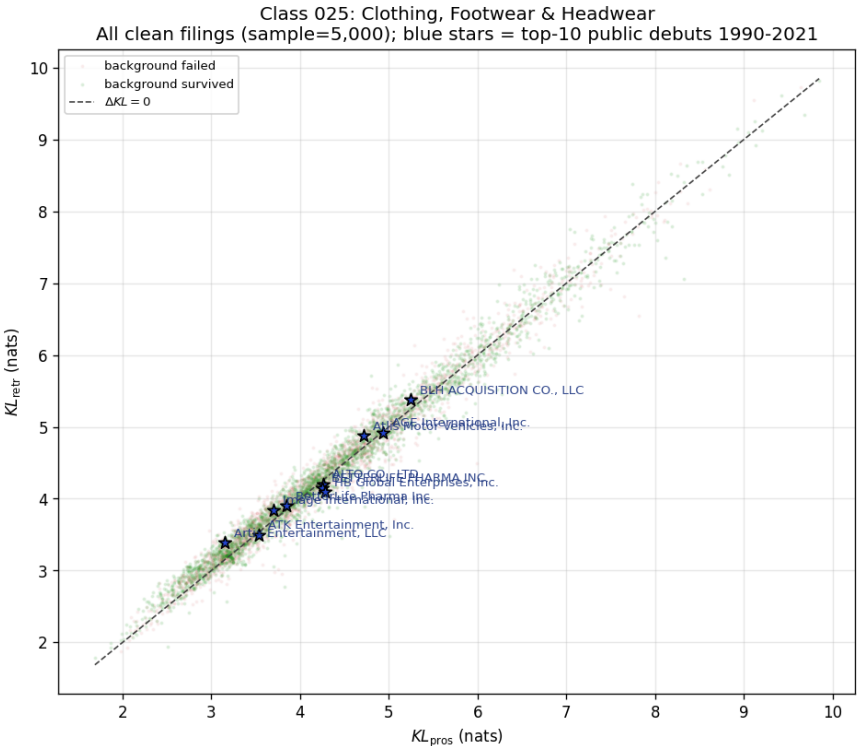
token	n in stratum	n in class	intra-class survival
textile	4,941	49,505	34%
bed	4,795	50,531	38%
towels	4,508	45,030	39%
covers	4,222	45,164	35%
blankets	3,963	42,694	38%
use	3,758	32,565	38%
clothing	3,293	27,312	40%
sheets	3,266	35,674	40%
bags	3,139	28,143	37%
table	3,139	30,422	35%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ATK Inc.	Entertainment,	AALAM	2020	2.00	-0.02	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
American Retail Corporation		JUST DREAM MATTRESS GALLERY	2013	4.67	+0.18	Bed covers; Bed pads; Mattress covers; Mattress pads; Pillow cases; Pillow covers; Sheet sets Beds; Box springs; Mattresses
Empirical LLC	Ventures,	HOLLYBERRY & COMPANY	2014	6.58	+0.34	Christmas tree skirts; Christmas tree stand covers [Table cloth of textile]
Kabushiki Kaisha Sunrise		G-SAVIOUR	1996	3.92	-0.16	textiles, namely, napkins, placemats, handkerchiefs, towels, bed covers, and table covers not of paper
WAL-MART STORES, INC.		NOBO NO BOUNDARIES	2003	6.08	-0.58	Household linens for bed
Wal-Mart Stores, Inc		DAILY'S	2000	2.56	-0.05	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.		AMERICAN RESIDENCE	1992	3.50	+0.10	shams, comforters, plastic and cloth shower curtains, bed sheets, dust ruffles, table cloths not of paper, window coverings; namely, curtains, and fabric; namely, cotton and polycotton
Wal-mart Stores, Inc.		BACKYARD GR LL	2011	5.44	+0.15	Barbecue smokers; Barbecues and grills; Ceramic briquettes for use in barbecue grills; Charcoal grills; Gas grills; Lighters primarily for lighting grills, fireplaces and candles; Roasters powered by
McKesson Corporation		MCKESSON	2005	4.11	+0.10	[Cadaver bags;] bags for medical linen and laundry Sitz baths House mark for a full line of medical and surgical products, equipment and supplies, namely patient apparel consisting of bibs, gowns, c
MICROSOFT CORPORATION		MICROSOFT	1991	5.94	-0.17	towels and blankets

Class 025: Clothing, Footwear & Headwear

Clean filings (1990–2021): 668,593
Mean KL_{pros} / KL_{retr} / ΔKL : 4.36 / 4.40 / -0.041 nats
5y survival rate: 51.6%
Share with SEC-listed owner: 1.39%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
mental	4.74×	741	45%
cannabis	4.73×	842	65%
speaking	3.88×	902	45%
facilitating	3.81×	679	50%
webinars	3.70×	671	40%
brand	3.69×	1,251	68%
podcasts	3.67×	3,407	46%
leadership	3.67×	935	43%
wellness	3.59×	2,449	52%
blogs	3.57×	3,274	46%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
mens	$3.85\times$	2,339	67%
womens	$3.76\times$	1,983	67%
childrens	$3.27\times$	1,354	68%
girl	$3.18\times$	1,482	55%
boys	$2.87\times$	3,811	54%
enhancing	$2.66\times$	767	62%
intimate	$2.64\times$	2,463	56%
wearing	$2.54\times$	10,771	55%
girls	$2.37\times$	4,983	55%
cheerleading	$2.32\times$	1,342	58%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

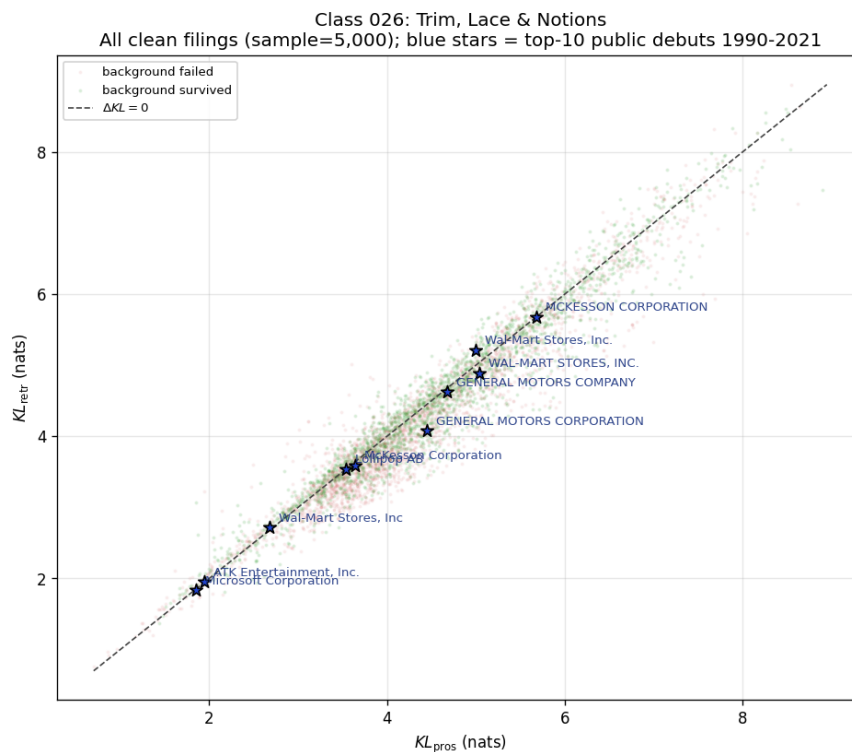
token	n in stratum	n in class	intra-class survival
shirts	47,595	452,457	51%
clothing	40,201	403,438	51%
hats	39,722	358,104	52%
jackets	39,691	370,707	51%
pants	38,278	356,666	52%
t-shirts	32,486	325,880	53%
caps	28,253	242,240	51%
shorts	25,313	260,271	54%
footwear	24,482	205,166	50%
tops	21,513	209,252	51%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.93	+0.01	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ALTO CO., LTD.	BELLAVIENTE	2021	4.26	+0.06	Winter gloves of leather; neckties; scarves; headwear; belts, namely, clothing belts made from imitation leather, and clothing belts of textile; clothing, namely, casual shirts, dress shirts, pants, s
ATK Entertainment, Inc.	ASAHD TUCK KHALED	2018	3.54	+0.04	Clothing, namely, shirts, pants, hats, caps, headwear, socks
Artix Entertainment, LLC	ADVENTUREQUEST	2008	3.16	-0.23	Clothing, namely, hats, caps, golf shirts, t-shirts, sweaters, sweatshirts, jackets, socks, pajamas, underwear and Halloween costumes
Atlis Motor Vehicles, Inc.	AMV	2021	4.72	-0.15	Clothing, namely, hats; baseball caps; beanie hats; t-shirts; baseball t-shirts; polo shirts; tank tops; jersey t-shirt; hoodies; and sweatshirts
BETTERLIFE PHARMA INC.	MOVE BET-TERX	2019	4.25	+0.10	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BLH ACQUISITION CO., LLC	DRINK POURN	2017	5.24	-0.14	Headwear, namely, caps and hats; Wearable garments and clothing, namely, shirts Bar and restaurant services
BetterLife Pharma Inc.	LIVE BET-TERX	2020	3.84	-0.06	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
HB Global Enterprises, Inc.	SOUL RECI-PROCITY	2020	4.29	+0.19	Retail store services featuring subscription boxes containing household items, personal care items, clothing, jewelry and gifts; Retail shops featuring gifts, clothing and jewelry Educational and ente
Image International, Inc.	NEW DEFINITIONS	1993	3.70	-0.14	wearing apparel; namely, boys' suits, pants, shirts, jackets, topcoats, neck wear, sweat shirts, sweat pants and shorts

Class 026: Trim, Lace & Notions

Clean filings (1990–2021): 42,323
Mean KL_{pros} / KL_{retr} / ΔKL : 4.47 / 4.34 / +0.124 nats
5y survival rate: 40.4%
Share with SEC-listed owner: 1.76%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
cabbages	$4.52\times$	84	12%
lei	$4.18\times$	146	23%
topiaries	$4.03\times$	387	16%
lemons	$3.70\times$	119	7%
orchids	$3.61\times$	316	17%
pre-lit	$3.46\times$	324	22%
botanicals	$3.44\times$	697	18%
ferns	$3.36\times$	330	18%
apples	$3.36\times$	137	7%
tulips	$3.31\times$	368	14%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
programme	$5.53\times$	103	37%
ordered	$4.87\times$	76	28%
simulcasting	$4.65\times$	226	37%
law	$4.58\times$	201	31%
risk	$4.57\times$	94	29%
languages	$4.52\times$	188	29%
alerts	$4.32\times$	213	31%
voices	$4.11\times$	246	37%
curricula	$3.98\times$	138	32%
scores	$3.96\times$	182	38%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

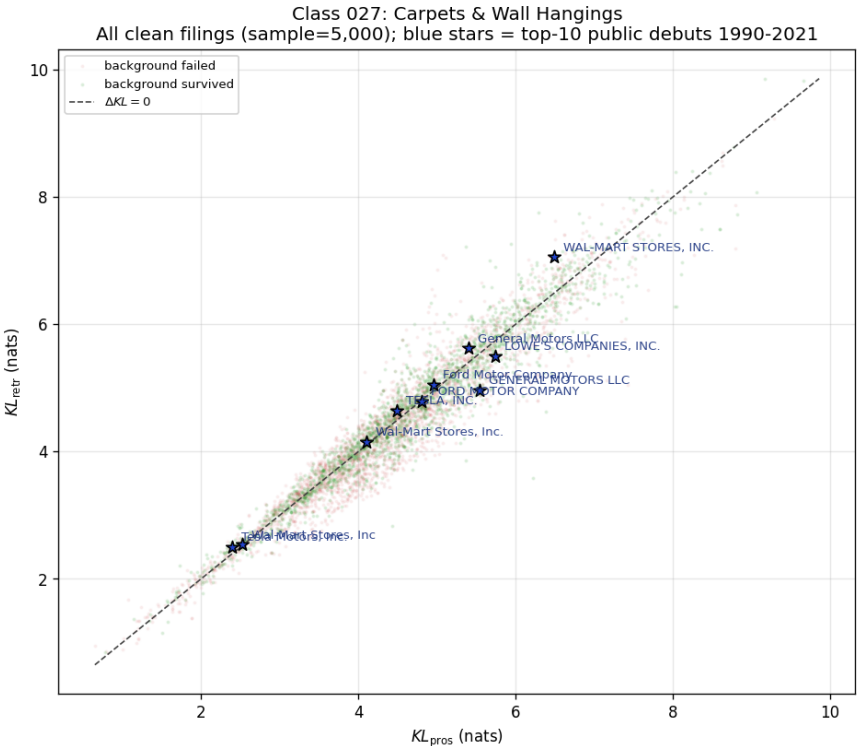
token	n in stratum	n in class	intra-class survival
hair	1,635	25,003	35%
clothing	1,394	15,603	40%
pins	1,092	12,971	37%
accessories	1,015	14,563	39%
ribbons	958	11,145	31%
bands	944	11,719	33%
buttons	819	9,739	38%
bags	800	6,509	44%
clips	794	10,251	37%
caps	785	9,151	35%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ATK Inc.	Entertainment,	AALAM KHALED	2020	1.95	-0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Lollipop AB		LOLLIPOP	1996	3.54	+0.00	stationery, paper materials for packing paper and cardboard boxes, paper and plastic bags for merchandise packaging, photo and memory albums and scrap-books, calendars and diaries, bookmarkers braid, h
WAL-MART STORES, INC.		GIRL CONNECTION	2003	5.04	+0.15	Hair ornaments Handbags Clothing, namely, hats, shoes and belts
Wal-Mart Stores, Inc		DAILY'S	2000	2.68	-0.04	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.		COLOR TREND BASICS	1994	5.00	-0.21	purses belts hair ornaments
MCKESSON CORPORATION		FOOD GUARD	2001	5.69	+0.01	[hair nets] disposable latex and vinyl gloves for use in the food service industry [protective clothing for use in the food service area, namely, bouffant caps, beard covers, aprons, sleeve covers, an
McKesson Corporation		SUNMARK	2003	3.64	+0.04	humidifiers and vaporizers House mark for a line of personal care products, namely, tweezers, nail clippers, nail and cuticle scissors, nail files, razors and razor blades and eyelash curlers House ma
Microsoft Corporation			2015	1.85	+0.02	[Household or kitchen utensils and] containers [, namely, rolling pins, spatulas, whisks, and containers; chinaware, namely, plates and] mugs; beverage glassware; porcelain, pottery and earthenware
GENERAL MOTORS COMPANY		H2	2001	4.67	+0.05	Belt buckles for clothing not made of precious metal, embroidered emblems, hat pins, zipper pulls
GENERAL MOTORS CORPORATION			1991	4.45	+0.37	belt buckles not of precious metal

Class 027: Carpets & Wall Hangings

Clean filings (1990–2021): 26,727
Mean KL_{pros} / KL_{retr} / ΔKL : 4.55 / 4.49 / +0.053 nats
5y survival rate: 40.9%
Share with SEC-listed owner: 3.14%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
logos	3.68×	133	28%
anti-breakage	3.68×	76	28%
mushiro	3.55×	124	21%
voices	3.40×	50	14%
hana-mushiro	3.36×	143	24%
slopes	3.32×	698	16%
resemble	3.27×	52	2%
slippage	3.27×	912	30%
fire-resistant	3.15×	746	16%
multiplayer	3.15×	54	0%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
wallcovering	$6.26\times$	254	30%
sample	$4.51\times$	206	43%
voices	$4.20\times$	50	14%
smooth	$4.11\times$	56	30%
resemble	$4.04\times$	52	2%
underarm	$3.96\times$	53	13%
generated	$3.89\times$	54	4%
run	$3.86\times$	57	2%
gravity	$3.82\times$	55	13%
multiplayer	$3.70\times$	54	0%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

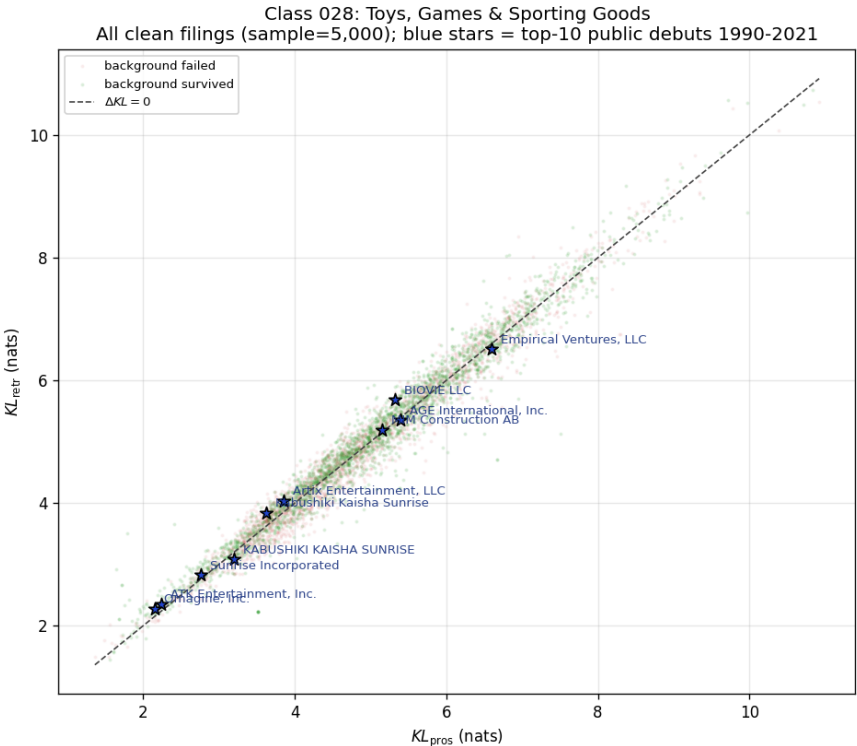
token	n in stratum	n in class	intra-class survival
mats	1,236	16,348	37%
floor	1,041	14,362	35%
rugs	871	11,505	37%
use	807	9,061	36%
textile	746	8,724	34%
wall	723	9,083	35%
carpets	717	10,155	31%
coverings	717	10,148	33%
bath	672	7,736	35%
plastic	630	6,553	37%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
WAL-MART STORES, INC.	HOMETREND	2003	6.49	-0.57	wall borders
Wal-Mart Stores, Inc	DAILY'S	2000	2.53	-0.01	Cooking appliances, namely, electric Dutch ovens, food steamers, electric skillets, fryers, microwaves, toasters, breadmakers, popcorn poppers, waffle irons, electric coffee pots, coffee makers; elect
Wal-Mart Stores, Inc.	AMERICAN RESI-DENCE	1992	4.10	-0.05	ceramic toothbrush holders, ceramic soap dishes, ceramic cups, and trash cans rugs, carpets, and wallpaper toss pillows, plastic shower curtain hooks, shower curtain rods, and window coverings; namely
GENERAL MOTORS LLC	SATURN	1992	5.55	+0.58	carpeting and floor mats for motor vehicles
General Motors LLC	Z-SPEC	2010	5.40	-0.23	Automotive aftermarket accessories, namely, pedal kits consisting of a gas pedal and a brake pedal, grills for vehicles, spoilers for vehicles, wheels for vehicles, badges for vehicles, automotive int
FORD MOTOR COMPANY	MERCURY	1993	4.81	+0.02	vehicle floor mats, [door mats,] carpet
Ford Motor Company	FORD OUT-FITTERS	1999	4.96	-0.09	floor mats for vehicles and rubber mats
LOWE'S COMPANIES, INC.	FASTLOCK	2001	5.75	+0.25	Floor covering material consisting of fiberboard or other wood-based material with a laminated decorative surface
TESLA, INC.	T	2009	4.49	-0.15	[printed maintenance manuals for others for use in maintaining and repairing automobiles] book bags; school bags [mats] Brake calipers for land vehicles; anti-skid chains, roof-racks, shock absorb
Tesla Motors, Inc.	TESLA T	2009	2.40	-0.10	vehicle axles; brake pads and brake linings, all for land vehicles; brake calipers, braking systems, bumper bars, clutches, differential gears, drive gears, drive shafts, gear change selectors, transm

Class 028: Toys, Games & Sporting Goods

Clean filings (1990–2021): 325,521
Mean KL_{pros} / KL_{retr} / ΔKL : 4.94 / 4.94 / -0.006 nats
5y survival rate: 47.1%
Share with SEC-listed owner: 3.83%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
scoring	$4.69\times$	1,464	87%
loose-leaf	$3.79\times$	1,771	76%
itself	$3.48\times$	2,536	23%
securely	$3.48\times$	2,534	23%
trimmer	$3.47\times$	2,552	23%
palm	$3.27\times$	1,212	19%
twisting	$3.15\times$	907	28%
grippers	$3.14\times$	1,896	24%
developers	$3.10\times$	984	22%
noisemaker	$3.08\times$	1,676	18%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
ride-in	$4.16\times$	358	93%
scoring	$3.33\times$	1,464	87%
squirt	$3.12\times$	528	88%
convertible	$2.78\times$	644	43%
loose-leaf	$2.73\times$	1,771	76%
headcovers	$2.64\times$	356	72%
windup	$2.64\times$	478	78%
molding	$2.61\times$	959	69%
chambers	$2.59\times$	425	48%
compound	$2.54\times$	965	73%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

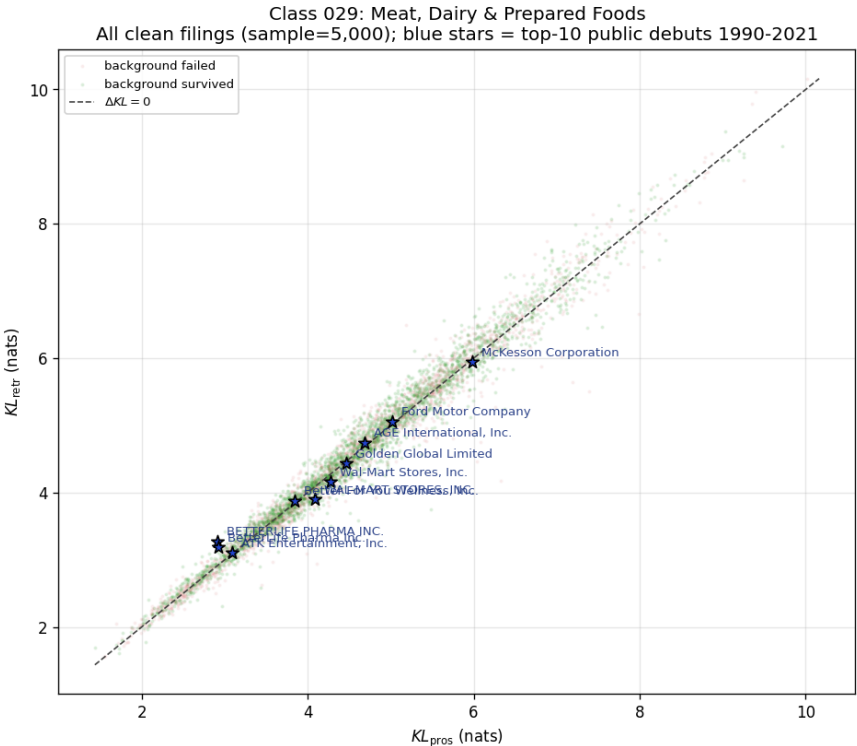
token	n in stratum	n in class	intra-class survival
games	9,844	104,843	46%
toys	9,796	122,074	44%
toy	8,750	109,103	42%
equipment	7,002	86,517	44%
game	6,629	72,060	47%
use	6,397	73,369	41%
playing	5,820	60,296	47%
accessories	5,017	58,390	52%
electronic	4,977	52,730	48%
balls	4,824	58,268	43%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	5.39	+0.04	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM	2020	2.24	-0.12	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
Artix Entertainment, LLC	ADVENTUREQUEST	2008	3.85	-0.19	Toys, namely, toy figures, dolls, stuffed plush toys, board games and puzzles
BIOVIE LLC	CELAFORME	2021	5.32	-0.36	Balance boards for improving strength, toning, conditioning, balance, and proprioception; Manually-operated exercise equipment
Empirical LLC	HOLLYBERRY & COMPANY	2014	6.60	+0.10	Christmas tree skirts; Christmas tree stand covers [Table cloth of textile]
KABUSHIKI KAISHA SUNRISE	MOBILE SUIT GUNDAM SEED ECLIPS	2021	3.19	+0.11	Toys, namely, toy figures, toy robots and stuffed toys; Toy construction sets; Video game machines; Arcade video game machines; Battery-powered computer game with LCD screen; Game controllers for comp
Kabushiki Kaisha Sunrise	G-SAVIOUR	1996	3.63	-0.21	television games and software for television games, namely, interactive video games of virtual reality comprised of computer hardware and software; automatic video games in the nature of amusement mac
MIM Construction AB	MIMSAFE	2016	5.15	-0.04	Cages for household pets Storage containers adapted for use in vehicle boots; vehicle partitions for cargo storage; fitted liners for the cargo area of vehicles; dog guards for use in vehicles; safety
Omagine, Inc.	OMAGINE	2008	2.15	-0.12	Address books; Almanacs; Appliques in the form of decals; Appointment books; Art prints; Arts and craft paint kits; Autograph books; Baby books; Ball point pens; Baseball cards; Binders; Bookends; Boo
Sunrise Incorporated	CRUSH GEAR	2002	2.75	-0.07	Accessories for dolls clothes, Action figures, Bubble making wand and solution sets, Doll accessories, Doll cases, Doll clothing, Doll furniture, Doll house furnishings, Dolls; Kites, Musical toys, Pa

Class 029: Meat, Dairy & Prepared Foods

Clean filings (1990–2021): 134,683
Mean KL_{pros} / KL_{retr} / ΔKL : 4.63 / 4.65 / -0.020 nats
5y survival rate: 51.8%
Share with SEC-listed owner: 3.42%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
relieving	$5.46\times$	141	59%
mood	$5.40\times$	150	61%
transdermal	$5.13\times$	185	73%
depression	$5.06\times$	170	60%
salves	$4.82\times$	303	79%
relaxation	$4.55\times$	191	60%
fatigue	$4.54\times$	196	51%
cannabidiol	$4.52\times$	168	89%
anxiety	$4.48\times$	203	62%
emollients	$4.40\times$	150	66%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
least	$4.19\times$	172	42%
franks	$4.04\times$	166	48%
arranged	$3.48\times$	187	37%
pasteurized	$3.27\times$	394	51%
hams	$3.08\times$	523	44%
designs	$3.04\times$	250	35%
roasts	$2.96\times$	186	47%
item	$2.94\times$	163	61%
percent	$2.91\times$	354	73%
delta-	$2.89\times$	408	75%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

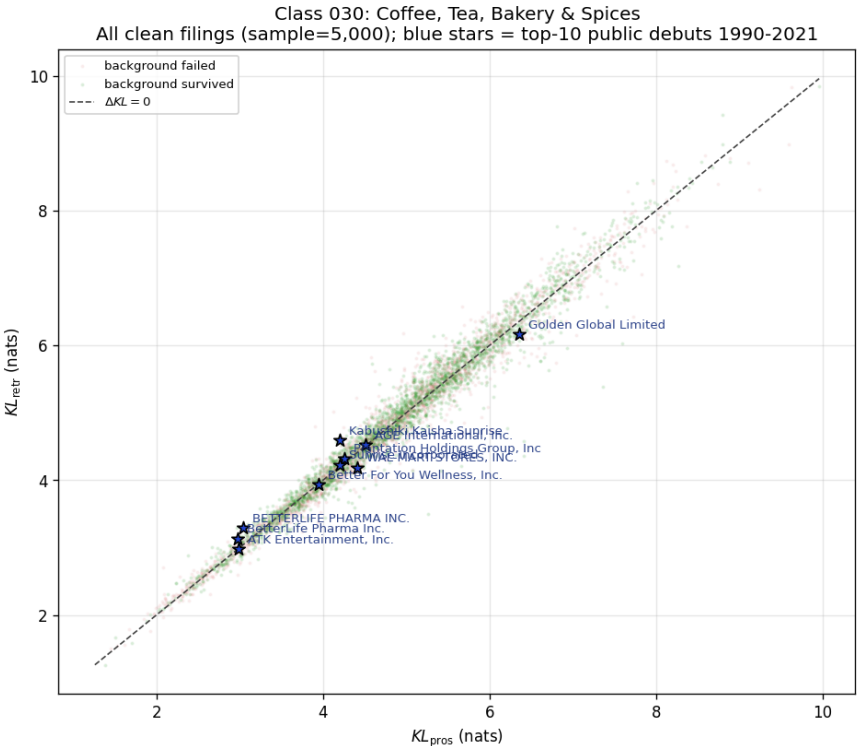
token	n in stratum	n in class	intra-class survival
food	6,117	43,355	50%
vegetables	5,426	39,964	49%
fruit	5,230	35,249	50%
processed	5,133	40,733	50%
meat	4,745	37,723	49%
frozen	4,680	36,038	49%
dried	4,594	29,899	45%
snack	4,475	28,644	52%
fruits	4,160	28,277	47%
products	4,142	32,292	47%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.68	-0.06	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM KHALED	2020	3.09	-0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.	BETTERX	2019	2.92	-0.36	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
Better For You Wellness, Inc.	BETTER FOR YOU WELLNESS	2021	3.85	-0.03	Protein bars, namely, protein supplements formed and packaged as bars Advisory and consultation services in the fields of agronomy, agriculture, and horticulture; agronomic consulting services; agrono
BetterLife Pharma Inc.	LIVE BET-TERX	2020	2.92	-0.26	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Golden Global Limited	O MY YO-GURT	2010	4.46	+0.03	Drinks based on yogurt; Yogurt; Yogurt drinks; Yogurt-based beverages
WAL-MART STORES, INC.	SAM'S CHOICE	1991	4.08	+0.19	[processed meats, processed edible seeds and fruit preserves] breakfast and processed cereals; candy; [condiments; namely, mustards] fruit drinks
Wal-Mart Stores, Inc.	GREAT VALUE	1993	4.27	+0.12	processed fish and poultry; processed edible seeds;] and soups and soup mixes
McKesson Corporation	INSTAPRO	2006	5.99	+0.05	Protein for use as a food additive
Ford Motor Company	CHARACTERS POWERED BY FORD	2009	5.02	-0.04	Yogurt, fruit-based snack food, fruit leathers

Class 030: Coffee, Tea, Bakery & Spices

Clean filings (1990–2021): 205,442
Mean KL_{pros} / KL_{retr} / ΔKL : 4.90 / 4.91 / -0.011 nats
5y survival rate: 51.3%
Share with SEC-listed owner: 2.72%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
cannabidiol	4.00×	285	91%
transdermal	3.95×	205	71%
cartomizers	3.72×	234	63%
salves	3.58×	427	78%
anxiety	3.35×	257	65%
well-being	3.26×	353	58%
refill	3.19×	1,314	58%
smokeless	3.17×	347	69%
relaxation	3.12×	260	62%
fatigue	3.07×	215	49%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
e-liquid	$4.87\times$	1,024	60%
refill	$4.33\times$	1,314	58%
propylene	$4.14\times$	249	67%
cartomizers	$4.06\times$	234	63%
glycol	$3.97\times$	257	66%
least	$3.60\times$	361	48%
designs	$3.38\times$	314	34%
glycerin	$3.23\times$	288	66%
vaporizer	$3.22\times$	351	74%
smokeless	$3.17\times$	347	69%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

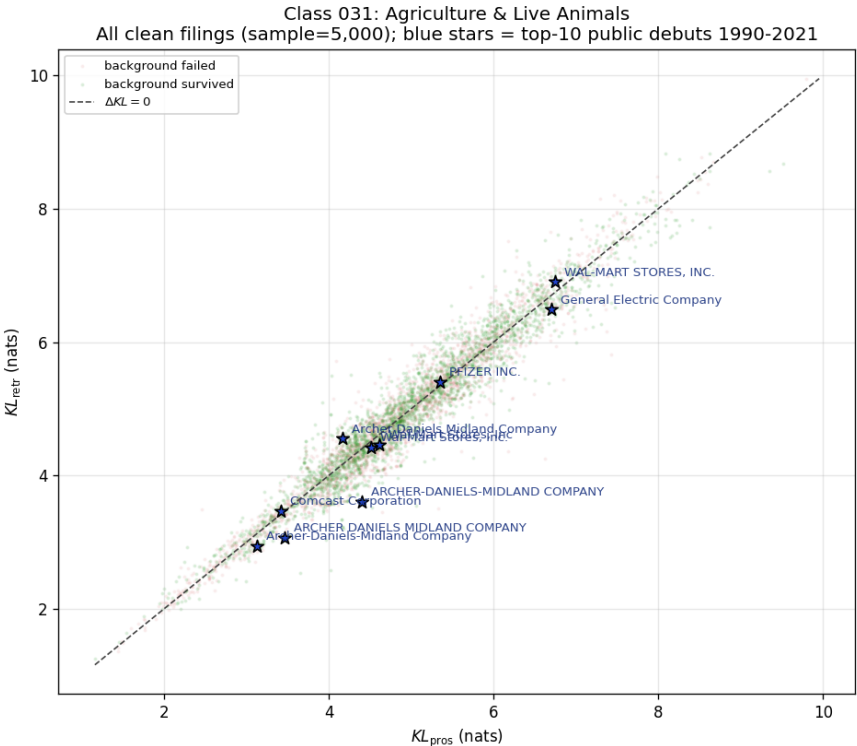
token	n in stratum	n in class	intra-class survival
food	8,009	56,718	50%
beverages	6,030	42,857	48%
coffee	5,960	44,886	47%
tea	5,617	39,383	46%
fruit	5,609	37,759	49%
snack	5,589	34,248	51%
chocolate	5,310	35,356	46%
foods	5,196	32,925	51%
frozen	5,145	37,251	51%
products	5,048	35,916	48%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.51	-0.01	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM	2020	2.99	+0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.	SLEEP BET-TERX	2019	3.05	-0.25	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
Better For You Wellness, Inc.	BETTER FOR YOU WELLNESS	2021	3.95	+0.01	Protein bars, namely, protein supplements formed and packaged as bars Advisory and consultation services in the fields of agronomy, agriculture, and horticulture; agronomic consulting services; agrono
BetterLife Pharma Inc.	LIVE BET-TERX	2020	2.98	-0.16	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Golden Global Limited	O MY YO-GURT	2010	6.36	+0.19	Frozen yogurt; Frozen yogurt confections
Kabushiki Kaisha Sunrise	GUNDAM	1996	4.21	-0.38	candies, bubble gum, chewing gum and frozen confections
Plantation Holdings Group, Inc.	THE PLANTATION SHOP	2004	4.26	-0.06	Gourmet cookies, coffee, wafers, biscuits, snack nut mix consisting primarily of chocolate covered nuts and pralines, candy, toffee and cheese straws Jewelry Retail store services and retail shop at h
Sunrise Incorporated	OUTLAW STAR	2002	4.20	-0.02	Candy, chocolate, bubble gum and chewing gum, cookies, ice cream, sorbets and confections, namely, frozen confections and confectioners sugar
WAL-MART STORES, INC.	SAM'S CHOICE	1991	4.41	+0.23	[processed meats, processed edible seeds and fruit preserves] breakfast and processed cereals; candy; [condiments; namely, mustards] fruit drinks

Class 031: Agriculture & Live Animals

Clean filings (1990–2021): 64,135
Mean KL_{pros} / KL_{retr} / ΔKL : 4.92 / 4.92 / -0.004 nats
5y survival rate: 49.8%
Share with SEC-listed owner: 2.21%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
opioid	$8.48\times$	79	65%
comestibles	$8.12\times$	80	62%
epilepsy	$7.74\times$	93	66%
addiction	$7.39\times$	88	60%
mood	$6.49\times$	111	62%
depression	$6.03\times$	126	59%
pre-baked	$5.78\times$	147	52%
fatigue	$5.68\times$	118	57%
glosses	$5.60\times$	91	53%
relieving	$5.57\times$	122	69%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
deciduous	$4.72\times$	106	45%
percent	$2.41\times$	555	72%
nectarines	$2.38\times$	181	47%
delta-	$2.35\times$	639	73%
sod	$2.33\times$	476	53%
less	$2.31\times$	91	79%
tetrahydrocannabinol	$2.28\times$	522	74%
amounts	$2.27\times$	75	79%
wholesalers	$2.25\times$	71	70%
concentration	$2.25\times$	632	72%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

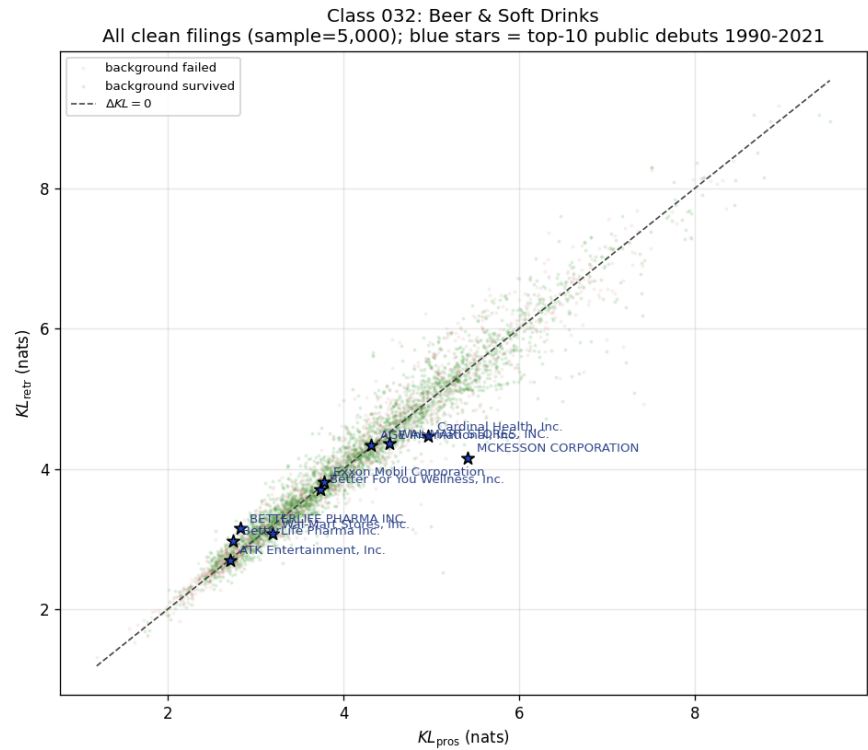
token	n in stratum	n in class	intra-class survival
fresh	1,900	19,830	47%
vegetables	1,512	13,668	47%
food	1,480	18,269	48%
seeds	1,280	12,362	46%
fruit	1,224	10,604	48%
fruits	1,209	10,998	47%
live	1,164	13,107	42%
plants	1,163	12,612	45%
products	1,007	9,554	46%
use	960	9,176	46%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
WAL-MART STORES, INC.	GARDEN CLASSICS	1991	6.75	-0.16	bagged and potted plants
Wal-Mart Stores, Inc	WAY TO CELEBRATE	2017	4.61	+0.14	Easter blow mold bunnies, namely, unlit molded plastic toy Easter bunny figurines; Easter egg coloring dye kits Easter blow mold bunnies, namely, prelit illuminated molded Easter bunny figurines of p
Wal-Mart Stores, Inc.	CAT FOOD	1991	4.51	+0.09	cat food
Comcast Corporation	COMCAST	2021	3.42	-0.05	Candy; Coffee; Cookies; Chocolate candies; Chocolate confections; Chocolate covered pretzels; Chocolates and chocolate-based ready to eat candies and snacks; Kettle corn; Lollipops; Hot chocolate mixe
General Electric Company	GE	2005	6.71	+0.22	Fluid filters, liquid filtering separators, and gas-liquid filtering separators; apparatus in the nature of filters for removing solids, liquids, and vapors from gases
PFIZER INC.	FLAVASURE	1998	5.35	-0.05	Meat, meat extracts, preserved meat, and sausages Live animals
ARCHER DANIELS MIDLAND COMPANY	SUPERMARKET TO THE WORLD	1990	3.47	+0.40	barges, vans and railway cars cosmetic preparations - namely face powder and body powder gin and grain alcohol natural agriculture products; namely, raw peanuts, unprocessed malt, dog and cat food, oa
ARCHER-DANIELS-MIDLAND COMPANY	MOORTEC	1990	4.40	+0.79	[animal feed for livestock, poultry and game animals; namely, feed supplements, medicated feeds and mineral feeds; worm expeller preparations; insecticide, pesticide, and rodenticide for domestic and
Archer Daniels Midland Company	TRYPTOSINE	1995	4.16	-0.39	animal feed premix
Archer-Daniels-Midland Company	ADM	2001	3.13	+0.19	Barges, vans, trucks and railway cars Staple foods, namely, corn germ, corn syrups, flour, corn starch, cornmeal, grits, spaghetti, sugar, noodles, and macaroni flours, meals and grits; processed semo

Class 032: Beer & Soft Drinks

Clean filings (1990–2021): 100,951
Mean KL_{pros} / KL_{retr} / ΔKL : 4.18 / 4.20 / -0.014 nats
5y survival rate: 56.6%
Share with SEC-listed owner: 3.11%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
smokeless	$4.11\times$	124	73%
sugar-based	$4.00\times$	130	47%
cannabidiol	$3.86\times$	272	86%
mood	$3.85\times$	161	66%
transdermal	$3.75\times$	192	74%
relieving	$3.73\times$	150	57%
depression	$3.63\times$	157	71%
e-liquid	$3.63\times$	135	83%
glosses	$3.62\times$	163	56%
emollients	$3.58\times$	173	68%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
bock	$4.66\times$	277	67%
perishable	$4.59\times$	146	86%
lager	$3.54\times$	7,562	56%
stout	$3.33\times$	5,990	57%
porter	$3.32\times$	6,431	57%
ale	$3.25\times$	10,702	55%
pasteurized	$3.22\times$	214	76%
shandy	$3.09\times$	3,031	58%
shelf-stable	$3.06\times$	121	85%
craft	$3.00\times$	2,030	49%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

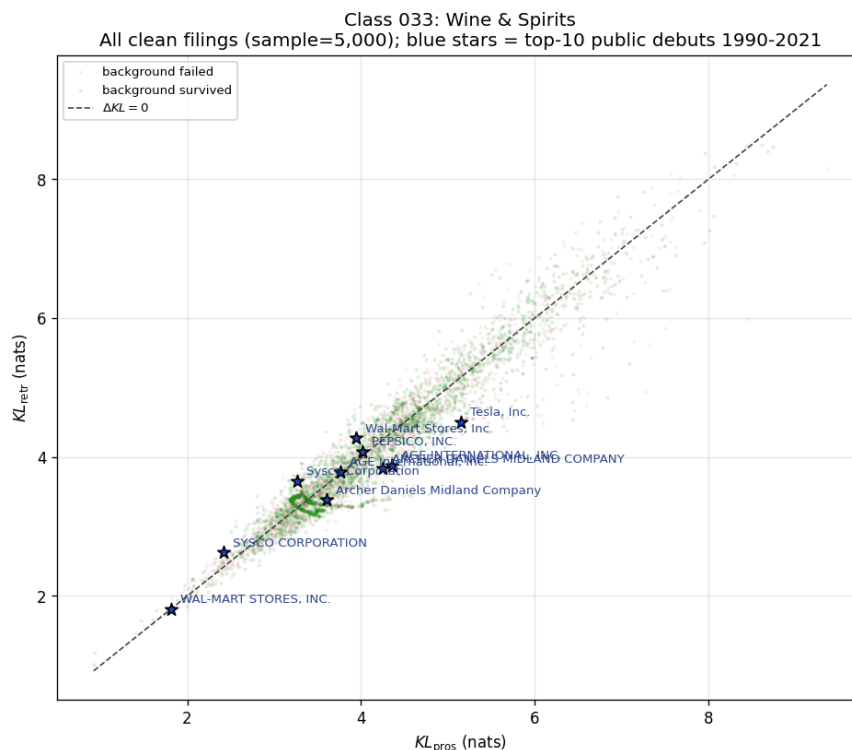
token	n in stratum	n in class	intra-class survival
beverages	5,855	53,054	55%
fruit	5,207	46,913	53%
drinks	5,032	51,807	55%
non-alcoholic	4,669	40,713	53%
juices	3,853	33,108	52%
water	3,412	30,730	57%
making	2,663	22,491	47%
soft	2,548	26,662	54%
mineral	2,525	20,232	46%
juice	2,373	19,839	54%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	4.31	-0.03	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ATK Entertainment, Inc.	AALAM TUCK KHALED	2020	2.71	+0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
BETTERLIFE PHARMA INC.	MOVE BET-TERX	2019	2.83	-0.32	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
Better For You Wellness, Inc.	BETTER FOR YOU WELLNESS	2021	3.73	+0.02	Protein bars, namely, protein supplements formed and packaged as bars Advisory and consultation services in the fields of agronomy, agriculture, and horticulture; agronomic consulting services; agrono
BetterLife Pharma Inc.	LIVE BET-TERX	2020	2.74	-0.23	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
WAL-MART STORES, INC.	SAM'S CHOICE	1991	4.52	+0.16	[processed meats, processed edible seeds and fruit preserves] breakfast and processed cereals; candy; [condiments; namely, mustards] fruit drinks
Wal-Mart Stores, Inc.	CONTE	1992	3.20	+0.13	soft drinks
Exxon Mobil Corporation	CRYO	2007	3.78	-0.04	Tea based cold beverages, coffee based cold beverages Fruit flavored beverages, nonalcoholic carbonated beverages, sports beverages, fruit flavored syrups, nonalcoholic frozen carbonated beverages, no
MCKESSON CORPORATION	THE WATERSTOP	1990	5.41	+1.26	bottled drinking water
Cardinal Health, Inc.	CARDINAL BERGEN HEALTH	1997	4.97	+0.50	computer software for inventory management, and preparation and administration of advertising and ordering, for drug stores, pharmacies, hospitals, clinics, alternate care centers, and other healthcar

Class 033: Wine & Spirits

Clean filings (1990–2021): 68,232
Mean KL_{pros} / KL_{retr} / ΔKL : 4.09 / 4.06 / +0.038 nats
5y survival rate: 53.1%
Share with SEC-listed owner: 0.90%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
blue	$5.76\times$	2,203	53%
agave	$5.19\times$	3,090	55%
weber	$4.65\times$	71	54%
tequilana	$4.65\times$	71	55%
mexico	$3.92\times$	153	65%
region	$3.53\times$	541	31%
certain	$3.52\times$	88	5%
seltzers	$3.45\times$	116	49%
provence	$3.42\times$	76	17%
accordance	$3.39\times$	882	33%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
viticultural	4.33×	134	50%
napa	4.32×	148	47%
valley	3.93×	168	51%
called	3.59×	145	17%
american	3.56×	177	50%
county	3.45×	84	55%
advertised	3.38×	449	57%
labeled	3.38×	450	58%
guaranteed	3.30×	103	11%
grown	3.20×	531	56%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

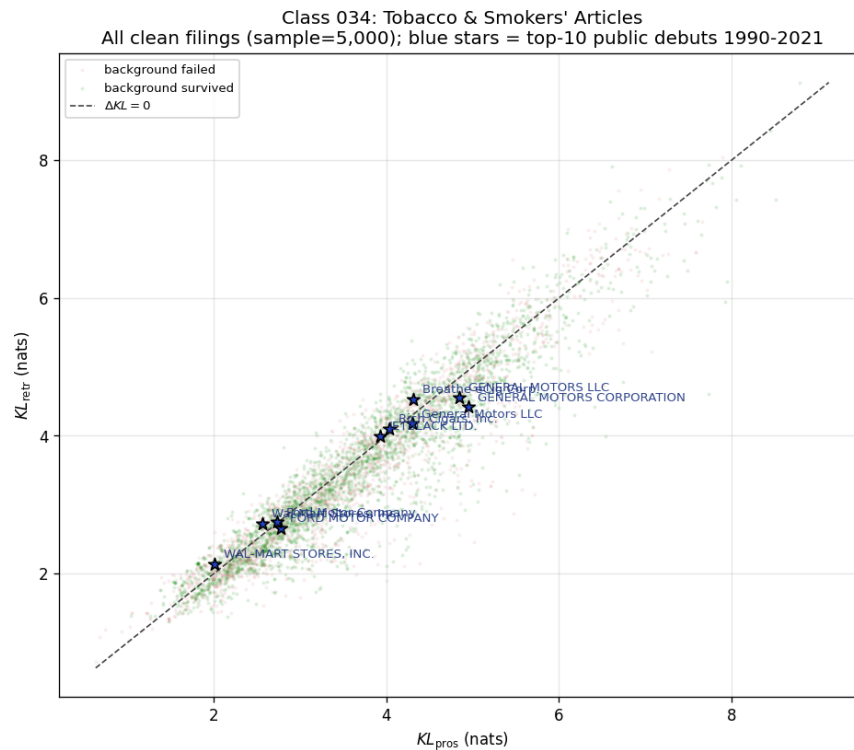
token	n in stratum	n in class	intra-class survival
alcoholic	3,789	47,399	52%
beverages	3,485	43,642	51%
except	2,408	29,046	48%
beers	2,036	24,003	50%
wine	1,591	17,805	52%
spirits	1,583	20,106	51%
wines	1,271	16,851	44%
fruit	1,211	11,721	46%
beer	1,113	12,622	49%
distilled	1,047	16,141	54%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE INTERNATIONAL, INC.	BLANTON'S	1990	4.35	+0.47	alcoholic beverages; namely, bourbon whiskey
AGE International, Inc.	BLANTON'S	2021	3.77	-0.02	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
WAL-MART STORES, INC.	CURIOUSLY J	2017	1.81	+0.00	Dyes for use in the manufacture of textiles and fabrics; food coloring; paints; finger paint; inks, toners, and dyes for use with or on textiles and fabrics; edible food coloring Alcoholic cocktail mi
Wal-Mart Stores, Inc.	MEMBERS MARK	1998	3.94	-0.33	Aperitif wines, Aperitifs with a wine base, Cooking wine, wine, wine cocktail, wine coolers and wine punches
ARCHER DANIELS MIDLAND COMPANY	SUPERMARKET TO THE WORLD	1990	4.25	+0.41	barges, vans and railway cars cosmetic preparations - namely face powder and body powder gin and grain alcohol natural agriculture products; namely, raw peanuts, unprocessed malt, dog and cat food, oa
Archer Daniels Midland Company	AMERICA'S CORN TRADITION	1994	3.60	+0.22	alcoholic beverages, namely vodka and distilled spirits
PEPSICO, INC.	ROCKARITA	2017	4.01	-0.07	Brewed malt-based alcoholic beverages in the nature of beers; flavored malt beverages, namely, brewed malt-based beers Alcoholic beverages produced from a brewed malt base with natural flavors; flavor
Tesla, Inc.	TESLAQUILA	2018	5.15	+0.65	Distilled agave liquor; Distilled blue agave liquor
SYSCO CORPORATION	SYSCO	1998	2.42	-0.21	Electric light bulbs, heat lamp bulbs and flood lamps; plumbing supplies, namely, urinal screens; grease and shortening filters for use in range hoods over stoves, fryers, and other cooking equipment
Sysco Corporation	IMPERIAL DYNASTY	1994	3.26	-0.39	food colors for domestic use fresh fruits, namely green bell peppers, jalapeno peppers, pasilla chili peppers, purple bell peppers, red bell peppers, serrano peppers, pineapple, bananas, quince, banan

Class 034: Tobacco & Smokers' Articles

Clean filings (1990–2021): 36,678
Mean KL_{pros} / KL_{retr} / ΔKL : 3.66 / 3.50 / +0.163 nats
5y survival rate: 55.4%
Share with SEC-listed owner: 1.42%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
comestibles	$8.37\times$	80	64%
opioid	$7.64\times$	89	67%
chronic	$7.10\times$	62	55%
epilepsy	$6.67\times$	108	70%
depression	$6.07\times$	122	75%
addiction	$5.95\times$	116	57%
fatigue	$5.90\times$	122	64%
enhancement	$5.62\times$	121	71%
melts	$5.38\times$	78	74%
mood	$5.35\times$	129	75%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
ride	$7.36\times$	53	19%
suitcase	$6.73\times$	52	19%
cabarets	$6.49\times$	77	23%
museum	$6.45\times$	76	20%
velvet	$6.36\times$	55	7%
blackboards	$6.35\times$	63	19%
nameplates	$6.33\times$	90	33%
half-boots	$6.32\times$	57	25%
detachable	$6.30\times$	54	17%
terracotta	$6.29\times$	97	20%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

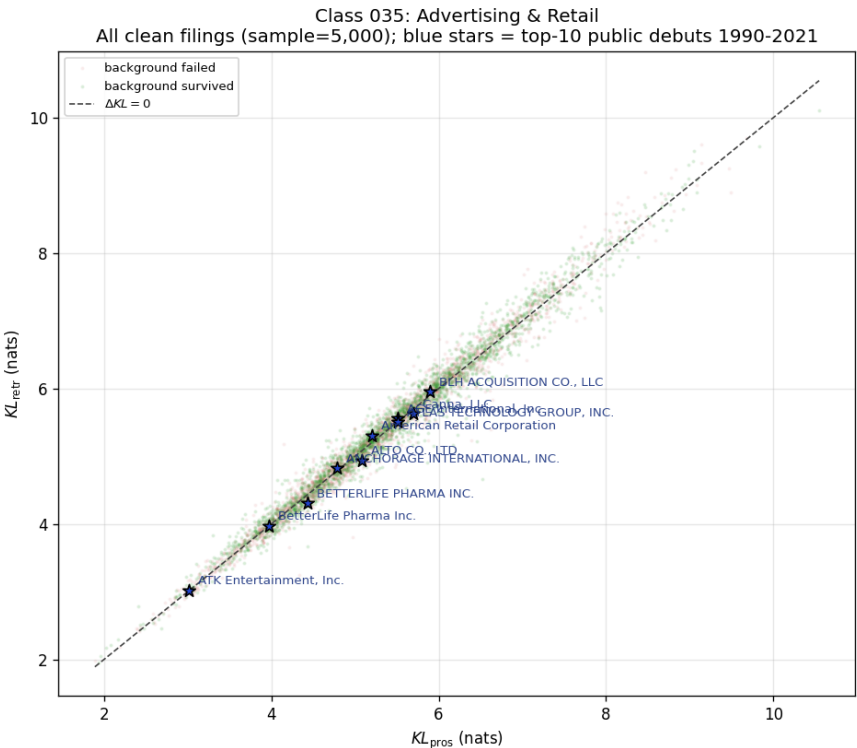
token	n in stratum	n in class	intra-class survival
cigarette	1,110	21,549	52%
tobacco	956	18,056	53%
cigarettes	950	19,333	54%
electronic	844	17,025	52%
smokers	689	12,430	49%
lighters	620	11,907	53%
smoking	592	11,616	58%
cases	565	9,079	48%
cigars	556	10,468	56%
liquid	550	10,534	51%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Breathe eCig Corp.	BREATHE ECIGS	2015	4.31	-0.22	Electronic cigarettes for use as an alternative to traditional cigarettes
JETBLACK LTD.	SIERRA	2015	3.93	-0.07	Smokeless cigar vaporizer pipes; Smokeless cigarette vaporizer pipe
Rich Cigars, Inc.	RICH CIGARS	2015	4.04	-0.06	Cigars, cigar wraps, humidors, cigar boxes, cigar cases, ash trays, match boxes, cigar holders, cigar tubes, lighters for tobacco products, match containers, and cigar cutters
WAL-MART STORES, INC.	CURIOSLY J	2017	2.01	-0.13	Dyes for use in the manufacture of textiles and fabrics; food coloring; paints; finger paint; inks, toners, and dyes for use with or on textiles and fabrics; edible food coloring Alcoholic cocktail mi
Wal-Mart Stores, Inc.	M	1998	2.56	-0.16	Ashtrays not of precious metal, Chewing tobacco, Cigar cutters, Cigarette cases not of precious metal, Cigarette holders not of precious metal, Cigarette lighters not for land vehicles, Cigarette ligh
GENERAL MOTORS CORPORATION	CADILLAC	1990	4.95	+0.54	cigarette lighters and matches
GENERAL MOTORS LLC	HUMMER	2008	4.84	+0.29	Lighters for cigars and cigarettes
General Motors LLC		2013	4.30	+0.13	Lighters for cigars and cigarettes not for land vehicles, cigar holders
FORD MOTOR COMPANY	AEROSTAR	1993	2.78	+0.12	cigarette lighters not of precious metal, cigarette cases not of precious metal, matches and ashtrays not of precious metal
Ford Motor Company	THINK	2000	2.73	-0.02	smokers' articles, namely, cigarette and cigar lighters not of precious metal, ashtrays not of precious metal, matches, cigarette cases not of precious metal and cigar cutters not of precious metal

Class 035: Advertising & Retail

Clean filings (1990–2021): 996,559
Mean KL_{pros} / KL_{retr} / ΔKL : 5.38 / 5.40 / -0.019 nats
5y survival rate: 47.0%
Share with SEC-listed owner: 3.65%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
blockchain	$5.73\times$	1,717	53%
worlds	$5.66\times$	1,167	63%
cryptocurrency	$5.28\times$	1,395	53%
delta-	$4.16\times$	1,621	61%
tetrahydrocannabinol	$4.06\times$	1,376	62%
thc	$4.01\times$	1,719	61%
percent	$3.88\times$	1,491	59%
cbd	$3.75\times$	1,793	68%
hemp	$3.65\times$	2,978	61%
currencies	$3.54\times$	1,023	44%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
chamber	$3.27\times$	2,109	37%
price-comparison	$3.03\times$	3,355	52%
page	$2.56\times$	3,999	50%
earned	$2.45\times$	1,100	61%
accessed	$2.45\times$	6,596	57%
placing	$2.42\times$	13,179	50%
dealership	$2.40\times$	2,143	48%
billboard	$2.35\times$	2,537	54%
rebates	$2.32\times$	8,195	51%
county	$2.31\times$	1,138	35%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

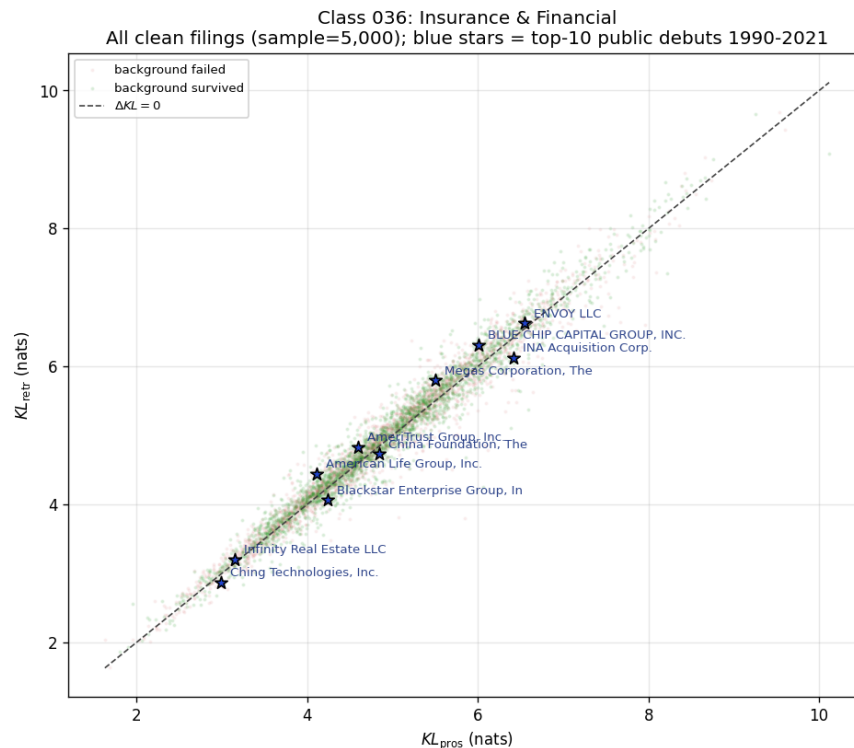
token	n in stratum	n in class	intra-class survival
featuring	49,481	364,651	44%
field	46,826	317,057	42%
business	46,425	320,210	42%
retail	43,302	328,625	46%
management	38,743	249,195	42%
information	37,357	258,493	45%
others	35,624	273,004	45%
store	35,602	259,918	44%
on-line	29,783	228,501	46%
online	28,460	190,528	44%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AGE International, Inc.	BLANTON'S	2021	5.51	-0.05	wall mounted coat rack; wooden display stands for bottle stoppers, bottles and tasting glasses; wall art made of wood reclaimed from bourbon barrels face coverings being masks for protection against v
ALTO CO., LTD.	ALTA	2012	5.08	+0.14	Optical goods, namely, optical reflectors, optical lenses; actinometers; electrical terminal blocks; electric light dimmers regulators; ballasts for fluorescent lamps; connections for electric lines;
ANCHORAGE INTERNATIONAL, INC.	NETEQUIPMENT	1999	4.78	-0.04	cooperative advertising and marketing via a global computer network and print media
ATK Entertainment, Inc.	AALAM KHALED	2020	3.01	-0.01	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
ATLAS TECHNOLOGY GROUP, INC.	IGE	2006	5.51	+0.01	Entertainment services, namely, providing on-line information via the Internet about video games; providing on-line news and information via the Internet on video games; providing on-line databases fo
American Retail Corporation	WATSON'S	2008	5.20	-0.10	Retail store services and on-line retail store services featuring swimming pools and accessories therefor, hot tubs, spas, saunas and accessories therefor, fireplaces, fireplace inserts, woodburning s
BETTERLIFE PHARMA INC.	LOVE BETTERX	2019	4.43	+0.12	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BLH ACQUISITION CO., LLC	BE PAID TO PARTY LOUIE NATION	2019	5.89	-0.06	Providing incentive award programs through issuance and processing of loyalty points for purchase of a company's goods and services
BetterLife Pharma Inc.	LIVE BETTERX	2020	3.97	-0.00	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Canna, LLC	CANNA ADVISORS DESIGN—BUILD—	2016 192	5.69	+0.07	Business development consulting services

Class 036: Insurance & Financial

Clean filings (1990–2021): 399,896
Mean KL_{pros} / KL_{retr} / ΔKL : 4.92 / 4.96 / -0.042 nats
5y survival rate: 47.3%
Share with SEC-listed owner: 6.69%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
decentralized	7.76×	817	61%
blockchain-based	7.49×	929	62%
cryptographic	7.18×	890	62%
token	7.11×	908	57%
blockchain	6.78×	2,507	59%
crypto	6.39×	638	57%
cryptocurrency	6.10×	3,954	59%
transferrable	5.90×	402	68%
cryptocurrencies	5.87×	828	57%
validating	5.64×	456	52%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
guaranty	4.10×	459	45%
calling	3.19×	3,618	49%
annuity	2.67×	5,246	47%
timeshare	2.40×	1,151	60%
universal	2.39×	615	46%
inland	2.36×	445	49%
malpractice	2.30×	556	53%
underwriting	2.26×	42,087	44%
fidelity	2.25×	636	46%
casualty	2.24×	10,709	46%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

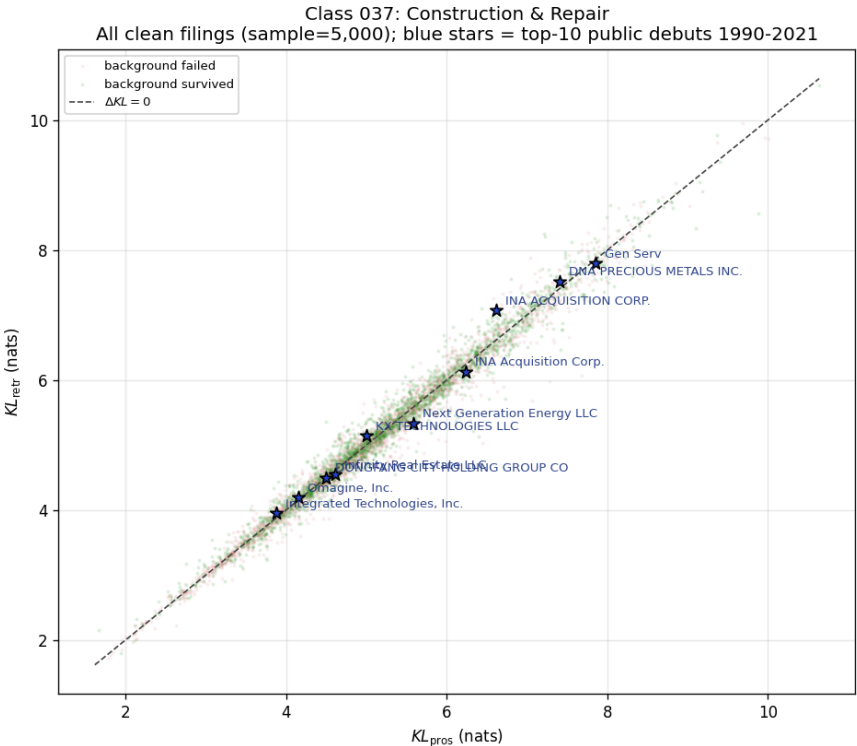
token	n in stratum	n in class	intra-class survival
financial	19,537	172,088	47%
management	16,051	130,568	45%
information	12,538	102,892	45%
field	12,115	105,707	44%
investment	10,544	92,365	47%
estate	8,787	88,964	46%
real	8,386	85,253	46%
others	8,316	70,650	45%
business	8,123	58,442	43%
electronic	7,133	56,743	48%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AmeriTrust Group, Inc.	AMERITRUST	2018	4.60	-0.23	Insurance services, namely, administration, brokerage, and underwriting in the fields of commercial property, casualty, umbrella, excess and general liability; insurance claims administration and clai
American Life Group, Inc.	AMERICAN LIFE GROUP, INC.	1995	4.11	-0.33	insurance underwriting services in the fields of life insurance and annuity products
BLUE CHIP CAPITAL GROUP, INC.	RAISEWISE	2020	6.01	-0.30	Crowdfunding services in the nature of accepting and administering monetary contributions from a group of individuals
Blackstar Enterprise Group, Inc.	BLOCKCHAIN FIRST	2021	4.24	+0.18	Financial services, namely, providing a securities exchange on an immutable block chain that prevents the disruption of order flow of customer trades; transaction processing and recording, namely, fin
China Foundation, The	CHINA FOUNDATION	2008	4.85	+0.12	Arranging and conducting educational conferences; organizing conferences in the field of public health and medical science Charitable services, namely, organizing and conducting volunteer programs and
Ching Technologies, Inc.	CANAL CIRCLE	2018	2.98	+0.12	Computer services for providing on-line, non-downloadable, hosted software applications for credit scoring and data analytics about credit scoring of persons or businesses lacking or having limited ve
ENVOY LLC	CHAMBERLIN EDMONDS	2004	6.55	-0.08	provider of revenue recovery services, namely, securing government reimbursement for uncompensated medical care expenses
INA Acquisition Corp.	INSITUFORM FINANCIAL SERVICES	2006	6.42	+0.30	Financial planning; Fiscal assessments of funding in the field of repair and reconstruction of existing sewer pipes, water pipes, gas mains and industrial pipelines; Financial forecasting needs of fun
Infinity Real Estate LLC	INFINITY GROUP	2019	3.15	-0.05	Assessment and management of real estate; Commercial and residential real estate agency services; Financial planning and investment advisory services; Investment advice; Land acquisition, namely, real
Megas Corporation, The	MEGAS	1995 1998	5.50	-0.30	Development and management of associations, and development and marketing of benefits for association members marketing and administration of insurance

Class 037: Construction & Repair

Clean filings (1990–2021): 178,756
Mean KL_{pros} / KL_{retr} / ΔKL : 5.18 / 5.18 / +0.001 nats
5y survival rate: 42.5%
Share with SEC-listed owner: 4.84%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
charging	3.45×	2,326	38%
recharging	3.10×	678	40%
rootering	2.96×	412	35%
microorganisms	2.83×	205	40%
suitable	2.58×	329	35%
spread	2.48×	222	32%
waiting	2.09×	187	30%
renewable	2.07×	2,872	37%
solar-powered	2.03×	276	34%
swimming-pool	2.00×	200	28%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
arcades	$4.08\times$	201	70%
lounges	$3.81\times$	194	63%
casinos	$3.76\times$	250	67%
geologic	$3.66\times$	227	44%
dialing	$3.64\times$	198	41%
convention	$3.61\times$	524	57%
dinner	$3.59\times$	217	57%
formations	$3.54\times$	268	43%
galleries	$3.35\times$	200	66%
housewares	$3.33\times$	183	67%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

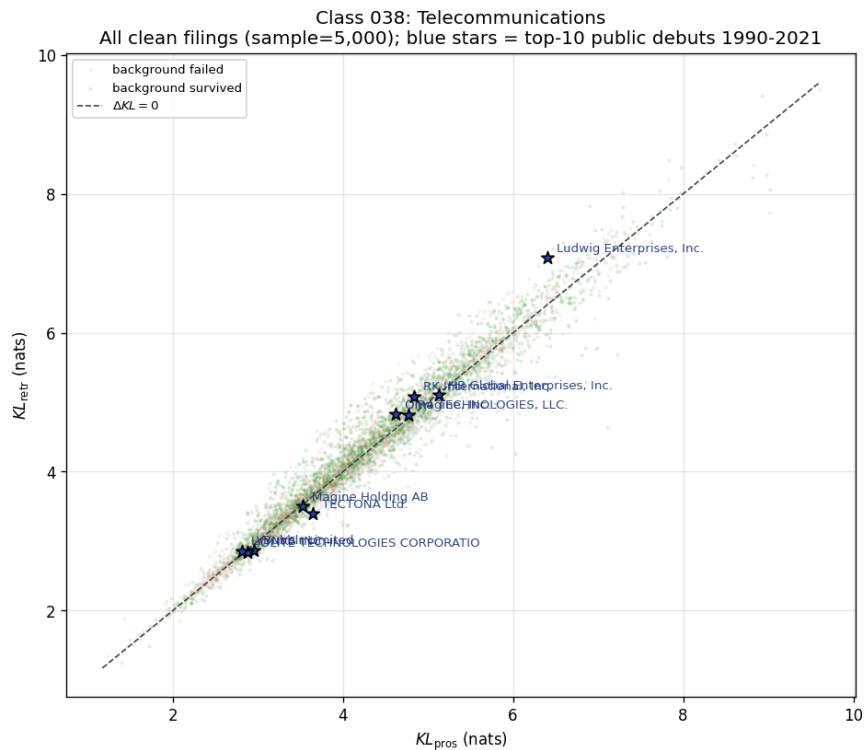
token	n in stratum	n in class	intra-class survival
repair	13,046	85,741	39%
maintenance	11,787	76,986	39%
installation	10,028	66,270	38%
construction	8,381	53,612	40%
field	7,737	46,645	37%
systems	6,480	40,587	37%
equipment	6,363	38,385	37%
management	6,213	37,024	40%
building	5,852	33,623	38%
development	5,798	36,545	40%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
DNA PRECIOUS METALS INC.	DNA PRECIOUS METALS	2013	7.40	-0.12	mining extraction services mining exploration services
DONGFANG CITY HOLDING GROUP COMPANY LIMITED	DONGFANGCITY	2021	4.50	+0.00	Real estate development; Real estate development and construction of commercial, residential and hotel property; Real estate site selection; Building construction supervision services for real estate
Gen Serv	GENSERV	2020	7.85	+0.06	We are a world wide Generator Service company in the Emergency Response company. We work for the DOD and all companies hotels Etc
INA ACQUISITION CORP.	INSITUFORM TECHNOLOGIES	1993	6.62	-0.45	repair or reconstruction of pipes such as sewer pipes or other conduits by inserting liners therewithin
INA Acquisition Corp.	CLEAN WATER FOR THE WORLD	2006	6.23	+0.11	Repair and reconstruction of existing sewer pipes, water pipes, gas mains and industrial pipelines by installing liners within the existing conduit and reinstatement of existing services
Infinity Real Estate LLC		2019	4.61	+0.06	Real estate development Assessment and management of real estate; Commercial and residential real estate agency services; Financial planning and investment advisory services; Investment advice; Land a
Integrated Technologies, Inc.	INTEGRATED TECHNOLOGIES, INC.	2016	3.88	-0.07	Consulting services in the field of factory construction; consulting services in the field of renovation planning; construction of buildings Engineering services; design of buildings; engineering desi
KX TECHNOLOGIES LLC	THE SCIENCE BEHIND BETTER WATER	2007	5.00	-0.14	[Installation, repair and maintenance of water treatment equipment] Water treatment equipment and parts thereof, namely, water softening and conditioning units, water filters, reverse-osmosis system
Next Generation Energy LLC	SUNGALOW	2007	5.59	+0.25	On-line retail store services featuring housing and storage structures designed and constructed to enhance solar energy efficiency, namely, mobile homes, transportable homes, prefabricated homes, modu
Omage, Inc.	IMAGINATION INSPIRATION. ILLU	2008	4.15	-0.04	Real estate services, namely, listing, leasing and financing of residential and commercial properties; Property management services, namely, management of residential and commercial properties Broadca

Class 038: Telecommunications

Clean filings (1990–2021): 170,826
Mean KL_{pros} / KL_{retr} / ΔKL : 4.32 / 4.37 / -0.050 nats
5y survival rate: 51.5%
Share with SEC-listed owner: 7.36%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
blockchain-based	5.53×	206	44%
blockchain	4.71×	567	44%
cryptocurrency	4.66×	496	42%
decentralized	4.32×	192	40%
currencies	3.44×	488	41%
token	3.30×	376	44%
cryptographic	2.86×	357	41%
ledger	2.65×	339	36%
wallet	2.45×	469	44%
avatars	2.41×	191	64%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
pager	$3.64\times$	744	50%
intrastate	$3.17\times$	287	56%
long	$2.96\times$	10,732	49%
interstate	$2.85\times$	382	54%
dot	$2.78\times$	230	36%
microfiche	$2.75\times$	233	36%
dial-up	$2.71\times$	1,145	47%
distance	$2.71\times$	11,827	47%
prepaid	$2.69\times$	3,601	48%
calling	$2.67\times$	6,096	53%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

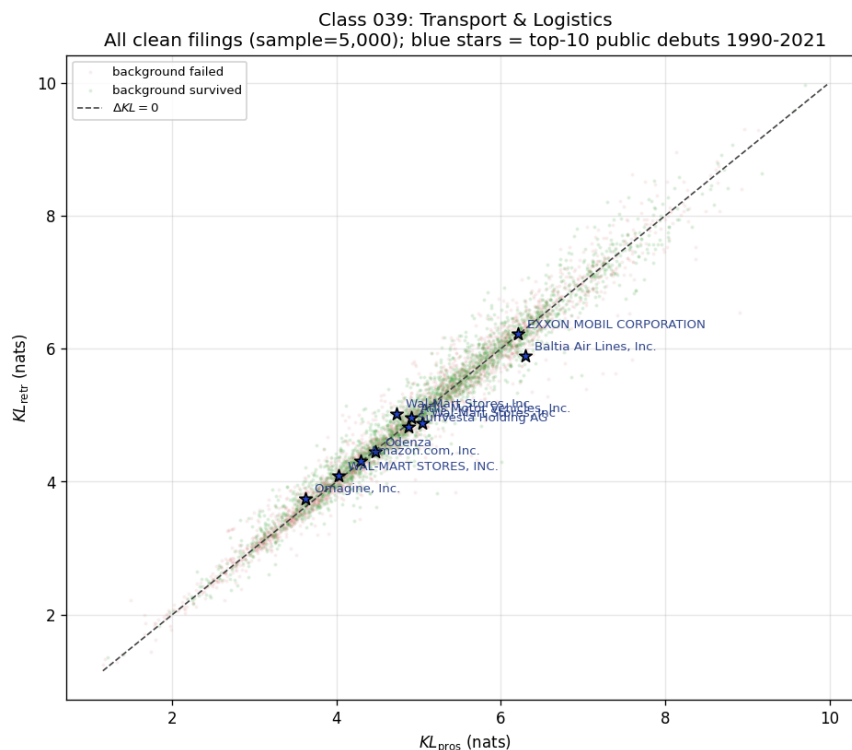
token	n in stratum	n in class	intra-class survival
computer	12,603	98,267	48%
transmission	11,558	97,943	47%
electronic	11,439	89,931	47%
information	10,920	82,632	47%
data	10,279	81,950	46%
internet	10,056	74,844	46%
software	9,004	57,208	44%
network	8,604	68,598	49%
field	8,558	56,975	42%
networks	8,336	59,701	44%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Bubblr Limited	C J CITI-ZENS JOURNALIST	2019	2.95	+0.09	Television and radio broadcasting services; audio and video broadcasting services, namely, broadcasting of news in the fields of current affairs, sports and topics of general interest over the internet
HB Global Enterprises, Inc.	SOUL RECI-PROCITY NETWORK	2021	5.13	+0.03	Broadcasting of programmes via the internet; Streaming of audio, video and audiovisual material on the Internet; Video broadcasting
IOLITE TECHNOLOGIES CORPORATION	SOCIAL MOSAIC	2011	2.88	+0.04	Providing access to on-line computer databases and on-line searchable databases in the field of classifieds, commercial listings and advertisements of services and products of others, social networkin
LYYNKS INC.	LYYNKS	2012	2.81	-0.05	Audio and video broadcasting services over the Internet; Broadcasting of video and audio programming over the Internet; Broadcasting services and provision of telecommunication access to video and aud
Ludwig Enterprises, Inc.	THE ONE	2008	6.41	-0.67	Television and radio broadcasting and rebroadcasting services Equipment for receiving broadcast radio transmissions, namely, radios, tuners, decoders, demodulators, receivers and antennas
Magine Holding AB	HEADWEB	2010	3.53	+0.03	Printed matter, namely, newsletters about movies, cinematic films, the entertainment and showbiz industry Broadcasting and streaming of cinematographic films and programs through a computer network; v
Omage, Inc.	OMAGINE	2008	4.62	-0.21	Broadcasting of radio and television programs
R4 TECHNOLOGIES, LLC.	THUMBPASS	2010	4.77	-0.05	Computer software for controlling and managing access to vending machines Providing user access to a global computer network to access vending machine accounts Providing an on-line portal that provide
RK International, Inc.	RK LIVE-TV	1997	4.84	-0.24	providing a telecommunications service and software support for the Internet related network computer boxes for the television marketing and selling Internet related network computer boxes for the tel
TECTONA Ltd.	BORED CO.	2021 201	3.64	+0.25	Telecommunications services in the nature of television broadcasting, television transmission, broadcasting, streaming and providing telecommunication access to films Technological services

Class 039: Transport & Logistics

Clean filings (1990–2021):	120,417
Mean KL_{pros} / KL_{retr} / ΔKL :	5.03 / 5.05 / -0.020 nats
5y survival rate:	45.4%
Share with SEC-listed owner:	6.90%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
dumpsters	4.54×	427	35%
payloads	3.50×	143	56%
roll-off	3.43×	169	34%
renewable	3.21×	1,456	37%
vicinity	3.15×	197	57%
debris	3.04×	963	35%
biofuels	2.86×	241	41%
recharging	2.85×	267	34%
regenerative	2.85×	123	20%
on-street	2.83×	212	47%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
business-to-business	$3.35\times$	164	23%
accompanied	$2.53\times$	154	58%
detailing	$2.32\times$	250	30%
tangible	$2.19\times$	151	23%
pvc	$2.05\times$	132	29%
occasion	$1.99\times$	241	50%
dances	$1.94\times$	263	57%
microcomputers	$1.87\times$	123	42%
reconditioning	$1.87\times$	284	18%
emerging	$1.84\times$	196	36%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

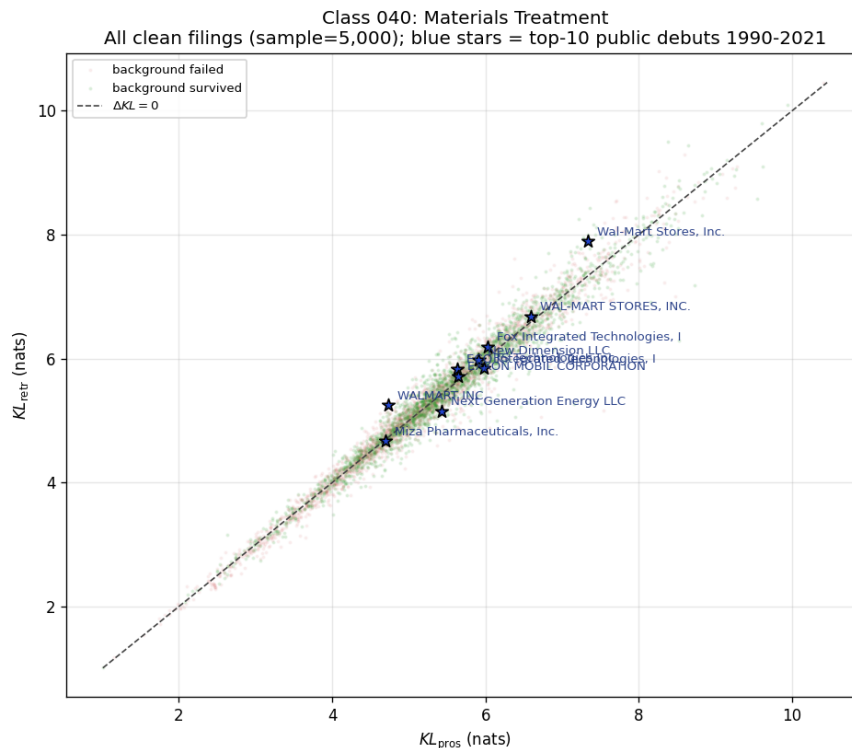
token	n in stratum	n in class	intra-class survival
transportation	5,604	51,832	43%
information	4,698	36,418	42%
others	4,567	34,665	41%
storage	4,466	34,843	41%
field	4,075	29,124	40%
transport	3,666	25,720	35%
delivery	3,637	29,995	43%
travel	3,563	33,822	42%
management	3,403	22,987	39%
computer	3,081	23,387	43%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
Atlis Motor Vehicles, Inc.	AMV	2021	4.91	-0.05	Subscription-based leasing services of vehicles; vehicle delivery services, namely, delivering vehicles to members of a subscription-based service; providing an online searchable database via the inte
Baltia Air Lines, Inc.	BALTIA	1990	6.30	+0.41	airline services; namely, transportation of people and property
Odenza	LET US TAKE YOU THERE	2015	4.47	+0.03	Travel agency services, namely, making reservations and bookings for transportation
Omage, Inc.	BASED ON A TRUE STORY	2008	3.62	-0.12	Travel agency services, namely, making reservations and booking for temporary lodging and dining; Hotel and resort hotel services; Resort lodging services; Temporary accommodation services; Restaurant
SunVesta Holding AG	SUNVESTA	2005	4.87	+0.05	Services for providing food and drink in cafes, cafeterias, restaurants and bars; hotel, motel, boarding house, restaurant, cafe, cafeteria and bar services; providing temporary guest accommodation; r
WAL-MART STORES, INC.	CURIO	2012	4.02	-0.06	Providing forums, electronic bulletin boards, RSS feeds and list servers for transmission of messages among computer users and subscribers concerning general merchandise and general consumer goods, pr
Wal-Mart Stores, Inc		2017	5.04	+0.16	Shipping and delivery of goods
Wal-Mart Stores, Inc.	SAM'S TRAVEL CLUB	1995	4.73	-0.29	travel club and information services
Amazon.com, Inc.	EAMAZON.CO	2000	4.29	-0.02	transport; travel arrangement; travel clubs; providing an interactive database featuring an automatically updating travel planner; delivery of goods; delivery services; rental services; distribution o
EXXON MOBIL CORPORATION	DENBURY	2012	6.21	-0.01	Transportation by pipeline Extraction of oil, gas, carbon dioxide and helium; Laying, constructing, maintaining and operating pipelines; Well drilling for oil, gas, helium and carbon dioxide Explo-rati

Class 040: Materials Treatment

Clean filings (1990–2021): 94,015
Mean KL_{pros} / KL_{retr} / ΔKL : 5.37 / 5.37 / -0.004 nats
5y survival rate: 44.6%
Share with SEC-listed owner: 5.75%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
vectors	$5.64\times$	101	53%
biomanufacturing	$5.00\times$	168	39%
cbd	$4.62\times$	147	76%
therapies	$4.62\times$	171	40%
delta-	$4.62\times$	210	66%
thc	$4.51\times$	204	67%
tetrahydrocannabinol	$4.49\times$	185	68%
autoimmune	$4.38\times$	130	32%
percent	$4.37\times$	192	65%
gene	$4.27\times$	295	44%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
snapshots	$5.67\times$	104	50%
taproom	$4.39\times$	123	64%
portraits	$3.56\times$	191	42%
brewing	$3.49\times$	1,037	54%
brewed	$3.36\times$	143	62%
digitizing	$3.31\times$	956	45%
retouching	$3.18\times$	1,036	44%
lager	$3.07\times$	114	67%
tee-shirt	$3.05\times$	95	82%
beer	$2.97\times$	1,318	52%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

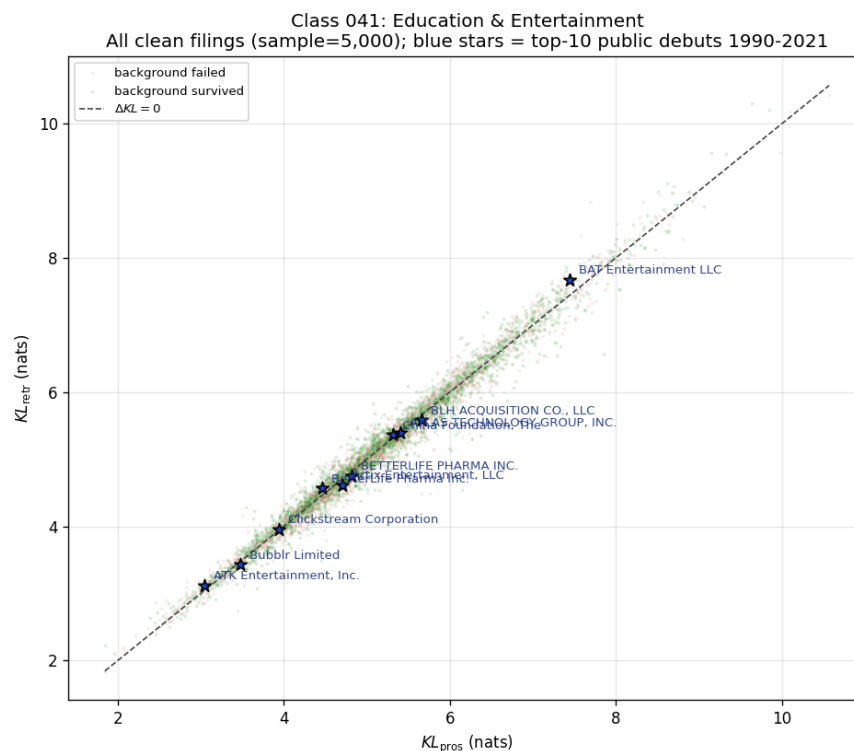
token	n in stratum	n in class	intra-class survival
others	4,275	37,163	41%
custom	3,709	32,029	41%
field	3,591	25,985	38%
use	2,907	21,005	38%
manufacture	2,835	24,294	40%
treatment	2,699	19,339	35%
design	2,559	17,922	35%
products	2,532	18,350	38%
materials	2,426	16,692	33%
manufacturing	2,289	17,273	39%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ECO Integrated Technologies, Inc.	ECO INTEGRATED TECHNOLOGIES	2015	5.63	-0.20	Environmental remediation services, namely, treatment of solid waste and water
Fox Integrated Technologies, Inc.	FOXCAM	1999	6.03	-0.16	Recycling services, namely, monitoring to allow verification of recycling of electronic components as an integral part of the recycling services
Miza Pharmaceuticals, Inc.	OPTOPICS	1999	4.70	+0.03	eye care products, namely, eye drops, eye washes, artificial tears, contact lens solutions and generic pharmaceutical drug solutions for use in the treatment of eye related diseases and disorders; and
New Dimension LLC	N NEW DIMENSION	2007	5.90	-0.07	digital printing services
Next Generation Energy LLC	NEXT GENERATION ENERGY	2006	5.42	+0.27	Develop new technology for others in the field of renewable energy related products and resources Contract manufacturing in the field of renewable energy related products
RS Technologies Inc.	POWERON	2021	5.98	+0.14	Non-metal utility poles; non-metal utility pole cross arms; non-metal covers for utility poles and utility pole cross arms Technical support services, namely, consulting relating to the use of polyure
WAL-MART STORES, INC.	WAL MART DIGITAL PHOTO CENTER	2004	6.60	-0.08	online photo processing services
WALMART INC.	ART.COM	1999	4.73	-0.52	custom framing services offered via computer networks and global communications networks; computerized on-line custom art framing services, consulting and advisory services on choosing art framing mat
Wal-Mart Stores, Inc.	DEVELOP YOUR BUSINESS	2003	7.34	-0.55	film developing services
EXXON MOBIL CORPORATION	DENBURY	2012	5.64	-0.07	Transportation by pipeline Extraction of oil, gas, carbon dioxide and helium; Laying, constructing, maintaining and operating pipelines; Well drilling for oil, gas, helium and carbon dioxide Explorati

Class 041: Education & Entertainment

Clean filings (1990–2021): 920,449
Mean KL_{pros} / KL_{retr} / ΔKL : 5.18 / 5.19 / -0.006 nats
5y survival rate: 47.0%
Share with SEC-listed owner: 3.06%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
blockchain	$6.41\times$	1,106	53%
cryptocurrency	$6.08\times$	957	50%
worlds	$3.63\times$	1,126	52%
wager	$3.33\times$	1,327	42%
accept	$3.26\times$	1,382	41%
tokens	$3.21\times$	2,330	43%
in-company	$3.17\times$	2,402	42%
currency	$3.05\times$	2,174	45%
keynote	$2.88\times$	3,167	39%
vlogs	$2.83\times$	2,230	32%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
instrumental	$3.76\times$	2,579	51%
pageants	$3.63\times$	2,304	57%
live-action	$2.75\times$	2,579	67%
batting	$2.66\times$	999	41%
skating	$2.63\times$	2,840	47%
band	$2.59\times$	21,828	49%
on-going	$2.53\times$	20,697	61%
highlight	$2.53\times$	924	54%
nightshirts	$2.53\times$	1,021	67%
hockey	$2.52\times$	5,857	49%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

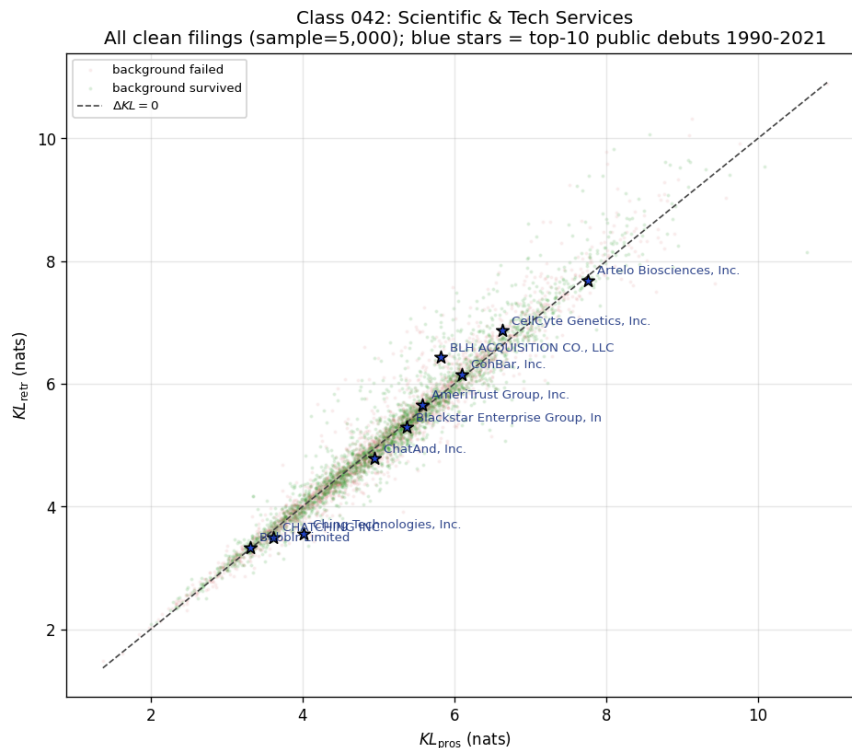
token	n in stratum	n in class	intra-class survival
field	60,199	419,101	42%
entertainment	49,261	407,659	50%
educational	44,270	309,765	42%
featuring	36,816	255,080	45%
information	36,171	230,417	45%
nature	36,114	267,431	45%
conducting	35,086	242,144	41%
training	30,092	202,842	40%
seminars	28,345	188,062	40%
education	26,981	176,329	42%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm		Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
ATK Inc.	Entertainment,	AALAM TUCK KHALED	2020	3.05	-0.08	Clothing, namely, shirts, t-shirts, pants, dresses, skirts, jeans, slacks, belts, hats, caps being headwear, gloves, ear muffs, headwear, neckwear, sweaters, fleece pullovers, jogging suits, warm-up s
ATLAS TECHNOLOGY GROUP, INC.		IGE	2006	5.41	+0.01	Entertainment services, namely, providing on-line information via the Internet about video games; providing on-line news and information via the Internet on video games; providing on-line databases fo
Artix LLC	Entertainment,	BATTLEON	2006	4.70	+0.09	Entertainment services, namely, providing an on-line computer game
BAT LLC	Entertainment	MISS INTERNATIONAL UNIVERSE	2018	7.44	-0.23	Entertainment in the nature of beauty pageants
BETTERLIFE PHARMA INC.		BETTER+	2019	4.81	+0.06	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
BLH ACQUISITION CO., LLC		ROCK THE TOT	2012	5.66	+0.08	Entertainment services, namely, arranging and conducting of eating competitions
BetterLife Pharma Inc.		LIVE BETTERX	2020	4.46	-0.10	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Bubblr Limited		C J CITIZENS JOURNALIST	2019	3.48	+0.05	Television and radio broadcasting services; audio and video broadcasting services, namely, broadcasting of news in the fields of current affairs, sports and topics of general interest over the interne
China Foundation, The		CHINA FOUNDATION	2008	5.31	-0.05	Arranging and conducting educational conferences; organizing conferences in the field of public health and medical science Charitable services, namely, organizing and conducting volunteer programs and
Clickstream Corporation		THINK QUIK, PLAY QUIK, WINQUIK	2020	3.94	-0.02	Entertainment services, namely, providing temporary use of non-downloadable interactive trivia games; conducting live entertainment in the nature of games featuring trivia questions and answers via th

Class 042: Scientific & Tech Services

Clean filings (1990–2021): 743,414
Mean KL_{pros} / KL_{retr} / ΔKL : 5.17 / 5.20 / -0.022 nats
5y survival rate: 46.1%
Share with SEC-listed owner: 5.61%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
non-fungible	8.30×	855	65%
blockchain-based	5.53×	1,341	57%
cryptocurrency	4.88×	3,183	54%
blockchain	4.42×	3,877	53%
decentralized	4.38×	1,252	55%
token	4.28×	1,178	51%
ledger	3.80×	1,398	52%
cryptographic	3.78×	1,424	53%
wallet	3.75×	1,618	51%
tokens	3.72×	3,652	48%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
carry-out	$7.92\times$	1,135	52%
sandwiches	$6.41\times$	1,234	52%
restaurant	$6.33\times$	18,456	55%
pizza	$6.31\times$	932	50%
motel	$6.10\times$	1,201	49%
bakery	$5.93\times$	1,397	46%
resort	$5.67\times$	2,116	56%
nightclub	$5.64\times$	948	66%
cafe	$5.38\times$	1,076	59%
lounge	$5.27\times$	1,595	54%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

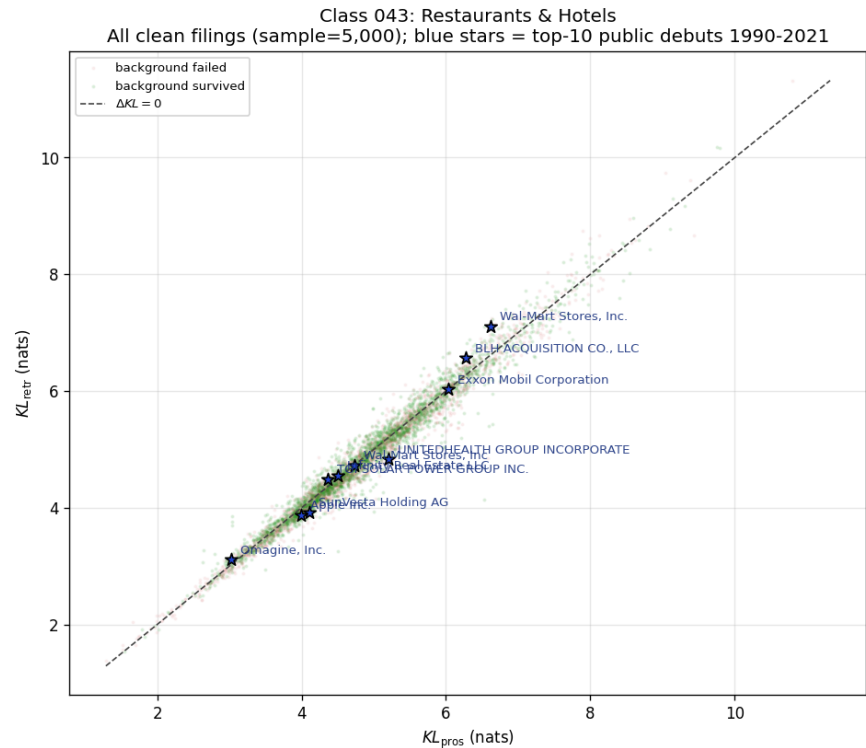
token	n in stratum	n in class	intra-class survival
software	46,113	355,301	43%
computer	41,094	345,425	46%
field	35,666	273,520	40%
information	31,871	244,099	45%
use	30,111	228,642	41%
design	27,704	202,311	38%
development	27,340	187,141	39%
featuring	26,369	209,628	41%
management	26,332	181,534	40%
others	26,085	208,844	42%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
AmeriTrust Group, Inc.	SIMPLE.FAST.DIRECT.	2010	5.58	-0.09	Insurance services, namely, administration, brokerage, and underwriting for workers' compensation; insurance claims administration and claims processing services; insurance agency services, namely, in
Artelo Biosciences, Inc.	ARTELO BIO-SCIENCES	2019	7.77	+0.08	Research and development of pharmaceuticals; namely, therapeutic agents that modulate the endocannabinoid system
BLH ACQUISITION CO., LLC	BAR LOUIE	1993	5.82	-0.62	tavern and restaurant services
Blackstar Enterprise Group, Inc.	INTERNET DIGITAL OFFERING	2021	5.37	+0.08	Downloadable financial services software for managing and hosting indications of interest for potential initial or secondary future offerings on an immutable block chain Financial services, namely, pr
Bubblr Limited	C J CITIZENS JOURNALIST	2019	3.31	-0.02	Television and radio broadcasting services; audio and video broadcasting services, namely, broadcasting of news in the fields of current affairs, sports and topics of general interest over the internet
CHATCHING INC.	USE IT, OWN IT	2011	3.62	+0.12	Internet based social networking and social introduction services; On-line social networking services; Online social networking services provided through a community website; Social introduction agency
CellCyte Genetics, Inc.	CELLCYTE	2005	6.63	-0.24	Research and development of therapeutic and diagnostic products for others related to a wide variety of diseases, disorders and conditions
ChatAnd, Inc.	CHAT&	2011	4.95	+0.17	Providing temporary use of online, non-downloadable computer software for web browsing and information sharing in the field of sales and conversion; providing temporary use of online, non-downloadable
Ching Technologies, Inc.	CHING WALLET	2018	4.01	+0.46	Software for use in electronically trading, storing, sending, receiving, accepting and transmitting digital currency, and managing digital currency payment and exchange transactions; services in the f
CohBar, Inc.	MITO+	2021	6.10	-0.05	Research and development of novel pharmaceutical products, namely, research and development of novel peptides for use as pharmaceuticals

Class 043: Restaurants & Hotels

Clean filings (1990–2021): 136,699
Mean KL_{pros} / KL_{retr} / ΔKL : 4.87 / 4.85 / +0.014 nats
5y survival rate: 48.4%
Share with SEC-listed owner: 2.80%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
truck	$3.11\times$	4,705	44%
vegan	$2.85\times$	652	45%
specified	$2.37\times$	156	76%
contest	$2.15\times$	149	81%
ccrcs	$2.14\times$	299	33%
proportioned	$2.06\times$	608	52%
plant-based	$2.02\times$	213	49%
properly	$2.02\times$	615	53%
brewpub	$1.99\times$	649	45%
facilitation	$1.98\times$	162	72%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
portions	$4.35\times$	170	78%
mixed-use	$3.88\times$	209	78%
puppet	$3.50\times$	177	59%
bistros	$3.16\times$	256	54%
contest	$3.15\times$	149	81%
drugs	$3.01\times$	219	69%
specified	$3.01\times$	156	76%
preferred	$2.93\times$	954	66%
facilitation	$2.78\times$	162	72%
carry-out	$2.75\times$	3,427	59%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

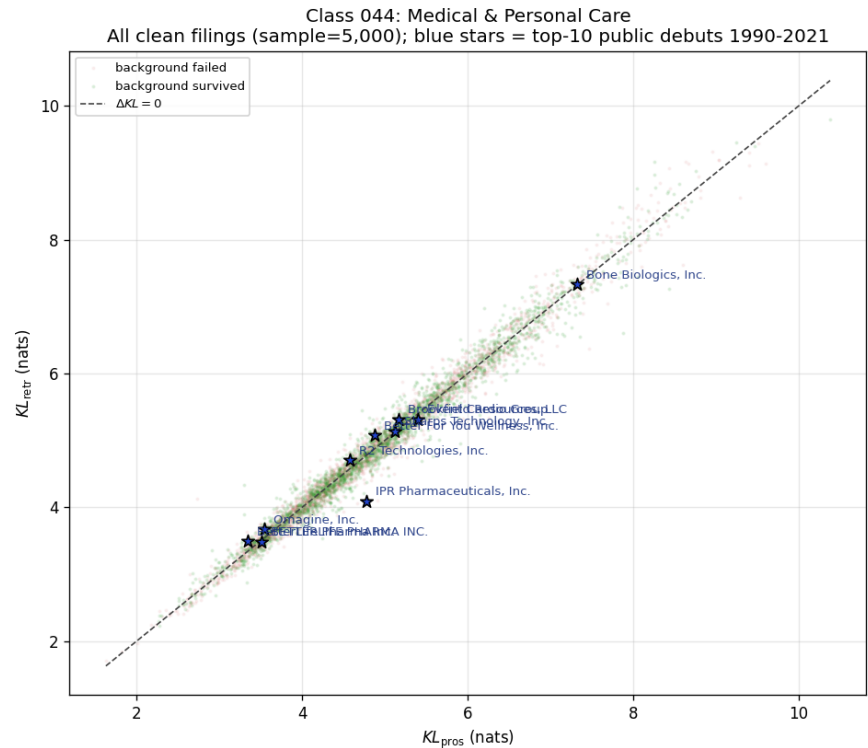
token	n in stratum	n in class	intra-class survival
restaurant	9,526	76,814	51%
food	7,062	55,402	46%
bar	5,865	41,470	51%
featuring	4,987	36,194	46%
catering	4,843	32,880	45%
beverages	3,598	23,790	45%
information	3,558	21,479	41%
temporary	3,541	23,817	40%
restaurants	3,511	25,081	49%
hotel	3,344	22,958	50%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
BLH ACQUISITION CO., LLC	DRINK POURN	2017	6.28	-0.29	Headwear, namely, caps and hats; Wearable garments and clothing, namely, shirts Bar and restaurant services
Infinity Real Estate LLC		2020	4.50	-0.05	Hotel services Real estate development Assessment and management of real estate; Commercial and residential real estate agency services; Financial planning and investment advisory services; Investment
Omagine, Inc.	OMAGINE	2008	3.03	-0.08	Travel agency services, namely, making reservations and booking for temporary lodging and dining; Hotel and resort hotel services; Resort lodging services; Temporary accommodation services; Restaurant
SunVesta Holding AG	SUNVESTA	2005	4.11	+0.19	Services for providing food and drink in cafes, cafeterias, restaurants and bars; hotel, motel, boarding house, restaurant, cafe, cafeteria and bar services; providing temporary guest accommodation; r
TGI SOLAR POWER GROUP INC.	ADVENT	2021	4.36	-0.13	Hotel and restaurant services; Providing hotel accommodation; Resort hotel services; Restaurant and bar services; Restaurant and café services; Restaurant and hotel services
Wal-Mart Stores, Inc	YUMPRINT	2017	4.74	+0.01	Providing a website featuring information about recipes, ingredients and cooking information; Providing an online computer database to consumers featuring information about recipes, ingredients and co
Wal-Mart Stores, Inc.	KID CONNECTION	2003	6.62	-0.49	restaurant services retail candy stores services
Exxon Mobil Corporation	BENGAL TRADERS GOURMET COFFEES	2004	6.04	+0.01	Coffee Preparation and serving of coffee for consumption on or off the premises
Apple Inc.		2012	3.99	+0.12	providing consumer product information for the purpose of selecting general consumer merchandise to meet the consumer's specifications; advisory and consultancy services relating to all the aforesaid
UNITEDHEALTH GROUP INCORPORATED	CARE24	2002	5.21	+0.37	Referral services in the field of financial information, employee assistance and work life programs, legal information, dependent care information, health care and mental health information Providing

Class 044: Medical & Personal Care

Clean filings (1990–2021): 188,794
Mean KL_{pros} / KL_{retr} / ΔKL : 4.96 / 4.95 / +0.005 nats
5y survival rate: 45.8%
Share with SEC-listed owner: 5.56%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
candidates	$3.21\times$	218	40%
inside	$3.20\times$	222	40%
hydration	$3.19\times$	700	42%
adding	$3.10\times$	203	36%
cell-based	$3.02\times$	212	53%
analyzed	$2.94\times$	204	33%
semiconductors	$2.86\times$	203	25%
intravenous	$2.69\times$	1,143	43%
abnormalities	$2.61\times$	280	60%
delta-	$2.59\times$	405	69%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
winery	$5.86\times$	341	64%
vineyard	$5.76\times$	481	59%
grapes	$4.99\times$	597	58%
condominiums	$4.90\times$	208	74%
stud	$4.36\times$	551	46%
lounges	$4.35\times$	209	66%
regimes	$4.15\times$	207	63%
percent	$3.96\times$	338	67%
shrub	$3.93\times$	252	40%
vacation	$3.84\times$	268	58%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

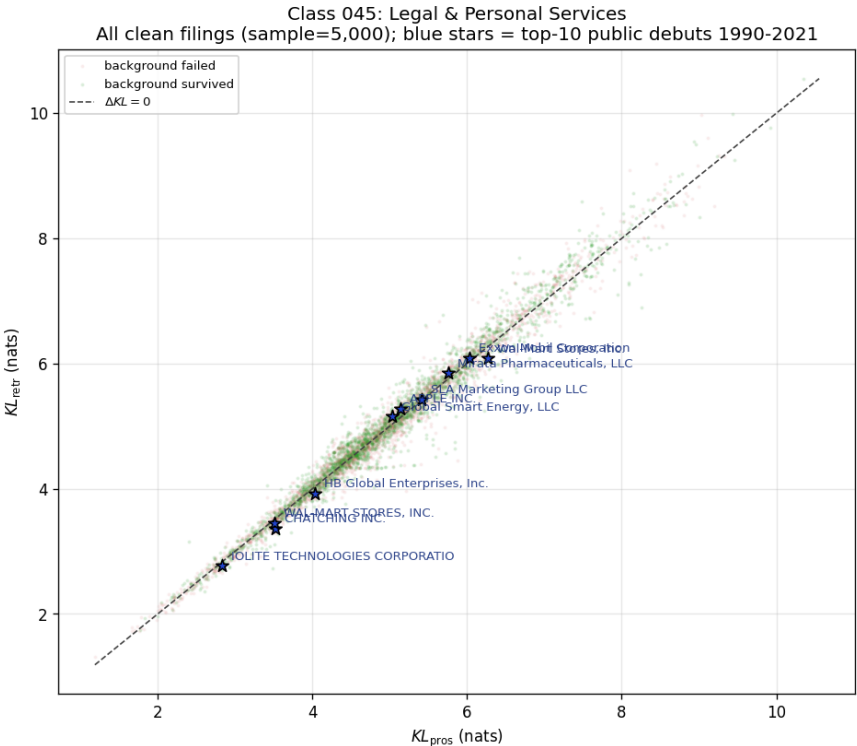
token	n in stratum	n in class	intra-class survival
health	10,877	79,196	45%
medical	10,847	79,310	44%
information	10,386	70,993	44%
care	10,206	71,534	45%
field	9,573	64,713	42%
treatment	6,025	40,540	42%
featuring	5,423	37,942	44%
wellness	4,779	33,138	45%
use	4,369	28,832	43%
purposes	4,284	28,697	40%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
BETTERLIFE PHARMA INC.	LOVE BET-TERX	2019	3.51	+0.02	Printed publications, namely, newsletters, brochures, reports and guides in the field of cannabis bath additives, namely, bath beads, bath bombs, bath crystals and bubble bath; bath herbs; bath oils;
Better For You Wellness, Inc.	BETTER FOR YOU WELLNESS	2021	4.88	-0.19	Protein bars, namely, protein supplements formed and packaged as bars Advisory and consultation services in the fields of agronomy, agriculture, and horticulture; agronomic consulting services; agrono
BetterLife Pharma Inc.	LIVE BET-TERX	2020	3.34	-0.15	Downloadable electronic publications, namely, newsletters, brochures, reports and guides in the field of cannabis Retail store and wholesale store services featuring industrial hemp food and body care
Bone Biologics, Inc.	BONE BIO-LOGICS	2006	7.32	-0.01	provider of orthobiologics, bone growth factors and scaffolds, bone graft substitutes, cartilage growth factors and scaffolds, spinal disc growth factors and scaffolds, articular cartilage growth factor
Brookfield Resources, LLC	BROOKFIELD RE-SOURCES, LLC BR	2004	5.17	-0.14	Consulting services in medical communications for the pharmaceutical and related industries
Event Cardio Group	EVENT CARDIO	2016	5.40	+0.09	Medical device, namely, a cardiac and cardiac event monitor having a sensor and capability to transmit cardiac and other physiological data of patients remotely Medical Services, namely, remote monito
IPR Pharmaceuticals, Inc.	RISING ABOVE THE STATIN QUO	2002	4.77	+0.69	Educational services, namely, conducting classes, conferences, workshops, seminars and symposia in the field of treatment of cardiovascular diseases, and cardiovascular health; and providing the above
Omage, Inc.	IMAGINATION. INSPIRATION. ILLU	2008	3.55	-0.13	Real estate services, namely, listing, leasing and financing of residential and commercial properties; Property management services, namely, management of residential and commercial properties Broadca
R2 Technologies, Inc.	CRYOAESTHETICS	2005	4.58	-0.13	Medical and cosmetic body care services using cryotherapy
Sharps Technology, Inc.	SHARPS PROVENSA	2018 219	5.12	-0.01	Online retail store and online retail services through direct solicitation by distributors directed to end-users featuring syringes and needles for medical use and for injection, disposable syringes a

Class 045: Legal & Personal Services

Clean filings (1990–2021): 126,420
Mean KL_{pros} / KL_{retr} / ΔKL : 4.96 / 4.99 / -0.022 nats
5y survival rate: 45.2%
Share with SEC-listed owner: 3.47%



Trending tokens (top-decile ΔKL) — with intra-class 5y survival rate.

token	lift	n filings	intra-class survival
authenticated	4.60×	126	55%
mediumship	4.58×	144	27%
blockchain	4.07×	381	52%
worlds	3.79×	153	44%
cryptocurrency	3.70×	305	49%
token	3.68×	174	45%
user-created	3.65×	178	56%
currencies	3.38×	207	41%
tarot	3.36×	393	34%
augmented	3.26×	368	45%

Ending tokens (bottom-decile ΔKL) — intra-class 5y survival rate.

token	lift	n filings	intra-class survival
localities	$6.04\times$	139	84%
retired	$5.42\times$	155	83%
practicing	$5.03\times$	163	71%
slogans	$4.70\times$	1,014	89%
coordinated	$4.21\times$	197	73%
cartoon	$4.06\times$	1,323	82%
occasion	$4.03\times$	513	50%
valuable	$3.87\times$	289	69%
evangelical	$3.86\times$	339	55%
acquaintances	$3.85\times$	187	50%

Flux-neutral most-frequent tokens ($|\Delta KL| \leq 0.02$).

token	n in stratum	n in class	intra-class survival
field	8,452	55,961	40%
information	8,089	54,797	42%
computer	4,972	33,473	44%
featuring	4,962	32,041	43%
personal	4,778	35,288	43%
management	4,663	26,810	38%
online	4,600	30,225	45%
others	4,562	29,050	41%
social	4,526	34,800	48%
nature	4,259	25,152	38%

Top-10 SEC-listed firms with debut in 1990–2021 (ranked by max observed revenue, debut filing g/s shown).

Firm	Mark	year	KL_{pros}	ΔKL	First-filing g/s excerpt
CHATCHING INC.	USE IT, OWN IT	2011	3.52	+0.16	Internet based social networking and social introduction services; On-line social networking services; Online social networking services provided through a community website; Social introduction agenc
Global Smart Energy, LLC	SMART GRID SPRING-BOARD SERVICE	2010	5.03	-0.13	Consulting services to utilities and electric power clients, namely, conducting energy audits of commercial and institutional facilities for the purpose of improving energy efficiency in the field of
HB Global Enterprises, Inc.	SOUL RECI- PROCITY	2020	4.03	+0.10	Retail store services featuring subscription boxes containing household items, personal care items, clothing, jewelry and gifts; Retail shops featuring gifts, clothing and jewelry Educational and ente
IOLITE TECHNOLOGIES CORPORATION	SOCIAL MOSAIC	2011	2.84	+0.07	Providing access to on-line computer databases and on-line searchable databases in the field of classifieds, commercial listings and advertisements of services and products of others, social networkin
Mirata Pharmaceuticals, LLC	MIRATA PHARMA- CEUTI- CALs	2020	5.76	-0.09	licensing of intellectual property pharmaceutical products, namely, gene therapies
SLA Marketing Group LLC	CB	2014	5.42	-0.01	Clothing, namely, scarves, dresses, tops, t-shirts, shirts, jackets, pants, hats Personal lifestyle consulting services; personal fashion consulting services Cosmetic brushes Various colored paper dia
WAL-MART STORES, INC.	JIG	2011	3.51	+0.06	computer software, computer software development tools, computer software application interfaces and application program interfaces (APIs), all for operating search engines and managing online service
Wal-Mart Stores, Inc.	WALMART TOGO	2011	6.27	+0.18	Personal shopping for others Retail and on-line grocery store services featuring home delivery service; Retail department store services
Exxon Mobil Corporation	EXXONMOBIL TRAVEL CLUB	2009	6.03	-0.05	Reimbursement of emergency travel expenses, of ambulance service expenses, and of unexpected transport expenses Automobile club services, namely, providing members with protection against auto theft b
APPLE INC.		2004	5.14	-0.14	(Based on Use in Commerce) providing a selection of on-line electronic greeting cards; [(Based on 44(e)) computer dating services; providing a selection